Polylogarithms and their Arithmetic Bridges

Volume 3: Hurwitz Jets, Stieltjes Calculus and Gamma Correlations

ProveIt Contributors

Collective manuscript, October 2026

About this volume

Spectral zeta jets unify primitives and parameter derivatives. Canonical finite parts, convolution and contact terms support Gamma correlations and twisted identities; positive kernels explain Stieltjes zero geometry.

The four volumes form one collective manuscript. They share definitions, notation, source evidence and a bibliography. Chapter, theorem and equation numbers are retained from the unified manuscript so references in research reports remain stable. Cross-volume references link to the companion PDF; keep the four PDF files together when downloading them.

Volume 1 Polylogarithm Identities and Arithmetic Reductions
Volume 2 Signed Kernels, Zero Geometry and Experimental Discovery
Volume 3 Hurwitz Jets, Stieltjes Calculus and Gamma Correlations
Volume 4 Gamma Grids, CM Periods and Herglotz Arithmetic

A rigorous analytic proof, a finite exact certificate and an independently repeated numerical diagnostic carry different evidence. Computation drives conjecture formation and exposes missing structure; successful numerical recognition does not itself prove an identity or arithmetic independence. The source inventory and editorial ledger record the accepted corrections and the scope of material still undergoing analytic reconciliation.

Contents

About this volume **i**

8 Zeta jets, Stieltjes antiderivatives and negative polygamma **1**

8.1 Below the digamma: $\psi^{(-2)}$ at rationals . . . 1
8.2 The general log-gamma integral and the fifth-point obstruction . . . 2
8.3 Binet's second formula and Malmsten's integrals: the Lambert-kernel face . . . 3
8.4 The log-gamma moment ladder . . . 3
8.5 Log-gamma moments: poles, late coefficients, and effective remainders . . . 4
8.6 Sharp truncation of the log-gamma moment expansion . . . 13
8.7 Reflected log-gamma moments and their late coefficients . . . 18
8.8 The Stieltjes layer . . . 25
8.9 The common Hurwitz-jet calculus . . . 26
8.10 Underived Stieltjes grids and character coordinates . . . 27
8.11 The ladder . . . 27
8.12 Zeta jets and the conversion dictionary . . . 28
8.13 Reflection formulas . . . 29
8.14 Moments . . . 30
8.15 Explicit values of Γ_n . . . 31
8.16 The negative-order polygamma floor . . . 34
8.17 Fourier conventions and the Hurwitz semigroup . . . 40
8.18 Canonical finite parts of the Stieltjes functions . . . 41
8.19 All-index convolution closure and exact normal forms . . . 42
8.20 Ordinary integral realizations and explicit Stieltjes identities . . . 44
8.21 Conventions and endpoint normalization . . . 45
8.22 The shifted Hurwitz correlation . . . 46
8.23 An all-order linear Stieltjes closure theorem . . . 48
8.24 Coincident endpoints and canonical collision subtraction . . . 51
8.25 Ordinary integrals of all antiderivative orders . . . 54
8.26 Explicit log-gamma and mixed identities . . . 56
8.27 Rational shifts and exact arithmetic coordinates . . . 58
8.28 All circular convolution powers of log Gamma . . . 59
8.29 Nonintegral twists: Lerch jets, contact terms, and the half-twist . . . 61
8.30 Unequal twists through beta mixtures and finite resolvent identities . . . 70
8.31 Gamma coefficients prove every harmonic-order identity . . . 72
8.32 Mixed Gamma derivatives and exact harmonic tail identities . . . 73
8.33 All centered trigamma-square moments and a polylogarithmic generator . . . 77
8.34 Digamma powers, harmonic spectral data and exact finite parts . . . 81

8.35 The universal contour polynomial . . . 83
8.36 A finite-part formula for every positive integer power . . . 84
8.37 A finite algorithm in strict harmonic words . . . 88
8.38 The fifth power with every finite correction . . . 90
8.39 Normalized primitives and polygamma identities . . . 96
8.40 Differential polynomials and the disappearance of subtractions . . . 97
8.41 Canonical integration by parts and Gamma parity . . . 100

9 The parameter-derivative tower **107**
9.1 The uniform master identity . . . 107
9.2 Differentiation and the contact-term correction . . . 108
9.3 A two-jet collapse for every polygamma convolution . . . 110
9.4 Contact terms in translated derivative products . . . 111
9.5 Dilation, distribution traces and exact logarithmic contacts . . . 112
9.6 The uniform rational-argument landscape . . . 114
9.7 Weighted distribution modules and exact rational-grid ranks . . . 114
9.8 Integral distribution, reflection and modular specialization . . . 121
9.9 Primitive-grid determinants, local defects and finite descent . . . 142
9.10 Explicit primitive-grid reduction coefficients . . . 144
9.11 All-orders Hurwitz–Stieltjes identities . . . 147
9.12 Finite conductor descent for all Stieltjes layers . . . 148
9.13 Cyclotomic polylogarithm traces . . . 150
9.14 Explicit tables for the second constant . . . 154
9.15 The derivative $\leftrightarrow$ negapolygamma duality . . . 155
9.16 One normalized functional equation for all derivative bridges . . . 156
9.17 Experimental independence tests . . . 160
9.18 Elementary grids and the sixth-point completion . . . 160
9.19 Fixed-coordinate periodic collision contacts . . . 161
9.20 The off-collision kernel and its second finite part . . . 163
9.21 The all-index collision-contact theorem . . . 164
9.22 Exact convergent integral identities from finite Fourier data . . . 167
9.23 A Hurwitz kernel for all positive-integer spectral jets . . . 168
9.24 Polylogarithm boundary values and invisible local terms . . . 170
9.25 Exact primitives and the log-gamma bridge . . . 171

10 Positive kernels and the geometry of Stieltjes zeros **173**
10.1 A unified Stieltjes–Lerch derivative family . . . 173
10.2 Real-rootedness and a global zero bound . . . 175
10.3 An all-order concentration expansion . . . 176
10.4 Exactly n eventual zeros and their locations . . . 178
10.5 A recurrence for the next zero coefficient . . . 181
10.6 The classical half-unit bracket . . . 181
10.7 Exact low-index zero counts . . . 182
10.8 High parameter derivatives of Stieltjes constants . . . 184
10.9 The first classical failure of saturation: the fifth index . . . 191
10.10 Exact sixth- and seventh-index saturation thresholds . . . 195

Chapter 8

Zeta jets, Stieltjes antiderivatives and negative polygamma

The gamma function is the first spectral derivative of Hurwitz zeta at zero. Further derivatives give a natural language for integrating Stieltjes functions. Integration in the Hurwitz parameter moves the spectral argument downward, creating the negative-order polygamma ladder. This chapter develops that common mechanism before tabulating its moments and rational values.

8.1 Below the digamma: $\psi^{(-2)}$ at rationals

One integration below the digamma sits $\psi^{(-2)}(z) = \int_0^z \log\Gamma(t)\,dt$. At denominators $3, 4, 6$, reflection and distribution reduce the integral to Clausen values, $\zeta'(-1)$ and logarithms. The odd character part is Clausen-explicit at general rational points, but the even part can retain Dirichlet derivatives. The three original examples below are correct; their earlier extrapolation to every denominator was too broad.

Result 8.1 (log-gamma integrals in the Clausen lattice).

$$\int_0^{1/4} \log\Gamma(t)\,dt = \frac{3}{32} + \frac{\log 2}{8} + \frac{\log\pi}{8} + \frac{G}{4\pi} - \frac{9}{8}\,\zeta'(-1),$$

$$\int_0^{1/3} \log\Gamma(t)\,dt = \frac{1}{9} + \frac{\log 2}{6} - \frac{\log 3}{72} + \frac{\log\pi}{6} + \frac{\mathrm{Cl}_2(2\pi/3)}{4\pi} - \frac{4}{3}\,\zeta'(-1),$$

$$\int_0^{1/6} \log\Gamma(t)\,dt = \frac{5}{72} + \frac{7\log 2}{72} + \frac{\log 3}{144} + \frac{\log\pi}{12} + \frac{\mathrm{Cl}_2(\pi/3)}{4\pi} - \frac{5}{6}\,\zeta'(-1).$$

The structural punchline: the quarter-point integral carries Catalan ($\mathrm{Cl}_2(\pi/2) = G$) and the third-point integral carries $\mathrm{Cl}_2(2\pi/3) = \frac{2}{3}\,\mathrm{Cl}_2(\pi/3)$, two-thirds of Gieseking's constant — the "level-3 Catalan" that had surfaced independently in the repository's Eisenstein-ladder work — now arriving from a completely different direction.

Search observation 8.2 (recorded on purpose). (i) The half-point probe $\int_0^{1/2} \log\Gamma$ initially found no relation — because $\log\Gamma(1/2) = \frac{1}{2}\log\pi$ made the PSLQ basis $\mathbb{Q}$-dependent, and PSLQ returned that degeneracy instead of the target. Remove known dependencies or search modulo their known relation space, and require a nonzero target coefficient. Unknown arithmetic independence is not a prerequisite for searching or proving a target identity. (ii) A $\psi^{(-3)}(1/4)$ probe via Cauchy's repeated-integration kernel produced only a large-coefficient candidate (heights $\sim 10^6$) with a weak

canary (10^{-46} at 100 dps) — *rejected* as a PSLQ false positive. The weight-3 level needs a properly prepared basis (presumably $\zeta'(-2)$-equivalents plus Barnes-G-layer logarithms).

8.2 The general log-gamma integral and the fifth-point obstruction

Proposition 8.3 (An exact denominator-five correction)**.** *Let χ_5 be the quadratic character modulo 5, with values $(1,-1,-1,1)$ on residues $1,2,3,4$. Then*

$$\int_0^{1/5} \log\Gamma(t)\,dt = \frac{2}{25} + \frac{\log(2\pi)}{10} + \frac{\mathrm{Cl}_2(2\pi/5)}{4\pi} - \frac{6}{5}\zeta'(-1) + \frac{1}{20}L'(-1,\chi_5) - \frac{29}{1200}\log 5. \tag{8.1}$$

More generally, for $a > 0$,

$$\int_0^a \log\Gamma(t)\,dt = \zeta'(-1,a) - \zeta'(-1) + \frac{a-a^2}{2} + \frac{a}{2}\log(2\pi). \tag{8.2}$$

Proof. Differentiate $\partial_a\zeta(s,a) = -s\zeta(s+1,a)$ with respect to s at $s=-1$. Lerch's identity and $\zeta(0,a) = \frac{1}{2} - a$ give

$$\partial_a\zeta'(-1,a) = \zeta'(0,a) - \zeta(0,a) = \log\Gamma(a) - \tfrac{1}{2}\log(2\pi) - \tfrac{1}{2} + a.$$

The shift identity for Hurwitz zeta shows that $\zeta'(-1,a) \to \zeta'(-1)$ as $a \downarrow 0$, since the additional term is $-a\log a$. Integration proves (8.2). These are classical Hurwitz-zeta identities [49]; the integrated theory is also developed by Espinosa and Moll [54].

Put $Z_r = \zeta'(-1,r/5)$ and $Z = \zeta'(-1)$. The Fourier reflection formula at this weight is

$$\zeta'(-1,a) - \zeta'(-1,1-a) = \frac{\mathrm{Cl}_2(2\pi a)}{2\pi}, \qquad 0 < a < 1. \tag{8.3}$$

For completeness, subtract the two Hurwitz Fourier expansions

$$\zeta(1-s,a) - \zeta(1-s,1-a) = \frac{4\Gamma(s)}{(2\pi)^s}\sin\frac{\pi s}{2}\sum_{m\ge 1}\frac{\sin(2\pi m a)}{m^s}$$

and differentiate at $s=2$. Absolute local convergence of the series and its logarithmic derivative justifies the differentiation; the zero of the sine factor leaves precisely (8.3).

Differentiating distribution and the character decomposition at $s=-1$ gives

$$Z_1 + Z_2 + Z_3 + Z_4 = -\frac{4}{5}Z - \frac{\log 5}{60},$$
$$Z_1 - Z_2 - Z_3 + Z_4 = \frac{1}{5}\big(L'(-1,\chi_5) + \log 5\, L(-1,\chi_5)\big).$$

Here $L(-1,\chi_5) = -2/5$, obtained directly from $\zeta(-1,a) = -B_2(a)/2$. Combining these equations with $Z_1 - Z_4 = \mathrm{Cl}_2(2\pi/5)/(2\pi)$ yields

$$Z_1 = -\frac{1}{5}Z + \frac{1}{20}L'(-1,\chi_5) - \frac{29}{1200}\log 5 + \frac{\mathrm{Cl}_2(2\pi/5)}{4\pi}.$$

Insert this into (8.2) at $a = 1/5$. □

The correction concerns the scope of the available reduction laws. It does not prove that $L'(-1,\chi_5)$ is transcendental or independent of the other constants in (8.1). A precise replacement for the earlier assertion is: the odd reflection component is Clausen-explicit, while the even component requires the appropriate Dirichlet-L derivatives unless further relations have been proved. The general Hurwitz-jet ladder below agrees with this corrected scope.

8.3 Binet's second formula and Malmsten's integrals: the Lambert-kernel face

Binet's second formula

$$\log\Gamma(z) = \Big(z - \tfrac{1}{2}\Big)\log z - z + \tfrac{1}{2}\log(2\pi) + 2\int_0^\infty \frac{\arctan(t/z)}{e^{2\pi t} - 1}\, dt$$

is the integrated Abel–Plana formula with the *Lambert kernel* $1/(e^{2\pi t} - 1)$ — the same kernel whose moment sums are the Eisenstein q-series of the companion CM article. Malmsten/Vardi log-log integrals $\int_0^1 \ln\ln(1/x)\, R(x)\, dx$ are the *Mellin avatars* of Kummer's Fourier series: expanding the rational kernel R in a geometric series produces $L'(\chi, 1)$-type sums, and Kummer's formula converts odd-character $L'(1)$ into $\log\Gamma$ at rationals.

Result 8.4. Binet's formula replays exactly at $z = 1, \frac{1}{2}, \frac{1}{4}$ [historical experiment: residuals $\le 10^{-52}$], turning every Γ-store atom into a closed-form arctan–Lambert integral; e.g.

$$\int_0^\infty \frac{\arctan(4t)}{e^{2\pi t} - 1}\, dt = \frac{1}{2}\Big[\log\Gamma\big(\tfrac{1}{4}\big) - \tfrac{1}{2}\log 2 + \tfrac{1}{4} - \tfrac{1}{2}\log(2\pi)\Big].$$

Result 8.5 (Vardi's integral [5], re-derived blind). PSLQ over $\{\pi\gamma,\ \pi\log 2,\ \pi\log\pi,\ \pi\log\Gamma(1/4)\}$, with the γ atom *declined* (coefficient 0) [historical experiment: canary exact at 80 dps]:

$$\int_0^1 \frac{\ln\ln(1/x)}{1 + x^2}\, dx = \frac{\pi}{4}\,\ln\Big(\frac{4\pi^3}{\Gamma(1/4)^4}\Big).$$

The $q = 3$ Malmsten variant [historical experiment: canary 3.4×10^{-81}; γ declined again]:

$$\int_0^1 \frac{\ln\ln(1/x)}{1 + x + x^2}\, dx = \frac{\pi}{6\sqrt{3}}\,\ln\Big(\frac{2^8\pi^8}{3^3\,\Gamma(1/3)^{12}}\Big).$$

Both integrals carry exactly the harvested-Γ-store atoms: the log-log family is a systematic *integral-representation generator* for the identity stores — every odd Dirichlet character (each denominator q) yields one such evaluation through Kummer's formula, with the Γ-lattice reducer supplying the closed form mechanically.

8.4 The log-gamma moment ladder

Result 8.6 (Hurwitz ladder; classical (Whittaker–Watson [8])). For $\operatorname{Re} z > 0$, use the holomorphic branch of $\log\Gamma(z)$ that is real on the positive axis, and the principal logarithm of z. Then

$$\log\Gamma(z) = \Big(z - \tfrac{1}{2}\Big)\log z - z + \tfrac{1}{2}\log(2\pi) + \frac{1}{2}\sum_{n\ge 2}\frac{n-1}{n(n+1)}\,\zeta(n, z+1).$$

The series is absolutely and locally uniformly convergent in this half-plane. On $\operatorname{Re} z \ge \sigma > 0$, its Hurwitz terms are bounded by $\zeta(n, 1+\sigma)$, a geometrically decaying majorant. The domain matters: at $z = -1/2$, $\zeta(n, z+1) = (2^n - 1)\zeta(n)$ and the series terms do not tend to zero. Since $\zeta(n, z+1)$ is $\psi^{(n-1)}(z+1)$ up to factorials, this is a $\mathbb{Q}$-linear ladder between $\log\Gamma$ and the *entire polygamma tower at the same point* — a candidate certificate law binding the Γ and trigamma stores into one linear system.

Result 8.7 (second moment; classical (Espinosa–Moll [7])). With $L = \log(2\pi)$,

$$\int_0^1 \big[\log\Gamma(x)\big]^2 dx = \frac{\gamma^2}{12} + \frac{\pi^2}{48} + \frac{\gamma L}{6} + \frac{L^2}{3} - (\gamma + L)\frac{\zeta'(2)}{\pi^2} + \frac{\zeta''(2)}{2\pi^2}.$$

The quadratic-moment layer adds $\zeta''(2)$ to the atom inventory.

The numerical cubic moment originally suggested enlarging the constant basket. Its known Tornheim representation and the convergent certificate below settle that representation question; reduction to a specified smaller algebra remains separate.

8.5 Log-gamma moments: poles, late coefficients, and effective remainders

Let

$$M_n = \int_0^1 \log^n \Gamma(x)\, dx, \qquad L = \log(2\pi), \qquad A = \gamma + L.$$

The cubic moment is a test of whether an experiment has selected the right coordinate system. Bailey, D. Borwein and J. M. Borwein evaluated it in Tornheim derivatives [65, Theorem 6]. A smaller single-zeta basket can miss that structure; failure of such a search gives no independence theorem. We give the explicit representation and its convergence proof.

Similarly, the full fixed-order expansion of $M_n/n!$ into the exponential scales k^{-n} is Theorem 8.1 of Amdeberhan et al. [64]. The results proved below add an explicit late-coefficient asymptotic, identify the divergent character of that expansion, and give a uniform remainder bound when the number of retained terms grows linearly with n. These are refinements of a known expansion, not a claim to have discovered that expansion or the cubic Tornheim representation.

8.5.1 A compact form of the cubic identity

Define the holomorphic germ near $(1,1,1)$

$$T(r,s,t) = \sum_{m,n\geq 1} \frac{1}{m^r n^s (m+n)^t}.$$

A subscript j denotes partial differentiation in its jth argument; all unmarked Tornheim terms in the next formula are evaluated at $(1,1,1)$. The double sums and all the derivatives used here converge absolutely.

Theorem 8.8 (Cubic moment in one Tornheim differential operator)**.** *With $A = \gamma + L$,*

$$\begin{aligned} M_3 = {} & \frac{L^3}{8} + \frac{\pi^2 L}{32} + \frac{3\zeta(3)}{16} + \frac{3L}{4\pi^2}\big(A^2\zeta(2) - 2A\zeta'(2) + \zeta''(2)\big) \\ & + \frac{3}{8\pi^2}\left(A^2 T - 2AT_3 + 2T_{13} - T_{12}\right). \end{aligned} \tag{8.4}$$

This is an equivalent form of the known cubic evaluation in [65, Theorem 6].

Proof. Write the Kummer Fourier series as

$$\log\Gamma(x) = \frac{L}{2} + \frac{1}{2}C(x) + \frac{1}{\pi}S(x), \quad C(x) = \sum_{j\geq 1}\frac{\cos(2\pi jx)}{j}, \quad S(x) = \sum_{j\geq 1}\frac{A + \log j}{j}\sin(2\pi jx).$$

Reflection $x \mapsto 1-x$ makes C even and S odd. Parseval gives

$$\int_0^1 C^2 = \frac{1}{2}\zeta(2), \qquad \int_0^1 S^2 = \frac{1}{2}\big(A^2\zeta(2) - 2A\zeta'(2) + \zeta''(2)\big).$$

The cosine resonance gives $\int C^3 = 3\zeta(3)/2$. Equivalently, one can differentiate the beta integral for $\int_0^1 (2\sin \pi x)^u dx$ three times at zero and use $C = -\log(2\sin \pi x)$. For the only remaining cubic term, the elementary product rule for two sines and one cosine gives

$$\begin{aligned}\int_0^1 CS^2 &= \frac{1}{2}\sum_{m,n\ge 1} \frac{(A+\log m)(A+\log(m+n))}{mn(m+n)} \\ &\quad - \frac{1}{4}\sum_{m,n\ge 1} \frac{(A+\log m)(A+\log n)}{mn(m+n)} \\ &= \frac{1}{4}\left(A^2T - 2AT_3 + 2T_{13} - T_{12}\right).\end{aligned}$$

Expansion of the cube now proves the formula.

For the convergence justification, first put an Abel factor r^j in every Fourier series, $0 < r < 1$. All products can then be integrated termwise. The functions C and S belong to every finite $L^p(0,1)$, since the identities with $\log\Gamma(x)$ and $\log\Gamma(1-x)$ bound them by endpoint logarithms. Poisson convolution therefore converges to them in L^3, so the integrated cubic products converge. In the resulting resonance sums the logarithmic numerator is bounded by a constant times $(1+\log(m+n))^2$. Summing first over $m+n=q$ uses

$$\sum_{m=1}^{q-1} \frac{1}{m(q-m)q} = \frac{2H_{q-1}}{q^2},$$

which proves absolute convergence and legitimizes the Abel limit. □

The double-coordinate derivative in (8.4) can also be written as a positive sum. Put

$$\mathcal{R}_A = \sum_{m,n\ge 1} \frac{(A+\log(m+n))^2 - \log((m+n)/m)\log((m+n)/n)}{mn(m+n)}.$$

Symmetry in m,n proves $\mathcal{R}_A = A^2T - 2AT_3 + 2T_{13} - T_{12}$. The numerator is positive: after expansion it is $A^2 + 2A\log(m+n) + (\log m + \log n)\log(m+n) - \log m \log n$. It is bounded above by $(A+\log(m+n))^2$. Consequently, if $\mathcal{R}_{A,N}$ restricts $m+n \le N$ and $N \ge 3$ is an integer, then

$$0 < \mathcal{R}_A - \mathcal{R}_{A,N} \le \frac{2}{N}\big(P(\log N) + P'(\log N) + P''(\log N) + P'''(\log N)\big), \quad P(u) = (1+u)(A+u)^2. \tag{8.5}$$

Indeed $H_{q-1} \le 1 + \log q$, and $(1+\log x)(A+\log x)^2/x^2$ decreases for $x \ge 3$. Integration of the resulting tail gives (8.5). This bound is elementary and rigorous but converges only as $O(\log^3 N/N)$; it is useful as an independent sign and normalization check, not as the preferred high-precision algorithm.

8.5.2 A positive summand form and an explicit tail bound

The combination in (8.4) can be computed without cancellation between separate Tornheim constants. For $k = m + n$, define

$$R_A(m,n) = A^2 + 2A\log k + \log m \log k + \log n \log k - \log m \log n.$$

Since $A > 0$, $\log k \geq \log m, \log n \geq 0$, we have $R_A(m,n) > 0$. Writing $p = \log m, q = \log n, r = \log k$, one also has $r(p+q) - pq \leq r^2$: for fixed r, the expression is increasing separately in $p, q \in [0, r]$. Consequently

$$0 < R_A(m,n) \leq (A + \log k)^2.$$

Therefore the diagonal partial sums

$$C_N = \frac{3}{32} \sum_{m+n \leq N} \frac{1}{mn(m+n)} + \frac{3}{8\pi^2} \sum_{m+n \leq N} \frac{R_A(m,n)}{mn(m+n)}$$

increase to the centered cubic moment $M_3 - a^3 - 3aV$. Moreover,

$$0 < (M_3 - a^3 - 3aV) - C_N \leq \sum_{k>N} \frac{2(1+\log k)}{k^2} \left(\frac{3}{32} + \frac{3(A+\log k)^2}{8\pi^2} \right).$$

This gives a transparent computable enclosure, although its convergence is only $O(\log^3 N/N)$. An integral bound may be used once the explicit summand is decreasing; direct summation can handle the finite initial range. This positivity observation is an immediate corollary of the Fourier calculation, not a claim of a new classical-constant evaluation.

8.5.3 Exponentially convergent numerical reproduction

For $t > 0$, put

$$C(t) = \mathrm{Li}_1(e^{-t}) = -\log(1 - e^{-t}), \qquad F(t) = \partial_s \mathrm{Li}_s(e^{-t})\big|_{s=1}.$$

The usual Mellin representation of the Tornheim sum gives

$$T_{12} = \int_0^\infty F(t)^2\, dt, \quad T_{13} = \int_0^\infty F(t)C(t)(\log t + \gamma)\, dt, \quad T_3 = \int_0^\infty C(t)^2(\log t + \gamma)\, dt. \tag{8.6}$$

For $0 < t < 2\pi$, cancellation of the two poles in Jonquière's expansion produces

$$C(t) = -\log t + \sum_{k \geq 1} c_k t^k, \qquad c_k = \frac{(-1)^k \zeta(1-k)}{k!},$$

$$F(t) = -\alpha - \gamma \log t - \tfrac{1}{2}\log^2 t + \sum_{k \geq 1} b_k t^k, \qquad b_k = \frac{(-1)^k \zeta'(1-k)}{k!},$$

$$\alpha = \gamma_1 + \frac{\gamma^2}{2} + \frac{\pi^2}{12}.$$

Here the Stieltjes convention is $\zeta(1+z) = z^{-1} + \sum_{j \geq 0} (-1)^j \gamma_j z^j / j!$. For computational stability the coefficient formulas can be rewritten as

$$c_1 = \tfrac{1}{2}, \qquad b_1 = \tfrac{1}{2}\log(2\pi),$$

$$c_{2j} = \frac{(-1)^j \zeta(2j)}{j(2\pi)^{2j}}, \qquad b_{2j} = c_{2j}\left(\log(2\pi) - \psi(2j) - \frac{\zeta'(2j)}{\zeta(2j)} \right),$$

$$c_{2j+1} = 0 \quad (j \geq 1), \qquad b_{2j+1} = \frac{(-1)^{j+1}\pi\zeta(2j+1)}{(2j+1)(2\pi)^{2j+1}} \quad (j \geq 1).$$

The functional equation of the Riemann zeta function proves these identities. Elementary bounds $\zeta(k) < 2$, $|\zeta'(k)| < 2$, $H_{k-1} \le k-1$, $\gamma < 1$, $\log(2\pi) < 2$, and $2\pi > 6$ give, for $k \ge 2$,

$$|b_k| \le 12\,6^{-k}, \qquad |c_k| \le 2\,6^{-k}.$$

Thus after degree $K \ge 1$ their remainders on $0 < t \le 1$ are at most

$$\epsilon_b = \frac{12}{5}6^{-K}, \qquad \epsilon_c = \frac{2}{5}6^{-K}.$$

All integrals of the resulting polynomials in t and $\log t$ are evaluated using $\int_0^1 t^k \log^j t\,dt = (-1)^j j!/(k+1)^{j+1}$.

On $t \ge 1$, insert $C(t) = \sum_{n\ge1} e^{-nt}/n$ and $F(t) = -\sum_{n\ge1}(\log n)e^{-nt}/n$. The integral with the logarithmic factor uses

$$\int_1^\infty e^{-kt}(\log t + \gamma)\,dt = \frac{E_1(k) + \gamma e^{-k}}{k}.$$

Truncate this sum at $m + n \le N$. A common upper bound for the three omitted tails is

$$B_N = 4\sum_{k>N} ke^{-k} = \frac{4e^{-(N+1)}((N+1) - Ne^{-1})}{(1-e^{-1})^2}.$$

For the three integrals in the order T_{12}, T_{13}, T_3, safe total series-truncation bounds are respectively

$$\epsilon_b(12+\epsilon_b) + B_N, \qquad 5\epsilon_b + 15\epsilon_c + 2\epsilon_b\epsilon_c + B_N, \qquad 10\epsilon_c + 2\epsilon_c^2 + B_N.$$

For example the elementary estimates $C(t) \le -\log t + 1$ and $|F(t)| \le 4 - \log t + \frac{1}{2}\log^2 t$ on $(0,1]$ imply all the constants above. The sharper bound $\int_0^1 |F(t)|dt < 6$ follows directly from its coefficient expansion (one may use $0 < \alpha < 2$). An elementary verification of the latter constant bound is available: the defining limit for γ_1 and the total variation of $x \mapsto \log x/x$ on $[1,\infty)$ give $|\gamma_1| \le 2/e < 3/4$; combine this with $0 < \gamma < 3/5$ and $3 < \pi < \sqrt{12}$. This is an analytic truncation certificate; floating-point rounding must be handled separately to turn an implementation into interval arithmetic.

The independent implementation `verify_tornheim.py` evaluates the three partials and compares (8.4) against direct log-gamma quadrature. Its output explicitly labels the rounding qualification.

8.5.4 A sign check for an auxiliary identity

If the June 2012 preprint is used, an auxiliary sign should be checked before copying its Remark 3. Put

$$S_\gamma = \sum_{m,n\ge1} \frac{(\gamma + \log m)(\gamma + \log n)}{mn(m+n)}.$$

Directly expanding the numerator, with absolutely convergent sums, gives

$$T_{12} = S_\gamma + 2\gamma\big(\zeta'(2,1) + \zeta'(3)\big) - 2\gamma^2\zeta(3).$$

The coefficient of S_γ is positive. The displayed minus sign in equations (63) and (65) of the author-hosted June 2012 preprint would make their right side negative, although $T_{12} > 0$. This issue does not affect the cubic formula proved above. This note does not assert that the same typographical error occurs in the final journal typesetting.

8.5.5 Meromorphic moment generating function

For $\Re z < 1$, let

$$F(z) = \int_0^1 \Gamma(x)^z \, dx = \sum_{n \geq 0} \frac{M_n}{n!} z^n \quad (|z| < 1).$$

The power uses the real logarithm of $\Gamma(x) > 0$. Introduce the positive numbers

$$c_k = \frac{1}{k}[u^{k-1}]\Gamma(1-u)^k, \qquad k \geq 1. \tag{8.7}$$

Their first values are

$$c_1 = 1, \quad c_2 = \gamma, \quad c_3 = \frac{3\gamma^2 + \zeta(2)}{2}, \quad c_4 = \frac{8\gamma^3 + 6\gamma\zeta(2) + \zeta(3)}{3}.$$

These agree with the coefficients in [64, Theorem 8.1].

Proposition 8.9 (All moment poles and their residues)**.** *The function F extends meromorphically to the whole z-plane. Its only poles are simple poles at every positive integer, with*

$$\operatorname*{Res}_{z=k} F(z) = (-1)^k k c_k \neq 0.$$

Proof. Fix $0 < a < 1$ and expand

$$\Gamma(x)^z = x^{-z}\Gamma(1+x)^z = x^{-z} \sum_{j=0}^{J-1} b_j(z) x^j + x^{J-z} E_J(x, z).$$

The b_j are entire functions (in fact polynomials) of z, and E_J is uniformly bounded for $0 \leq x \leq a$ and z in compact subsets of the plane. Integration from zero to a gives

$$\sum_{j=0}^{J-1} \frac{b_j(z) a^{j+1-z}}{j+1-z} + \int_0^a x^{J-z} E_J(x, z) \, dx.$$

The second term is holomorphic for $\Re z < J + 1$; the integral from a to one is entire. Since J is arbitrary, these formulas give the claimed continuation. At $z = k$ the residue is $-b_{k-1}(k) = (-1)^k [u^{k-1}]\Gamma(1-u)^k$. Finally

$$\log \Gamma(1-u) = \gamma u + \sum_{j \geq 2} \frac{\zeta(j)}{j} u^j$$

has strictly positive coefficients, so (8.7) is positive. □

The pole at k contributes $(-1)^{k-1} c_k / k^n$ to the Taylor coefficient of F at zero. This explains the exponential scales in the known moment asymptotics, but an infinite partial-fraction sum cannot be inferred without controlling the growth of c_k. That growth is decisive.

8.5.6 Late coefficients and the inverse reciprocal gamma function

Set $\phi(u) = \Gamma(1-u)$ and let τ be the unique point in $(0,1)$ satisfying

$$-\tau\psi(1-\tau) = 1. \tag{8.8}$$

Equivalently $\psi(-\tau) = 0$. Define

$$\rho = \frac{\tau}{\Gamma(1-\tau)}, \qquad \alpha = -\log\rho, \qquad v = 1 + \tau^2\psi'(1-\tau), \qquad C = \frac{\tau}{\sqrt{2\pi v}}. \tag{8.9}$$

Numerically,

$$\begin{aligned}
\tau &= 0.50408300826445540925826930453330249895538518236858\ldots,\\
\rho &= 0.28211580900629804215379660062046372823230190757940\ldots,\\
\alpha &= 1.26543762211086561338203591116547399218544556554213\ldots,\\
v &= 2.27159994608893009911993485126846348661898954658806\ldots,\\
C &= 0.13342776130944528489602057053941782432703493711204\ldots.
\end{aligned}$$

These displayed approximations are numerical evaluations; the theorem uses the exact definitions (8.8)–(8.9).

Theorem 8.10 (Late-coefficient asymptotic)**.** *The coefficients* (8.7) *satisfy*

$$c_k = C\rho^{-k}k^{-3/2}\bigl(1 + O(k^{-1})\bigr). \tag{8.10}$$

In fact they have a full expansion in inverse powers of k. The power series $h(w) = \sum_{k\geq 1} c_k w^k$ has radius of convergence ρ, converges absolutely on $|w| \leq \rho$, and satisfies

$$h(w) = w\Gamma(1 - h(w)), \qquad h(\rho) = \tau.$$

For every fixed real n, the formal moment series $\sum_{k\geq 1}(-1)^{k-1}c_k/k^n$ diverges, since its terms do not tend to zero.

Proof. Let $D = u\,d/du$. The function

$$\mu(u) = D\log\phi(u) = -u\psi(1-u) = \gamma u + \sum_{j\geq 2}\zeta(j)u^j$$

is strictly increasing from zero to infinity on $(0,1)$. Hence (8.8) has exactly one solution. Moreover $\phi(u) > 1$ there, so $0 < \rho < 1$ and $\alpha > 0$.

Write $\phi(u) = \sum_{j\geq 0} a_j u^j$ and introduce an integer-valued random variable X by

$$\Pr(X = j) = \frac{a_j\tau^j}{\phi(\tau)}.$$

All its probabilities are positive, in particular those at 0 and 1; it has lattice span one. Analyticity of ϕ on $|u| < 1$ gives an exponential moment. Its mean is $\mu(\tau) = 1$ and its variance is $D^2\log\phi(\tau) = 1 + \tau^2\psi'(1-\tau) = v > 0$. For independent copies and $S_k = X_1 + \cdots + X_k$, coefficient extraction gives

$$c_k = \frac{\tau\rho^{-k}}{k}\Pr(S_k = k-1). \tag{8.11}$$

Fourier inversion gives

$$\Pr(S_k = k-1) = \frac{1}{2\pi}\int_{-\pi}^{\pi}\left(\frac{\phi(\tau e^{it})}{\phi(\tau)}e^{-it}\right)^k e^{it}\,dt.$$

For $t \neq 0$ modulo 2π, the modulus of the bracket is strictly less than one, because the distribution has positive mass at 0 and 1. On a small interval around zero its logarithm is $-vt^2/2 + O(t^3)$. Split off the complement of that interval, whose contribution is exponentially small, and put $t = y/\sqrt{k}$ in the remaining integral. Taylor expansion, dominated by a Gaussian on a shrinking central interval, gives

$$\Pr(S_k = k-1) = \frac{1}{\sqrt{2\pi v k}}\bigl(1 + O(k^{-1})\bigr).$$

For a fully quantitative split, use $|t| \le k^{-2/5}$, the band $k^{-2/5} < |t| \le \delta$, and the fixed outer band. On the middle band the modulus is at most $e^{-ck^{1/5}}$; on the outer band it is $O(q^k)$ for some $q < 1$. The nominal $k^{-1/2}$ correction on the central band is odd and integrates to zero. Taylor expansion to arbitrary even order with its analytic remainder gives the full inverse-power expansion. This elementary saddle argument proves (8.10) and avoids any assumption about unexamined complex critical values of the reciprocal gamma function.

Lagrange inversion gives the stated functional equation for h near zero. Equation (8.10) proves that its radius is ρ and that it converges absolutely at the boundary. On $[0, \tau]$, the function $f(u) = u/\phi(u)$ increases from zero to ρ, with positive derivative before τ, since $f'(u) = (1 - \mu(u))/\phi(u)$. Its inverse is the positive real branch of h, hence continuity at the boundary gives $h(\rho) = \tau$. Finally (8.10) proves the stated divergence. □

For completeness, the first correction in (8.10) is

$$c_k = C\rho^{-k}k^{-3/2}\left(1 + \frac{\beta_1}{k} + O(k^{-2})\right),$$

where, with $\kappa_j = D^j \log\phi(\tau)$,

$$\begin{aligned}\beta_1 &= -\frac{1}{2v} + \frac{\kappa_3}{2v^2} + \frac{\kappa_4}{8v^2} - \frac{5\kappa_3^2}{24v^3}\\ &= 0.30196282878120524112458668271647811378573305248468\ldots.\end{aligned}$$

This follows by retaining the quartic and squared cubic terms, together with the factor e^{it}, in the Fourier integral.

8.5.7 An effective remainder with a growing number of terms

Theorem 8.11 (Explicit truncation bound)**.** *For integers $n \ge 1$ and $K \ge 1$ with $K\alpha < n$, put*

$$R_{n,K} = \frac{M_n}{n!} - \sum_{k=1}^{K}\frac{(-1)^{k-1}c_k}{k^n}.$$

Then, writing $\Gamma(n,z) = \int_z^\infty t^{n-1}e^{-t}\,dt$,

$$|R_{n,K}| \le \frac{\alpha^n}{n!}\left(1 + \frac{n\tau}{n - K\alpha}\right) + \frac{\tau\rho^{-(K+1)}}{(K+1)^n}\frac{\Gamma(n,(K+1)\alpha)}{\Gamma(n)}. \tag{8.12}$$

In particular, whenever $n \geq \max\{2, 4\alpha^2\}$, the choice $K = \lfloor n/\alpha - \sqrt{n} \rfloor$ is positive and gives

$$|R_{n,K}| \leq \left(\frac{1}{\sqrt{2\pi n}} + \frac{\tau}{\alpha\sqrt{2\pi}} + \tau e^{2\alpha^2} \right) \left(\frac{e\alpha}{n} \right)^n. \tag{8.13}$$

In particular every integer $n \geq 7$ meets the displayed hypothesis. Indeed $\rho \geq 1/(2\sqrt{\pi})$ by evaluating $u/\Gamma(1-u)$ at $u = 1/2$, and therefore $\alpha \leq \log(2\sqrt{\pi}) < 13/10$, so $4\alpha^2 < 169/25 < 7$.

Proof. The function $\log \Gamma(x)$ decreases strictly from infinity to zero on $(0, 1]$, because $\psi'(x) > 0$ and $\psi(1) = -\gamma < 0$. Let $x(y) \in (0, 1]$ be its inverse. The layer-cake formula, or integration by parts after this monotone change of variables, gives

$$\frac{M_n}{n!} = \frac{1}{\Gamma(n)} \int_0^\infty y^{n-1} x(y)\, dy. \tag{8.14}$$

For $y \geq \alpha$, Lagrange inversion and continuity on the convergence circle give

$$x(y) = -h(-e^{-y}) = \sum_{k \geq 1} (-1)^{k-1} c_k e^{-ky}.$$

The identity initially holds for sufficiently large y and extends along the negative real inverse branch. This branch cannot fail before $w = -\rho$: for $0 < x \leq 1$, the derivative of $1/\Gamma(x)$ is $-\psi(x)/\Gamma(x) > 0$.

For $y \geq \alpha$, positivity and $\sum c_k \rho^k = \tau$ give

$$\left| x(y) - \sum_{k=1}^{K} (-1)^{k-1} c_k e^{-ky} \right| \leq \tau \rho^{-(K+1)} e^{-(K+1)y}.$$

Integrating this on $[\alpha, \infty)$ produces the second term of (8.12). For the interval $[0, \alpha]$ use $0 < x(y) \leq 1$ and, since $n > k\alpha$,

$$\int_0^\alpha y^{n-1} e^{-ky}\, dy \leq \frac{\alpha^n e^{-k\alpha}}{n - k\alpha}.$$

To see this inequality, differentiate $y^n e^{-ky}$ and use $n - ky \geq n - k\alpha$. Sum over $1 \leq k \leq K$ and bound $\sum c_k e^{-k\alpha} \leq \tau$ to obtain the first term.

For the stated choice of K, $n - K\alpha \geq \alpha\sqrt{n}$. Let $\theta = (K+1)\alpha/n$. Then $1 - \alpha/\sqrt{n} < \theta \leq 1$ and the hypothesis ensures $\theta \geq 1/2$. Since

$$\theta - 1 - \log\theta \leq 2(1-\theta)^2 \qquad (1/2 \leq \theta \leq 1),$$

we have

$$\rho^{-(K+1)} (K+1)^{-n} = \left(\frac{e\alpha}{n} \right)^n e^{n(\theta - 1 - \log\theta)} \leq e^{2\alpha^2} \left(\frac{e\alpha}{n} \right)^n.$$

Finally use $\Gamma(n, z)/\Gamma(n) \leq 1$ and the lower Stirling bound $n! \geq \sqrt{2\pi n}(n/e)^n$. □

For each fixed K, retaining one additional term in (8.12) recovers the usual asymptotic interpretation. The stronger content of (8.13) is that K may grow with n. The bound has the scale $(e\alpha/n)^n$. The late-coefficient theorem puts the least term near $(n + 3/2)/\alpha$ and gives its magnitude

$$C \left(\frac{e\alpha}{n + 3/2} \right)^{n+3/2} (1 + O(n^{-1})).$$

The rigorously proved remainder bound is within a polynomial factor of this scale. A least-term estimate alone would not prove a remainder bound; the inverse-function integral is what supplies that missing control.

The following values are numerical diagnostics from an independent integral in the variable $t = -\log x$, computed at 180 decimal digits. The rightmost column evaluates the explicit bound in (8.13); the inequality itself is proved above.

n	K	Observed $\lvert R_{n,K}\rvert$	Explicit upper bound
8	3	$3.1162 \cdot 10^{-5}$	$1.4837 \cdot 10^{-2}$
20	11	$2.1101 \cdot 10^{-18}$	$6.4881 \cdot 10^{-15}$
40	25	$2.8961 \cdot 10^{-46}$	$3.0208 \cdot 10^{-42}$
80	54	$1.8102 \cdot 10^{-113}$	$5.9717 \cdot 10^{-109}$

The displayed bound decimals have been rounded upward. They serve as a scale comparison; exact theorem constants are defined by (8.9). Full numerical data are in `moment_diagnostics.json`.

8.5.8 A 112-digit rational certificate for the cubic moment

The file `certify_cubic.py` implements a separate high-precision calculation using integer outward intervals. No floating-point special function is trusted. Put $a = 1/2$, $\ell = \log 2$ and

$$b_1 = \gamma, \qquad b_j = \frac{\zeta(j)}{j} \ (j \geq 2), \qquad B_M(x) = \sum_{j=1}^{M} b_j x^j.$$

On $0 < x \leq a$ we approximate $\log\Gamma(x)$ by $-\log x + B_M(-x)$; on the upper half of the interval, put $t = 1 - x$ and approximate $\log\Gamma(1-t)$ by $B_M(t)$. Both errors are at most

$$Dx^{M+1}, \qquad D = \frac{2\zeta(2)}{M+1},$$

with the corresponding variable x or t. This follows from the Taylor series and $\zeta(j) \leq \zeta(2)$ for $j \geq 2$. Define the exact elementary integrals

$$J_{j,p} = \int_0^{1/2} x^j (-\log x)^p dx = \frac{2^{-j-1} p!}{(j+1)^{p+1}} \sum_{r=0}^{p} \frac{((j+1)\ell)^r}{r!}.$$

If b_j^{*r} are the coefficients of B_M^r, the integrated polynomial approximation is

$$V_M = J_{0,3} + 3\sum_{j=1}^{M} (-1)^j b_j J_{j,2} + 3\sum_{j=2}^{2M} (-1)^j b_j^{*2} J_{j,1} + 2 \sum_{\substack{j \leq 3M \\ j \text{ even}}} b_j^{*3} J_{j,0}.$$

The last sum starts at $j = 4$. Since $0 \leq \log\Gamma(x) \leq -\log x$ on the lower half and $0 \leq \log\Gamma(1-t) \leq 1$ on the upper half,

$$\begin{aligned} |M_3 - V_M| \leq E_M := {} & 3D(J_{M+1,2} + J_{M+1,0}) \\ & + 3D^2(J_{2M+2,1} + J_{2M+2,0}) + 2D^3 J_{3M+3,0}. \end{aligned} \tag{8.15}$$

Here $\Gamma(1+x) \leq 1$ follows from log-convexity between 1 and 2; the upper-half bound follows from $\gamma < 1$, $\zeta(j) < 2$, and $\sum_{j \geq 2} (1/2)^j / j \leq 1/4$.

For reproducibility, the constant intervals come from the positive atanh series for $\log 2$, the Euler–Maclaurin formula for $\zeta(s)$ at $N = 128$, and the harmonic-number formula for γ. With $m = 96$, their respective analytic error bounds are

$$\frac{|B_{2m}|}{(2m)!}(s)_{2m-1}N^{-s-2m+1}, \qquad \frac{|B_{2m}|}{2mN^{2m}}.$$

The sharper first-omitted-term zeta bound uses the Stieltjes remainder property, not just an absolute periodic-Bernoulli estimate. Namely,

$$\frac{1}{1-e^{-t}} = \frac{1}{t} + \frac{1}{2} + 2t\sum_{r\geq 1}\frac{1}{t^2+4\pi^2r^2}.$$

Expanding each denominator geometrically through the term preceding B_{2m} gives a signed remainder bounded in absolute value by $|B_{2m}|t^{2m-1}/(2m)!$. Integrate this against $t^{s-1}e^{-Nt}/\Gamma(s)$ to obtain the stated bound. This argument also specifies exactly which Euler–Maclaurin terms are retained: $j = 1, \ldots, m-1$. The gamma bound follows by expanding the geometric factor in Binet's integral for the digamma function. For $s \geq 64$ the code instead brackets the tail of the direct positive sum by the two adjacent integrals. All rational operations round outwards at a fixed denominator 10^{130}. Exact Bernoulli numbers are provided by SymPy; no numerical Bernoulli approximation is involved.

At $M = 360$, the total interval has width less than $2.387 \cdot 10^{-113}$. In particular it certifies the displayed digits

$$M_3 = 5.74038880722947428001957168810246146296101300745487\ldots.$$

The full lower and upper rational endpoints, and the analytic error bound, are stored in `cubic_certificate.json`. An independent 150-digit mpmath quadrature lies inside this proved interval; that numerical check is a diagnostic, whereas (8.15) and the outward arithmetic supply the certificate.

8.6 Sharp truncation of the log-gamma moment expansion

Use the constants τ, ρ, C of the preceding late-coefficient theorem, and put $\Lambda = -\log\rho$. In this section write $b_k = c_k > 0$ for those positive coefficients, and $\widetilde{c}_k = (-1)^{k-1}b_k$. The real inverse $g(t)$ of $1/\Gamma(x)$ on $0 \leq x \leq 1$ satisfies $g(t) = -h(-t)$, where $h(w) = w\Gamma(1-h(w))$ is the earlier inverse germ. Thus $g(t) = \sum_{k\geq 1}\widetilde{c}_k t^k$ near zero. The moment integral is

$$m_n := M_n/n! = \frac{1}{\Gamma(n)}\int_0^\infty y^{n-1}g(e^{-y})\,dy.$$

The first relative coefficient d_1 below is the earlier β_1. Set $P = \psi'(-\tau)$, $Q = \psi''(-\tau)$, and $R = \psi^{(3)}(-\tau)$; equivalently $d_1 = R/(8P^2) - 5Q^2/(24P^3)$.

Lemma 8.12 (The dominant inverse singularity)**.** *The germ g continues to a slit disk of radius greater than ρ, with its only singularity on $|t| = \rho$ at $t = -\rho$, a square-root branch point.*

Proof. The positive inverse h has $h(\rho) = \tau$ and is absolutely convergent on that circle. Its positive coefficients of degrees one and two imply $|h(t)| < \tau$ at every other point of the circle. With $\Phi(u) = \Gamma(1-u)$, one then has $|t\Phi'(h(t))| \leq \rho\Phi'(|h(t)|) < \rho\Phi'(\tau) = 1$. The implicit-function theorem continues the inverse across those points. At $t = \rho$, local inversion at the simple critical point gives a convergent square-root expansion. Compactness supplies a slightly larger slit disk; replace t by $-t$ for g. This verifies the usual smooth implicit-function hypotheses directly.[79] □

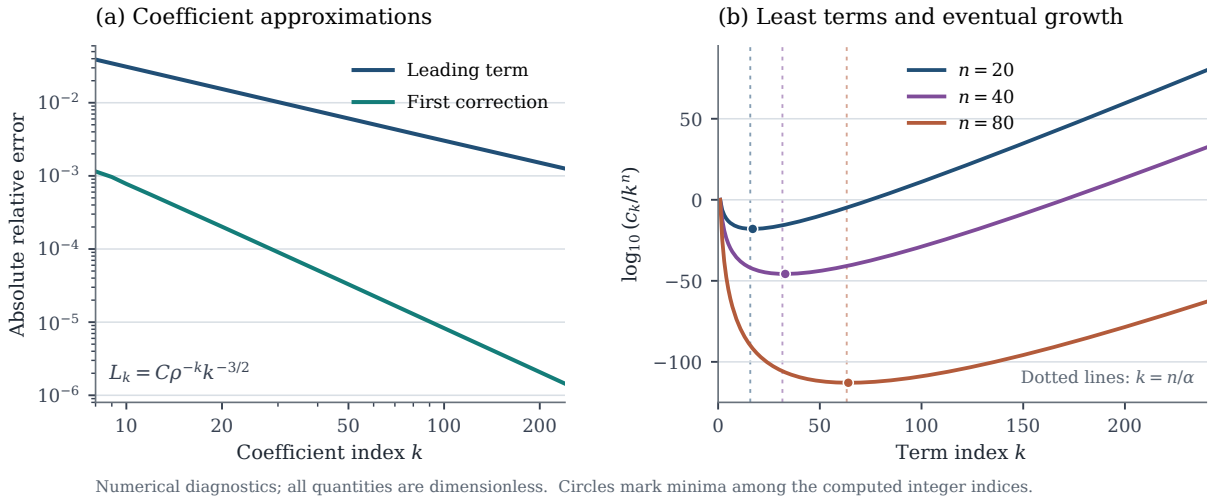


Figure 8.1: Numerical diagnostics for the proved coefficient and moment asymptotics. Left: absolute relative errors of $C\rho^{-k}k^{-3/2}$ and $C\rho^{-k}k^{-3/2}(1+\beta_1/k)$, using coefficients computed from their positive defining recurrence. Right: $\log_{10}(c_k/k^n)$ for $n = 20, 40, 80$. Dotted vertical lines mark $k = n/\alpha$; circles mark the sampled minima. All plotted quantities are dimensionless. The figure illustrates the theorems and supplies no independent error certificate.

8.6.1 A uniform Taylor-remainder lemma

The next statement is the key passage from fixed-order asymptotics to optimal truncation. It explicitly crosses the radius of convergence on the positive real axis; merely summing an infinite Taylor tail there would be invalid.

Lemma 8.13 (Uniform inverse remainder)**.** *For some $\delta > 0$, uniformly for $t \in [\rho e^{-\delta}, \rho e^{\delta}]$, put $r = t/\rho$. Then*

$$\frac{g(t) - \sum_{k=1}^{N-1} \widetilde{c}_k t^k}{\widetilde{c}_N t^N} = \frac{1}{1+r} + \frac{3r}{2N(1+r)^2} + O(N^{-2}). \tag{8.16}$$

On each of the remaining real intervals $0 < t \le \rho e^{-\delta}$ and $\rho e^{\delta} \le t \le 1$, the same normalized remainder is bounded uniformly in N sufficiently large.

Proof. The preceding theorem gives a slit disk of continuation for g, with its only singularity at $-\rho$. Choose its outer radius $R > \rho$, and then $\delta > 0$ so small that $\rho e^{\delta} < R$ and $\rho e^{\delta} < 1$. Cauchy's integral formula for $g(t)$, less its first $N-1$ Taylor terms, has kernel $t^N/(z^N(z-t))$. Deform its contour to the two banks of the negative cut and an outer contour. The coefficient $\widetilde{c}_N$ uses kernel z^{-N-1}. On the cut $z = -s$, the ratio of these kernels, after removing t^N, is

$$\frac{z}{z-t} = \frac{s}{s+t}.$$

The jump of g at $s = \rho$ is a nonzero constant times $(s-\rho)^{1/2}(1 + O(s-\rho))$, with a complete expansion. The coefficient integral is thus a Watson-type endpoint integral whose normalized variable $u = (s-\rho)/\rho$ has

$$\langle u \rangle = \frac{3}{2N} + O(N^{-2}), \qquad \langle u^2 \rangle = O(N^{-2}).$$

These formulas follow directly by comparing $\int_0^\epsilon u^{j+1/2}(1+u)^{-N-1}\,\mathrm{d}u$ with its beta integral extended to infinity. Terms outside a fixed small neighborhood of zero are exponentially smaller. Expanding

$$\frac{s}{s+t} = \frac{1}{1+r} + \frac{r}{(1+r)^2}u + O(u^2)$$

proves (8.16). The outer contour contributes $O(R^{-N})$ to the coefficient and $O((t/R)^N)$ to the remainder; after normalization these errors are exponentially small uniformly on the stated compact t-interval. This contour argument defines and estimates the actual analytic remainder on both sides of $t = \rho$.

It remains to bound the two outer real intervals. The estimate $b_k \asymp \rho^{-k}k^{-3/2}$ implies that, below $\rho e^{-\delta}$, the absolutely convergent tail is bounded by a constant times $b_N t^N$. Above ρe^{δ}, use

$$\sum_{k<N} \frac{b_k t^k}{b_N t^N} \ll \sum_{k<N} (N/k)^{3/2}(\rho/t)^{N-k}.$$

Splitting at $k = N/2$ bounds this uniformly by a geometric series plus an exponentially small term times a polynomial in N. Furthermore $0 \le g(t) \le 1$, whereas $b_N t^N$ grows exponentially uniformly on this interval. This proves the last claim. □

8.6.2 Sharp optimal truncation

Theorem 8.14 (Half the first omitted term)**.** *Let n, N tend to infinity through positive integers with $\eta = N\Lambda - n$ in a fixed bounded interval. Then*

$$\boxed{m_n - \sum_{k=1}^{N-1} \frac{(-1)^{k-1}b_k}{k^n} = \frac{(-1)^{N-1}b_N}{N^n}\left(\frac{1}{2} + \frac{3-2\eta}{8N} + O(N^{-2})\right).} \tag{8.17}$$

The O-constant is uniform for bounded η. The ratio in parentheses has a complete asymptotic expansion in $1/N$, with coefficients computable from the local inverse expansion and gamma moments. In particular, taking the nearest integer to $(n+3/2)/\Lambda$ gives a rigorously justified least-term approximation. The error eventually has the sign of the first omitted term and is asymptotic to one half of that term.

Proof. Subtract the finite sum under (8.14), using

$$\frac{1}{(n-1)!}\int_0^\infty y^{n-1}e^{-ky}\,\mathrm{d}y = k^{-n}.$$

With Y having the gamma density $N^n y^{n-1}e^{-Ny}/(n-1)!$, the error divided by $\widetilde{c}_N/N^n$ is the expectation of the normalized inverse remainder at $t = e^{-Y}$.

The gamma distribution concentrates at Λ:

$$\mathbb{E}Y = \frac{n}{N} = \Lambda - \frac{\eta}{N}, \qquad \operatorname{Var} Y = \frac{n}{N^2}.$$

For every fixed small $\delta > 0$, its probability outside $[\Lambda - \delta, \Lambda + \delta]$ is $O(e^{-a_\delta N})$. This follows from its moment-generating function $(1 - u/N)^{-n}$ and Chernoff's inequality. The uniform outer bounds in Lemma 8.13 therefore make the outer contribution exponentially small.

On the middle interval, the leading kernel is

$$L(Y-\Lambda)=\frac{1}{1+e^{\Lambda-Y}},\qquad L(v)=\frac{1}{2}+\frac{v}{4}-\frac{v^3}{48}+O(v^4).$$

Writing $V=Y-\Lambda$, its centered gamma moments give

$$\mathbb{E}V=-\frac{\eta}{N},\qquad \mathbb{E}V^3=O(N^{-2}),\qquad \mathbb{E}V^4=O(N^{-2}),$$

uniformly for bounded η. Thus $\mathbb{E}L(V)=1/2-\eta/(4N)+O(N^{-2})$. It is essential here to use the signed third moment; an absolute cubic bound alone would lose half a power. The correction kernel in (8.16) is

$$\frac{3e^{-V}}{2N(1+e^{-V})^2}.$$

Its value at zero is $3/(8N)$, and its first derivative vanishes there, so its expectation is $3/(8N)+O(N^{-2})$. Adding these terms proves (8.17). The contour calculation and the Taylor/gamma-moment calculation can both be continued to arbitrary fixed order. Gamma central moments are polynomials in n and N^{-1}; their parity and degree give integer powers of N^{-1}. □

The theorem strengthens the classical expansion in two ways. It allows N to grow linearly with n, and it estimates the actual error after truncation. The least-term estimate alone would not justify either conclusion. No global sign assertion for every n, N is needed.

Corollary 8.15 (Second correction)**.** *Under the hypotheses of Theorem 8.14, the normalized error has the more precise expansion*

$$\frac{m_n-\sum_{k<N}\widetilde{c}_k/k^n}{\widetilde{c}_N/N^n}=\frac{1}{2}+\frac{3-2\eta}{8N}+\frac{1}{N^2}\left(\frac{d_1}{4}+\frac{\Lambda(6\eta-13)}{96}\right)+O(N^{-3}).\tag{8.18}$$

Proof. Here are the additional local coefficients, so that the assertion is independently reproducible. Let $v=B-\tau$, and use P,Q,R from (8.12). The characteristic inverse satisfies

$$\log\frac{B/\Gamma(1-B)}{\rho}=-\frac{P}{2}v^2+\frac{Q}{6}v^3-\frac{R}{24}v^4+O(v^5).$$

Putting $w=\sqrt{1-t/\rho}$ and $\alpha_*=\sqrt{2/P}$, inversion gives

$$B(t)=\tau-\alpha_* w+\frac{Q}{3P^2}w^2+\mu w^3+O(w^4),\qquad D:=\frac{\mu}{\alpha_*}=-\frac{1}{4}+\frac{R}{12P^2}-\frac{5Q^2}{36P^3}.$$

Thus the positive cut density for B is $(\alpha_*/\pi)u^{1/2}(1+Du+O(u^2))$, where $u=t/\rho-1$. Transfer gives $d_1=3/8+3D/2$, proving (8.12) as well. The normalized endpoint moments are

$$\langle u\rangle=\frac{3}{2N}+\frac{9/4+3D/2}{N^2}+O(N^{-3}),\qquad \langle u^2\rangle=\frac{15}{4N^2}+O(N^{-3}),\qquad \langle u^3\rangle=O(N^{-3}).$$

They follow from beta integrals and the displayed cut density. The coefficient of N^{-2} in the inverse remainder at $r=t/\rho$ is therefore

$$K_2(r)=\frac{r}{(1+r)^2}\left(\frac{9}{4}+\frac{3D}{2}\right)-\frac{15r}{4(1+r)^3},\qquad K_2(1)=d_1/4.$$

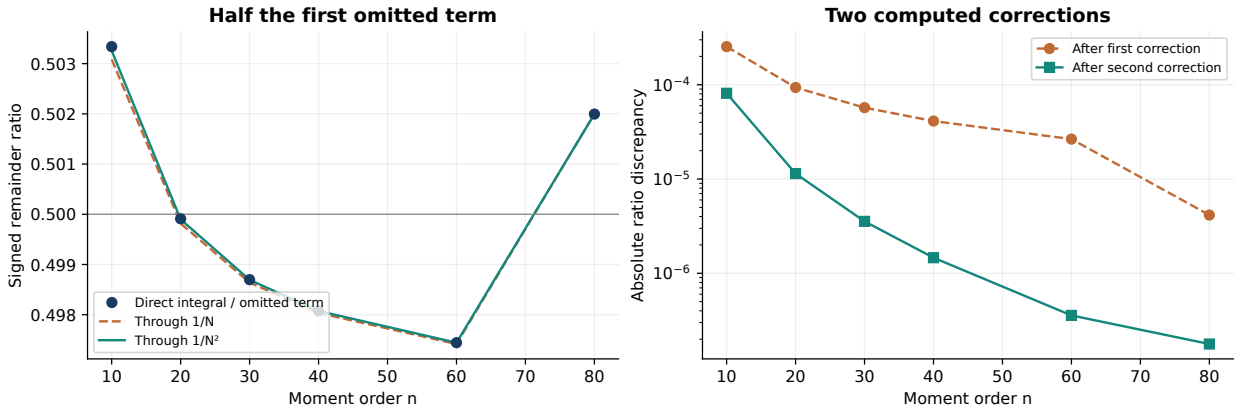


Figure 8.2: The signed error divided by the first omitted term, at a truncation near $(n+3/2)/\Lambda$. The first-correction formula tracks the small changes caused by rounding the truncation index. The plot is numerical corroboration; Theorem 8.14 supplies the proof.

For the gamma average, put $K_1(v) = 3e^{-v}/[2(1+e^{-v})^2]$ and $V = Y - \Lambda$. Taylor expansion through degree five, using $\mathbb{E}|V|^6 = O(N^{-3})$, gives

$$\mathbb{E}L(V) = \frac{1}{2} - \frac{\eta}{4N} + \frac{\Lambda(3\eta-2)}{48N^2} + O(N^{-3}), \qquad \mathbb{E}K_1(V) = \frac{3}{8} - \frac{3\Lambda}{32N} + O(N^{-2}).$$

For example $\mathbb{E}V^3 = \Lambda(2-3\eta)N^{-2} + O(N^{-3})$; the signed fifth moment is $O(N^{-3})$. Also $\mathbb{E}K_2(e^{-V}) = K_2(1) + O(N^{-1})$. The uniform outer bound again makes all outer contributions exponentially small. Combining the three terms proves (8.18). □

8.6.3 A convergent identity with an explicit tail bound

An asymptotic series should not be used as an infinite exact sum. The same inverse coordinate gives an exact convergent alternative.

Proposition 8.16 (Cutoff identity)**.** *For $n \geq 1$ and $Y > \Lambda$,*

$$m_n = \frac{1}{(n-1)!}\int_0^Y y^{n-1} g(e^{-y})\,dy + \sum_{k\geq 1}\frac{\widetilde{c}_k}{k^n}Q(n,kY), \qquad Q(n,z) = e^{-z}\sum_{j=0}^{n-1}\frac{z^j}{j!}. \tag{8.19}$$

The sum is absolutely convergent. If $Y > \log 4$, the absolute tail after $k = K$ is at most

$$\frac{(4e^{-Y})^{K+1}}{2(1-4e^{-Y})}\sum_{j=0}^{n-1}\frac{Y^j}{j!(K+1)^{n-j}}. \tag{8.20}$$

Proof. Split (8.14) at Y, substitute the convergent inverse series on the tail, and integrate term by term. The coefficient growth theorem proves absolute convergence. For the elementary explicit bound, note that

$$\frac{1/2}{\Phi(1/2)} = \frac{1}{2\sqrt{\pi}} > \frac{1}{4}.$$

The smaller solution to $B = t\Phi(B)$ at $t = 1/4$ is therefore less than $1/2$; equivalently $B(1/4) < 1/2$. Positivity gives $b_k \leq 4^k/2$. For $k > K$ and $j \leq n-1$, $k^{j-n} \leq (K+1)^{j-n}$. The geometric tail now gives (8.20). □

The numerical implementation uses ordinary high-precision arithmetic and explicitly distinguishes analytic truncation bounds from interval rounding bounds. Formula (8.19) is also suitable for a future ball-arithmetic evaluator: the finite inverse integral and all constant evaluations can be enclosed separately.

8.7 Reflected log-gamma moments and their late coefficients

Set $f(x) = \log\Gamma(x)$ for $0 < x < 1$, and define

$$M_{n,m} = \int_0^1 f(x)^n f(1-x)^m \, dx, \qquad n, m \in \mathbb{Z}_{\geq 0}.$$

These integrals already occur as $T_{n+m,n}$ in Amdeberhan et al. [64, Section 5]. Their symmetry $M_{n,m} = M_{m,n}$ and the identities obtained by expanding Euler's reflection formula are therefore classical. The asymptotic question here is different: keep the reflected exponent m fixed and let n tend to infinity. The factor $f(1-x)^m$ vanishes to order m at the endpoint that produces the factorial growth. This shifts the first exponential scale from 1 to $m+1$ without removing the eventual divergence of the full expansion. Theorem 8.18 identifies its late terms, including their signs and the first relative correction.

8.7.1 The meromorphic transform and all finite asymptotic orders

Put

$$B(x) = \log\Gamma(1-x) = \gamma x + \sum_{j\geq 2} \frac{\zeta(j)}{j} x^j, \qquad |x| < 1,$$

where γ is Euler's constant. For $k \geq 1$ define

$$A_{k,m} = \frac{1}{k}[x^{k-1}] \exp\{kB(-x)\} B(x)^m. \tag{8.21}$$

The convention $B(x)^0 = 1$ is in force. In particular $A_{k,m} = 0$ when $k \leq m$. Every coefficient is a finite polynomial over $\mathbb{Q}$ in $\gamma, \zeta(2), \ldots, \zeta(k-1)$.

Theorem 8.17 (Reflected moment expansion)**.** *For each fixed integer $m \geq 0$, the function*

$$F_m(z) = \int_0^1 \Gamma(x)^z f(1-x)^m \, dx$$

is holomorphic for $\operatorname{Re} z < m+1$, has Taylor expansion

$$F_m(z) = \sum_{n\geq 0} \frac{M_{n,m}}{n!} z^n \qquad (|z| < m+1),$$

and extends meromorphically to $\mathbb{C}$. Its only possible poles are simple poles at integers $k \geq m+1$, with

$$\operatorname*{Res}_{z=k} F_m(z) = -kA_{k,m}. \tag{8.22}$$

The first pole is present. For every fixed integer $K \geq m+1$,

$$\frac{M_{n,m}}{n!} = \sum_{k=m+1}^{K} \frac{A_{k,m}}{k^n} + O_{m,K}\big((K+1)^{-n}\big), \qquad n \longrightarrow \infty. \tag{8.23}$$

Consequently

$$M_{n,m} \sim \frac{\gamma^m n!}{(m+1)^{n+1}}. \tag{8.24}$$

Proof. As $x \downarrow 0$, $f(x) = -\log x + O(x)$ and $B(x) = \gamma x + O(x^2)$. At the other endpoint, $f(x) = O(1-x)$ and $f(1-x) = O(1 + |\log(1-x)|)$. These estimates prove absolute convergence and compact domination in the asserted half-plane, including after any number of differentiations in z. They also justify the Taylor expansion near zero.

Fix $0 < a < 1$ and an integer $J \geq 1$. Taylor expansion in x gives

$$\Gamma(1+x)^z B(x)^m = \sum_{j=0}^{J-1} b_{j,m}(z) x^j + x^J E_{J,m}(x,z),$$

where each $b_{j,m}$ is an entire function of z, and the remainder is uniformly bounded for $0 \leq x \leq a$ and z in compact sets. The integral over $[0,a]$ is therefore

$$\sum_{j=0}^{J-1} \frac{b_{j,m}(z) a^{j+1-z}}{j+1-z} + \int_0^a x^{J-z} E_{J,m}(x,z)\, dx.$$

The last integral is holomorphic for $\operatorname{Re} z < J+1$. The integral over $[a,1]$ is entire, since its logarithmic singularity at 1 is integrable uniformly for z in compact sets. Letting J increase gives the continuation. At $z = k$ its residue is $-b_{k-1,m}(k) = -kA_{k,m}$. Since $B(x)^m$ starts with $\gamma^m x^m$, the first nonzero coefficient is $A_{m+1,m} = \gamma^m/(m+1) > 0$.

For the asymptotic assertion, subtract the polar functions

$$\sum_{k=m+1}^{K+1} \frac{kA_{k,m}}{k-z}$$

from $F_m(z)$. The difference is holomorphic on $|z| < K+2$. Cauchy's estimate on any circle $K+1 < R < K+2$ bounds its nth Taylor coefficient by $O_{m,K,R}(R^{-n})$. Restoring the term at $K+1$ proves (8.23), even if that particular coefficient happens to vanish. Taking $K = m+1$ proves (8.24). □

The first three coefficients are explicit. Write $k = m+3$ in the last line:

$$\begin{aligned} A_{m+1,m} &= \frac{\gamma^m}{m+1}, \\ A_{m+2,m} &= \frac{\gamma^{m-1}}{m+2}\left(\frac{m\zeta(2)}{2} - (m+2)\gamma^2\right), \\ A_{m+3,m} &= \frac{\gamma^m}{k}\left\{\frac{m\zeta(3)}{3\gamma} + \frac{m(m-1)\zeta(2)^2}{8\gamma^2} + \frac{k(1-m)\zeta(2)}{2} + \frac{k^2\gamma^2}{2}\right\}. \end{aligned} \tag{8.25}$$

These displays also apply to $m = 0$, since $\gamma > 0$ and factors of m remove the superfluous terms. For example,

$$\begin{aligned} \frac{M_{n,1}}{n!} &= \frac{\gamma}{2^{n+1}} + \frac{\zeta(2)/2 - 3\gamma^2}{3^{n+1}} + \frac{\zeta(3)/3 + 8\gamma^3}{4^{n+1}} + O(5^{-n}), \\ \frac{M_{n,2}}{n!} &= \frac{\gamma^2}{3^{n+1}} + \frac{\gamma\zeta(2) - 4\gamma^3}{4^{n+1}} + O(5^{-n}). \end{aligned}$$

The second displayed correction is positive, whereas the corresponding correction for $m = 1$ is negative. Thus strict alternation from the first term cannot be imported from the unreflected family. The two signs can, for instance, be checked using the elementary enclosures $0.577 < \gamma < 0.578$ and $1.644 < \zeta(2) < 1.646$. Eventual alternation has a different, precise rule.

8.7.2 Late coefficients and a universal inverse-gamma scale

Let $\phi(u) = \Gamma(1-u)$ and $D = u\,d/du$. Define $\tau \in (0,1)$ by

$$D \log \phi(\tau) = -\tau\psi(1-\tau) = 1, \tag{8.26}$$

and put

$$\rho = \frac{\tau}{\phi(\tau)}, \quad v = D^2 \log \phi(\tau) = 1 + \tau^2\psi'(1-\tau), \quad C = \frac{\tau}{\sqrt{2\pi v}}, \quad b = -\log\Gamma(1+\tau). \tag{8.27}$$

The equation for τ has a unique solution: its left side equals $\gamma u + \sum_{j\ge 2}\zeta(j)u^j$, which is strictly increasing from zero to infinity. Also $0 < \rho < 1$ and $b > 0$, the latter by strict log-convexity of Γ between 1 and 2. Numerically, $\tau = 0.5040830082\ldots$ and $\rho = 0.2821158090\ldots$. Only the exact definitions are used below.

Theorem 8.18 (Late reflected coefficients)**.** *For every fixed nonnegative integer m,*

$$A_{k,m} = (-1)^{k+m-1} C b^m \rho^{-k} k^{-3/2} \left(1 + \frac{\beta_m}{k} + O_m(k^{-2})\right). \tag{8.28}$$

There is a full asymptotic expansion in integer powers of $1/k$. To specify the first correction, set $G(u) = \log\Gamma(1+u)$, $H_m(u) = G(u)^m$, and $\kappa_j = D^j \log\phi(\tau)$. All quantities on the right below are evaluated at $u = \tau$:

$$\beta_m = -\frac{(D+1)^2 H_m}{2vH_m} + \frac{\kappa_3(D+1)H_m}{2v^2 H_m} + \frac{\kappa_4}{8v^2} - \frac{5\kappa_3^2}{24v^3}. \tag{8.29}$$

In particular, every sufficiently large possible pole in Theorem 8.17 is present. More explicitly,

$$\operatorname{Res}_{z=k} F_m(z) = (-1)^{k+m} C b^m \rho^{-k} k^{-1/2} \left(1 + \frac{\beta_m}{k} + O_m(k^{-2})\right),$$

so the eventual residue sign is $(-1)^{k+m}$. For every fixed real n, the formal series $\sum_{k\ge m+1} A_{k,m}/k^n$ diverges because its terms do not tend to zero.

Proof. Replacing x by $-u$ in (8.21) gives

$$A_{k,m} = \frac{(-1)^{k-1}}{k}[u^{k-1}]\phi(u)^k H_m(u).$$

The function H_m is holomorphic for $|u| < 1$. If $\phi(u) = \sum_{j\ge 0} a_j u^j$, all the a_j are strictly positive, since $\log\phi$ has positive coefficients. Thus

$$p_j = \frac{a_j \tau^j}{\phi(\tau)}$$

is a probability distribution with exponential moments, mean one, variance $v > 0$, and positive mass at both 0 and 1. It follows that the function

$$Q(t) = \frac{\phi(\tau e^{it})}{\phi(\tau)} e^{-it}$$

has modulus strictly less than one for $0 < |t| \le \pi$, while

$$\log Q(t) = -\frac{v}{2}t^2 - \frac{i\kappa_3}{6}t^3 + \frac{\kappa_4}{24}t^4 + O(t^5)$$

near zero. Cauchy's coefficient formula becomes

$$A_{k,m} = \frac{(-1)^{k-1}\tau\rho^{-k}}{2\pi k} \int_{-\pi}^{\pi} Q(t)^k e^{it} H_m(\tau e^{it})\, dt. \tag{8.30}$$

Fix a sufficiently small $\delta > 0$. On $\delta \le |t| \le \pi$ the integral is $O(q^k)$ for some $q < 1$. On $k^{-2/5} \le |t| \le \delta$ it is $O(e^{-ck^{1/5}})$, because $|Q(t)| \le e^{-ct^2}$. The analytic amplitude is bounded on the whole circle. On the remaining interval set $t = y/\sqrt{k}$. Gaussian domination and Taylor expansion to any fixed order, with an analytic remainder, give a full expansion. Odd terms integrate to zero, so only integer powers of $1/k$ occur. The leading integral is

$$H_m(\tau)\sqrt{\frac{2\pi}{vk}} = (-1)^m b^m \sqrt{\frac{2\pi}{vk}},$$

which proves the leading term in (8.28).

For the first correction, the amplitude derivatives at $t = 0$ are H_m, $i(D{+}1)H_m$, and $-(D{+}1)^2 H_m$. The Gaussian moments of degrees 2, 4, and 6 contribute, respectively, the amplitude's quadratic term, the amplitude–cubic cross term together with the quartic phase term, and the square of the cubic phase term. Their normalized sum is exactly (8.29). Expanding two further even orders gives a relative $O_m(k^{-2})$ remainder after the displayed correction. Since $Cb^m > 0$ and $\rho < 1$, the assertions about nonvanishing, signs, and divergence follow at once. □

This proof uses the circle $|u| = \tau$ and a strict modulus inequality; it does not presume a classification of all complex critical values of the reciprocal gamma function. The same circle governs every fixed reflected exponent. With $\alpha = -\log\rho$, the continuous minimum of the leading term magnitude $\rho^{-k}k^{-n-3/2}$ occurs at $k = (n + 3/2)/\alpha$. A least-term calculation alone does not bound the remainder. The next result supplies an explicit bound with a growing number of terms.

8.7.3 An effective remainder despite the initial sign changes

Theorem 8.19 (Explicit reflected truncation bound)**.** *Let $m \ge 0$, $n \ge 1$, and $K \ge 1$ be integers, put $\alpha = -\log\rho$ and $T_m = \tau B(\tau)^m$, and suppose $K\alpha < n$. With*

$$R_{n,K}^{(m)} = \frac{M_{n,m}}{n!} - \sum_{k=1}^{K} \frac{A_{k,m}}{k^n}, \qquad \Gamma(n,s) = \int_s^{\infty} t^{n-1}e^{-t}\, dt,$$

one has the entirely explicit estimate

$$|R_{n,K}^{(m)}| \le \frac{\alpha^n}{n!}\left(m! + \frac{nT_m}{n - K\alpha}\right) + \frac{T_m\rho^{-K-1}}{(K+1)^n}\frac{\Gamma(n,(K+1)\alpha)}{\Gamma(n)}. \tag{8.31}$$

If $n \ge \max\{2, 4\alpha^2\}$ and $K = \lfloor n/\alpha - \sqrt{n}\rfloor$, then $K \ge 1$ and

$$|R_{n,K}^{(m)}| \le \left(\frac{m!}{\sqrt{2\pi n}} + \frac{T_m}{\alpha\sqrt{2\pi}} + T_m e^{2\alpha^2}\right)\left(\frac{e\alpha}{n}\right)^n. \tag{8.32}$$

This bound holds for every m, but no useful uniform relative error is asserted when m grows with n.

Proof. The function f decreases from infinity to zero on $(0,1]$: $\psi'(x) > 0$ and $\psi(1) = -\gamma < 0$. Let $x(y)$ denote its inverse, and set $W_m(x) = \int_0^x B(t)^m\,dt$. Tonelli's theorem applied to $f(x)^n = n\int_0^{f(x)} y^{n-1}\,dy$ gives

$$\frac{M_{n,m}}{n!} = \frac{1}{\Gamma(n)}\int_0^\infty y^{n-1}W_m(x(y))\,dy. \tag{8.33}$$

Let $h(q)$ be the local solution of $h = q\phi(h)$ with $h(0) = 0$. Lagrange inversion gives positive coefficients

$$h(q) = \sum_{k\ge1} c_k q^k, \qquad c_k = \frac{1}{k}[u^{k-1}]\phi(u)^k.$$

The $m = 0$ case of Theorem 8.18 proves that its radius is ρ and that the series converges absolutely on $|q| \le \rho$. The map $u \mapsto u/\phi(u)$ increases strictly from 0 to ρ on $[0,\tau]$, so inversion along this interval and continuity imply $h(\rho) = \tau$. No other inverse branch is used.

Lagrange–Bürmann inversion now gives

$$W_m(x(y)) = \sum_{k\ge1} A_{k,m}e^{-ky} \qquad (y \ge \alpha).$$

Indeed $x(y) = -h(-e^{-y})$ initially for large y, and then throughout $y > \alpha$ by analytic continuation along the strictly monotone real inverse. The endpoint follows by absolute convergence. To prove that convergence and obtain its quantitative majorant, define

$$d_{k,m} = \frac{1}{k}[u^{k-1}]\phi(u)^k B(u)^m \ge 0.$$

Coefficientwise absolute values in $B(-u)^m$ are $B(u)^m$. Consequently $|A_{k,m}| \le d_{k,m}$, while another application of Lagrange–Bürmann gives

$$\sum_{k\ge1} d_{k,m}\rho^k = W_m(h(\rho)) = W_m(\tau) \le \tau B(\tau)^m = T_m.$$

Here B has positive Taylor coefficients and is increasing on $[0,\tau]$. Thus on $y \ge \alpha$,

$$\left|W_m(x(y)) - \sum_{k=1}^{K} A_{k,m}e^{-ky}\right| \le T_m\rho^{-K-1}e^{-(K+1)y}.$$

Its integral in (8.33) is the second term of (8.31).

On $0 \le y \le \alpha$, use $0 \le W_m(x(y)) \le M_{m,0} \le m!$. The last inequality follows from $0 \le \log\Gamma(x) \le -\log x$, since log-convexity gives $\Gamma(1+x) \le 1$ for $0 \le x \le 1$. Also, for $k \le K$,

$$\int_0^\alpha y^{n-1}e^{-ky}\,dy \le \frac{\alpha^n e^{-k\alpha}}{n - k\alpha}.$$

This follows by differentiating $y^n e^{-ky}$ and using $n - ky \ge n - k\alpha > 0$. Summing with $\sum |A_{k,m}|\rho^k \le T_m$ proves the first term of (8.31).

For the stated choice of K, $n - K\alpha \ge \alpha\sqrt{n}$ and $\theta = (K+1)\alpha/n$ lies in $[1/2, 1]$. The inequality $\theta - 1 - \log\theta \le 2(1-\theta)^2$ gives

$$\rho^{-K-1}(K+1)^{-n} \le e^{2\alpha^2}(e\alpha/n)^n.$$

Finally, $\Gamma(n,s) \le \Gamma(n)$ and $n! \ge \sqrt{2\pi n}(n/e)^n$ yield (8.32). □

8.7.4 Sharp truncation for every fixed reflected exponent

The reflected coefficient expansion and the sharp inverse remainder now combine. This statement uses the two incoming developments together; it is stronger than their separate absolute-bound and unreflected results. Write $\Lambda = -\log \rho$, and retain β_m from (8.29).

Corollary 8.20 (Half the first omitted reflected term)**.** *Fix an integer $m \geq 0$. Let $n, N \to \infty$ through positive integers, with $\eta = N\Lambda - n$ in a fixed bounded interval. Then*

$$\frac{M_{n,m}/n! - \sum_{k=1}^{N-1} A_{k,m}/k^n}{A_{N,m}/N^n} = \frac{1}{2} + \frac{3-2\eta}{8N} + \frac{1}{N^2}\left[\frac{\beta_m}{4} + \frac{\Lambda(6\eta-13)}{96}\right] + O_m(N^{-3}), \tag{8.34}$$

uniformly for bounded η. In particular the error eventually has sign $(-1)^{N+m-1}$ and is asymptotic to half the first omitted term. The denominator is nonzero for every sufficiently large N.

Proof. Set $W_m(x) = \int_0^x B(u)^m\, du$ and $G_m(t) = W_m(g(t))$, where g is the real inverse used in the preceding sharp-moment section. Lagrange–Bürmann inversion gives $G_m(t) = \sum_{k\geq 1} A_{k,m} t^k$ near zero. The layer-cake formula (8.33) becomes

$$M_{n,m}/n! = \frac{1}{\Gamma(n)} \int_0^\infty y^{n-1} G_m(e^{-y})\, dy.$$

Since W_m is holomorphic on $|x| < 1$, it can be composed with the slit-disk inverse of Lemma 8.12. At the dominant point $g(-\rho) = -\tau$ its derivative is

$$W_m'(-\tau) = B(-\tau)^m = \big(\log \Gamma(1+\tau)\big)^m = (-b)^m \neq 0.$$

Thus G_m has the same sole dominant square-root singularity, with a nonzero leading amplitude. Its late coefficients have the relative coefficient β_m established in Theorem 8.18. The Cauchy cut-contour argument in Lemma 8.13 therefore applies to G_m with the same leading kernel $1/(1+t/\rho)$ and the same $3/(2N)$ endpoint-moment correction. The sign of the leading jump cancels against $A_{N,m}$ in this ratio. The second normalized cut coefficient at $t = \rho$ is $\beta_m/4$, by the same square-root transfer and beta-moment calculation used in Corollary 8.15.

The outer real-interval bounds also persist. Above the convergence radius, the finite coefficient sum is bounded using $|A_{k,m}| \asymp_m \rho^{-k} k^{-3/2}$, while $0 \leq G_m(t) \leq m!$ for $0 \leq t \leq 1$. The finitely many exceptional early coefficients contribute only an exponentially small term after normalization. Below the radius, the absolute Taylor tail gives the corresponding uniform bound. Average the normalized remainder against the gamma density with shape n and rate N. Its signed moments are exactly those used in the unreflected calculation. The first two corrections are consequently (8.34), with d_1 replaced by β_m; the analytic cut expansion and moments through degree six give the stated uniform $O_m(N^{-3})$ remainder. Eventual nonvanishing and the sign follow from (8.28). □

This fixed-m corollary does not supply uniform relative estimates when m grows with n. That joint regime remains a separate problem.

8.7.5 An exact identity coupling all reflected residues

The coefficients across different m satisfy a finite identity in which Euler's constant and all odd zeta values cancel.

Proposition 8.21 (Reflection sum for the residues)**.** *For every integer $k \geq 1$,*

$$\sum_{m=0}^{k-1} \frac{k^m}{m!} A_{k,m} = \begin{cases} 0, & k \text{ even}, \\ \dfrac{\pi^{k-1}}{k\,2^{k-1}} \dbinom{k-1}{(k-1)/2}, & k \text{ odd}. \end{cases} \tag{8.35}$$

Proof. Because $B(x)^m$ has order m, extending the sum over m to infinity does not affect $[x^{k-1}]$. Euler's reflection formula gives

$$\sum_{m=0}^{k-1} \frac{k^m}{m!} A_{k,m} = \frac{1}{k}[x^{k-1}]\big(\Gamma(1+x)\Gamma(1-x)\big)^k = \frac{1}{k}[x^{k-1}]\left(\frac{\pi x}{\sin \pi x}\right)^k.$$

The coefficient is zero for even k. For odd k, compute it as the residue at $x = 0$ of $(\pi/\sin \pi x)^k\,dx$. The local change of variable $t = \sin \pi x$ gives

$$[x^{k-1}]\left(\frac{\pi x}{\sin \pi x}\right)^k = \pi^{k-1}[t^{k-1}](1-t^2)^{-1/2},$$

and the binomial expansion proves (8.35). □

For example, the cases $k = 2, 3$ read $A_{2,0} + 2A_{2,1} = 0$ and $A_{3,0} + 3A_{3,1} + \frac{9}{2}A_{3,2} = \pi^2/6$. They are exact identities among the asymptotic coefficients, rather than convergent sums of the moment expansions. No summation over the late index k is used.

8.7.6 A recorded log-sine conjecture has a formal proof

There is also a small but useful correction to the logical status of a claim in the literature. The author preprint of [64] labels a coefficient identity as Conjecture 5.5 and motivates it using transcendence assumptions on $\log 2$ and $\log \pi$. For the canonical formal polynomials supplied by the beta integral, the coefficient identity needs none of those assumptions. Under the preprint's assumptions these formal coefficients coincide with its uniquely specified numerical coefficients, so the result also proves that stated implication. We do not assert uniqueness of arbitrary numerical expressions in zeta values and logarithms, nor priority over any later treatment.

Define polynomials $\mathcal{P}_N(X)$ by

$$\sum_{N \geq 0} \mathcal{P}_N(X) \frac{z^N}{N!} = \exp\left\{ Xz + \sum_{j \geq 2} \frac{(1 - 2^{1-j})\zeta(j)}{j} z^j \right\}. \tag{8.36}$$

Proposition 8.22 (Unconditional coefficient and translation identities)**.** *For $N \geq 0$,*

$$\int_0^1 [-\log \sin(\pi x)]^N\,dx = \mathcal{P}_N(\log 2), \tag{8.37}$$

$$[X^j]\mathcal{P}_N(X) = \binom{N}{j} \mathcal{P}_{N-j}(0), \qquad 0 \leq j \leq N,$$

$$\sum_{j=0}^{N} \binom{N}{j} M_{j,N-j} = \mathcal{P}_N(\log(2\pi)).$$

Moreover, the coefficients admit the finite partition formula

$$\mathcal{P}_N(X) = N! \sum_{r_1 + 2r_2 + \cdots + N r_N = N} \frac{X^{r_1}}{r_1!} \prod_{j=2}^{N} \frac{1}{r_j!} \left(\frac{(1 - 2^{1-j})\zeta(j)}{j}\right)^{r_j}. \tag{8.38}$$

Proof. For $\operatorname{Re} z < 1$, the beta integral gives

$$\int_0^1 \sin(\pi x)^{-z}\,dx = \frac{\Gamma((1-z)/2)}{\sqrt{\pi}\,\Gamma(1-z/2)}.$$

Expanding its logarithm at zero gives the right side of (8.36) with $X = \log 2$; its degree-j coefficient for $j \geq 2$ is $(1-2^{1-j})\zeta(j)/j$. Extracting Taylor coefficients proves (8.37). The factor e^{Xz} gives $[X^j]\mathcal{P}_N = \binom{N}{j}\mathcal{P}_{N-j}(0)$ by formal coefficient extraction. Euler's reflection formula turns the sum of mixed moments into the integral of $(\log\pi - \log\sin\pi x)^N$, which translates X by $\log\pi$. Expanding each exponential factor separately proves (8.38). All formal operations in a fixed degree involve only finitely many coefficients. □

The constant-column generating function has the additional factorization

$$\sum_{N\geq 0} \mathcal{P}_N(0)\frac{z^N}{N!} = \sqrt{\frac{\tan(\pi z/2)}{\pi z/2}}\exp\left\{\sum_{\substack{j\geq 3\\ j\text{ odd}}} \frac{(1-2^{1-j})\zeta(j)}{j} z^j\right\}, \tag{8.39}$$

where the square root is the formal branch with constant term one. Indeed, multiplying the left side by its value at $-z$ and applying the reflection formula yields $\tan(\pi z/2)/(\pi z/2)$. The even logarithmic part is therefore the displayed square root. This makes the multiplicative factorization of its formal odd-zeta monomial coefficients explicit, while the even-zeta contribution has an elementary trigonometric generating function.

Verification and remaining questions. The accompanying script `code/verify_reflected_moments.py` checks the three coefficient formulas by exact symbolic arithmetic, checks finite reflection sums and the formal Appell formulas, and compares independent quadrature with the fixed-order expansion. It also compares the late coefficients with (8.28) and (8.29). The numerical calculations are diagnostics; the preceding analytic and formal arguments are the proofs. The fixed-reflected-exponent signed remainder is now supplied by Corollary 8.20. Uniformity when m increases with n remains a substantive next question. In the former regime the endpoint approximation $B(x)^m \approx \gamma^m x^m$ need not be uniform, so the fixed-m theorem does not settle the simultaneous large-exponent problem.

8.8 The Stieltjes layer

Differentiating $L(s,\chi) = q^{-s}\sum_k \chi(k)\,\zeta(s,k/q)$ at $s=1$ gives the exact ladder

$$L'(\chi,1) = -\ln q\; L(1,\chi) - \frac{1}{q}\sum_k \chi(k)\,\gamma_1\Big(\frac{k}{q}\Big),$$

so every Malmsten log-log integral is a statement about *first generalized Stieltjes constants* $\gamma_1(k/q)$ — Blagouchine's program, equivalent to Deninger's $\zeta''(0,a)$ layer via the functional equation.

Result 8.23 (even-character Malmsten integral). [historical experiment: verified to 10^{-51}; even character mod 5; φ the golden ratio]

$$\int_0^1 \ln\ln\frac{1}{x}\cdot\frac{(1-x)(1-x^2)}{1-x^5}\,dx = -(\gamma+\ln 5)\,\frac{2\ln\varphi}{\sqrt{5}} - \frac{\gamma_1(\frac{1}{5}) - \gamma_1(\frac{2}{5}) - \gamma_1(\frac{3}{5}) + \gamma_1(\frac{4}{5})}{5}.$$

Result 8.24 (the parity dichotomy). The dichotomy is empirically sharp. The *odd* γ_1-combination reduces exactly as differentiated-Kummer predicts [historical experiment: verified blind]:

$$\gamma_1\tfrac{1}{3} - \gamma_1\tfrac{2}{3} = -\frac{\pi}{\sqrt{3}}\Big(\gamma + 4\ln 2 - \tfrac{1}{2}\ln 3 + 4\ln\pi - 6\ln\Gamma\tfrac{1}{3}\Big),$$

which is *Malmsten's 1846 reflection identity* [4] (long misattributed to Almkvist–Meurman). The even combination did not reduce in the tested standard basket [historical experiment: PSLQ junk at coefficient height 10^8 — honest negative]; pushing Blagouchine's rational-arguments theorem [6] through it collapses everything except a ζ''-block [historical experiment: derived and verified to 10^{-50}]:

$$\gamma_1\tfrac{1}{5} - \gamma_1\tfrac{2}{5} - \gamma_1\tfrac{3}{5} + \gamma_1\tfrac{4}{5} = \sqrt{5}\left[\zeta''\tfrac{1}{5} - \zeta''\tfrac{2}{5} - \zeta''\tfrac{3}{5} + \zeta''\tfrac{4}{5}\right] - 2\sqrt{5}\,\ln\varphi\,(\gamma + \ln 10\pi).$$

The canonical atom for the even layer is thus the even-character $\zeta''(0, l/q)$ combination (Deninger's layer [9]); Malmsten integrals, even γ_1-combinations, and even $L'(\chi, 1)$ are all elementary-equivalent to it. Two practical records: Blagouchine's particular value $\gamma_1(1/4)$ was re-verified exactly against `mpmath`'s generalized Stieltjes constants, and the reduction coefficients live in $\mathbb{Q}(\sqrt{q})$ — rational-coefficient PSLQ baskets *miss* the relation, so a $\sqrt{q}$-weighted basket or direct derivation is required.

8.9 The common Hurwitz-jet calculus

The generalized Stieltjes constants $\gamma_n(x)$ are defined by the Laurent expansion of the Hurwitz zeta function at its pole,

$$\zeta(s, x) = \frac{1}{s-1} + \sum_{n\geq 0} \frac{(-1)^n}{n!}\,\gamma_n(x)\,(s-1)^n, \tag{8.40}$$

so that $\gamma_0(x) = -\psi(x)$ and $\gamma_n(1) = \gamma_n$ are the classical Stieltjes constants, $\gamma_0 = \gamma$. Although universally called *constants*, the $\gamma_n(x)$ are real-analytic functions of the argument on $(0, \infty)$, and it is this function-theoretic point of view we adopt. Following Blagouchine [22, 6] we write

$$\Gamma_n(p) \;:=\; \int_1^p \gamma_n(x)\,dx, \qquad \text{so that} \;\; \Gamma_n(1) = 0,$$

for the antiderivative normalized at 1. (The notation is Blagouchine's; the clash with the Euler Γ-function is resolved by the subscript, and indeed $\Gamma_0(p) = -\ln\Gamma(p)$, as Theorem 8.25 makes plain.) Throughout, $\zeta^{(k)}(s_0, a) := \partial_s^k \zeta(s, a)\big|_{s=s_0}$ and $\zeta^{(k)}(s_0) := \zeta^{(k)}(s_0, 1)$ denote derivative jets with respect to the *order*; $\Lambda_q := \gamma + \ln(2\pi q)$; $H_n = \sum_{i\leq n} 1/i$ and $H_n^{(2)} = \sum_{i\leq n} 1/i^2$ are harmonic numbers; A is the Glaisher–Kinkelin constant, $\ln\mathsf{A} = \frac{1}{12} - \zeta'(-1)$; and

$$S_j\Big(\frac{p}{q}\Big) := \sum_{k=1}^{q-1} \sin\frac{2\pi kp}{q}\,\zeta^{(j)}\Big(0, \frac{k}{q}\Big), \qquad S_0\Big(\frac{p}{q}\Big) = \tfrac{1}{2}\cot\frac{\pi p}{q}, \quad S_1\Big(\frac{p}{q}\Big) = \sum_{k=1}^{q-1} \sin\frac{2\pi kp}{q}\,\ln\Gamma\Big(\frac{k}{q}\Big)$$

(the closed forms for S_0, S_1 use $\zeta(0, a) = \frac{1}{2} - a$, $\zeta'(0, a) = \ln\Gamma(a) - \frac{1}{2}\ln(2\pi)$, and $\sum_k \sin(2\pi kp/q) = 0$).

The starting observation is elementary but organizing: differentiating (8.40) in x against $\partial_x \zeta(s, x) = -s\,\zeta(s+1, x)$ identifies every derivative and antiderivative of every γ_n with a derivative jet of the Hurwitz zeta function at an *integer* value of s. Antiderivatives live at $s = 0$; the functions themselves at $s = 1$; their x-derivatives at $s = 2$ [25] — a single ladder.

The preceding gamma-integral examples are the first rung of this calculus. We now prove the integration identity, build its spectral jets, and use reflection and distribution to compute rational arguments. All source residuals in this chapter are historical; locally rerun checks are recorded separately in the verification directory.

8.10 Underived Stieltjes grids and character coordinates

Expanding the Hurwitz distribution identity at $s = 1$ gives

$$\sum_{j=0}^{d-1} \gamma_1((x+j)/d) = d\gamma_1(x) + d\log d\,\psi(x) - \frac{d}{2}\log^2 d,$$
$$\sum_{j=0}^{d-1} \gamma_2((x+j)/d) = d\left[\gamma_2(x) - 2\log d\,\gamma_1(x) - \log^2 d\,\psi(x) + \frac{1}{3}\log^3 d\right].$$

For a nonprincipal character,

$$L'(1,\chi) = -\log q\,L(1,\chi) - \frac{1}{q}\sum_{p=1}^{q}\chi(p)\gamma_1(p/q).$$

For a real quadratic character χ_D, $\zeta_{\mathbb{Q}(\sqrt{D})}(s) = \zeta(s)L(s,\chi_D)$ shows $L'(1,\chi_D)/L(1,\chi_D) = \gamma_{\mathbb{Q}(\sqrt{D})} - \gamma$, where the first term is the field's Euler–Kronecker constant. The source's exact finite matrices for $3 \le q \le 30$ left $\varphi(q)/2 - 1$ coordinates after distribution and first-constant reflection. Their negative searches constrain the selected baskets, without proving independence of Euler–Kronecker constants. The reflection formulas below retain the conductor factor $\gamma + \log(2\pi q)$: the earlier printed $q = 5$ specialization without $\log 5$ was corrected in the source reports.

8.11 The ladder

Theorem 8.25 (antiderivative ladder)**.** *For $n \ge 0$ and $p > 0$,*

$$\frac{d}{dp}\,\zeta^{(n+1)}(0,p) = (-1)^{n+1}(n+1)\,\gamma_n(p), \tag{8.41}$$

and consequently

$$\Gamma_n(p) = \frac{(-1)^{n+1}}{n+1}\Big[\zeta^{(n+1)}(0,p) - \zeta^{(n+1)}(0)\Big]. \tag{8.42}$$

Proof. Differentiating $\partial_x\zeta(s,x) = -s\,\zeta(s+1,x)$ is legitimate termwise for $\operatorname{Re} s > 1$ and extends by analytic continuation. Expand both sides around $s = 0$: on the right, $-s\,\zeta(s+1,x) = -s\big[\frac{1}{s} + \sum_k \frac{(-1)^k}{k!}\gamma_k(x)s^k\big] = -1 - \sum_k \frac{(-1)^k}{k!}\gamma_k(x)s^{k+1}$, whose s^{n+1}-coefficient is $(-1)^{n+1}\gamma_n(x)/n!$. On the left the s^{n+1}-coefficient is $\partial_x\zeta^{(n+1)}(0,x)/(n+1)!$. Equating gives (8.41); integrating from 1 to p gives (8.42). □

The differential form (8.41) is Lemma 2 of Chakraborty–Kanemitsu–Kuzumaki [23] (for $n = 1$ it is Deninger's R'-law [9]); the integrated form (8.42) is eq. (2.16) of Connon [26]. For $n = 0$, $\zeta'(0,p) = \ln\Gamma(p) - \frac{1}{2}\ln(2\pi)$ recovers $\Gamma_0(p) = -\ln\Gamma(p)$.

Theorem 8.26 (interpolation of power sums of logarithms)**.** *For $k \ge 1$ and integer $N \ge 0$,*

$$\Gamma_{k-1}(N+1) \;=\; -\frac{1}{k}\sum_{j=1}^{N}\ln^k j.$$

Proof. From $\zeta(s,N+1) = \zeta(s) - \sum_{j=1}^{N} j^{-s}$, $\zeta^{(k)}(0,N+1) = \zeta^{(k)}(0) - \sum_j(-\ln j)^k = \zeta^{(k)}(0) - (-1)^k\sum_j \ln^k j$. Insert into (8.42) with $n = k-1$. □

Thus Γ_{k-1} is an analytic interpolant of the power sums $\sum \ln^k j$ (up to the factor $-1/k$), exactly as $\ln\Gamma$ interpolates $\ln N!$: the heuristic "differentiate $1^r+\cdots+N^r=\zeta(-r)-\zeta(-r,N+1)$ k times at $r=0$" becomes precise through Theorem 8.25.

Corollary 8.27 (unit translates)**.** *For $k\ge 0$ and integer $n\ge 1$,*

$$\int_n^{n+1}\gamma_k(x)\,dx=-\frac{\ln^{k+1}n}{k+1};\qquad \textit{in particular}\quad \int_1^2\gamma_k(x)\,dx=0.$$

(The case $n=1$ is eqs. (2.18)–(2.19) of [26].)

8.12 Zeta jets and the conversion dictionary

Making (8.42) explicit requires the constants $\zeta^{(k)}(0)$ and, for the moment evaluations of Section 8.14, the jets at negative integers. All of them follow mechanically from the functional equation $\zeta(s)=2^s\pi^{s-1}\sin(\pi s/2)\,\Gamma(1-s)\,\zeta(1-s)$ by expanding $\Gamma(1-s)=\exp\bigl(\gamma s+\sum_{k\ge2}\zeta(k)s^k/k\bigr)$ and the Laurent series of $\zeta(1-s)$ at the pole.

Proposition 8.28 (jets at 0)**.** *With $L:=\ln(2\pi)$,*

$$\begin{aligned}
\zeta'(0)&=-\tfrac{L}{2},\\
\zeta''(0)&=\tfrac{\gamma^2}{2}+\gamma_1-\tfrac{\pi^2}{24}-\tfrac{L^2}{2},\\
\zeta'''(0)&=\gamma^3+3\gamma\gamma_1+\tfrac{3}{2}\gamma_2-\zeta(3)+\tfrac{3}{2}\gamma^2L+3\gamma_1L-\tfrac{\pi^2}{8}L-\tfrac{L^3}{2},\\
\zeta^{(4)}(0)&=\tfrac{3}{2}\gamma^4+4\gamma^3L+6\gamma^2\gamma_1+3\gamma^2L^2+\tfrac{\pi^2}{4}\gamma^2+12\gamma\gamma_1L+6\gamma\gamma_2+6\gamma_1L^2+\tfrac{\pi^2}{2}\gamma_1\\
&\quad+6\gamma_2L+2\gamma_3-\tfrac{L^4}{2}-\tfrac{\pi^2}{4}L^2-4\zeta(3)L-\tfrac{19\pi^4}{480}.
\end{aligned}$$

Proposition 8.29 (fifth jet at 0)**.** *With $L:=\ln(2\pi)$,*

$$\begin{aligned}
\zeta^{(5)}(0)&=2\gamma^5+\tfrac{15}{2}\,\gamma^4L+\gamma^3\Bigl(10L^2+\tfrac{5\pi^2}{6}\Bigr)+\gamma^2\Bigl(5L^3+\tfrac{5\pi^2}{4}\,L+10\,\zeta(3)\Bigr)\\
&\quad+\gamma\gamma_1\Bigl(30L^2+\tfrac{5\pi^2}{2}\Bigr)+\gamma_1\Bigl(10\gamma^3+30\gamma^2L+10L^3+\tfrac{5\pi^2}{2}\,L+20\,\zeta(3)\Bigr)\\
&\quad+\gamma_2\Bigl(15\gamma^2+30\gamma L+15L^2+\tfrac{5\pi^2}{4}\Bigr)+\gamma_3\,(10\gamma+10L)+\tfrac{5}{2}\,\gamma_4\\
&\quad-\zeta(3)\Bigl(10L^2+\tfrac{5\pi^2}{6}\Bigr)-12\,\zeta(5)-\tfrac{L^5}{2}-\tfrac{5\pi^2}{12}\,L^3-\tfrac{19\pi^4}{96}\,L.
\end{aligned}$$

The values $\zeta''(0)$ and $\zeta'''(0)$ are classical (Ramanujan [21]; Apostol [19]; Choudhury [24]). Apostol's Theorem 3 [19] in fact gives a closed form of $\zeta^{(n)}(0)$ at every order, in terms of the Laurent coefficients of $\Gamma(s)\zeta(s)$ at $s=1$, and prints the example $n=4$ in that vocabulary; we have not located the fully expanded forms above — nor any explicit form of $\zeta^{(5)}(0)$ — in print. The pattern continues to every order by the same expansion, which packages into the following closed recurrence.

Remark 8.30 (explicit recurrence for the jets at 0). In $\zeta(s)=\chi(s)\,\zeta(1-s)$ with $\chi(s)=2^s\pi^{s-1}\sin(\pi s/2)\,\Gamma(1-s)$, the simple zero of χ at $s=0$ cancels the pole of $\zeta(1-s)$; factor it out as $\chi(s)=\frac{s}{2}\,e^{g(s)}$ with $g(s):=\ln\bigl(2\chi(s)/s\bigr)=\ln\bigl[(2\pi)^s\,\frac{\sin(\pi s/2)}{\pi s/2}\,\Gamma(1-s)\bigr]$, analytic on $|s|<1$ and real on $(-1,1)$ (the bracket is positive there), $g(0)=0$. From $\ln\Gamma(1-s)=\gamma s+\sum_{k\ge2}\zeta(k)s^k/k$ and $\ln\bigl[\sin(\pi s/2)/(\pi s/2)\bigr]=-\sum_{m\ge1}\zeta(2m)(s/2)^{2m}/m$,

$$g(s)=(\gamma+L)\,s+\sum_{k\ge2}c_k s^k,\qquad c_k=\frac{\zeta(k)}{k}\cdot\begin{cases}1, & k\text{ odd},\\ 1-2^{1-k}, & k\text{ even},\end{cases}$$

so $c_2 = \frac{\pi^2}{24}$, $c_3 = \frac{\zeta(3)}{3}$, $c_4 = \frac{7\pi^4}{2880}$, $c_5 = \frac{\zeta(5)}{5}$. Since $s\,\zeta(1-s) = -1 + \sum_{m\ge1} \gamma_{m-1}\, s^m/(m-1)!$ is entire, the Leibniz rule gives

$$\zeta^{(n)}(0) = \frac{1}{2}\sum_{k=0}^{n} \binom{n}{k} B_k\big(1!\,c_1, 2!\,c_2, \dots, k!\,c_k\big)\, h_{n-k}, \qquad h_0 = -1, \quad h_m = m\,\gamma_{m-1}\ (m \ge 1),$$

where $c_1 = \gamma + L$ and B_k is the complete Bell polynomial, $B_k\big(g'(0), \dots, g^{(k)}(0)\big) = \big(e^g\big)^{(k)}(0)$: $B_0 = 1$, $B_1 = c_1$, $B_2 = c_1^2 + 2c_2$, $B_3 = c_1^3 + 6c_1c_2 + 6c_3$. The cases $n = 0, 1$ recover $\zeta(0) = -\frac{1}{2}$ and $\zeta'(0) = -\frac{L}{2}$; $n = 2, \dots, 5$ reproduce the two propositions verbatim.

Proposition 8.31 (jets at -1 and -2)**.** *Expanding the functional equation around $s = -1$ (where $\zeta(2-\varepsilon)$ is regular) and $s = -2$ (a trivial zero) gives, with $L = \ln(2\pi)$:*

$$\zeta(-1) = -\tfrac{1}{12}, \qquad \zeta'(-1) = \frac{\zeta'(2)}{2\pi^2} + \frac{1-\gamma-L}{12}, \tag{8.43}$$

$$\zeta''(-1) = -\frac{\zeta''(2)}{2\pi^2} + \frac{\zeta'(2)}{\pi^2}(\gamma + L - 1) - \frac{(\gamma+L)^2}{12} + \frac{\gamma+L}{6} + \frac{\pi^2}{144}, \tag{8.44}$$

$$\zeta(-2) = 0, \qquad \zeta'(-2) = -\frac{\zeta(3)}{4\pi^2}, \qquad \zeta''(-2) = \frac{\zeta(3)}{4\pi^2}\,(3 - 2\gamma - 2L) + \frac{\zeta'(3)}{2\pi^2}. \tag{8.45}$$

Remark 8.32 (Glaisher spelling). Since $\ln \mathsf{A} = \frac{1}{12} - \zeta'(-1)$, eq. (8.43) is equivalent to

$$\zeta'(2) = \zeta(2)\big(\gamma + \ln 2 + \ln \pi - 12 \ln \mathsf{A}\big),$$

the form returned by Mathematica's `FunctionExpand[Zeta'[2]]`. Either spelling may be used in all the moment formulas below; we keep $\zeta'(2), \zeta''(2), \zeta'(3)$ as the canonical second-layer constants.

8.13 Reflection formulas

Hurwitz's formula (see, e.g., [20, Ch. 12]), for $0 < p/q < 1$,

$$\zeta\Big(1-s, \frac{p}{q}\Big) = \frac{2\,\Gamma(s)}{(2\pi q)^s} \sum_{k=1}^{q} \Big[\cos\Big(\frac{\pi s}{2} - \frac{2\pi k p}{q}\Big)\Big] \zeta\Big(s, \frac{k}{q}\Big),$$

expanded around $s = 0$, expresses every $\gamma_n(p/q)$ through the jets $\zeta^{(j)}(0, k/q)$, $j \le n+1$. Taking the part odd under $p \mapsto q - p$ isolates the sine sums S_j and yields reflection formulas. Write $E(s) := \exp\big(-\Lambda_q s + \sum_{k\ge2} (-1)^k \zeta(k) s^k/k\big) = \sum_i e_i s^i$ for the prefactor expansion $(2\Gamma(s)(2\pi q)^{-s} = \frac{2}{s}\,E(s))$.

Theorem 8.33 (reflection, general mechanism)**.** *For $n \ge 1$ and $0 < p < q$,*

$$\gamma_n\Big(\frac{p}{q}\Big) - \gamma_n\Big(1 - \frac{p}{q}\Big) \;=\; 4\,n! \sum_{i+2l+1+j\,=\,n+1} e_i\, \frac{(-1)^l (\pi/2)^{2l+1}}{(2l+1)!}\, \frac{S_j(p/q)}{j!}\,.$$

Explicitly, with $\Lambda = \Lambda_q = \gamma + \ln 2\pi q$ (the letter distinguishes it from the Glaisher constant A):

$$\gamma_2\Big(\tfrac{p}{q}\Big) - \gamma_2\Big(1 - \tfrac{p}{q}\Big) = 2\pi S_2 - 4\pi\Lambda\, S_1 + \Big(2\Lambda^2 + \tfrac{\pi^2}{6}\Big)\pi S_0, \tag{8.46}$$

$$\gamma_3\Big(\tfrac{p}{q}\Big) - \gamma_3\Big(1 - \tfrac{p}{q}\Big) = 2\pi S_3 - 6\pi\Lambda\, S_2 + \Big(6\Lambda^2 + \tfrac{\pi^2}{2}\Big)\pi S_1 - \Big(2\Lambda^3 + \tfrac{\pi^2}{2}\Lambda + 4\zeta(3)\Big)\pi S_0, \tag{8.47}$$

$$\begin{aligned}\gamma_4\Big(\tfrac{p}{q}\Big) - \gamma_4\Big(1 - \tfrac{p}{q}\Big) &= 2\pi S_4 - 8\pi\Lambda\, S_3 + \big(12\Lambda^2 + \pi^2\big)\pi S_2 - \big(8\Lambda^3 + 2\pi^2\Lambda + 16\zeta(3)\big)\pi S_1 \\ &\quad + \Big(2\Lambda^4 + \pi^2\Lambda^2 + 16\Lambda\,\zeta(3) + \tfrac{19\pi^4}{120}\Big)\pi S_0.\end{aligned} \tag{8.48}$$

Proof. Split the cosine, $\cos(\pi s/2-\theta)=\cos(\pi s/2)\cos\theta+\sin(\pi s/2)\sin\theta$. Under $p\mapsto q-p$ the cosine sums are even and the sine sums odd, so the odd part of $\zeta(1-s,p/q)$ equals $\frac{2}{s}\,E(s)\sin(\pi s/2)\sum_j S_j\,s^j/j!$. By (8.40), $\zeta(1-s,p/q)=-\frac{1}{s}+\sum_n\gamma_n(p/q)\,s^n/n!$, whose odd part has s^n-coefficient $\frac{1}{2\,n!}\big[\gamma_n(p/q)-\gamma_n(1-p/q)\big]$. Matching coefficients of s^n (i.e. s^{n+1} of $E\cdot\sin\cdot\sum S_j s^j/j!$) gives the general formula; instantiating $e_0=1$, $e_1=-\Lambda$, $e_2=\frac{\Lambda^2}{2}+\frac{\pi^2}{12}$, $e_3=-\frac{\Lambda^3}{6}-\frac{\pi^2}{12}\Lambda-\frac{\zeta(3)}{3}$, $e_4=\frac{\Lambda^4}{24}+\frac{\pi^2}{24}\Lambda^2+\frac{\zeta(3)}{3}\Lambda+\frac{\pi^4}{288}+\frac{\pi^4}{360}$ yields (8.46)–(8.48). □

Formula (8.46) is eq. (89) of Blagouchine [6], where a footnote records that it also appears in unpublished work of D. Connon; (8.47) and (8.48) appear to be new. The same expansion, one order down, recovers Malmsten's reflection formula for γ_1 [6]; an elementary alternative proof of that $n=1$ case, via Kummer's Fourier expansion of $\ln\Gamma$ rather than Hurwitz's functional equation, is given by Kouba [30]. For $q=3$ and $q=4$ the sums collapse ($S_j(\frac13)=\frac{\sqrt3}{2}\big[\zeta^{(j)}(0,\frac13)-\zeta^{(j)}(0,\frac23)\big]$, $S_j(\frac14)=\zeta^{(j)}(0,\frac14)-\zeta^{(j)}(0,\frac34)$), so (8.46)–(8.48) can be *inverted* to express the odd jet differences through the Stieltjes differences $\gamma_n(p)-\gamma_n(1-p)$ — the key to Section 8.15.

8.14 Moments

Theorem 8.34 (Hurwitz transform and variants)**.** *For $m\ge1$ and $\operatorname{Re}s<1$,*

$$\int_0^1 x^m\,\zeta(s,x)\,dx=-\sum_{j=1}^{m}\frac{m!}{(m-j+1)!}\,\frac{\zeta(s-j)}{(s-1)(s-2)\cdots(s-j)},\tag{8.49}$$

and $\int_0^1\zeta(s,x)\,dx=0$. The same recursion with boundary terms $\zeta(\sigma,\frac12)=(2^\sigma-1)\zeta(\sigma)$ and $\zeta(\sigma,2)=\zeta(\sigma)-1$ gives the analogous closed forms for $\int_{1/2}^1$ and $\int_1^2$.

Proof. Since $\zeta(s,x)=-\partial_x\zeta(s-1,x)/(s-1)$, integration by parts gives $I_m(s)=-\zeta(s-1)/(s-1)+\frac{m}{s-1}I_{m-1}(s-1)$ for $m\ge1$. In the stated half-plane the factor x^m kills the lower boundary. For the base case $m=0$, the shift relation gives equal limits of $\zeta(s-1,x)$ at $x=0+$ and $x=1$ when $\operatorname{Re}s<1$; these generally nonzero values cancel, giving $I_0(s)=0$. Unrolling yields (8.49), which then extends meromorphically in s. □

For the unit interval, (8.49) is Theorem 3.7 of Espinosa–Moll [28]. Expanding (8.49) at $s=1+t$ against $\int_0^1 x^m\zeta(1+t,x)\,dx=\frac{1}{(m+1)t}+M_0-M_1t+\frac{M_2}{2}t^2-\cdots$, where $M_k=\int_0^1 x^m\gamma_k(x)\,dx$, reads off every Stieltjes moment. The expansion $1/[(t-1)\cdots(t-j+1)]=\frac{(-1)^{j-1}}{(j-1)!}\big(1+H_{j-1}t+\frac{H_{j-1}^2+H_{j-1}^{(2)}}{2}t^2+\cdots\big)$ gives the coefficients in closed form:

Theorem 8.35 (general first moment)**.** *For every integer $m\ge1$,*

$$\int_0^1 x^m\,\gamma_1(x)\,dx=\sum_{j=1}^{m}(-1)^{j-1}\binom{m}{j-1}\left[\frac{\zeta''(1-j)}{2}+H_{j-1}\,\zeta'(1-j)+\frac{H_{j-1}^2+H_{j-1}^{(2)}}{2}\,\zeta(1-j)\right].$$

The γ_n-moment carries the analogous formula with the order-$(n+1)$ jet leading and complete homogeneous symmetric functions of $1,\frac12,\dots,\frac{1}{j-1}$ as coefficients.

Combining Theorems 8.34 and 8.35 with the dictionary of Proposition 8.31 produces the following table. Items (8.50)–(8.53) are restated from the corpus that motivated this work; the remaining

evaluations appear to be new.

$$\int_{1/2}^{1}\gamma_1(x)\,dx=\frac{\gamma_1}{2}+\frac{\gamma^2}{4}-\frac{\pi^2}{48}+\frac{\ln^2 2}{2}-\frac{\ln^2\pi}{4}, \tag{8.50}$$

$$\int_{1}^{2}x\,\gamma_1(x)\,dx=1+\frac{\gamma_1}{2}+\frac{\gamma^2}{4}-\frac{\pi^2}{48}-\frac{\ln^2 2\pi}{4}, \tag{8.51}$$

$$\int_{0}^{1}x^2\,\gamma_1(x)\,dx=\frac{\gamma_1}{2}+\frac{\gamma^2}{3}-\frac{\pi^2}{36}-\frac{\ln^2 2\pi}{6}+\frac{\gamma}{6}\ln 2\pi-\frac{\zeta'(2)}{\pi^2}\,(\gamma+\ln 2\pi)+\frac{\zeta''(2)}{2\pi^2}, \tag{8.52}$$

$$\begin{aligned}\int_{1/2}^{1}x\,\gamma_1(x)\,dx=&\frac{\gamma_1}{2}+\frac{5\gamma^2}{16}-\frac{5\pi^2}{192}+\frac{\ln^2 2}{8}-\frac{\ln 2\,\ln\pi}{6}-\frac{3\ln^2\pi}{16}+\gamma\Big(\frac{\ln 2}{12}+\frac{\ln\pi}{8}\Big)\\&-\frac{\zeta'(2)}{4\pi^2}\,(3\gamma+2\ln 2+3\ln\pi)+\frac{3\,\zeta''(2)}{8\pi^2},\end{aligned} \tag{8.53}$$

$$\int_{0}^{1}x\,\gamma_1(x)\,dx=\frac{\zeta''(0)}{2}=\frac{\gamma^2}{4}+\frac{\gamma_1}{2}-\frac{\pi^2}{48}-\frac{\ln^2 2\pi}{4}, \tag{8.54}$$

$$\begin{aligned}\int_{0}^{1}x^3\,\gamma_1(x)\,dx=&\frac{3\gamma^2}{8}+\frac{\gamma_1}{2}-\frac{\pi^2}{32}+\frac{\gamma\ln 2\pi}{4}-\frac{\ln^2 2\pi}{8}-\frac{3\,\big(\zeta(3)+2\zeta'(2)\big)}{4\pi^2}\,(\gamma+\ln 2\pi)\\&+\frac{3\,\big(\zeta'(3)+\zeta''(2)\big)}{4\pi^2},\end{aligned} \tag{8.55}$$

$$\int_{1}^{2}x^2\,\gamma_1(x)\,dx=\frac{9}{4}+\frac{3\gamma_1}{2}+\frac{5\gamma^2}{6}-\frac{5\pi^2}{72}+\frac{\gamma\ln 2\pi}{6}-\frac{2\ln^2 2\pi}{3}-\frac{\zeta'(2)}{\pi^2}(\gamma+\ln 2\pi)+\frac{\zeta''(2)}{2\pi^2}, \tag{8.56}$$

$$\int_{0}^{1}x\,\gamma_2(x)\,dx=-\frac{\zeta'''(0)}{3}=\frac{\zeta(3)}{3}-\frac{\gamma^3}{3}-\gamma\gamma_1-\frac{\gamma_2}{2}-\frac{\gamma^2}{2}\ln 2\pi-\gamma_1\ln 2\pi+\frac{\pi^2}{24}\ln 2\pi+\frac{\ln^3 2\pi}{6}, \tag{8.57}$$

$$\int_{1}^{2}x\,\gamma_2(x)\,dx=\int_{0}^{1}x\,\gamma_2(x)\,dx-2,\qquad\int_{1/2}^{1}\gamma_2(x)\,dx=-\Gamma_2\big(\tfrac12\big). \tag{8.58}$$

At $m=3$ the constants $\zeta(3)$ and $\zeta'(3)$ enter through (8.45), and each further unit of m opens one more column of jets at the next negative integer; the second moment of γ_1 over $[1/2,1]$ and the higher γ_2-moments are equally mechanical but bulkier, and we omit them.

8.15 Explicit values of Γ_n

8.15.1 The first antiderivative

Theorem 8.36 (Γ_1 in closed form)**.** *For all $p>0$,*

$$\Gamma_1(p)=c+\tfrac12\,\zeta''(0,p),\qquad c:=-\tfrac12\zeta''(0)=\frac{\pi^2}{48}+\frac{\ln^2 2\pi}{4}-\frac{\gamma^2}{4}-\frac{\gamma_1}{2}.$$

Moreover $\int_0^1\Gamma_1(x)\,dx=c$, equivalently (8.54) (since $\int_0^1\zeta''(0,x)\,dx=0$ [9, 25]).

Theorem 8.37 (half-integral and even-sum values)**.**

$$\Gamma_1\big(\tfrac12\big) = \frac{\pi^2}{48} - \frac{\gamma^2}{4} - \frac{\gamma_1}{2} - \frac{\ln^2 2}{4} - \frac{\ln 2\,\ln 2\pi}{2} + \frac{\ln^2 2\pi}{4},$$

$$\Gamma_1\big(\tfrac13\big) + \Gamma_1\big(\tfrac23\big) = 2c - \frac{\ln^2 3}{4} - \frac{\ln 3\,\ln 2\pi}{2}, \qquad \Gamma_1\big(\tfrac14\big) + \Gamma_1\big(\tfrac34\big) = 2c - \frac{3\ln^2 2}{4} - \frac{\ln 2\,\ln 2\pi}{2},$$

$$\Gamma_1\big(\tfrac16\big) + \Gamma_1\big(\tfrac56\big) = 2c - \frac{\ln 2\,\ln 3}{2}.$$

Proof. $\zeta(s,\frac12) = (2^s-1)\zeta(s)$ gives $\zeta''(0,\frac12) = -\frac12\ln^2 2 - \ln 2\,\ln(2\pi)$; the multiplication identities $\zeta(s,\frac13)+\zeta(s,\frac23) = (3^s-1)\zeta(s)$, $\zeta(s,\frac14)+\zeta(s,\frac34) = (4^s-2^s)\zeta(s)$, $\zeta(s,\frac16)+\zeta(s,\frac56) = (6^s-3^s-2^s+1)\zeta(s)$, twice differentiated at $s=0$, give the even sums (for $q=6$ the $\zeta(0)$-coefficient $\ln^2 6 - \ln^2 3 - \ln^2 2 = 2\ln 2\ln 3$ and the $\zeta'(0)$-coefficient $\ln 6 - \ln 3 - \ln 2 = 0$ produce the strikingly short third identity). Apply Theorem 8.36. □

Theorem 8.38 (third and quarter values)**.**

$$\begin{aligned}\Gamma_1\big(\tfrac13\big) &= \frac{\pi^2}{72} - \frac{\ln^2 2\pi}{3} + \frac{\ln^2 3}{24} - \frac{2}{3}\ln 3\,\ln 2\pi - \frac{\gamma}{12}\big(4\gamma + 8\ln 2\pi - \ln 3\big) + \big(\gamma + \ln 6\pi\big)\ln\Gamma\big(\tfrac13\big)\\ &\quad - \frac{\gamma_1}{2} + \frac{\sqrt3}{12\pi}\Big(\gamma_2\big(\tfrac13\big) - \gamma_2\big(\tfrac23\big)\Big),\\ \Gamma_1\big(\tfrac14\big) &= \frac{\pi^2}{96} - \frac{3}{8}\ln^2\pi - \frac{9}{4}\ln 2\,\ln 2\pi - \frac{\gamma}{8}\big(3\gamma + 8\ln 2 + 6\ln\pi\big) + \big(\gamma + \ln 8\pi\big)\ln\Gamma\big(\tfrac14\big)\\ &\quad - \frac{\gamma_1}{2} + \frac{1}{8\pi}\Big(\gamma_2\big(\tfrac14\big) - \gamma_2\big(\tfrac34\big)\Big),\end{aligned}$$

and $\Gamma_1(\frac23)$, $\Gamma_1(\frac34)$ *follow from Theorem 8.37. Furthermore, with the abbreviation* $D_j := \zeta^{(j)}(0,\frac13) - \zeta^{(j)}(0,\frac23)$,

$$\Gamma_1\big(\tfrac16\big) - \Gamma_1\big(\tfrac56\big) = \frac{\ln 2\,\ln 18}{6} + \ln 2\Big(2\ln\Gamma\big(\tfrac13\big) + \frac{\ln 3}{6} - \ln 2 - \ln\pi\Big) + D_2,$$

which together with the even sum of Theorem 8.37 gives $\Gamma_1(\frac16)$ *and* $\Gamma_1(\frac56)$ *explicitly — in the* same *obstruction atom* $\gamma_2(\frac13) - \gamma_2(\frac23)$*, since inverting* (8.46) *at* $q=3$ *expresses* $D_2 = \frac{2}{\sqrt3}S_2(\frac13)$ *through it.*

Proof. By Theorem 8.36 everything reduces to the single jets $\zeta''(0,p)$ at $p = \frac13, \frac14, \frac16$. Their even parts are in Theorem 8.37; their odd parts are the inversions of (8.46) at $q = 3, 4$:

$$S_2 = \frac{1}{2\pi}\Big[\gamma_2\big(\tfrac pq\big) - \gamma_2\big(1-\tfrac pq\big) + 4\pi\Lambda\,S_1 - \Big(2\Lambda^2 + \tfrac{\pi^2}{6}\Big)\pi S_0\Big],$$

with $S_1(\frac13) = \frac{\sqrt3}{2}\big(2\ln\Gamma(\frac13) - \ln\frac{2\pi}{\sqrt3}\big)$, $S_0(\frac13) = \frac{1}{2\sqrt3}$, $S_1(\frac14) = 2\ln\Gamma(\frac14) - \ln(\pi\sqrt2)$, $S_0(\frac14) = \frac12$. For $q=6$, the Dirichlet character χ_{-3} gives $\zeta(s,\frac16) - \zeta(s,\frac56) = (6^s+3^s)\,L(s,\chi_{-3})$ while $\zeta(s,\frac13) - \zeta(s,\frac23) = 3^s\,L(s,\chi_{-3})$; eliminating the jets of $L(s,\chi_{-3})$ at $s=0$ ($L(0,\chi_{-3}) = \frac13$, $L'(0,\chi_{-3}) = 2\ln\Gamma(\frac13) + \frac{\ln 3}{6} - \ln 2 - \ln\pi$) yields the displayed transport identity. □

8.15.2 The second and third antiderivatives

Theorem 8.39 (Γ_2 values)**.**

$$\Gamma_2\big(\tfrac12\big) = \frac{\gamma_2}{2} + \frac{\gamma^3}{3} - \frac{\zeta(3)}{3} + \ln^3 2 - \frac{\ln^3\pi}{6} + \frac{\ln\pi}{24}\big(12\gamma^2 - \pi^2 + 24\ln^2 2\big) + (\gamma + \ln\pi)\,\gamma_1,$$

and at the third and quarter points, with $Z_3(\frac{1}{q}) := \zeta'''(0,\frac{1}{q}) + \zeta'''(0,1-\frac{1}{q})$ *and* $D_3(\frac{1}{q}) := \zeta'''(0,\frac{1}{q}) - \zeta'''(0,1-\frac{1}{q})$,

$$\Gamma_2\big(\tfrac{1}{q}\big) = -\frac{1}{3}\Big[\frac{Z_3 + D_3}{2} - \zeta'''(0)\Big], \qquad q \in \{3,4\},$$

where the even parts are elementary,

$$Z_3\big(\tfrac{1}{3}\big) = -\tfrac{\ln^3 3}{2} - \tfrac{3\ln^2 3\,\ln 2\pi}{2} + 3\ln 3\,\zeta''(0), \qquad Z_3\big(\tfrac{1}{4}\big) = \partial_s^3\big[(4^s - 2^s)\zeta(s)\big]_{s=0},$$

and the odd parts are the inversions of (8.47)*:* $D_3 = \frac{2}{\sqrt{3}} S_3$ *at* $q=3$, $D_3 = S_3$ *at* $q=4$, *with*

$$S_3 = \frac{1}{2\pi}\Big[\gamma_3\big(\tfrac{1}{q}\big) - \gamma_3\big(1-\tfrac{1}{q}\big) + 6\pi\Lambda\, S_2 - \Big(6\Lambda^2 + \tfrac{\pi^2}{2}\Big)\pi S_1 + \Big(2\Lambda^3 + \tfrac{\pi^2}{2}\Lambda + 4\zeta(3)\Big)\pi S_0\Big]$$

and S_2 *as in the proof of Theorem 8.38. Thus* $\Gamma_2(\frac{1}{3})$ *and* $\Gamma_2(\frac{1}{4})$ *are explicit in the constants* $\big\{\gamma, \gamma_1, \gamma_2, \zeta(3), \ln 2, \ln 3, \ln\pi, \ln\Gamma(\frac{1}{q}),\ \gamma_2(\frac{1}{q}) - \gamma_2(1-\frac{1}{q}),\ \gamma_3(\frac{1}{q}) - \gamma_3(1-\frac{1}{q})\big\}$, *the two Stieltjes differences being the only non-classical ingredients.*

Proof. $\Gamma_2(p) = -\frac{1}{3}[\zeta'''(0,p) - \zeta'''(0)]$ by Theorem 8.25. At $p = \frac{1}{2}$ use $\zeta'''(0,\frac{1}{2}) = \partial_s^3[(2^s - 1)\zeta(s)]_{s=0} = \ln^3 2\,\zeta(0) + 3\ln^2 2\,\zeta'(0) + 3\ln 2\,\zeta''(0)$ and Proposition 8.28; at $p = \frac{1}{3}, \frac{1}{4}$ split into even and odd parts as displayed. □

Corollary 8.40 (Γ_2 even sums)**.** *With* $L = \ln 2\pi$,

$$\begin{aligned}
\Gamma_2\big(\tfrac{1}{3}\big) + \Gamma_2\big(\tfrac{2}{3}\big) &= \tfrac{2}{3}\,\zeta'''(0) - \ln 3\,\zeta''(0) + \tfrac{L}{2}\ln^2 3 + \tfrac{\ln^3 3}{6},\\
\Gamma_2\big(\tfrac{1}{4}\big) + \Gamma_2\big(\tfrac{3}{4}\big) &= \tfrac{2}{3}\,\zeta'''(0) - \ln 2\,\zeta''(0) + \tfrac{3L}{2}\ln^2 2 + \tfrac{7\ln^3 2}{6},\\
\Gamma_2\big(\tfrac{1}{6}\big) + \Gamma_2\big(\tfrac{5}{6}\big) &= \tfrac{2}{3}\,\zeta'''(0) + L\,\ln 2\,\ln 3 + \tfrac{\ln 2\,\ln 3\,\ln 6}{2},
\end{aligned}$$

so that $\Gamma_2(\frac{2}{3})$, $\Gamma_2(\frac{3}{4})$ *follow from Theorem 8.39; as in Theorem 8.37, the* $\zeta''(0)$*-term vanishes at* $q=6$ *because* $\ln 6 - \ln 3 - \ln 2 = 0$. *(Generally, with the inclusion–exclusion combination* $\zeta(s,\frac{1}{q}) + \zeta(s,1-\frac{1}{q}) = \big(\sum_i c_i m_i^s\big)\zeta(s)$, *the even sum equals* $\frac{2}{3}\zeta'''(0) + \frac{1}{6}\sum_i c_i \ln^3 m_i + \frac{L}{2}\sum_i c_i \ln^2 m_i - \zeta''(0)\sum_i c_i \ln m_i$.*)*

Theorem 8.41 (Γ_3 values)**.**

$$\begin{aligned}
\Gamma_3\big(\tfrac{1}{2}\big) = {}& -\frac{7\ln^4 2}{4} + \frac{\ln^4\pi}{8} - \frac{3\gamma^4}{8} - \frac{\gamma_3}{2} + \frac{19\pi^4}{1920} + \gamma_1\Big(3\ln^2 2 - \frac{3\gamma^2}{2} - \frac{3\ln^2\pi}{2} - \frac{\pi^2}{8}\Big)\\
& - \frac{3\gamma_2}{2}(\gamma + \ln\pi) + \zeta(3)\ln\pi + \gamma^2\Big(\frac{3\ln^2 2}{2} - \frac{3\ln^2\pi}{4} - \frac{\pi^2}{16}\Big) - \gamma^3\ln\pi - 3\gamma\gamma_1\ln\pi\\
& - 3\ln\pi\,\ln^3 2 - \frac{3\ln^2 2\,\ln^2\pi}{2} - \frac{\pi^2\ln^2 2}{8} + \frac{\pi^2\ln^2\pi}{16},
\end{aligned}$$

and $\Gamma_3(\frac{1}{3})$, $\Gamma_3(\frac{1}{4})$ *are given by* $\Gamma_3(\frac{1}{q}) = \frac{1}{4}[\zeta^{(4)}(0,\frac{1}{q}) - \zeta^{(4)}(0)]$ *with the even part* $\partial_s^4[(q\text{-orbit})\,\zeta(s)]_{s=0}$ *as in Theorem 8.39 and the odd part inverted from* (8.48)*; the resulting expressions are explicit in the same vocabulary augmented by* $\gamma_4(\frac{1}{q}) - \gamma_4(1-\frac{1}{q})$.

8.15.3 The fourth antiderivative at $1/2$

Theorem 8.42 ($\Gamma_4(1/2)$)**.**

$$\begin{aligned}\Gamma_4\tfrac{1}{2} = {}& \frac{\gamma_4}{2} + 3\ln^5 2 - \frac{12\,\zeta(5)}{5} - \frac{\ln^5\pi}{10} + \frac{2\gamma^5}{5} + \frac{3\gamma^4\ln\pi}{2} \\ & + \gamma_1\Big(2\gamma^3 - 12\ln^3 2 + 2\ln^3\pi + 4\zeta(3) + \frac{\pi^2\ln\pi}{2} - 12\ln\pi\,\ln^2 2 + 6\gamma^2\ln\pi\Big) \\ & + \gamma\gamma_1\Big(\frac{\pi^2}{2} - 12\ln^2 2 + 6\ln^2\pi\Big) + \gamma_2\Big(3\gamma^2 - 6\ln^2 2 + 3\ln^2\pi + \frac{\pi^2}{4} + 6\gamma\ln\pi\Big) \\ & + 2\gamma_3\,(\gamma + \ln\pi) + \zeta(3)\Big(4\ln^2 2 - 2\ln^2\pi - \frac{\pi^2}{6}\Big) \\ & + \gamma^2\Big(\ln^3\pi - 6\ln^3 2 + 2\zeta(3) - 6\ln\pi\,\ln^2 2 + \frac{\pi^2\ln\pi}{4}\Big) + \gamma^3\Big(2\ln^2\pi - 4\ln^2 2 + \frac{\pi^2}{6}\Big) \\ & + \frac{\pi^2\ln^3 2}{2} + 2\ln^2 2\,\ln^3\pi + 6\ln^3 2\,\ln^2\pi + 7\ln\pi\,\ln^4 2 \\ & - \frac{19\pi^4\ln\pi}{480} - \frac{\pi^2\ln^3\pi}{12} + \frac{\pi^2\ln\pi\,\ln^2 2}{2}.\end{aligned}$$

Proof. $\Gamma_4(\frac{1}{2}) = -\frac{1}{5}[\zeta^{(5)}(0,\frac{1}{2}) - \zeta^{(5)}(0)]$, with

$$\zeta^{(5)}\big(0,\tfrac{1}{2}\big) = \partial_s^5\big[(2^s-1)\zeta(s)\big]_{s=0} = \sum_{k=0}^{4}\binom{5}{k}\ln^{5-k}2\;\zeta^{(k)}(0)$$

(the $k=5$ term carries the factor $2^0 - 1 = 0$) and Proposition 8.29. □

The pattern of Theorems 8.39–8.42 is general: each rung Γ_n adds the next Stieltjes constant γ_n and the functional-equation zeta terms, and one degree to the logarithmic monomials at $p = \frac{1}{2}$; at $p = \frac{1}{3}, \frac{1}{4}$ it adds the single new obstruction atom $\gamma_{n+1}(p) - \gamma_{n+1}(1-p)$, delivered by the order-$(n+1)$ reflection formula of Theorem 8.33. At denominators such as $5, 8, 12$ the even character decomposition introduces additional coordinates (the multiplication identities no longer close over the principal character), and the values $\Gamma_n(p/q)$ additionally involve the even-character combinations of $\zeta^{(n+1)}(0,\cdot)$ — the obstruction layer studied, at $n=1$, in [6].

8.16 The negative-order polygamma floor

The ladder of Section 8.11 integrates the coefficient functions of the Laurent expansion at $s=1$ in their argument. The same Hurwitz-jet calculus has a second, equally natural direction, and this section — a second main act rather than an appendix — follows it downward. Integrating in the *argument* at the s-jet floor produces the polygamma functions of negative integer order ("negapolygamma", a term Adamchik [18] credits to Gosper): for real $z > 0$,

$$\psi^{(-1)}(z) = \ln\Gamma(z), \qquad \psi^{(-n)}(z) = \int_0^z \psi^{(-(n-1))}(t)\,dt = \frac{1}{(n-2)!}\int_0^z (z-t)^{n-2}\ln\Gamma(t)\,dt,$$

which is precisely Mathematica's `PolyGamma[-n, z]` (all identities below are stated on $0 < z < 1$, where the rational arguments live, and extend to $z > 0$ by the same formulas wherever both sides are defined). Everything built for the Stieltjes side transfers wholesale: the ladder identity has a one-line analogue, the reflection mechanism of Section 8.13 reappears at the jet floor with bare

Clausen values in place of the sine sums S_j, the obstruction constants again organize along Dirichlet characters — and the bridge identities for those constants close a circle the Stieltjes side leaves open. The tools: a one-line ladder, reflection and multiplication laws, bridge identities, and explicit rational-argument values.

8.16.1 The ladder

Theorem 8.43. *For $n \geq 2$ and $z > 0$,*

$$\psi^{(-n)}(z) = \frac{\zeta'(1-n,z) - \zeta'(1-n)}{(n-1)!} + Q_n(z),$$

where Q_n is an explicit polynomial with coefficients in $\mathbb{Q}\big[\ln 2\pi,\, \zeta'(-1), \ldots, \zeta'(2-n)\big]$:

$$\begin{aligned}
Q_2(z) &= \tfrac{z}{2}\ln 2\pi + \tfrac{z(1-z)}{2},\\
Q_3(z) &= \tfrac{z^2}{4}\ln 2\pi - \tfrac{z^3}{4} + \tfrac{3z^2}{8} - \tfrac{z}{24} - z\,\zeta'(-1),\\
Q_4(z) &= \tfrac{z^3}{12}\ln 2\pi - \tfrac{11z^4}{144} + \tfrac{11z^3}{72} - \tfrac{5z^2}{144} - \tfrac{z^2}{2}\zeta'(-1) - \tfrac{z}{2}\zeta'(-2),
\end{aligned}$$

and in general Q_n has leading term $-H_{n-1}z^n/n!$ and $\ln 2\pi$-coefficient $z^{n-1}/(2\,(n-1)!)$.

Proof. $\int_0^z \zeta(s,t)\,dt = -[\zeta(s-1,z) - \zeta(s-1)]/(s-1)$ for $\operatorname{Re} s < 1$; differentiating at $s = -k$,

$$\int_0^z \zeta'(-k,t)\,dt = \frac{\zeta'(-k-1,z) - \zeta'(-k-1)}{k+1} + \frac{\zeta(-k-1,z) - \zeta(-k-1)}{(k+1)^2},$$

with $\zeta(-m,z) = -B_{m+1}(z)/(m+1)$. Iterate from $\psi^{(-1)} = \zeta'(0,z) + \frac{1}{2}\ln 2\pi$: each step shifts the single ζ'-atom by one and deposits Bernoulli and constant terms into the polynomial. □

An equivalent expansion is Adamchik's Proposition 2 [18]; the spelling above isolates the one z-dependent transcendental.

8.16.2 Reflection

Theorem 8.44 (reflection at the jet floor)**.** *For every integer $k \geq 0$ and $0 < z < 1$,*

$$\zeta'(-k,z) + (-1)^k\,\zeta'(-k,1-z) = (-1)^{\lfloor k/2\rfloor}\,\frac{k!}{(2\pi)^k}\,\operatorname{Cl}_{k+1}(2\pi z),$$

where $\operatorname{Cl}_{2m}(\theta) = \operatorname{Im}\operatorname{Li}_{2m}(e^{i\theta})$, $\operatorname{Cl}_{2m+1}(\theta) = \operatorname{Re}\operatorname{Li}_{2m+1}(e^{i\theta})$.

Proof. In Hurwitz's formula, written as $\zeta(1-s,z) = \frac{2\Gamma(s)}{(2\pi)^s}[\cos(\frac{\pi s}{2})C_s(z) + \sin(\frac{\pi s}{2})S_s(z)]$ with C_s, S_s the cosine/sine Dirichlet series, the combination even (odd) under $z \mapsto 1-z$ keeps only the C (S) term. At the integer $s = k+1$, exactly one of $\cos\frac{\pi s}{2}$, $\sin\frac{\pi s}{2}$ vanishes, and it is the one attached to the parity-matching combination; differentiating in s there, the product rule retains only the term in which d/ds falls on the vanishing trigonometric factor, so a single *bare* Clausen value survives and no derivative of the Dirichlet series enters. Collecting constants gives the display. □

For $k \geq 1$ this is the real part of Adamchik's Proposition 1 [18], stated there for every positive integer order with the Bernoulli term as the imaginary part (equivalently Miller–Adamchik [31], eq. (21)); the bare-Clausen real spelling, the $k = 0$ inclusion, and the mechanism note are the reformulation here. $k = 0$ is Kummer-flavoured ($\operatorname{Cl}_1(2\pi z) = -\ln(2\sin\pi z)$). The parity-*opposite*

combination does not close: it produces the series $\sum \ln n \,\{\cos \,|\, \sin\}(2\pi n z)/n^{k+1}$ — the generalized-Glaisher atoms of §8.16.4.

Propagating Theorem 8.44 through the ladder yields reflection formulas at the $\psi^{(-n)}$-level — a relation which Espinosa and Moll [27] record as not known to them ("*We have been unable to find a generalization of the other well-known functional relation for the polygamma function*"):

Theorem 8.45 (negapolygamma reflection)**.**

$$\begin{aligned}
\psi^{(-2)}(z) - \psi^{(-2)}(1-z) &= \frac{\mathrm{Cl}_2(2\pi z)}{2\pi} + \Big(z - \tfrac{1}{2}\Big)\ln 2\pi,\\
\psi^{(-3)}(z) + \psi^{(-3)}(1-z) &= -\frac{\mathrm{Cl}_3(2\pi z)}{4\pi^2} + \Big(\tfrac{z^2}{2} - \tfrac{z}{2} + \tfrac{1}{4}\Big)\ln 2\pi - \zeta'(-1) - \zeta'(-2) + \tfrac{1}{12},\\
\psi^{(-4)}(z) - \psi^{(-4)}(1-z) &= -\frac{\mathrm{Cl}_4(2\pi z)}{8\pi^3} + \Big(\tfrac{z^3}{6} - \tfrac{z^2}{4} + \tfrac{z}{4} - \tfrac{1}{12}\Big)\ln 2\pi\\
&\quad + \Big(\tfrac{1}{2} - z\Big)\big(\zeta'(-1) + \zeta'(-2)\big) + \tfrac{z}{12} - \tfrac{1}{24},\\
\psi^{(-5)}(z) + \psi^{(-5)}(1-z) &= \frac{\mathrm{Cl}_5(2\pi z)}{16\pi^4} + \Big(\tfrac{z^4}{24} - \tfrac{z^3}{12} + \tfrac{z^2}{8} - \tfrac{z}{12} + \tfrac{1}{48}\Big)\ln 2\pi\\
&\quad + \Big(\tfrac{z - z^2}{2}\Big)\big(\zeta'(-1) + \zeta'(-2)\big)\\
&\quad - \tfrac{\zeta'(-1)}{6} - \tfrac{\zeta'(-2)}{4} - \tfrac{\zeta'(-3)}{6} - \tfrac{\zeta'(-4)}{12} + \tfrac{z^2 - z}{24} + \tfrac{49}{4320}.
\end{aligned}$$

8.16.3 Multiplication

Differentiating the Hurwitz distribution relation at $s = -k$ gives, for all $k \geq 0$, $m \geq 2$,

$$\sum_{j=0}^{m-1} \zeta'\Big(-k,\, z + \frac{j}{m}\Big) = m^{-k}\Big[\zeta'(-k, mz) - \ln m\, \frac{B_{k+1}(mz)}{k+1}\Big] \tag{8.59}$$

($k = 0$ is Gauss's multiplication theorem for $\ln\Gamma$), and through the ladder,

$$\sum_{j=0}^{m-1} \psi^{(-n)}\Big(z + \frac{j}{m}\Big) = m^{1-n}\Big[\psi^{(-n)}(mz) - \ln m\, \frac{B_n(mz)}{n!}\Big] + \rho_{n,m}(z)$$

with elementary $\rho_{n,m}$; for $n = 2$,

$$\rho_{2,m}(z) = \frac{(m-1)(2z+1)}{4}\ln 2\pi + \frac{m^2 - 1}{12m} - \frac{m^2 - 1}{m}\,\zeta'(-1),$$

so only the *rational* part of the remainder is $(m^2 - 1)/(12m)$. The ψ-level law is, in the "balanced" convention of their negapolygamma [29], Theorem 3.2 of Espinosa–Moll [27]; the form above is its transcription to the Wolfram convention.

8.16.4 Bridges

At rational arguments the layers $\zeta'(-k, p/q)$ decompose along Dirichlet characters, and two laws organize the resulting constants.

Proposition 8.46 (trivial-zero bridges)**.** *Let χ be primitive or principal, $k \geq 1$. If $\chi(-1) = (-1)^k$ (so that $L(-k,\chi) = 0$, since $B_{k+1}(\chi) \neq 0$ exactly when $\chi(-1) = (-1)^{k+1}$), then $L'(-k,\chi)$ is an elementary multiple of $L(k+1,\bar{\chi})$. For the principal character and χ_{-4}:*

$$\zeta'(-2k) = \frac{(-1)^k (2k)!}{2(2\pi)^{2k}}\,\zeta(2k+1), \qquad \beta'(1-2k) = (-1)^{k+1}(2k-1)!\,\frac{2^{2k-1}}{\pi^{2k-1}}\,\beta(2k),$$

so in particular $\beta'(-1) = 2G/\pi$ (G Catalan's constant) and $\beta'(-3) = -48\,\beta(4)/\pi^3$.

Theorem 8.47 (harmonic-number bridge)**.** *Let χ be a primitive character mod q with $\chi(-1) = (-1)^{k+1}$ (so that $L(-k,\chi) \neq 0$; for χ principal read ζ and require k odd). Then*

$$\frac{L'(k+1,\chi)}{L(k+1,\chi)} + \overline{\left(\frac{L'(-k,\chi)}{L(-k,\chi)}\right)} = \gamma + \ln\frac{2\pi}{q} - H_k.$$

For real χ the conjugation bar is invisible; for complex χ it is essential — equivalently, the bridge pairs χ at $s = k+1$ with $\bar{\chi}$ at $s = -k$, and the un-conjugated sum has the purely imaginary defect $2i\,\mathrm{Im}[L'(-k,\chi)/L(-k,\chi)]$.

Proof. Write the completed function $\Lambda(s) = (q/\pi)^{(s+\delta)/2}\,\Gamma\big(\frac{s+\delta}{2}\big)L(s,\chi)$ with $\delta = (1-\chi(-1))/2$, so $\Lambda(s) = \varepsilon\overline{\Lambda}(1-\bar{s})$, whose logarithmic derivative at real s gives $\Lambda'/\Lambda(s) = -\overline{\Lambda'/\Lambda(1-s)}$; at $s = k+1$,

$$\frac{L'(k+1,\chi)}{L(k+1,\chi)} + \overline{\left(\frac{L'(-k,\chi)}{L(-k,\chi)}\right)} = -\ln\frac{q}{\pi} - \frac{1}{2}\,\psi\Big(\frac{k+1+\delta}{2}\Big) - \frac{1}{2}\,\psi\Big(\frac{\delta-k}{2}\Big).$$

Since $\chi(-1) = (-1)^{k+1}$ means $\delta \equiv k+1 \pmod 2$, the argument $(\delta-k)/2$ is a (possibly negative) half-integer, where $\cot \pi x = 0$; the reflection formula thus gives $\psi\big(\frac{\delta-k}{2}\big) = \psi\big(\frac{k+2-\delta}{2}\big)$, and $\big\{\frac{k+1+\delta}{2}, \frac{k+2-\delta}{2}\big\}$ is a duplication pair $\{x, x+\frac{1}{2}\}$ with $2x = k+1$. Hence the two digammas sum to $2\psi(k+1) - 2\ln 2 = 2(H_k - \gamma) - 2\ln 2$, and the display follows. □

Consequently, at each level $-k$ every character direction of the grid $\{\zeta'(-k, j/q)\}$ either reduces (trivial-zero bridge: to the $\zeta(k+1)$/Catalan-family values) or carries exactly one new constant (the parity-opposite L'), the harmonic-number bridge converting freely between its $s = k+1$ and $s = -k$ spellings. Exact rational row-reduction of the full known-relation lattice (all reflection rows and all-divisor multiplication rows) makes this quantitative:

Corollary 8.48 (Uniform negative-jet ranks)**.** *For all $q \geq 3$ and $k \geq 1$, the formal negative-jet grid modulo the specified reflection, distribution and endpoint rows has residual dimension $\varphi(q)/2 - 1$ for odd k, and $\varphi(q)/2$ for even k.*

The exact character proof is given in Corollary 9.17. The original 112-cell finite experiment supported the result; the polynomial normal form now proves it for every denominator and order. This does not assert arithmetic independence of the remaining L-derivative coordinates.

8.16.5 Rational values

Theorem 8.49 (the half-argument tower)**.** *For every $n \geq 2$,*

$$\psi^{(-n)}\big(\tfrac{1}{2}\big) = \frac{(2^{1-n}-2)\,\zeta'(1-n) - 2^{1-n}\ln 2 \cdot B_n/n}{(n-1)!} + Q_n\big(\tfrac{1}{2}\big);$$

in lowest cases

$$\psi^{(-2)}\big(\tfrac{1}{2}\big) = \frac{3}{2}\ln \mathsf{A} + \frac{5\ln 2}{24} + \frac{\ln \pi}{4} \quad \textit{(classical)}, \qquad \psi^{(-3)}\big(\tfrac{1}{2}\big) = \frac{7\,\zeta(3)}{32\pi^2} + \frac{\ln \mathsf{A}}{2} + \frac{\ln 2\pi}{16}.$$

Theorem 8.50 (third- and quarter-argument values at $n = 3, 4$)**.**

$$\psi^{(-3)}\big(\tfrac{1}{4}\big) = \frac{35\,\zeta(3)}{256\pi^2} + \frac{\beta'(3)}{4\pi^3} + \frac{\ln \mathsf{A}}{4} + \frac{\ln 2\pi}{128} - \frac{\gamma}{128},$$
$$\psi^{(-3)}\big(\tfrac{1}{3}\big) = \frac{13\,\zeta(3)}{72\pi^2} + \frac{\sqrt{3}\,L'(3,\chi_{-3})}{8\pi^3} + \frac{\ln \mathsf{A}}{3} + \frac{7\ln 2\pi}{324} - \frac{\gamma}{162},$$

via the exact spellings $\beta'(-2) = \frac{1}{4}\big(3 - 2\gamma - 2\ln\frac{\pi}{2}\big) + 16\beta'(3)/\pi^3$ *and* $L'(-2,\chi_{-3}) = \frac{2}{9}\big(\ln\frac{3}{2\pi} + \frac{3}{2} - \gamma\big) + \frac{9\sqrt{3}}{2\pi^3}L'(3,\chi_{-3})$*; and at the next level the parity alternates, so that* $\psi^{(-4)}(\frac{1}{4})$ *is explicit with* no *new constant:*

$$\psi^{(-4)}\big(\tfrac{1}{4}\big) = \frac{\zeta'(-3,\frac{1}{4}) - \zeta'(-3)}{6} + Q_4\big(\tfrac{1}{4}\big), \qquad \zeta'\big(-3,\tfrac{1}{4}\big) = \frac{Z + D}{2}$$

with the even part $Z = (4^{-3} - 2^{-3})\zeta'(-3) + (4^{-3}\ln 4 - 2^{-3}\ln 2)\zeta(-3)$ *and the odd part* $D = -48\,\beta(4)/(64\pi^3)$ *from Proposition 8.46.*

Remark 8.51. $\psi^{(-3)}(\frac{1}{2})$ appears in Adamchik [18]; the flattened quarter- and third-values appear to be new — notably, Mathematica's `FunctionExpand` reduces `PolyGamma[-3, 1/2]` fully but leaves the jet $\zeta'(-2,\frac{1}{4})$ unevaluated inside `PolyGamma[-3, 1/4]`: the atom $\beta'(3)$ is exactly what is missing. The same atom resolves a naive integer-relation search over $\{\zeta(3)/\pi^2, \ln \mathsf{A}, \text{logs}\}$, which must fail for this value.

Corollary 8.52 (reflection and transport companions)**.** *With the same atoms as Theorem 8.50,*

$$\psi^{(-3)}\big(\tfrac{2}{3}\big) = \frac{13\,\zeta(3)}{72\pi^2} - \frac{\sqrt{3}\,L'(3,\chi_{-3})}{8\pi^3} + \frac{2\ln \mathsf{A}}{3} + \frac{19\ln 2\pi}{162} + \frac{\gamma}{162},$$
$$\psi^{(-3)}\big(\tfrac{3}{4}\big) = \frac{35\,\zeta(3)}{256\pi^2} - \frac{\beta'(3)}{4\pi^3} + \frac{3\ln \mathsf{A}}{4} + \frac{19\ln 2\pi}{128} + \frac{\gamma}{128},$$
$$\psi^{(-3)}\big(\tfrac{1}{6}\big) = \frac{\zeta(3)}{12\pi^2} + \frac{5\sqrt{3}\,L'(3,\chi_{-3})}{32\pi^3} + \frac{\ln \mathsf{A}}{6} - \frac{\ln 2\pi}{1296} - \frac{\ln 2}{648} - \frac{5\gamma}{648},$$
$$\psi^{(-3)}\big(\tfrac{5}{6}\big) = \frac{\zeta(3)}{12\pi^2} - \frac{5\sqrt{3}\,L'(3,\chi_{-3})}{32\pi^3} + \frac{5\ln \mathsf{A}}{6} + \frac{235\ln 2\pi}{1296} + \frac{\ln 2}{648} + \frac{5\gamma}{648}.$$

Proof. The first two from Theorem 8.45 with $\mathrm{Cl}_3(2\pi/3) = -\frac{4}{9}\zeta(3)$ and $\mathrm{Cl}_3(\pi/2) = -\frac{3}{32}\zeta(3)$; the sixths from the transport $\zeta(s,\frac{1}{6}) - \zeta(s,\frac{5}{6}) = (6^s + 3^s)\,L(s,\chi_{-3})$ differentiated at $s = -2$, the even part being $\zeta'(-2,\frac{1}{6}) + \zeta'(-2,\frac{5}{6}) = \frac{2}{3}\zeta'(-2)$ by inclusion–exclusion. The entire sixth family thus rides on the one atom $L'(3,\chi_{-3})$. □

8.16.6 Denominators 5, 8, 12: quadratic and quartic atoms

The denominators 5, 8, 12 are the first where Proposition 8.48 forces genuinely new constants beyond the χ_{-3}/χ_{-4} families: at $k = 1$ each grid carries one new atom (the even real quadratic direction), and at $k = 2$, $q = 5$ the residual space is two-dimensional (a conjugate quartic pair).

Theorem 8.53 (the $k = 1$ grids)**.** *Write* $Z_1 = \zeta'(-1)$ *and let* χ_q *be the even real quadratic primitive character mod* $q \in \{5, 8, 12\}$ *(Legendre mod* 5*; Kronecker* $(8|\cdot)$*,* $(12|\cdot)$*), with the exact*

values $L(-1,\chi_5)=-\frac{2}{5}$, $L(-1,\chi_8)=-1$, $L(-1,\chi_{12})=-2$. *Then for* $\gcd(p,q)=1$,

$$\zeta'\big(-1,\tfrac{p}{5}\big)=-\frac{Z_1}{5}+\frac{\mathrm{Cl}_2(2\pi p/5)}{4\pi}+\chi_5(p)\Big[\frac{L'(-1,\chi_5)}{20}-\frac{\ln 5}{50}\Big]-\frac{\ln 5}{240},$$
$$\zeta'\big(-1,\tfrac{p}{8}\big)=-\frac{Z_1}{32}+\frac{\mathrm{Cl}_2(\pi p/4)}{4\pi}+\chi_8(p)\Big[\frac{L'(-1,\chi_8)}{32}-\frac{3\ln 2}{32}\Big]+\frac{\ln 2}{384},$$
$$\zeta'\big(-1,\tfrac{p}{12}\big)=\frac{Z_1}{24}+\frac{\mathrm{Cl}_2(\pi p/6)}{4\pi}+\chi_{12}(p)\Big[\frac{L'(-1,\chi_{12})}{48}-\frac{\ln 2}{12}-\frac{\ln 3}{24}\Big]+\frac{\ln 3}{576},$$

whence $\psi^{(-2)}(p/q)=\zeta'(-1,p/q)-Z_1+Q_2(p/q)$ *through the ladder; for instance*

$$\psi^{(-2)}\big(\tfrac{1}{5}\big)=-\frac{6Z_1}{5}+\frac{\mathrm{Cl}_2(2\pi/5)}{4\pi}+\frac{L'(-1,\chi_5)}{20}-\frac{29\ln 5}{1200}+\frac{\ln 2\pi}{10}+\frac{2}{25}.$$

Proof. On each grid the two reflection rows (Theorem 8.44), the multiplication rows (8.59) for every divisor (non-coprime entries reduced through the sub-grids $q\in\{2,3,4,6\}$), and the character row $\sum_p \chi_q(p)\,\zeta'(-1,\frac{p}{q})=\frac{1}{q}\big[L'(-1,\chi_q)+\ln q\,L(-1,\chi_q)\big]$ form a system of full rank $\varphi(q)=4$; solving it and reducing the composite-argument Clausen values ($\mathrm{Cl}_2(\frac{3\pi}{4})=\mathrm{Cl}_2(\frac{\pi}{4})-\frac{G}{2}$, $\mathrm{Cl}_2(\frac{\pi}{6})=\frac{2}{3}G+\frac{3}{8}\,\mathrm{Cl}_2(\frac{2\pi}{3})$) gives the display. For the composite q the known rows are one short of independent — the single dependency is itself a Clausen duplication law in disguise (at $q=8$: $\mathrm{refl}(\frac{1}{8})-\mathrm{refl}(\frac{3}{8})=m_2(\frac{1}{8})-m_2(\frac{3}{8})$ is equivalent to $\mathrm{Cl}_2(\frac{3\pi}{4})=\mathrm{Cl}_2(\frac{\pi}{4})-\frac{G}{2}$) — and the quadratic character row supplies the missing dimension, exactly as Proposition 8.48 predicts ($r_{q,1}=1$ for $q\in\{5,8,12\}$). □

Theorem 8.54 (the quartic floor: $k=2$, $q=5$)**.** *Let* χ *be the odd quartic character mod* 5 *with* $\chi(2)=i$, *for which* $L(-2,\chi)=-\frac{4+2i}{5}$ *exactly, and write* $L'(-2,\chi)=R+iI$. *Then*

$$\zeta'\big(-2,\tfrac{1}{5}\big),\ \zeta'\big(-2,\tfrac{4}{5}\big)=-\frac{\mathrm{Cl}_3(2\pi/5)}{4\pi^2}\pm\Big[\frac{R}{50}-\frac{2\ln 5}{125}\Big],$$
$$\zeta'\big(-2,\tfrac{2}{5}\big),\ \zeta'\big(-2,\tfrac{3}{5}\big)=-\frac{\mathrm{Cl}_3(4\pi/5)}{4\pi^2}\pm\Big[\frac{I}{50}-\frac{\ln 5}{125}\Big],$$

and, flattening through the ladder with $\mathrm{Cl}_3(\frac{2\pi}{5}),\mathrm{Cl}_3(\frac{4\pi}{5})=-\frac{6}{25}\zeta(3)\pm\frac{\sqrt{5}}{4}L(3,\chi_5)$,

$$\psi^{(-3)}\big(\tfrac{1}{5}\big)=\frac{31\,\zeta(3)}{200\pi^2}-\frac{\sqrt{5}\,L(3,\chi_5)}{32\pi^2}+\frac{R}{100}-\frac{\ln 5}{125}+\frac{\ln 2\pi}{100}+\frac{7}{1500}-\frac{Z_1}{5},$$
$$\psi^{(-3)}\big(\tfrac{2}{5}\big)=\frac{31\,\zeta(3)}{200\pi^2}+\frac{\sqrt{5}\,L(3,\chi_5)}{32\pi^2}+\frac{I}{100}-\frac{\ln 5}{250}+\frac{\ln 2\pi}{25}+\frac{41}{1500}-\frac{2Z_1}{5},$$
$$\psi^{(-3)}\big(\tfrac{3}{5}\big)=\frac{31\,\zeta(3)}{200\pi^2}+\frac{\sqrt{5}\,L(3,\chi_5)}{32\pi^2}-\frac{I}{100}+\frac{\ln 5}{250}+\frac{9\ln 2\pi}{100}+\frac{7}{125}-\frac{3Z_1}{5},$$
$$\psi^{(-3)}\big(\tfrac{4}{5}\big)=\frac{31\,\zeta(3)}{200\pi^2}-\frac{\sqrt{5}\,L(3,\chi_5)}{32\pi^2}-\frac{R}{100}+\frac{\ln 5}{125}+\frac{4\ln 2\pi}{25}+\frac{59}{750}-\frac{4Z_1}{5}.$$

Remark 8.55. The remaining derivative coordinates in these formulas are the quartic pair $\{R,I\}$, matching $r_{5,2}=\varphi(5)/2=2$ of Proposition 8.48; the multiplication row $\sum_p \zeta'(-2,\frac{p}{5})=\frac{6}{25}\zeta(3)/\pi^2$ is *dependent* on the reflections — it is the Clausen multiplication law $\mathrm{Cl}_3(\frac{2\pi}{5})+\mathrm{Cl}_3(\frac{4\pi}{5})=-\frac{12}{25}\zeta(3)$ in disguise. Through the conjugated harmonic-number bridge (Theorem 8.47 with $k=2$, $H_2=\frac{3}{2}$) the pair $\{R,I\}$ is interchangeable with $\{\mathrm{Re},\mathrm{Im}\}$ of $L'(3,\chi)$.

The single-function jet calculus extends to products once the endpoint normalization is fixed. The four independent continuations [108, 109, 110, 111] agree on the scale-one finite parts used below. Their common proof is organized through Fourier normalization, singular Laurent coefficients, convolution closure, translated products and normalized primitive lifting. The contact terms belong to the differentiated extension in the next chapter.

8.17 Fourier conventions and the Hurwitz semigroup

We work on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, with normalized Lebesgue measure. For a periodic distribution T, use the convention

$$\widehat{T}(k) = \langle T, e^{-2\pi i k x} \rangle, \qquad k \in \mathbb{Z}.$$

The constant function 1 and the point mass δ_0 both have mean one, but their nonzero Fourier coefficients are respectively 0 and 1. Thus

$$\mathsf{I} = \delta_0 - 1$$

is the convolution identity on the space of mean-zero distributions. In particular, $\mathsf{I} * T = T$ whenever $\widehat{T}(0) = 0$.

The logarithm used throughout is

$$\lambda_k = \log(2\pi i k) = \log(2\pi|k|) + \frac{\pi i}{2} \operatorname{sgn} k, \qquad k \neq 0. \tag{8.60}$$

All powers below refer to this fixed branch.

Definition 8.56. *For $\alpha \in \mathbb{C}$, define R_α by*

$$\widehat{R}_\alpha(0) = 0, \qquad \widehat{R}_\alpha(k) = e^{-\alpha \lambda_k} \quad (k \neq 0). \tag{8.61}$$

Equivalently, its Fourier series is $\sum_{k \neq 0} (2\pi i k)^{-\alpha} e^{2\pi i k x}$ in the distribution sense.

Lemma 8.57 (Entire distribution family)**.** *The family $\alpha \mapsto R_\alpha$ is entire as a periodic-distribution family. It is represented by an absolutely convergent Fourier series for $\operatorname{Re}\alpha > 1$. For $\operatorname{Re}\alpha > 0$, it is the locally integrable periodic extension of*

$$R_\alpha(x) = \frac{\zeta(1-\alpha, x)}{\Gamma(\alpha)}, \qquad 0 < x < 1. \tag{8.62}$$

Proof. On each compact set of parameters, the coefficients in (8.61), together with any fixed number of parameter derivatives, have at most polynomial growth in $|k|$. Fourier coefficients of a smooth test function decay faster than every polynomial. Hence the paired Fourier series and all its parameter derivatives converge locally uniformly. This proves the first assertion; the absolute-convergence assertion follows immediately from $\sum k^{-\operatorname{Re}\alpha}$.

For $\operatorname{Re}\alpha > 1$, equation (8.62) is precisely Hurwitz's Fourier formula. To continue to $\operatorname{Re}\alpha > 0$, use

$$\zeta(1-\alpha, x) = x^{\alpha-1} + \zeta(1-\alpha, 1+x).$$

On compact parameter subsets of that half-plane, the first term and its parameter derivatives are dominated by integrable powers of x times powers of $|\log x|$. The second term is regular in $x \in [0,1]$; any apparent parameter singularity at $\alpha = 0$ is outside the stated open half-plane. The two holomorphic distribution families agree in an open subregion, and hence agree throughout the half-plane. □

Theorem 8.58 (Semigroup and differential ladder)**.** *For all $\alpha, \beta \in \mathbb{C}$,*

$$R_\alpha * R_\beta = R_{\alpha+\beta}, \qquad D R_\alpha = R_{\alpha-1}, \qquad R_0 = \mathsf{I}. \tag{8.63}$$

If $J_{\alpha,r} = \partial_\alpha^r R_\alpha$, then

$$J_{\alpha,r} * J_{\beta,s} = J_{\alpha+\beta,r+s}, \qquad D^k J_{\alpha,r} = J_{\alpha-k,r}. \tag{8.64}$$

Proof. Periodic distributions can be convolved: equivalently, their polynomially growing Fourier coefficients can be multiplied. For $k \neq 0$,

$$e^{-\alpha\lambda_k} e^{-\beta\lambda_k} = e^{-(\alpha+\beta)\lambda_k}, \qquad (2\pi i k) e^{-\alpha\lambda_k} = e^{-(\alpha-1)\lambda_k}.$$

Both sides have zero zeroth coefficient. Fourier uniqueness proves (8.63). Differentiating the locally uniformly paired Fourier series gives (8.64). □

For real $\alpha, \beta > 1$, the first identity, expressed using (8.62), is the nonsingular law in Sun [107, Remark 3.2]. The new use here is at its singular integer parameters and their spectral jets. At positive integers,

$$R_m(x) = -\frac{B_m(\{x\})}{m!} \quad (m \geq 1),$$

up to immaterial point values of the sawtooth for $m = 1$. At negative integers,

$$R_{-k} = \delta_0^{(k)} \quad (k \geq 1). \tag{8.65}$$

The quotient in (8.62) vanishes pointwise at $\alpha = -k$ away from the integers, while the distribution is nonzero. This is the basic reason contact terms must be retained.

8.18 Canonical finite parts of the Stieltjes functions

The generalized Stieltjes functions are defined for $x > 0$ by

$$\zeta(1 - z, x) = -\frac{1}{z} + \sum_{n=0}^{\infty} \gamma_n(x) \frac{z^n}{n!}. \tag{8.66}$$

Thus $\gamma_0(x) = -\psi(x)$. The Hurwitz shift relation gives

$$\gamma_n(x) = \frac{\log^n x}{x} + \gamma_n(1 + x). \tag{8.67}$$

Consequently, periodic extension of γ_n is not locally integrable at 0; a convention is necessary.

Definition 8.59 (Canonical periodic finite part)**.** *For a smooth periodic test function ϕ, define*

$$\langle G_n, \phi \rangle = \int_0^1 \left(\gamma_n(x)\phi(x) - \frac{\log^n x}{x} \phi(0) \right) \mathrm{d}x \tag{8.68}$$

$$= \lim_{\epsilon \downarrow 0} \left\{ \int_\epsilon^1 \gamma_n(x)\phi(x)\,\mathrm{d}x + \frac{\log^{n+1} \epsilon}{n+1} \phi(0) \right\}. \tag{8.69}$$

The integrand in (8.68) is integrable: its singular part is $\log^n x\,[\phi(x) - \phi(0)]/x$, which is $O(|\log x|^n)$. No arbitrary finite multiple of δ_0 is added.

Lemma 8.60 (The distributional Laurent expansion)**.** *Let*

$$Z(s) = \Gamma(1 - s) R_{1-s}.$$

It is a meromorphic periodic-distribution family, agreeing with the periodic locally integrable Hurwitz zeta for $\operatorname{Re} s < 1$*. Near $z = 0$,*

$$Z(1 - z) = \frac{\mathsf{I}}{z} + \sum_{n=0}^{\infty} G_n \frac{z^n}{n!}. \tag{8.70}$$

Every G_n has mean zero.

Proof. The meromorphic continuation follows from Lemma 8.57. For $\operatorname{Re} z > 0$, split the pairing using $\zeta(1-z,x) = x^{z-1} + \zeta(1-z,1+x)$. The first term has expansion

$$\int_0^1 x^{z-1}\phi(x)\,\mathrm{d}x = \frac{\phi(0)}{z} + \sum_{n\geq 0}\frac{z^n}{n!}\int_0^1 \frac{\log^n x}{x}[\phi(x)-\phi(0)]\,\mathrm{d}x.$$

The second term contributes $-z^{-1}\int_0^1 \phi$ and the coefficients $\int_0^1 \gamma_n(1+x)\phi(x)\,\mathrm{d}x$. These expansions converge locally near zero after pairing, by the same endpoint domination used above. Combining them proves (8.70) with the exact subtraction in Definition 8.59. Since $\widehat{Z}(s)(0) = 0$, every coefficient has mean zero. Equivalently, direct integration over $[1,2]$ gives $\int_1^2 \zeta(1-z,x)\,\mathrm{d}x = -1/z$, whose regular coefficients vanish. □

Remark 8.61 (Two different residues). The scalar function in (8.66) has residue -1 as a function of z, whereas the periodic-distribution family has residue $\delta_0 - 1$. Analytic continuation across a nonintegrable endpoint adds a point-supported residue. Treating the scalar residue as the whole periodic residue gives wrong convolution constants.

8.19 All-index convolution closure and exact normal forms

Write $\mathcal{G}(z) = \sum_{n\geq 0} G_n z^n/n!$, holomorphic as a distribution family for $|z| < 1$. By (8.70),

$$\mathsf{I} + z\mathcal{G}(z) = \Gamma(1+z)R_z. \tag{8.71}$$

Let $X = G_0$. For a complete exponential Bell polynomial we use

$$\exp\left(\sum_{j\geq 1} b_j \frac{z^j}{j!}\right) = \sum_{r\geq 0}\mathcal{B}_r(b_1,\dots,b_r)\frac{z^r}{r!}. \tag{8.72}$$

When its arguments are distributions, all products in this definition are convolution products and its scalar unit is I.

Theorem 8.62 (One-generator convolution algebra)**.** *For every $n \geq 0$,*

$$(n+1)G_n = \mathcal{B}_{n+1}(b_1,\dots,b_{n+1}), \qquad b_1 = X, \quad b_j = (-1)^j(j-1)!\zeta(j)\mathsf{I}\ (j\geq 2). \tag{8.73}$$

The span of $\mathsf{I}, G_0, G_1, \dots$ is a convolution algebra isomorphic to the polynomial algebra $\mathbb{C}[X]$. These distributions form a basis of that algebra, with the unit listed first.

Proof. For $k \neq 0$, the Fourier transform of (8.71) is

$$1 + z\widehat{\mathcal{G}}(z)(k) = \Gamma(1+z)e^{-z\lambda_k} = \exp\left[(-\gamma-\lambda_k)z + \sum_{j\geq 2}\frac{(-1)^j\zeta(j)}{j}z^j\right].$$

The linear coefficient is $\widehat{X}(k) = -\gamma - \lambda_k$. Comparing with (8.72) proves (8.73); the zero Fourier coefficient of each distributional expression vanishes. The leading term in G_n is $X^{*(n+1)}/(n+1)$, so the listed distributions span exactly the polynomial algebra generated by X.

There are no polynomial relations among its powers. Indeed, if $P(X) = 0$ distributionally, then $P(-\gamma - \log(2\pi k) - i\pi/2) = 0$ for every positive integer k. These are infinitely many distinct complex numbers, and hence P is the zero polynomial. This also proves the basis assertion. □

This algebraic freeness concerns distributions as functions of the variable x. It is not an independence theorem about their values at a fixed rational point, nor about zeta or Stieltjes constants as numbers.

The first Bell forms are

$$\begin{aligned} 2G_1 &= X^{*2} + \zeta(2)\mathsf{I}, \\ 3G_2 &= X^{*3} + 3\zeta(2)X - 2\zeta(3)\mathsf{I}, \\ 4G_3 &= X^{*4} + 6\zeta(2)X^{*2} - 8\zeta(3)X + [3\zeta(2)^2 + 6\zeta(4)]\mathsf{I}. \end{aligned} \tag{8.74}$$

They already imply many exact integrals. An efficient all-index multiplication rule is supplied by the following Gamma quotient.

Theorem 8.63 (Bivariate multiplication law)**.** *Define, near* $(u, v) = (0, 0)$,

$$\begin{aligned} B(u,v) &= \frac{\Gamma(1+u)\Gamma(1+v)}{\Gamma(1+u+v)} \\ &= \exp\left(\sum_{j\geq 2} \frac{(-1)^j \zeta(j)}{j}[u^j + v^j - (u+v)^j]\right). \end{aligned} \tag{8.75}$$

Then

$$\begin{aligned} uv\,\mathcal{G}(u) * \mathcal{G}(v) = [B(u,v) - 1]\mathsf{I} \\ + B(u,v)(u+v)\mathcal{G}(u+v) - u\mathcal{G}(u) - v\mathcal{G}(v). \end{aligned} \tag{8.76}$$

The right side is divisible by uv *as a convergent distribution-valued power series. In particular,*

$$G_m * G_n - \frac{m+n+2}{(m+1)(n+1)} G_{m+n+1} \in \operatorname{span}_{\mathbb{C}}(\{\mathsf{I}\} \cup \{G_j : 0 \leq j < m+n\}), \tag{8.77}$$

where the lower Stieltjes span is empty when $m + n = 0$*. The index* $m + n$ *is absent because the Gamma quotient has no linear term. Every coefficient is a polynomial over* $\mathbb{Q}$ *in zeta values at positive integers. Its complete value is obtained by multiplying the coefficient of* $u^{m+1}v^{n+1}$ *in* (8.76) *by* $m!n!$*.*

Proof. Convolve (8.71) at parameters u and v and apply Theorem 8.58. The result is

$$[\mathsf{I} + u\mathcal{G}(u)] * [\mathsf{I} + v\mathcal{G}(v)] = B(u,v)[\mathsf{I} + (u+v)\mathcal{G}(u+v)].$$

Subtract the unit and the two linear summands. Both restrictions $u = 0$ and $v = 0$ vanish identically, proving divisibility. Coefficient extraction is legitimate in a sufficiently small bidisc. The largest-index term comes from $(u+v)\mathcal{G}(u+v)$ with the constant coefficient of B; its coefficient is $\frac{m!n!}{(m+n+1)!}\binom{m+n+2}{m+1}$, which equals the number in (8.77). The exponential expansion of B proves the zeta-polynomial assertion. □

Four useful multiplication rules are

$$G_0 * G_0 = 2G_1 - \zeta(2)\mathsf{I}, \tag{8.78}$$

$$G_0 * G_1 = \frac{3}{2}G_2 - \zeta(2)G_0 + \zeta(3)\mathsf{I}, \tag{8.79}$$

$$G_0 * G_2 = \frac{4}{3}G_3 - 2\zeta(2)G_1 + 2\zeta(3)G_0 - 2\zeta(4)\mathsf{I}, \tag{8.80}$$

$$G_1 * G_1 = G_3 - 2\zeta(2)G_1 + 2\zeta(3)G_0 - \frac{1}{4}\zeta(4)\mathsf{I}. \tag{8.81}$$

The last simplification uses $\zeta(2)^2 = 5\zeta(4)/2$. The verification catalog retains independent formal symbols for zeta values until such a specific relation is explicitly applied.

8.20 Ordinary integral realizations and explicit Stieltjes identities

The singular support of G_n is contained in $\{0\}$. Thus its convolution with another G_m is smooth away from 0: in a local convolution integral at $0 < x < 1$, the two singular factors occur at the distinct integration points $t = 0$ and $t = x$. We now give a direct formula that also proves this local realization without requiring a product of distributions at the same point.

Theorem 8.64 (Absolutely convergent subtracted integrals)**.** *Put $a_n(t) = \log^n(t)/t$. For $m, n \geq 0$ and $0 < x < 1$, define*

$$\begin{aligned} C_{m,n}(x) = &\int_0^x \big[\gamma_m(t)\gamma_n(x-t) - \gamma_n(x)a_m(t) - \gamma_m(x)a_n(x-t)\big]\,\mathrm{d}t \\ &+ \int_x^1 \gamma_m(t)\gamma_n(1+x-t)\,\mathrm{d}t \\ &+ \gamma_n(x)\frac{\log^{m+1} x}{m+1} + \gamma_m(x)\frac{\log^{n+1} x}{n+1}. \end{aligned} \tag{8.82}$$

Both displayed integrals are absolutely convergent, and $C_{m,n}(x) = (G_m * G_n)(x)$. In particular, every expression (8.82) has a finite exact reduction using Theorem 8.63.

Proof. Near $t = 0$, the shift relation (8.67) leaves a possible singularity

$$a_m(t)[\gamma_n(x-t) - \gamma_n(x)] = O(|\log t|^m).$$

The other endpoint is the same argument with m, n interchanged. The second integral is nonsingular. These observations prove absolute convergence, locally uniformly with derivatives in x after using endpoint-fixed local coordinates.

For the normalization, approximate G_m by the cutoff function $\mathbf{1}_{[\epsilon,1)}\gamma_m$ plus $\log^{m+1}\epsilon\,\delta_0/(m+1)$, and similarly for G_n. On a compact subinterval of $0 < x < 1$, their convolution consists of

$$\int_\epsilon^{x-\epsilon} \gamma_m(t)\gamma_n(x-t)\,\mathrm{d}t + \int_x^1 \gamma_m(t)\gamma_n(1+x-t)\,\mathrm{d}t$$

plus the two logarithmic counterterms multiplying $\gamma_n(x)$ and $\gamma_m(x)$. The product of the two delta counterterms is supported at 0, outside that compact subinterval. Integrating the two subtracted a-terms explicitly and taking the limit gives (8.82). The locally uniform endpoint argument identifies this limit with the restriction of the distributional convolution. □

Corollary 8.65 (Four scalar identity families)**.** *For $0 < x < 1$,*

$$C_{0,0}(x) = 2\gamma_1(x) + \zeta(2), \tag{8.83}$$

$$C_{0,1}(x) = \frac{3}{2}\gamma_2(x) - \zeta(2)\gamma_0(x) - \zeta(3), \tag{8.84}$$

$$C_{0,2}(x) = \frac{4}{3}\gamma_3(x) - 2\zeta(2)\gamma_1(x) + 2\zeta(3)\gamma_0(x) + 2\zeta(4), \tag{8.85}$$

$$C_{1,1}(x) = \gamma_3(x) - 2\zeta(2)\gamma_1(x) + 2\zeta(3)\gamma_0(x) + \frac{1}{4}\zeta(4). \tag{8.86}$$

Proof. Restrict (8.78)–(8.81) to $0 < x < 1$, where G_n agrees with γ_n and I agrees with the constant -1. □

8.20.1 A concrete digamma integral

Since $\gamma_0 = -\psi$, the first case reads

$$\int_0^x \left[\psi(t)\psi(x-t) + \psi(x)\left(\frac{1}{t} + \frac{1}{x-t}\right)\right] \mathrm{d}t \tag{8.87}$$
$$+ \int_x^1 \psi(t)\psi(1+x-t)\,\mathrm{d}t - 2\psi(x)\log x = 2\gamma_1(x) + \frac{\pi^2}{6}.$$

The two divergent-looking summands inside the first bracket must be integrated together; their sum is integrable. Equation (8.87) is not an assertion that either unregularized convolution integral exists.

At $x = 1/2$, the Hurwitz identity $\zeta(s, 1/2) = (2^s - 1)\zeta(s)$ implies

$$\gamma_1(1/2) = \gamma_1 - 2\gamma\log 2 - \log^2 2.$$

Thus the left side of (8.87) at $1/2$ equals

$$2\gamma_1 - 4\gamma\log 2 - 2\log^2 2 + \frac{\pi^2}{6}. \tag{8.88}$$

The same method at any rational argument turns the full multiplication table into exact relations among rational-argument Stieltjes functions, with no integer-relation search.

8.21 Conventions and endpoint normalization

Throughout,

$$0 < a < 1, \qquad b = 1 - a, \qquad \ell = \log(2\pi), \qquad \kappa = \gamma + \ell, \qquad B_2(a) = a^2 - a + \frac{1}{6}.$$

The fractional part $\{x + a\}$ is interpreted periodically; its value at the single jump point does not affect ordinary integrals. The notation

$$\zeta^{[j]}(s, x) = \partial_s^j \zeta(s, x), \qquad \mathrm{Li}_s^{[j]}(z) = \partial_s^j \mathrm{Li}_s(z)$$

always means *spectral* differentiation. In contrast, $\gamma_m^{(r)}(x)$ and $\psi^{(r)}(x)$ mean derivatives in the real argument. Euler's constant is $\gamma = \gamma_0(1)$, not a new parameter.

The generalized Stieltjes constants are normalized by

$$\zeta(1+u, x) = \frac{1}{u} + R(u, x), \qquad R(u, x) = \sum_{m \geq 0} \frac{(-1)^m}{m!}\gamma_m(x)u^m. \tag{8.89}$$

Thus $\gamma_0(x) = -\psi(x)$. The Hurwitz shift identity gives the exact local decomposition

$$\gamma_m(x) = \frac{\log^m x}{x} + \gamma_m(1+x), \qquad x > 0. \tag{8.90}$$

Every logarithm of a positive real argument is the real logarithm. For $z = e^{2\pi i a} \neq 1$, $\mathrm{Li}_s(z)$ and $\mathrm{Li}_s(z^{-1})$ denote their usual boundary values from the unit disk, continued in s; these are entire in s. For real s they are conjugate, but we retain them separately when s is complex.

Definition 8.66 (Canonical shifted Stieltjes finite part)**.** *For nonnegative integers m, n, define*

$$J_{mn}(a) = \lim_{\epsilon\downarrow 0}\Bigg\{ \int_\epsilon^b \gamma_m(x)\gamma_n(x+a)\,\mathrm{d}x + \int_\epsilon^a \gamma_m(b+t)\gamma_n(t)\,\mathrm{d}t + \frac{\gamma_n(a)}{m+1}\log^{m+1}\epsilon + \frac{\gamma_m(b)}{n+1}\log^{n+1}\epsilon\Bigg\}. \tag{8.91}$$

Equivalently, this is denoted $\mathrm{FP}\int_0^1 \gamma_m(x)\gamma_n(\{x+a\})\,\mathrm{d}x$.

The two cutoffs can tend to zero independently: each counterterm belongs to its own local endpoint. The coordinates are exactly x and $t = x - b$, with no rescaling or additional subtraction length. A different finite-part convention can change local constants and therefore change the identities.

Lemma 8.67 (Existence and an ordinary-integral version)**.** *The limit* (8.91) *exists and is equivalent to*

$$\begin{aligned} J_{mn}(a) = {} & \int_0^b \left[\gamma_m(x)\gamma_n(x+a) - \gamma_n(a)\frac{\log^m x}{x}\right]\mathrm{d}x \\ & + \int_0^a \left[\gamma_m(b+t)\gamma_n(t) - \gamma_m(b)\frac{\log^n t}{t}\right]\mathrm{d}t \\ & + \frac{\gamma_n(a)}{m+1}\log^{m+1} b + \frac{\gamma_m(b)}{n+1}\log^{n+1} a. \end{aligned} \tag{8.92}$$

Both integrals on the right are ordinary improper integrals. Moreover, $J_{mn}(a) = J_{nm}(1-a)$.

Proof. By (8.90), subtracting the singular term in the first integral leaves $O(1+|\log x|^m)$ near zero: the other factor differs from $\gamma_n(a)$ by $O(x)$. The second integral has the same property. Integrating $\log^m x/x$ explicitly yields (8.92). Interchanging the two pieces and replacing a by b proves the symmetry. □

We use analogous finite parts for polygamma and derivative products: expand each smooth factor at the distinct singular point, integrate the resulting powers and logarithms, and retain the constant term after deleting all divergent cutoff terms. Section 9.2 makes this prescription precise through periodic distributions.

8.22 The shifted Hurwitz correlation

Proposition 8.68 (Classical Fourier method, shifted form)**.** *For $\Re s < 1$, $\Re t < 1$, define the ordinary integral*

$$C(s,t;a) = \int_0^1 \zeta(s,x)\zeta(t,\{x+a\})\,\mathrm{d}x. \tag{8.93}$$

It is jointly holomorphic in this domain. Put $w = 2-s-t$ and $z = e^{2\pi\mathrm{i}a}$. Then

$$C(s,t;a) = \frac{\Gamma(1-s)\Gamma(1-t)}{(2\pi)^w}\Big[e^{-\pi\mathrm{i}(s-t)/2}\,\mathrm{Li}_w(z) + e^{\pi\mathrm{i}(s-t)/2}\,\mathrm{Li}_w(z^{-1})\Big]. \tag{8.94}$$

It also has the meromorphic representation

$$C(s,t;a) = B(1-s,s+t-1)\zeta(s+t-1,a) + B(1-t,s+t-1)\zeta(s+t-1,b). \tag{8.95}$$

At exceptional arguments of the beta factors, the sum *in* (8.95) *is continued before evaluation. Apparent poles of separate summands need not be poles of* C*.*

Proof. Split the integral at b. At $x = 0$ only the first factor is singular and has the form x^{-s} plus a smooth function. At $x = b+$ only the second factor is singular. Compact subsets of $\Re s < 1$, $\Re t < 1$ therefore admit integrable dominating functions, including arbitrary spectral derivatives, because powers of $|\log x|$ remain integrable against $x^{-\sigma}$ for $\sigma < 1$. This proves convergence and holomorphy. There is no additional condition $\Re(s+t) < 1$ for a nonzero shift.

For $\Re s < 0$, the Hurwitz Fourier expansion [93, Eq. 25.11.9] is absolutely convergent and its nonzero coefficients are

$$\widehat{\zeta}_s(k) = \Gamma(1-s)(2\pi|k|)^{s-1}\exp\left(-\frac{\pi\mathrm{i}}{2}(1-s)\operatorname{sgn} k\right), \qquad k \neq 0. \tag{8.96}$$

The zero coefficient vanishes. For $\Re s, \Re t < 0$, multiply the two absolutely convergent series and integrate. Orthogonality gives

$$C(s,t;a) = \sum_{k\neq 0} \widehat{\zeta}_s(k)\widehat{\zeta}_t(-k)e^{-2\pi\mathrm{i}ka}.$$

The positive and negative frequencies give exactly (8.94). Both sides are holomorphic on the larger connected domain stated in the proposition; the identity theorem extends the equality there.

To obtain (8.95), apply the same Fourier formula to $\zeta(1-w,a)$ and $\zeta(1-w,b)$, initially away from gamma poles. Euler reflection gives $\Gamma(w)\Gamma(1-w) = \pi/\sin\pi w$ and $1/\Gamma(t) = \Gamma(1-t)\sin\pi t/\pi$. The coefficient of $\mathrm{Li}_w(z)$ reduces using

$$\frac{\sin(\pi t)e^{-\pi\mathrm{i}w/2} + \sin(\pi s)e^{\pi\mathrm{i}w/2}}{\sin(\pi w)} = e^{-\pi\mathrm{i}(s-t)/2}, \qquad w = 2-s-t.$$

The other coefficient is obtained by reversing the phase. This proves equality on an open set and hence meromorphically everywhere. □

Corollary 8.69 (All ordinary spectral-jet correlations)**.** *For integers* $p, q \geq 0$ *and* $\Re s, \Re t < 1$*,*

$$\int_0^1 \zeta^{[p]}(s,x)\zeta^{[q]}(t,\{x+a\})\,\mathrm{d}x = \partial_s^p\partial_t^q C(s,t;a).$$

Thus (8.94) *generates ordinary identities at nonpositive spectral integers, while continuation of* (8.95) *generates finite-part identities at positive integers.*

Proof. Differentiate under the integral using the compact domination established above. The formulas on the right are finite applications of the product and chain rules. □

The distinction between nonzero and zero shifts is essential. At $a = 0$, the singularities coincide and the original convergence region changes to include a constraint on $s+t$. One must not substitute $a = 0$ directly into a separated-singularity finite-part formula. The continuous ordinary log-gamma correlation will have a legitimate limit at zero, but that is a different assertion.

Proposition 8.70 (Coefficient extraction at the double pole)**.** *For $a \in (0,1)$,*

$$J_{mn}(a) = (-1)^{m+n} m!n![u^m v^n] C(1+u, 1+v; a). \tag{8.97}$$

The brackets mean the regular Laurent coefficient at the specified bidegree. In particular, the double-pole coefficient is $-1/(uv)$, not $1/(uv)$.

Proof. Write $Z_s = Z(s)$ for the periodic family of Lemma 8.60. Its raising-parameter expansion is

$$Z_{1+u} = -\mathsf{I}/u + \sum_{m \geq 0} (-1)^m G_m u^m / m!. \tag{8.98}$$

The singular supports of $Z_{1+u}(x)$ and $Z_{1+v}(x+a)$ are 0 and b, which are distinct. Choose disjoint neighborhoods of these points and a smooth partition of unity. On each neighborhood only one factor is a distribution and the other is smooth, so their product and integral are defined without any product of distributions at a common singularity. This gives the meromorphic continuation of the original integral.

The product of the regular coefficients in (8.98), integrated with this partition, is exactly (8.91). The pole terms give

$$\begin{aligned} C(1+u, 1+v; a) = &-\frac{1}{uv} - \frac{R(v,a)}{u} - \frac{R(u,b)}{v} \\ &+ \sum_{m,n \geq 0} \frac{(-1)^{m+n}}{m!n!} J_{mn}(a) u^m v^n. \end{aligned} \tag{8.99}$$

The integrated product of $1 - \delta_0$ and $1 - \delta_b$ is $1 - 1 - 1 = -1$ away from $a = 0$, and each G_m has mean zero. This proves the statement. □

Corollary 8.71 (Polylogarithmic generator)**.** *Set $E(u) = \Gamma(1-u)(2\pi)^u$ and*

$$\begin{aligned} \mathcal{H}(u,v;a) = E(u)E(v)\big[&e^{-\pi \mathrm{i}(u-v)/2} \operatorname{Li}_{-u-v}(z) \\ &+ e^{\pi \mathrm{i}(u-v)/2} \operatorname{Li}_{-u-v}(z^{-1})\big]. \end{aligned} \tag{8.100}$$

Then

$$J_{mn}(a) = (-1)^{m+n} m!n![u^{m+1} v^{n+1}] \mathcal{H}(u,v;a). \tag{8.101}$$

Proof. In (8.94), set $s = 1+u$, $t = 1+v$ and use $\Gamma(-u) = -\Gamma(1-u)/u$. Thus $C = \mathcal{H}/(uv)$, and (8.97) applies. □

This representation explains how polylogarithm *order derivatives*, rather than only argument derivatives, enter the exact Stieltjes product calculus.

8.23 An all-order linear Stieltjes closure theorem

The beta representation is particularly effective near $(s,t) = (1,1)$. Define

$$A(u,v) = \frac{\Gamma(1-u)\Gamma(1+u+v)}{\Gamma(1+v)}, \qquad T(u,v) = \frac{A(u,v) - 1}{u(u+v)}. \tag{8.102}$$

The apparent singularities in T are removable: $A(0,v)=1$ and $A(u,-u)=1$. Its Taylor coefficients belong to the polynomial algebra generated by positive integer zeta values. More explicitly,

$$\log A(u,v)=\sum_{k\geq 2}\frac{\zeta(k)}{k}\left[u^k+(-1)^k\big((u+v)^k-v^k\big)\right]. \tag{8.103}$$

Euler's constant cancels identically. This series converges in a neighborhood of the origin; it can also be used as an exact formal power series.

Theorem 8.72 (All-index finite-part closure)**.** *Let $m,n\geq 0$ and $d=m+n$. Define the analytic germ*

$$\begin{aligned}\mathcal{F}(u,v;a)&=T(u,v)+T(v,u)\\&\quad+(u+v)\big[T(u,v)R(u+v,a)+T(v,u)R(u+v,b)\big].\end{aligned} \tag{8.104}$$

Then

$$\boxed{J_{mn}(a)=\frac{\gamma_{d+1}(a)}{m+1}+\frac{\gamma_{d+1}(b)}{n+1}-(-1)^d m!n![u^m v^n]\mathcal{F}(u,v;a).} \tag{8.105}$$

Every coefficient is obtained by a finite calculation. Apart from the two leading terms displayed explicitly, the right side contains only $\gamma_j(a)$ and $\gamma_j(b)$ with $0\leq j\leq d-1$, and a constant term. Its coefficients lie in $\mathbb{Q}[\zeta(2),\dots,\zeta(d+2)]$. There are no products of generalized Stieltjes functions and no term involving γ_d.

Proof. Since $B(-u,1+u+v)=-A(u,v)/u$, equation (8.95) becomes

$$C(1+u,1+v;a)=-\frac{A(u,v)}{u}\zeta(1+u+v,a)-\frac{A(v,u)}{v}\zeta(1+u+v,b). \tag{8.106}$$

Substitute $A(u,v)=1+u(u+v)T(u,v)$ and the Laurent expansion (8.89). The result is

$$C(1+u,1+v;a)=-\frac{1}{uv}-\frac{R(u+v,a)}{u}-\frac{R(u+v,b)}{v}-\mathcal{F}(u,v;a). \tag{8.107}$$

Here the apparent diagonal pole at $u+v=0$ has canceled exactly.

The regular coefficient of $-R(u+v,a)/u$ at $u^m v^n$ is

$$-\frac{(-1)^{d+1}\gamma_{d+1}(a)}{(d+1)!}\binom{d+1}{m+1}=\frac{(-1)^d\gamma_{d+1}(a)}{(m+1)!n!}.$$

The other term similarly contributes $(-1)^d\gamma_{d+1}(b)/(m!(n+1)!)$. Now apply (8.97) to get (8.105).

To obtain a coefficient of total degree d in T, only terms of total degree at most $d+2$ in A are required. Formula (8.103) proves the asserted zeta coefficient algebra. In the product $(u+v)TR$, a Stieltjes index j occupies degree $j+1$ before multiplication by T, so $j\leq d-1$. All appearances of R are linear. These observations prove the remaining claims. □

Corollary 8.73 (Formal weight conservation)**.** *Assign weight $j+1$ to $\gamma_j(a)$ and $\gamma_j(b)$ and weight k to $\zeta(k)$. Every monomial in (8.105) has weight $m+n+2$.*

Proof. The degree-k component of $\log A$ has coefficient weight k. Thus a coefficient of total degree r in T has weight $r+2$. A contribution from $(u+v)TR$ to total degree d has $r+j+1=d$ and weight $(r+2)+(j+1)=d+2$. The leading Stieltjes terms have the same weight. □

This is a grading of the displayed symbolic algebra, not a theorem that the evaluated constants have no additional relations.

8.23.1 The first four nontrivial identities

The beginning of the universal kernel is

$$T(u,v)=\zeta(2)-\zeta(3)v+\frac{\zeta(4)+\zeta(2)^2}{2}(u^2+uv)+\zeta(4)v^2+O((|u|+|v|)^3). \tag{8.108}$$

Write $S_j(a)=\gamma_j(a)+\gamma_j(b)$. Coefficient extraction gives

$$J_{00}(a)=S_1(a)-2\zeta(2), \tag{8.109}$$
$$J_{10}(a)=\tfrac{1}{2}\gamma_2(a)+\gamma_2(b)+\zeta(2)S_0(a)-\zeta(3), \tag{8.110}$$
$$J_{11}(a)=\tfrac{1}{2}S_3(a)+2\zeta(2)S_1(a)+\zeta(3)S_0(a)-\zeta(4)-\zeta(2)^2, \tag{8.111}$$
$$J_{20}(a)=\tfrac{1}{3}\gamma_3(a)+\gamma_3(b)+2\zeta(2)S_1(a)+2\zeta(3)\gamma_0(b) - 3\zeta(4)-\zeta(2)^2. \tag{8.112}$$

These are exact analytic identities under Definition 8.66, not conjectured PSLQ relations. The asymmetric coefficients in J_{10} and J_{20} are necessary; exchanging a and b exchanges the two Laurent indices.

Because $\gamma_0=-\psi$, the first identity takes the particularly simple form

$$\operatorname{FP}\int_0^1 \psi(x)\psi(\{x+a\})\,\mathrm{d}x=\gamma_1(a)+\gamma_1(b)-\frac{\pi^2}{3}. \tag{8.113}$$

An equivalent identity containing only ordinary integrals is

$$\int_0^b\left[\psi(x)\psi(x+a)+\frac{\psi(a)}{x}\right]\mathrm{d}x+\int_0^a\left[\psi(b+t)\psi(t)+\frac{\psi(b)}{t}\right]\mathrm{d}t = \gamma_1(a)+\gamma_1(b)-\frac{\pi^2}{3}+\psi(a)\log b+\psi(b)\log a. \tag{8.114}$$

The endpoint singularities cancel inside each bracket. This version is useful when a conventional improper integral is preferable to distribution notation.

8.23.2 A finite exact coefficient algorithm

For target total index d, truncate (8.103) at degree $d+2$. Exponentiate by

$$A=\sum_{j=0}^{\lfloor (d+2)/2\rfloor}\frac{(\log A)^j}{j!}\pmod{\text{total degree}>d+2}.$$

Exact polynomial division of $A-1$ by $u(u+v)$ has zero remainder. Let $T(u,v)=\sum t_{ij}u^iv^j$. The correction coefficient needed in (8.105) is

$$[u^mv^n]\mathcal{F}=t_{mn}+t_{nm} + \sum_{\substack{i\le m,\ j\le n\\ k=d-i-j-1\ge 0}} t_{ij}\frac{(-1)^k}{k!}\binom{k+1}{m-i}\gamma_k(a) + \sum_{\substack{j\le m,\ i\le n\\ k=d-i-j-1\ge 0}} t_{ij}\frac{(-1)^k}{k!}\binom{k+1}{m-j}\gamma_k(b). \tag{8.115}$$

Only polynomial arithmetic over rational numbers and formal zeta symbols is used. The implementation does not perform floating-point relation recognition. Exact divisibility, reflection symmetry, linearity, and weight conservation provide independently checkable invariants of the generated formulas.

For the collision comparison, write the already proved linear closure as

$$J_{mn}(a) = e_{mn} + \sum_j (c_{mn,j}\gamma_j(a) + d_{mn,j}\gamma_j(b)).$$

Use the local germs $\mathcal{A}(u,v) = A(v,u)$ and $\mathcal{B}(u,v) = A(u,v)$ from (8.102), and put $w = u + v$. The beta formula then reads

$$K(u,v;a) := uvC(1+u,1+v;a) = -u\mathcal{A}(u,v)\zeta(1+w,b) - v\mathcal{B}(u,v)\zeta(1+w,a). \tag{8.116}$$

Put $Q(u,v) = (u\mathcal{A} + v\mathcal{B})/w$. Its numerator is divisible by w: both germs equal one at $v = -u$. This removes the apparent diagonal pole and supplies the finite coefficient construction used below.

8.24 Coincident endpoints and canonical collision subtraction

A translated correlation has two separated logarithmic endpoint divergences. At zero translation they merge into an additional x^{-2} singularity. We now show that the same coefficient table still applies, but only after the merged singularity is subtracted explicitly.

For an integer $r \geq 0$, put

$$E_r(L) = \sum_{k=0}^{r} \frac{r!}{k!} L^k. \tag{8.117}$$

The identity $E_r - E_r' = L^r$ implies

$$\int_\epsilon^1 \frac{\log^r x}{x^2}\,\mathrm{d}x = \frac{E_r(\log\epsilon)}{\epsilon} - r!. \tag{8.118}$$

Definition 8.74 (Coincident-endpoint finite part)**.** *Write $\gamma_j = \gamma_j(1)$ and define*

$$J_{mn}^{\mathrm{coll}} = \lim_{\epsilon\downarrow 0}\Bigg\{ \int_\epsilon^1 \gamma_m(x)\gamma_n(x)\,\mathrm{d}x - \frac{E_{m+n}(\log\epsilon)}{\epsilon} + \frac{\gamma_n}{m+1}\log^{m+1}\epsilon + \frac{\gamma_m}{n+1}\log^{n+1}\epsilon \Bigg\}. \tag{8.119}$$

Theorem 8.75 (Coincident closure)**.** *The finite part (8.119) exists. With exactly the coefficients of Theorem 8.72,*

$$J_{mn}^{\mathrm{coll}} = e_{mn} + \sum_{j=0}^{m+n+1} (c_{mn,j} + d_{mn,j})\gamma_j. \tag{8.120}$$

Equivalently it is the same mixed coefficient of

$$K_0(u,v) = -(u\mathcal{A}(u,v) + v\mathcal{B}(u,v))\zeta(1+u+v). \tag{8.121}$$

In particular,

$$J_{00}^{\mathrm{coll}} = 2\gamma_1 - \frac{\pi^2}{3}. \tag{8.122}$$

Proof. By (8.90), the cutoff in Definition 8.74 converges to the absolutely convergent expression

$$\begin{aligned} J^{\mathrm{coll}}_{mn} = & -(m+n)! \\ & + \int_0^1 \left\{ \frac{\log^m x}{x}[\gamma_n(1+x)-\gamma_n] + \frac{\log^n x}{x}[\gamma_m(1+x)-\gamma_m] \right. \\ & \qquad \left. + \gamma_m(1+x)\gamma_n(1+x) \right\} \mathrm{d}x. \end{aligned} \tag{8.123}$$

The two bracketed differences vanish linearly at zero, and (8.118) supplies the constant $-(m+n)!$.

To identify the coefficients, start where the ordinary coincident Hurwitz integral converges, for example $\operatorname{Re} s, \operatorname{Re} t < 0$. Fourier orthogonality gives

$$C_0(s,t) = 2\Gamma(1-s)\Gamma(1-t)(2\pi)^{s+t-2}\cos\left(\frac{\pi(s-t)}{2}\right)\zeta(2-s-t).$$

The Riemann functional equation, equivalently the specialization of the Fourier calculation in Theorem 8.68, gives $uvC_0(1+u,1+v) = K_0(u,v)$. Meromorphic continuation is justified by subtracting the local terms as above.

More explicitly, the holomorphic generating expression near zero is

$$\begin{aligned} R_0(u,v) = & -\frac{1}{1+u+v} \\ & + \int_0^1 \left\{ x^{-1-u}[R(v,1+x)-R(v,1)] + x^{-1-v}[R(u,1+x)-R(u,1)] \right. \\ & \qquad \left. + R(u,1+x)R(v,1+x) \right\} \mathrm{d}x. \end{aligned}$$

The first term is the analytic continuation of $\int_0^1 x^{-2-u-v}\,\mathrm{d}x$, not a discarded divergence. The remaining integrals converge locally uniformly near zero. On a starting convergence region, and hence by continuation,

$$R_0 = \frac{K_0 + 1 + uR(u,1) + vR(v,1)}{uv}.$$

Its Taylor coefficients are (8.123). Decomposing (8.121) as $-Q - (u\mathcal{A} + v\mathcal{B})R(u+v,1)$ now gives (8.120) by the identical finite algebra used earlier. □

For the first case, an ordinary integral version is worth recording:

$$\int_0^1 \left\{ \psi(1+x)^2 - \frac{2[\psi(1+x)+\gamma]}{x} \right\} \mathrm{d}x = 1 + 2\gamma_1 - \frac{\pi^2}{3}. \tag{8.124}$$

The apparently singular numerator in the second term vanishes at zero. There is no positivity contradiction in the negative value of J^{coll}_{00}: it is a finite part of a positive divergent square, not its ordinary nonnegative integral.

8.24.1 An exact convergent expansion at collision

The relationship between the separated and coincident finite parts can be expressed by an equality of convergent series, not merely an asymptotic prescription.

Theorem 8.76 (Collision polynomial and convergent regular part)**.** *For $m, n \geq 0$ define a polynomial*

$$\mathcal{P}_{mn}(L) = (-1)^{m+n+1} m! n! [u^{m+1} v^n] \mathcal{B}(u, v) e^{-(u+v)L}. \tag{8.125}$$

It has degree $m+n+1$, leading coefficient $1/(m+1)$, and coefficients in positive integer zeta values. There is an exact expansion

$$J_{mn}(a) = \frac{\mathcal{P}_{mn}(\log a)}{a} + J_{mn}^{\mathrm{coll}} + \sum_{k \geq 1} h_{mn,k} a^k, \qquad 0 < a < 1, \tag{8.126}$$

whose regular power series converges locally uniformly for $|a| < 1$. Its coefficients are

$$\begin{aligned} h_{mn,k} = \frac{(-1)^{m+n+1} m! n!}{k!} &[u^{m+1} v^{n+1}] \\ &\big(u\mathcal{A}(u, v) + (-1)^k v \mathcal{B}(u, v)\big)(1 + u + v)_k \zeta(1 + u + v + k). \end{aligned} \tag{8.127}$$

For $m = n$, all odd-indexed $h_{mm,k}$ vanish. In particular,

$$\lim_{a \downarrow 0} \left\{ J_{mn}(a) - \frac{\mathcal{P}_{mn}(\log a)}{a} \right\} = J_{mn}^{\mathrm{coll}}. \tag{8.128}$$

Proof. The classical parameter Taylor series at one gives, locally uniformly in the spectral variable,

$$\zeta(1 + w, 1 - a) = \sum_{k \geq 0} \frac{(1 + w)_k}{k!} \zeta(1 + w + k) a^k,$$

while the shift law gives

$$\zeta(1 + w, a) = a^{-1-w} + \sum_{k \geq 0} \frac{(1 + w)_k}{k!} \zeta(1 + w + k)(-a)^k.$$

These are meromorphic in w at zero, with the $k = 0$ pole explicitly visible. Substitution in (8.116) yields

$$\begin{aligned} K(u, v; a) = \; &- v \mathcal{B}(u, v) a^{-1-w} \\ &- \sum_{k \geq 0} \frac{u\mathcal{A}(u, v) + (-1)^k v \mathcal{B}(u, v)}{k!} (1 + w)_k \zeta(1 + w + k) a^k. \end{aligned} \tag{8.129}$$

The $k = 0$ summand is exactly K_0; its apparent pole at $w = 0$ is removable. The remaining series is holomorphic for small u, v and $|a| < 1$, uniformly on compact subsets. For example, on a fixed small spectral polydisc its terms and spectral derivatives grow at most polynomially in k times powers of $\log k$, whereas $|a|^k$ decays geometrically on compact subdiscs. Coefficient extraction is therefore legitimate. The singular first term gives (8.125), the constant gives Theorem 8.75, and the other terms give (8.127).

The highest power of L comes from $B_0 = 1$ and has degree $m + n + 1$; a binomial coefficient gives its stated leading coefficient. For $m = n$ and odd k, the factor $u\mathcal{A} - v\mathcal{B}$ is antisymmetric under $u \leftrightarrow v$, while every other factor in (8.127) is symmetric. Its diagonal mixed coefficient is zero. Finally the regular series has zero limit at $a = 0$, proving (8.128). □

The first collision polynomials are

$$\mathcal{P}_{00}(L) = L, \qquad \mathcal{P}_{10}(L) = \frac{L^2}{2} + \zeta(2), \qquad \mathcal{P}_{01}(L) = L^2 + \zeta(2). \tag{8.130}$$

The asymmetry of the last two is required: the factor that becomes singular at $a = 0$ has changed its index. In particular,

$$J_{00}(a) - \frac{\log a}{a} \longrightarrow 2\gamma_1 - \frac{\pi^2}{3}.$$

There is no arbitrary additional finite constant in this limit. It is fixed by the initial endpoint subtraction and agrees with the coincident prescription.

8.25 Ordinary integrals of all antiderivative orders

The finite-part identities have a convergent counterpart. We first choose mean-zero periodic primitives, rather than implicitly choosing constants of integration at a singular endpoint.

Definition 8.77 (Normalized primitive tower)**.** *For integers $m \geq 0$ and $r \geq 1$, put*

$$\mathfrak{h}_j^{(r-1)} = [u^j] \prod_{\nu=1}^{r-1} (1 - u/\nu)^{-1},$$
$$U_{m,r}(x) = \frac{(-1)^{m+1} m!}{(r-1)!} \sum_{j=0}^{m+1} \mathfrak{h}_{m+1-j}^{(r-1)} \frac{\zeta^{[j]}(1-r, x)}{j!}. \tag{8.131}$$

The empty product convention gives $\mathfrak{h}_0^{(0)} = 1$ and $\mathfrak{h}_j^{(0)} = 0$ for $j > 0$. Write $P_m = U_{m,1}$ and $Q_m = U_{m,2}$. Explicitly,

$$P_m(x) = \frac{(-1)^{m+1}}{m+1} \zeta^{[m+1]}(0, x), \tag{8.132}$$

$$Q_m(x) = (-1)^{m+1} m! \sum_{j=0}^{m+1} \frac{\zeta^{[j]}(-1, x)}{j!}. \tag{8.133}$$

These are normalized Hurwitz-jet primitives. The repository's primitive based at 1 is $P_m(x) - P_m(1)$; it is not $P_m(x)$ itself. Likewise, zero-based negative polygamma conventions must be converted explicitly before using the following correlations.

Lemma 8.78 (Primitive identities and regularity)**.** *Each $U_{m,r}$ has mean zero on $(0, 1)$. One has*

$$DP_m = G_m, \qquad DU_{m,r} = U_{m,r-1} \quad (r \geq 2), \qquad D^r U_{m,r} = G_m$$

as periodic distributions. The functions P_m lie in $L^2(\mathbb{T})$, and for $r \geq 2$, $U_{m,r}$ is a periodic C^{r-2} function.

Proof. Consider the spectral primitive

$$V_r(u, x) = \frac{(-1)^r \zeta(1 + u - r, x)}{(1 + u - r)_r} = -\frac{\zeta(1 + u - r, x)}{(r-1)! u} \prod_{\nu=1}^{r-1} (1 - u/\nu)^{-1}. \tag{8.134}$$

Its regular coefficient $(-1)^m m![u^m]V_r$ is (8.131). Pointwise differentiation gives $\partial_x V_r = V_{r-1}$ for $r \geq 2$, and $\partial_x V_1 = \zeta(1+u,x)$. Comparing regular coefficients proves the pointwise primitive relations.

For $r = 1$, the shift identity gives

$$P_m(x) = \frac{\log^{m+1} x}{m+1} + P_m(1+x).$$

This proves square integrability. Integration by parts against a periodic test function, with a lower cutoff, yields the boundary term $\phi(0)\log^{m+1}\epsilon/(m+1)$ and hence $DP_m = G_m$ by (8.59). For $r \geq 2$, the singular term in $\zeta^{[j]}(1-r,x)$ is $x^{r-1}(-\log x)^j$, which tends to zero at $x = 0$. The endpoint values at $0+$ and 1 consequently match. Iterating the pointwise derivative relations shows periodic C^{r-2} regularity and no extra jump terms. This proves the distributional relations. Finally, $\int_0^1 \zeta(s,x)\,\mathrm{d}x = 0$ for $\Re s < 1$, and its spectral derivatives also integrate to zero. Apply this to each term in (8.131). □

Theorem 8.79 (All-order convergent lifting)**.** *Suppose the closure formula for fixed m, n is written as*

$$J_{mn}(a) = \sum_j c_j \gamma_j(a) + \sum_j d_j \gamma_j(b) + c, \tag{8.135}$$

where all sums are finite and the coefficients are independent of a. For $r, k \geq 1$, $N = r + k$, define the ordinary integral

$$W_{m,n}^{r,k}(a) = \int_0^1 U_{m,r}(x) U_{n,k}(\{x+a\})\,\mathrm{d}x.$$

Then

$$\begin{aligned} W_{m,n}^{r,k}(a) = (-1)^r \sum_j c_j U_{j,N}(a) + (-1)^k \sum_j d_j U_{j,N}(b) \\ + \frac{(-1)^r c}{N!} B_N(a), \end{aligned} \tag{8.136}$$

where B_N is the ordinary Bernoulli polynomial. Thus every such integral reduces to finitely many Hurwitz-zeta jets at the single spectral integer $1 - r - k$ and zeta-valued coefficients. The highest required spectral derivative is $m + n + 2$.

Proof. The integral exists by the L^2 property and Cauchy–Schwarz. Its mean in a is zero by Fubini and the mean-zero normalization of both factors. In the distributional sense, periodic integration by parts gives

$$\frac{d^N}{da^N} W_{m,n}^{r,k}(a) = (-1)^r J_{mn}(a), \qquad 0 < a < 1. \tag{8.137}$$

The right side is smooth there.

We need enough endpoint regularity to fix all integration constants. From (8.96) and (8.98), for $q \neq 0$,

$$\begin{aligned} \widehat{G}_m(q) = (-1)^{m+1} m! [u^{m+1}] \exp\Bigg[\left(\kappa + \log|q| + \frac{\pi \mathrm{i}}{2} \operatorname{sgn} q \right) u \\ + \sum_{j \geq 2} \frac{\zeta(j)}{j} u^j \Bigg]. \end{aligned} \tag{8.138}$$

In particular $\widehat{G}_m(q) = O_m((1+\log|q|)^{m+1})$. The primitive relations imply $\widehat{U}_{m,r}(q) = \widehat{G}_m(q)/(2\pi i q)^r$. The Fourier coefficients of W are therefore $O((1+\log|q|)^{m+n+2}/|q|^N)$. After $N-2$ derivatives the series is still absolutely convergent. Hence W is periodic C^{N-2}.

Let $F(a)$ be the proposed right side of (8.136). It also has mean zero and is periodic C^{N-2}, using Lemma 8.78 and the endpoint identities for Bernoulli polynomials. Its Nth derivative on $(0,1)$ is $(-1)^r J_{mn}$: the reflected primitive contributes $(-1)^N$, and $B_N^{(N)} = N!$. Thus $W - F$ is a polynomial of degree at most $N-1$ on $(0,1)$. Matching its endpoint derivatives of orders $0,\dots,N-2$ forces every nonconstant coefficient to vanish, starting with the leading one. Its zero mean then forces the remaining constant to vanish.

The closure uses indices at most $m+n+1$, and $U_{j,N}$ uses spectral derivatives at most $j+1$. This gives the final derivative-order assertion. □

8.25.1 The first-primitive correlation

For $K_{mn}(a) = W^{1,1}_{m,n}(a)$, the theorem simplifies to

$$K_{mn}(a) = -\sum_j c_j Q_j(a) - \sum_j d_j Q_j(b) - \frac{c}{2}B_2(a), \qquad K''_{mn}(a) = -J_{mn}(a). \tag{8.139}$$

Equivalently, ordinary differentiation of the master integral at $(s,t) = (0,0)$ gives

$$K_{mn}(a) = (-1)^{m+n} m!n![s^{m+1}t^{n+1}]C(s,t;a). \tag{8.140}$$

The signs and factorials follow from (8.132); no finite-part operation occurs in this integral.

For example, (8.111) gives the fully explicit convergent identity

$$\begin{aligned} K_{11}(a) = {} & -\tfrac{1}{2}\big(Q_3(a)+Q_3(b)\big) - 2\zeta(2)\big(Q_1(a)+Q_1(b)\big) \\ & -\zeta(3)\big(Q_0(a)+Q_0(b)\big) + \tfrac{1}{2}\big(\zeta(4)+\zeta(2)^2\big)B_2(a). \end{aligned} \tag{8.141}$$

This evaluates a product of two second spectral derivatives at zero using derivatives at -1 of order at most four.

An example with unequal integration orders is

$$\int_0^1 Q_0(x)P_0(\{x+a\})\,\mathrm{d}x = U_{1,3}(a) - U_{1,3}(b) - \frac{\zeta(2)}{3}B_3(a), \tag{8.142}$$

where

$$U_{1,3}(x) = \frac{7}{8}\zeta(-2,x) + \frac{3}{4}\zeta^{[1]}(-2,x) + \frac{1}{4}\zeta^{[2]}(-2,x).$$

Thus the same finite closure gives exact identities involving successive antiderivatives rather than merely a formal recurrence for integration constants.

8.26 Explicit log-gamma and mixed identities

Let

$$f(x) = \log\Gamma(x) - \frac{\ell}{2}, \qquad \mathcal{K}(a) = \int_0^1 f(x)f(\{x+a\})\,\mathrm{d}x.$$

Lerch's identity gives $P_0 = -f$, so $\mathcal{K} = K_{00}$. No regularization is used in this definition: logarithmic singularities are square integrable.

Theorem 8.80 (Translated log-gamma correlation)**.** *For* $0 < a < 1$,

$$\mathcal{K}(a) = (1+\zeta(2))B_2(a) - \zeta^{[1]}(-1,a) - \zeta^{[1]}(-1,b) - \frac{1}{2}\big[\zeta^{[2]}(-1,a) + \zeta^{[2]}(-1,b)\big]. \tag{8.143}$$

Equivalently, for $z = e^{2\pi i a}$,

$$\mathcal{K}(a) = \frac{1}{2\pi^2}\Re\left[\mathrm{Li}_2^{[2]}(z) - 2\kappa\,\mathrm{Li}_2^{[1]}(z) + \left(\kappa^2 + \frac{\pi^2}{4}\right)\mathrm{Li}_2(z)\right]. \tag{8.144}$$

For the uncentered product,

$$\int_0^1 \log\Gamma(x)\log\Gamma(\{x+a\})\,\mathrm{d}x = \frac{\ell^2}{4} + \mathcal{K}(a). \tag{8.145}$$

Finally,

$$\mathcal{K}''(a) = \frac{\pi^2}{3} - \gamma_1(a) - \gamma_1(b). \tag{8.146}$$

Proof. Apply (8.139) to $J_{00} = \gamma_1(a) + \gamma_1(b) - 2\zeta(2)$. Since

$$Q_1(x) = \zeta(-1,x) + \zeta^{[1]}(-1,x) + \tfrac{1}{2}\zeta^{[2]}(-1,x)$$

and $\zeta(-1,x) = -B_2(x)/2$, the result is (8.143).

For an independent Fourier derivation of (8.144), differentiate (8.96) at $s = 0$. The positive-frequency coefficient of f is

$$\widehat{f}(n) = \frac{1}{4n} - \frac{\mathrm{i}(\kappa + \log n)}{2\pi n}, \qquad n \geq 1.$$

The negative-frequency coefficient is its conjugate. These coefficients also give the classical Kummer Fourier expansion. The L^2 correlation therefore equals the absolutely convergent series

$$\mathcal{K}(a) = \frac{1}{2\pi^2}\sum_{n\geq 1}\frac{\cos(2\pi n a)}{n^2}\left[(\kappa+\log n)^2 + \frac{\pi^2}{4}\right]. \tag{8.147}$$

Taking the first two spectral derivatives of the absolutely convergent polylogarithm series at order 2 proves (8.144). The mean of f is zero, so expanding the uncentered product proves (8.145). Equation (8.146) is the $m = n = 0$ case of (8.139). □

Corollary 8.81 (A mixed log-gamma–digamma identity)**.** *The same finite-part convention gives*

$$\mathrm{FP}\int_0^1 \log\Gamma(x)\psi(\{x+a\})\,\mathrm{d}x = \frac{1}{2}\big[\zeta^{[2]}(0,b) - \zeta^{[2]}(0,a)\big] + \zeta(2)(2a-1). \tag{8.148}$$

It is unchanged when $\log\Gamma(x)$ *is replaced by* $f(x)$.

Proof. Since $DP_0 = G_0$ with no contact term at this first primitive stage, $\mathcal{K}'(a) = \mathrm{FP}\int f(x)\psi(\{x+a\})\,\mathrm{d}x$. Differentiate (8.143) using

$$\partial_x\zeta^{[j]}(-1,x) = \zeta^{[j]}(0,x) - j\zeta^{[j-1]}(0,x).$$

The first spectral derivatives at zero cancel, and the terms from $\zeta(0,x) = 1/2 - x$ simplify to $\zeta(2)(2a-1)$. This gives (8.148). The canonical periodic finite part of ψ has mean zero, so subtracting $\ell/2$ from the log-gamma factor does not change the answer. □

For example, its left side with the centered function equals the entirely convergent expression

$$\int_0^b f(x)\psi(x+a)\,\mathrm{d}x + \int_0^a \left[(f(b+t)-f(b))\psi(t)+f(b)\psi(1+t)\right]\,\mathrm{d}t - f(b)\log a.$$

The recurrence $\psi(1+t)=\psi(t)+1/t$ removes the endpoint pole explicitly. This form also provides a direct numerical check independent of differentiating a finite-part integral.

8.26.1 Half shift and the zero-shift limit

At $a=1/2$, use $\mathrm{Li}_s(-1)=(2^{1-s}-1)\zeta(s)$. With $L_2=\log 2$,

$$\mathcal{K}(1/2) = -\frac{\zeta^{[2]}(2)}{4\pi^2}+\frac{\kappa-L_2}{2\pi^2}\zeta^{[1]}(2) + \frac{L_2^2+2\kappa L_2-\kappa^2-\pi^2/4}{24}. \tag{8.149}$$

This formula concerns a *translated* correlation, not the reflected product $f(x)f(1-x)$.

The absolutely convergent series (8.147) is continuous at $a=0$. Its value there recovers the classical second log-gamma moment [28]:

$$\int_0^1 \log^2\Gamma(x)\,\mathrm{d}x = \frac{\ell^2}{4}+\frac{\zeta^{[2]}(2)}{2\pi^2}-\frac{\kappa\zeta^{[1]}(2)}{\pi^2}+\frac{\kappa^2}{12}+\frac{\pi^2}{48}. \tag{8.150}$$

Equivalently, taking the continuous limit in (8.143) gives

$$\mathcal{K}(0)=\frac{1+\zeta(2)}{6}-2\zeta^{[1]}(-1)-\zeta^{[2]}(-1).$$

These are useful consistency checks, not priority claims. In contrast, $J_{00}(a)=\log a/a+O(1)$ as $a\downarrow 0$, by (8.90). The finite-part product has no continuous zero-shift specialization. The ordinary correlation remains continuous but its second derivative develops the expected singularity.

8.27 Rational shifts and exact arithmetic coordinates

For integers $q\ge 2$ and $1\le p<q$, residue-class decomposition gives the classical finite Fourier identity

$$\mathrm{Li}_s(e^{2\pi\mathrm{i}p/q})=q^{-s}\sum_{r=1}^{q}e^{2\pi\mathrm{i}pr/q}\zeta(s,r/q). \tag{8.151}$$

It is proved by absolutely convergent series for $\Re s>1$ and then by analytic continuation. Spectral differentiation yields

$$\mathrm{Li}_s^{[j]}(e^{2\pi\mathrm{i}p/q})=q^{-s}\sum_{r=1}^{q}e^{2\pi\mathrm{i}pr/q}\sum_{h=0}^{j}\binom{j}{h}(-\log q)^{j-h}\zeta^{[h]}(s,r/q). \tag{8.152}$$

At $s=1$ the poles of the finite sum cancel because the roots sum to zero. One must combine the pole terms before evaluating; separately substituting $\zeta(1,r/q)$ is invalid.

At a rational shift, Theorem 8.72 reduces all Stieltjes correlations to rational-argument Stieltjes constants and ordinary zeta values. Theorem 8.79 places every primitive correlation at a negative integer spectral argument. Equation (8.152) instead evaluates the polylogarithmic side using positive-order Hurwitz jets. The two routes give independent choices of arithmetic coordinates for numerical replay and exact symbolic comparison.

For a Dirichlet character modulo q, including an imprimitive one with its actual modulus, the identity

$$\sum_{r=1}^{q} \chi(r)\zeta(s, r/q) = q^s L(s, \chi)$$

and its spectral derivatives transform the character components into Dirichlet-L jets. This gives a reduction framework, not a claim of numerical independence or minimality of the resulting coordinates.

Corollary 8.82 (Small-denominator digamma correlations)**.** *Write* $\gamma_1 = \gamma_1(1)$*. Then*

$$\Pi_{0,0}(1/2) = 2\gamma_1 - 4\gamma \log 2 - 2\log^2 2 - \frac{\pi^2}{3}, \tag{8.153}$$

$$\Pi_{0,0}(1/3) = 2\gamma_1 - 3\gamma \log 3 - \frac{3}{2}\log^2 3 - \frac{\pi^2}{3}, \tag{8.154}$$

$$\Pi_{0,0}(1/4) = 2\gamma_1 - 6\gamma \log 2 - 7\log^2 2 - \frac{\pi^2}{3}. \tag{8.155}$$

Proof. Expand at $s = 1$ the exact identities

$$\begin{aligned} \zeta(s, 1/2) &= (2^s - 1)\zeta(s), \\ \zeta(s, 1/3) + \zeta(s, 2/3) &= (3^s - 1)\zeta(s), \\ \zeta(s, 1/4) + \zeta(s, 3/4) &= (4^s - 2^s)\zeta(s). \end{aligned}$$

Compare the coefficient of $s - 1$ with (8.89) and insert the resulting sums into (8.113). In the first line the correlation contains two copies of $\gamma_1(1/2)$, which accounts for its different coefficients. □

Higher indices require no new search: expand these finite exponential factors to the needed order and use (8.105). At larger denominators, retaining a nontrivial character block is legitimate; failing to reduce that block does not establish transcendence or independence.

The canonical finite-part and primitive identities above are audited against the independent coefficient engines and their separately labeled numerical replays under `verification/eighth-replay/`. The full contact law is proved in the next chapter before derivatives of these finite-part identities are used. The singular same-point finite part and the continuous ordinary log-Gamma autocorrelation have different zero-shift behavior.

8.28 All circular convolution powers of log Gamma

At the integrated end no finite part is needed. Put

$$L = \log(2\pi), \qquad Z_j(x) = \zeta^{[j]}(0, x), \qquad g(x) = Z_1(x) = \log \Gamma(x) - \frac{L}{2}. \tag{8.156}$$

Each Z_j has mean zero. It is locally a polynomial in $\log x$ plus a smooth function near zero, and hence belongs to every finite $L^p(0, 1)$.

Theorem 8.83 (All mixed integrated jets)**.** *Let $d \geq 2$ and $r_1, \dots, r_d \geq 0$ be integers. Write $S = s_1 + \cdots + s_d$. Then, as an identity of continuous functions after circular convolution,*

$$Z_{r_1} * \cdots * Z_{r_d}(x) = \partial_{s_1}^{r_1} \cdots \partial_{s_d}^{r_d} \left[\frac{\prod_{i=1}^{d} \Gamma(1-s_i)}{\Gamma(d-S)} \zeta(1-d+S, x) \right] \Bigg|_{s_1 = \cdots = s_d = 0}. \tag{8.157}$$

It holds for $0 < x < 1$ and at the integer point by continuous extension. Thus any mixed convolution of the centered primitives A_n reduces to finitely many jets at the single integer $1-d$.

Proof. Before differentiation, write each Hurwitz factor as $\zeta(s_i, x) = \Gamma(1-s_i) R_{1-s_i}(x)$ and apply Theorem 8.58 repeatedly. For parameters sufficiently close to zero, each factor is holomorphic as an L^2-valued family: the endpoint singularities are bounded by $x^{-\eta}(1 + |\log x|^M)$ with $2\eta < 1$ on each fixed derivative order and compact parameter set. Convolution of two L^2 functions is continuous, by Cauchy–Schwarz and translation continuity; further convolution with L^1 factors preserves continuity. These same estimates justify differentiating all the convolutions. The right side is continuous at the integer point since $\mathrm{Re}(1-d+S) < 0$ in a small parameter neighborhood. This proves the formula and its endpoint extension. □

The case with every $r_i = 1$ admits a compact all-d formula. Define $H_m^{(j)} = \sum_{a=1}^{m} a^{-j}$, and, for fixed $d \geq 2$, set

$$b_1 = H_{d-1}, \qquad b_j = (j-1)!\big[H_{d-1}^{(j)} - \zeta(j)\big] \quad (j \geq 2). \tag{8.158}$$

Theorem 8.84 (All convolution powers of centered log-gamma)**.** *For every integer $d \geq 2$,*

$$g^{*d}(x) = \frac{1}{(d-1)!} \sum_{j=0}^{d} \binom{d}{j} \mathcal{B}_{d-j}(b_1, \dots, b_{d-j}) \zeta^{[j]}(1-d, x), \tag{8.159}$$

where $\mathcal{B}_0 = 1$ and the b_j are scalar quantities in (8.158).

Proof. In (8.157), differentiate once in each s_i. Only the squarefree coefficient $s_1 \cdots s_d$ matters. Modulo $s_i^2 = 0$, one has

$$\prod_{i=1}^{d} \Gamma(1-s_i) = \prod_{i=1}^{d} (1 + \gamma s_i) = \exp(\gamma S).$$

Consequently the required mixed derivative equals the ordinary dth derivative at $S = 0$ of $e^{\gamma S} \zeta(1-d+S, x)/\Gamma(d-S)$. Now

$$\log \frac{e^{\gamma S} \Gamma(d)}{\Gamma(d-S)} = H_{d-1} S + \sum_{j \geq 2} \frac{H_{d-1}^{(j)} - \zeta(j)}{j} S^j.$$

This follows directly from $\Gamma(d-S) = \Gamma(1-S) \prod_{a=1}^{d-1} (a-S)$ and the Taylor series of $\log \Gamma(1-S)$. Apply (8.72) and the Leibniz rule; $\Gamma(d) = (d-1)!$ supplies the prefactor. □

The first two nontrivial cases are

$$g * g = \zeta''(-1, x) + 2\zeta'(-1, x) + [2 - \zeta(2)] \zeta(-1, x), \tag{8.160}$$

$$\begin{aligned} g^{*3} = {} & \frac{1}{2} \zeta'''(-2, x) + \frac{9}{4} \zeta''(-2, x) + \left[\frac{21}{4} - \frac{3}{2} \zeta(2) \right] \zeta'(-2, x) \\ & + \left[\frac{45}{8} - \frac{9}{4} \zeta(2) - \zeta(3) \right] \zeta(-2, x). \end{aligned} \tag{8.161}$$

The first left side is explicitly

$$\int_0^x g(t)g(x-t)\,\mathrm{d}t + \int_x^1 g(t)g(1+x-t)\,\mathrm{d}t.$$

For uncentered $f = \log\Gamma$, mean-zero convolution annihilates all mixed constant/nonconstant terms, so

$$f^{*d}(x) = \left(\frac{L}{2}\right)^d + g^{*d}(x). \tag{8.162}$$

Remark 8.85 (Convolution powers are not pointwise moments). The value $g^{*3}(0)$ is a two-dimensional circular-convolution integral. It is not $\int_0^1 g(t)^3\,\mathrm{d}t$. Thus (8.161) does not eliminate the Tornheim derivative coordinates of the ordinary cubic log-gamma moment discussed in the repository. Likewise, the mixed reflected pointwise moments in `07-reflected-moments.tex` are different objects. This distinction prevents a spurious claim of resolving that separate reduction problem.

8.29 Nonintegral twists: Lerch jets, contact terms, and the half-twist

The convolution-algebra continuation explicitly asks for a finite-part Stieltjes normal form for nonintegral frequency shifts, with an all-index half-twist multiplication table that includes its point-supported counterterms [108, "Twists, characters, and alternate endpoint normalizations"]. The twisted continuation [119] supplies the full proof below. The underlying Lerch function is classical [115, §25.14]; at the half-twist its order jets are the modified Stieltjes constants of Hu and Kim [117]. An ordinary, nonsingular alternating Hurwitz convolution is already given by Zhu, Hu, and Kim [118, Theorem 1.13]. The additional content below is the continuation through the singular orders, including the exact contact terms, covariant primitives, and degeneration to the untwisted normalization. In particular, twisting changes the special-function jets and the convolution identity; the positive-integer zeta values in the Gamma correction remain the same.

8.29.1 The meromorphic family and its endpoint normalization

Fix $0 < \theta < 1$, put $\omega = 2\pi\theta$ and $z = e^{-i\omega}$, and retain the Fourier and convolution conventions of Section 8.17. Any other real nonintegral twist $\theta = m + \vartheta$, $m \in \mathbb{Z}$, $0 < \vartheta < 1$, is obtained by the periodic modulation $U(x) \mapsto e^{-2\pi i m x}U(x)$. Thus the chosen interval loses no real nonintegral shifts. Define

$$\lambda_{n,\theta} = 2\pi i(n+\theta), \qquad L_{n,\theta} = \log(2\pi|n+\theta|) + \frac{i\pi}{2}\operatorname{sgn}(n+\theta), \qquad n \in \mathbb{Z}. \tag{8.163}$$

Every power of $\lambda_{n,\theta}$ uses this logarithm. There is no zero frequency to remove. We write

$$D_\theta = D + 2\pi i\theta, \qquad \widehat{D_\theta U}(n) = \lambda_{n,\theta}\widehat{U}(n).$$

Here D_θ acts on ordinary periodic distributions, rather than on functions satisfying a different boundary condition. The nonperiodic factor $e^{-i\omega x}$ below is used only for ordinary representatives on $0 < x < 1$ and for local endpoint Taylor coefficients. It is not used as a global multiplier on periodic distributions; the Fourier or cutoff definitions fix those extensions. Explicitly, for $m \in \mathbb{Z}$ the extension to shifted intervals is $W_s^{\theta+m}(x) = e^{-2\pi i m x}W_s^\theta(x)$, with the same identity for Z_s, U_n, and P_k after their definitions below.

Theorem 8.86 (Twisted Lerch family)**.** *The Fourier coefficients*

$$\widehat{W_s^\theta}(n) = \lambda_{n,\theta}^{s-1}, \qquad n \in \mathbb{Z}, \tag{8.164}$$

define an entire distribution-valued function of s. The family

$$Z_s^\theta = \Gamma(1-s) W_s^\theta \tag{8.165}$$

is meromorphic, with simple poles precisely at $s = k+1$, $k \geq 0$, and

$$\operatorname{Res}_{s=k+1} Z_s^\theta = \frac{(-1)^{k+1}}{k!} D_\theta^k \delta_0. \tag{8.166}$$

For $\operatorname{Re} s < 1$, it is the locally integrable periodic extension of

$$f_s^\theta(x) = e^{-i\omega x} \Phi(z, s, x), \qquad 0 < x < 1. \tag{8.167}$$

For every s, the restriction of the meromorphic distribution to $(0,1)$ is the entire function of s on the right of (8.167). The convolution and differential identities are

$$W_s^\theta * W_t^\theta = W_{s+t-1}^\theta, \qquad W_1^\theta = \delta_0, \tag{8.168}$$

$$D_\theta W_s^\theta = W_{s+1}^\theta, \qquad D_\theta Z_s^\theta = -s Z_{s+1}^\theta. \tag{8.169}$$

Consequently,

$$Z_s^\theta * Z_t^\theta = \frac{\Gamma(1-s)\Gamma(1-t)}{\Gamma(2-s-t)} Z_{s+t-1}^\theta \tag{8.170}$$

as a meromorphic identity, interpreted through (8.168) at intersections of polar divisors.

Proof. On compact sets of s, the coefficients in (8.164), and each fixed number of their order derivatives, have a common polynomial growth bound in $|n|$. Pairing their Fourier series with the rapidly decreasing Fourier coefficients of a smooth test function therefore gives normal convergence. This proves entire dependence. The Fourier multiplier of $D_\theta^k \delta_0$ is $\lambda_{n,\theta}^k$; the residues of $\Gamma(1-s)$ prove (8.166). These distributions are nonzero, so every displayed pole is present.

For completeness, the Lerch identification can be proved without assuming a boundary-value Fourier formula. Initially let $0 < \operatorname{Re} s < 1$ and $\epsilon > 0$. Abel damping gives

$$f_{s,\epsilon}^\theta(x) = \sum_{m \geq 0} e^{-(\epsilon + i\omega)(m+x)} (m+x)^{-s}.$$

Absolute convergence permits unfolding its nth Fourier coefficient:

$$\begin{aligned} \int_0^1 f_{s,\epsilon}^\theta(x) e^{-2\pi i n x}\, dx &= \int_0^\infty t^{-s} e^{-(\epsilon + \lambda_{n,\theta})t}\, dt \\ &= \Gamma(1-s)(\epsilon + \lambda_{n,\theta})^{s-1}. \end{aligned}$$

Since $z \neq 1$, the partial sums of z^m are bounded. Summation by parts shows that the tail starting at $m = 1$ converges uniformly in $x \in [0,1]$, and locally uniformly for s in this strip, as $\epsilon \downarrow 0$. The remaining term is bounded by an integrable multiple of $x^{-\operatorname{Re} s}$. Thus the limit can be taken in $L^1(0,1)$ and in its Fourier coefficients, giving (8.167). Repeated summation by parts, or finite differences of $(m+x)^{-s}$, continues the Lerch tail holomorphically to every s, locally uniformly when x stays away from zero. Near zero its only nonsmooth term is

$$f_s^\theta(x) = e^{-i\omega x} x^{-s} + e^{-i\omega x} z \Phi(z, s, 1+x). \tag{8.171}$$

This proves L^1 holomorphy throughout $\operatorname{Re} s < 1$ and the asserted continuation away from the endpoint. The remaining identities follow by multiplying the Fourier coefficients; the Gamma recurrence gives the last identity of (8.169). □

The pointwise function in (8.167) has no pole in s. Every pole in (8.165) is an endpoint distribution. This is different from the untwisted Hurwitz family, whose pole at $s = 1$ also has an ordinary constant contribution away from the endpoint.

For $n \geq 0$ define ordinary Lerch jets and their periodic finite parts by

$$c_n^\theta(x) = (-1)^n \, \partial_s^n f_s^\theta(x)\Big|_{s=1}, \qquad 0 < x < 1, \tag{8.172}$$

$$Z_{1+u}^\theta = -\frac{\delta_0}{u} + \sum_{n\geq 0} \frac{(-1)^n}{n!} U_n^\theta u^n. \tag{8.173}$$

Set, also,

$$P_k^\theta = (-1)^{k+1} k! \, \mathrm{FP}_{s=k+1} \, Z_s^\theta, \qquad k \geq 0. \tag{8.174}$$

In particular $P_0^\theta = -U_0^\theta$. Off the endpoint it equals $p_k^\theta(x) = (-1)^{k+1} k! f_{k+1}^\theta(x)$.

Proposition 8.87 (Explicit singular normalization)**.** *For a smooth periodic test function ϕ, put $h(x) = e^{-i\omega x}\phi(x)$ near zero. Then*

$$\langle U_n^\theta, \phi \rangle = \lim_{\epsilon \downarrow 0} \left\{ \int_\epsilon^1 c_n^\theta(x)\phi(x)\, dx + \frac{\phi(0)}{n+1} \log^{n+1} \epsilon \right\}. \tag{8.175}$$

If $F_k^\theta = \mathrm{FP}_{s=k+1} \, Z_s^\theta$ and $k \geq 1$, then

$$\langle F_k^\theta, \phi \rangle = \lim_{\epsilon \downarrow 0} \left\{ \int_\epsilon^1 f_{k+1}^\theta(x)\phi(x)\, dx - \sum_{j=0}^{k-1} \frac{h^{(j)}(0)}{j!} \frac{\epsilon^{j-k}}{k-j} + \frac{h^{(k)}(0)}{k!} \log \epsilon \right\}. \tag{8.176}$$

These normalizations retain the zero Fourier coefficient; no mean-zero condition is imposed.

Proof. Write the second term in (8.171) as $r_s^\theta(x)$. After subtracting the degree-N Taylor polynomial of h, the continuation of the test-function pairing is

$$\begin{aligned} \langle Z_s^\theta, \phi \rangle = {} & \int_0^1 x^{-s} \left(h(x) - \sum_{j=0}^{N} \frac{h^{(j)}(0)}{j!} x^j \right) dx + \sum_{j=0}^{N} \frac{h^{(j)}(0)}{j!(j+1-s)} \\ & + \int_0^1 r_s^\theta(x)\phi(x)\, dx. \end{aligned} \tag{8.177}$$

The first integral is holomorphic for $\mathrm{Re}\, s < N + 2$, and the last integral is entire. At $s = 1 + u$, expand $x^{-1-u} = x^{-1} \sum_n (-1)^n u^n \log^n x/n!$ after the constant subtraction. This gives (8.175). At $s = k + 1$, take the constant Laurent coefficient and integrate each subtracted monomial explicitly; this gives (8.176). The residue computed this way agrees with (8.166), because

$$\langle D_\theta^k \delta_0, \phi \rangle = (-1)^k h^{(k)}(0).$$

This also fixes the signs of all the lower derivatives of δ_0 inside $D_\theta^k \delta_0$. □

8.29.2 All-index multiplication and covariant contact terms

Let

$$B_\theta(u) = -uZ_{1+u}^\theta = \Gamma(1-u) W_{1+u}^\theta, \qquad G_\theta(u) = \sum_{n\geq 0} \frac{(-1)^n}{n!} U_n^\theta u^n. \tag{8.178}$$

Thus $B_\theta(u) = \delta_0 - uG_\theta(u)$. All the following equalities are analytic germs at zero in the algebra of periodic distributions with its usual convolution identity δ_0.

Theorem 8.88 (Twisted Stieltjes normal form)**.** *Put* $C(u,v)=\Gamma(1-u)\Gamma(1-v)/\Gamma(1-u-v)$. *Then*

$$B_\theta(u)*B_\theta(v)=C(u,v)B_\theta(u+v), \tag{8.179}$$

$$B_\theta(u)=\exp_*\left(uP_0^\theta+\sum_{r\ge2}\frac{\zeta(r)}{r}u^r\delta_0\right). \tag{8.180}$$

In particular,

$$U_n^\theta=\frac{(-1)^{n+1}}{n+1}\mathcal{B}_{n+1}^*\big(P_0^\theta,\zeta(2)\delta_0,2!\zeta(3)\delta_0,\dots,n!\zeta(n+1)\delta_0\big). \tag{8.181}$$

An explicit all-index multiplication table is furnished by the removable coefficient formula

$$\begin{aligned}G_\theta(u)*G_\theta(v)=\frac{1}{uv}\big\{&uG_\theta(u)+vG_\theta(v)-(u+v)C(u,v)G_\theta(u+v)\\&+(C(u,v)-1)\delta_0\big\}.\end{aligned} \tag{8.182}$$

More precisely, $U_m^\theta*U_n^\theta$ *is* $(-1)^{m+n}m!n!$ *times the coefficient of* u^mv^n *on the right. This is a finite combination of* $\delta_0,U_0^\theta,\dots,U_{m+n+1}^\theta$*, with coefficients polynomial in ordinary zeta values.*

Proof. The first identity follows from (8.168). At every Fourier mode, including $n=0$,

$$\widehat{P_0^\theta}(n)=\gamma+L_{n,\theta},\qquad\widehat{B_\theta(u)}(n)=\exp\left((\gamma+L_{n,\theta})u+\sum_{r\ge2}\frac{\zeta(r)}{r}u^r\right).$$

This proves (8.180) and the Bell formula, with normal convergence after test-function pairing. Substitute $B_\theta(u)=\delta_0-uG_\theta(u)$ into (8.179) and solve for $G_\theta(u)*G_\theta(v)$ to obtain (8.182). Its numerator vanishes when either variable is zero, so the quotient is removable. Extracting a fixed coefficient uses only finitely many coefficients of the Gamma quotient and of G_θ. □

For example, the full identities, including their contact masses, are

$$U_0^\theta*U_0^\theta=2U_1^\theta-\zeta(2)\delta_0, \tag{8.183}$$

$$U_0^\theta*U_1^\theta=\tfrac{3}{2}U_2^\theta-\zeta(2)U_0^\theta+\zeta(3)\delta_0, \tag{8.184}$$

$$U_0^\theta*U_2^\theta=\tfrac{4}{3}U_3^\theta-2\zeta(2)U_1^\theta+2\zeta(3)U_0^\theta-2\zeta(4)\delta_0, \tag{8.185}$$

$$U_1^\theta*U_1^\theta=U_3^\theta-2\zeta(2)U_1^\theta+2\zeta(3)U_0^\theta-\tfrac{1}{4}\zeta(4)\delta_0. \tag{8.186}$$

The presence of δ_0, rather than δ_0-1, matters even when these formulas are eventually evaluated away from zero.

Theorem 8.89 (All covariant contact corrections)**.** *For* $k\ge0$*, with* $H_0=0$*,*

$$\widehat{P_k^\theta}(n)=\lambda_{n,\theta}^k(\gamma+L_{n,\theta}-H_k), \tag{8.187}$$

$$D_\theta P_k^\theta=P_{k+1}^\theta+\frac{1}{k+1}D_\theta^{k+1}\delta_0, \tag{8.188}$$

$$Q_k^\theta:=D_\theta^kP_0^\theta=P_k^\theta+H_kD_\theta^k\delta_0. \tag{8.189}$$

Hence the all-index twisted polygamma product is

$$P_k^\theta*P_\ell^\theta=D_\theta^{k+\ell}\big\{2U_1^\theta+(H_k+H_\ell)U_0^\theta+(H_kH_\ell-\zeta(2))\delta_0\big\}. \tag{8.190}$$

More generally, let $R^{\theta}_{n,k}$ be the raw Hadamard finite part of the ordinary function $D^k_\theta c^\theta_n$ on $(0,1)$, using Taylor subtraction of its powers of x and $\log x$ at zero and no additional endpoint mass. Then

$$D^k_\theta U^\theta_n = R^\theta_{n,k} - h_{k,n} D^k_\theta \delta_0, \qquad h_{k,n} = (-1)^n n! [u^{n+1}] \prod_{j=1}^{k} \left(1 + \frac{u}{j}\right). \tag{8.191}$$

In particular $h_{k,0} = H_k$, and $h_{k,n} = 0$ for $n \geq k$.

Proof. At $s = k + 1 + u$,

$$\Gamma(-k-u) = \frac{(-1)^{k+1}}{k!u} + \frac{(-1)^k}{k!}(H_k - \gamma) + O(u).$$

Multiply this by $\lambda^k_{n,\theta} e^{uL_{n,\theta}}$ and take the finite part. This proves (8.187). The two derivative formulas follow by multiplication by $\lambda_{n,\theta}$ and $H_{k+1} - H_k = 1/(k+1)$. Multiplying the two affine polynomials $\gamma + L - H_k$ and $\gamma + L - H_\ell$ and using (8.183) gives (8.190).

For clarity, the raw finite part in (8.191) can equivalently be defined by the coefficientwise constant cutoff term of $D^k_\theta f^\theta_{1+u}$. The Mellin subtraction (8.177) identifies the regular Laurent coefficients of Z^θ_{k+1+u} with those cutoff finite parts of its order derivatives. Repeated application of (8.169) gives

$$D^k_\theta Z^\theta_{1+u} = (-1)^k (1+u)_k Z^\theta_{k+1+u}.$$

The contribution from the pole on the right is exactly

$$-\frac{1}{u} \prod_{j=1}^{k} \left(1 + \frac{u}{j}\right) D^k_\theta \delta_0.$$

Its constant and higher coefficients supply all the difference between derivative-compatible and raw finite parts. Equating the coefficients of $(-1)^n u^n/n!$ proves (8.191). □

For example, $D^2_\theta \delta_0 = \delta_0'' + 4\pi i\theta \delta_0' - (2\pi\theta)^2 \delta_0$. Thus (8.189) specifies the lower point-supported terms as well as the highest derivative. Writing only $H_k \delta_0^{(k)}$ would give the wrong normalization for a nonzero twist.

8.29.3 Rational twists and ordinary special functions

Proposition 8.90 (Finite Hurwitz reduction)**.** *Let $\theta = p/q$, where $1 \leq p < q$ are relatively prime, and put $z = e^{-2\pi i p/q}$. Then*

$$f^{p/q}_s(x) = \frac{e^{-2\pi i p x/q}}{q^s} \sum_{r=0}^{q-1} z^r \zeta\left(s, \frac{x+r}{q}\right), \qquad 0 < x < 1. \tag{8.192}$$

The sum is entire in s: its Hurwitz residues cancel. Consequently,

$$c^{p/q}_n(x) = \frac{e^{-2\pi i p x/q}}{q} \sum_{r=0}^{q-1} z^r \sum_{j=0}^{n} \binom{n}{j} (\log q)^{n-j} \gamma_j \left(\frac{x+r}{q}\right), \tag{8.193}$$

$$p^{p/q}_k(x) = \frac{e^{-2\pi i p x/q}}{q^{k+1}} \sum_{r=0}^{q-1} z^r \psi^{(k)}\left(\frac{x+r}{q}\right), \qquad k \geq 0. \tag{8.194}$$

These ordinary functions, extended by (8.175) and (8.176), give the finite-part distributions in Theorems 8.88 and 8.89.

Proof. For $\operatorname{Re} s > 1$, split the defining Lerch series into its residue classes modulo q. This gives (8.192); analytic continuation proves it everywhere. Since $\sum_r z^r = 0$, the common Hurwitz pole vanishes before the factor q^{-s} is expanded. Differentiating the resulting Taylor expansion at $s = 1$ gives (8.193). At positive integers use the Hurwitz–polygamma identity; its limiting case at $s = 1$ uses $\gamma_0 = -\psi$ and the same pole cancellation. This proves (8.194) also for $k = 0$. □

Thus additive rational twists require finite colored combinations of Hurwitz jets, while the coefficients governing their convolution remain the ordinary $\zeta(r)$ from $\log\Gamma(1-u)$. No replacement by Dirichlet eta values is needed in (8.181) or (8.182).

8.29.4 The half-twist and convergent alternating identities

At $\theta = \frac{1}{2}$, set

$$\zeta_E(s,x) = \Phi(-1,s,x) = 2^{-s}\left\{\zeta\left(s,\frac{x}{2}\right) - \zeta\left(s,\frac{x+1}{2}\right)\right\}, \tag{8.195}$$

$$\widetilde{\gamma}_n(x) = (-1)^n\,\partial_s^n\zeta_E(s,x)|_{s=1}\,. \tag{8.196}$$

These modified Stieltjes constants are the established functions of Hu and Kim [117]. The explicit conversion is

$$\widetilde{\gamma}_n(x) = \frac{1}{2}\sum_{j=0}^{n}\binom{n}{j}(\log 2)^{n-j}\left\{\gamma_j\left(\frac{x}{2}\right) - \gamma_j\left(\frac{x+1}{2}\right)\right\}, \qquad c_n^{1/2}(x) = e^{-\pi i x}\widetilde{\gamma}_n(x). \tag{8.197}$$

In particular, the Nielsen beta function is

$$\beta(x) = \widetilde{\gamma}_0(x) = \frac{1}{2}\left\{\psi\left(\frac{x+1}{2}\right) - \psi\left(\frac{x}{2}\right)\right\}. \tag{8.198}$$

The elementary shift relation

$$\widetilde{\gamma}_n(x) + \widetilde{\gamma}_n(x+1) = \frac{\log^n x}{x} \tag{8.199}$$

fixes its endpoint singularity, including every logarithmic order.

Theorem 8.91 (All-index alternating finite-part convolution)**.** *For $0 < a < 1$ and $m, n \geq 0$, the following expression consists of absolutely convergent integrals:*

$$\begin{aligned}\mathcal{A}_{mn}(a) = &\int_0^a \left\{\widetilde{\gamma}_m(x)\widetilde{\gamma}_n(a-x) - \widetilde{\gamma}_n(a)\frac{\log^m x}{x} - \widetilde{\gamma}_m(a)\frac{\log^n(a-x)}{a-x}\right\}dx \\ &- \int_a^1 \widetilde{\gamma}_m(x)\widetilde{\gamma}_n(1+a-x)\,dx \\ &+ \frac{\widetilde{\gamma}_n(a)}{m+1}\log^{m+1} a + \frac{\widetilde{\gamma}_m(a)}{n+1}\log^{n+1} a.\end{aligned} \tag{8.200}$$

It equals $e^{\pi i a}(U_m^{1/2} * U_n^{1/2})(a)$, where the convolution is smooth on $(0,1)$. Every $\mathcal{A}_{mn}$ is therefore the finite linear combination of $\widetilde{\gamma}_0, \ldots, \widetilde{\gamma}_{m+n+1}$ obtained from (8.182) by discarding its δ_0 term. In particular,

$$\mathcal{A}_{00}(a) = 2\widetilde{\gamma}_1(a), \tag{8.201}$$

$$\mathcal{A}_{01}(a) = \tfrac{3}{2}\widetilde{\gamma}_2(a) - \zeta(2)\widetilde{\gamma}_0(a), \tag{8.202}$$

$$\mathcal{A}_{11}(a) = \widetilde{\gamma}_3(a) - 2\zeta(2)\widetilde{\gamma}_1(a) + 2\zeta(3)\widetilde{\gamma}_0(a). \tag{8.203}$$

Proof. For integrable orders, convolution on the circle splits into $0 < x < a$ and $a < x < 1$. The phases of the two factors have product $e^{-\pi i a}$ in the first interval and $e^{-\pi i(1+a)} = -e^{-\pi i a}$ in the second. This is the source of the alternating sign in (8.200), and agrees with the nonsingular alternating convolution of [118, Theorem 1.13].

To continue to the jets, use the normalized cutoff distributions in (8.175). On a compact subinterval of $0 < a < 1$, the two endpoint singularities in the first integral occur at distinct points, $x = 0$ and $x = a$. Their only divergent terms are the two explicit subtractions in (8.200). For instance, at zero, (8.199) writes the first factor as $\log^m x/x$ plus a smooth function, while $\widetilde{\gamma}_n(a-x) - \widetilde{\gamma}_n(a) = O(x)$. The subtracted integrand is therefore $O(1 + |\log x|^m)$ there, and has the analogous integrable behavior at a. The wrapped integral is smooth. Integration of the two subtracted singular monomials gives exactly the two finite $\log a$ terms shown. This verifies the cutoff expression and permits all parameter coefficients to be taken under the subtracted integrals. Equivalently it is the restriction of the meromorphic convolution identity (8.170). The identities (8.201)–(8.203) now follow from (8.183), (8.184), and (8.186). □

For ordinary digamma functions this gives the concrete identity

$$\begin{aligned}\int_0^a \left\{\beta(x)\beta(a-x) - \beta(a)\left(\frac{1}{x} + \frac{1}{a-x}\right)\right\} dx - \int_a^1 \beta(x)\beta(1+a-x)\,dx + 2\beta(a)\log a \\ = \gamma_1\left(\frac{a}{2}\right) - \gamma_1\left(\frac{a+1}{2}\right) + 2\log 2\,\beta(a).\end{aligned} \tag{8.204}$$

The full distributional identity behind it still contains $-\zeta(2)\delta_0$. Omitting that mass would preserve (8.204) on $0 < a < 1$ but would destroy the Fourier coefficients and every identity involving endpoint tests.

Corollary 8.92 (A quarter-Gamma evaluation)**.** *At $a = \frac{1}{2}$, the absolutely convergent integral in* (8.204) *has the exact value*

$$\mathcal{A}_{00}\left(\tfrac{1}{2}\right) = \gamma_1\left(\tfrac{1}{4}\right) - \gamma_1\left(\tfrac{3}{4}\right) + \pi\log 2 = \pi\left\{4\log\Gamma\left(\tfrac{1}{4}\right) - \gamma - 3\log(2\pi)\right\}. \tag{8.205}$$

Proof. Distinguish the Dirichlet beta function $\beta_{\mathrm{D}}(s) = \sum_{r\ge 0}(-1)^r(2r+1)^{-s}$ from the Nielsen function $\beta(x)$ in (8.198). The identity $\zeta_E(s, \frac{1}{2}) = 2^s\beta_{\mathrm{D}}(s)$ gives

$$2\widetilde{\gamma}_1\left(\tfrac{1}{2}\right) = -\pi\log 2 - 4\beta_{\mathrm{D}}'(1),$$

since $\beta_{\mathrm{D}}(1) = \pi/4$. The completed Dirichlet functional equation [116, §25.15(i)] is, for this character,

$$\Lambda(s) = \left(\frac{4}{\pi}\right)^{(s+1)/2}\Gamma\left(\frac{s+1}{2}\right)\beta_{\mathrm{D}}(s), \qquad \Lambda(s) = \Lambda(1-s).$$

Also

$$\beta_{\mathrm{D}}(s) = 4^{-s}\left\{\zeta\left(s, \tfrac{1}{4}\right) - \zeta\left(s, \tfrac{3}{4}\right)\right\}, \quad \beta_{\mathrm{D}}(0) = \tfrac{1}{2}, \quad \beta_{\mathrm{D}}'(0) = \log\Gamma\left(\tfrac{1}{4}\right) - \log\Gamma\left(\tfrac{3}{4}\right) - \log 2.$$

Logarithmic differentiation of the completed functional equation at 0 and 1, followed by the Gamma reflection formula, yields

$$\beta_{\mathrm{D}}'(1) = \frac{\pi}{4}\left\{\gamma + 2\log 2 + 3\log\pi - 4\log\Gamma\left(\tfrac{1}{4}\right)\right\}.$$

Substitution proves (8.205); the first equality also uses $\beta(\frac{1}{2}) = \pi/2$ in (8.204). □

8.29.5 Reflection at the half-twist

For the half-twist, the map $n \mapsto -n-1$ reverses the shifted frequency $n+\frac{1}{2}$. Its periodic realization is

$$\mathcal{J}U(x) = e^{-2\pi i x}U(-x), \qquad \widehat{\mathcal{J}U}(n) = \widehat{U}(-n-1). \tag{8.206}$$

This is an involutive convolution automorphism with $\mathcal{J}\delta_0 = \delta_0$ and $D_{1/2}\mathcal{J} = -\mathcal{J}D_{1/2}$. Ordinary reflection alone does not reverse the half-shifted frequencies. Also define

$$\widehat{\mathcal{H}_{1/2}U}(n) = -i\,\operatorname{sgn}\left(n+\tfrac{1}{2}\right)\widehat{U}(n), \qquad n \in \mathbb{Z}.$$

Here $\mathcal{H}_{1/2}^2 = -\mathrm{Id}$ on all periodic distributions, because every shifted frequency is nonzero.

Corollary 8.93 (Reflected half-twist table)**.** *Writing $U_n = U_n^{1/2}$ and $B = B_{1/2}$ in this corollary,*

$$B(u) * \mathcal{J}B(v) = C(u,v)\big(\cos(\pi v)\,\mathrm{Id} + \sin(\pi v)\,\mathcal{H}_{1/2}\big)B(u+v). \tag{8.207}$$

Thus the coefficient prescription gives an all-index reflected table as well. Two explicit instances are

$$U_0 * \mathcal{J}U_1 = \tfrac{1}{2}U_2 + \mathcal{J}U_2 + \zeta(2)(U_0 + \mathcal{J}U_0) + \zeta(3)\delta_0, \tag{8.208}$$

$$U_1 * \mathcal{J}U_1 = \tfrac{1}{2}(U_3 + \mathcal{J}U_3) + 2\zeta(2)(U_1 + \mathcal{J}U_1) + \zeta(3)(U_0 + \mathcal{J}U_0) + \tfrac{7}{2}\zeta(4)\delta_0. \tag{8.209}$$

For the polygamma finite parts,

$$P_k^{1/2} * \mathcal{J}P_\ell^{1/2} = (-1)^\ell D_{1/2}^{k+\ell}\big\{U_1 + \mathcal{J}U_1 + H_\ell U_0 + H_k\mathcal{J}U_0 + (H_kH_\ell + 2\zeta(2))\delta_0\big\}. \tag{8.210}$$

Proof. At mode n put $\sigma = \operatorname{sgn}(n+\frac{1}{2})$. The logarithms of $\lambda_{n,1/2}$ and $-\lambda_{n,1/2}$ differ by $i\pi\sigma$. Dividing the Fourier coefficient of the left side of (8.207) by that of $B(u+v)$ gives $C(u,v)e^{-i\pi v\sigma}$, which is exactly the multiplier on the right. Coefficient extraction, with $\zeta(2)^2 = \frac{5}{2}\zeta(4)$, proves the two displayed instances. Multiplication of $\gamma + L_{n,1/2} - H_k$ and $\gamma + L_{-n-1,1/2} - H_\ell$ proves (8.210); the factor $(-1)^\ell$ comes from $\lambda_{-n-1,1/2}^\ell = (-\lambda_{n,1/2})^\ell$. □

The modulation in (8.206) is periodic specifically because $2\theta = 1$. For a general twist, frequency reversal connects θ with $1-\theta$; the same fixed-twist involution should not be asserted without changing the space or the twist.

8.29.6 Unique covariant primitives

Because $\lambda_{n,\theta} \neq 0$ for every n, the operator D_θ has a unique inverse on periodic distributions:

$$\widehat{D_\theta^{-q}U}(n) = \lambda_{n,\theta}^{-q}\widehat{U}(n), \qquad q \geq 1.$$

This removes the choice of an integration constant that occurs for ordinary periodic primitives.

Theorem 8.94 (All-order covariant antiderivatives)**.** *Define the complete homogeneous reciprocal-harmonic coefficients by*

$$\sum_{j\geq 0} h_j^{(q-1)}w^j = \prod_{r=1}^{q-1}\left(1-\frac{w}{r}\right)^{-1}; \qquad h_0^{(0)} = 1, \quad h_j^{(0)} = 0 \ (j>0). \tag{8.211}$$

Then for $q \geq 1$ and $n \geq 0$,

$$D_\theta^{-q} U_n^\theta = \frac{(-1)^{n+1} n!}{(q-1)!} \sum_{j=0}^{n+1} \frac{h_{n+1-j}^{(q-1)}}{j!} \, \partial_s^j Z_s^\theta \Big|_{s=1-q}. \tag{8.212}$$

Its ordinary value on $(0,1)$ is obtained by replacing Z_s^θ by f_s^θ. For $q \geq 2$ this is a periodic C^{q-2} function. For $q > k \geq 0$, setting $p = q - k$ gives

$$D_\theta^{-q} P_k^\theta = \frac{\partial_s Z_s^\theta\big|_{s=1-p} + (H_{p-1} - H_k) Z_{1-p}^\theta}{(p-1)!}, \tag{8.213}$$

$$D_\theta^{-q} Q_k^\theta = \frac{\partial_s Z_s^\theta\big|_{s=1-p} + H_{p-1} Z_{1-p}^\theta}{(p-1)!}. \tag{8.214}$$

At the half-twist, the first primitive has the explicit Gamma form

$$D_{1/2}^{-1} U_0^{1/2}(x) = e^{-\pi i x} \left\{ \log \Gamma\left(\frac{x+1}{2}\right) - \log \Gamma\left(\frac{x}{2}\right) + \frac{1}{2} \log 2 \right\}, \qquad 0 < x < 1. \tag{8.215}$$

The displayed additive term is fixed by the periodic distributional normalization.

Proof. The spectral definition and the Gamma recurrence give

$$D_\theta^{-q} Z_{1+u}^\theta = \frac{Z_{1-q+u}^\theta}{(-u)_q}, \qquad \frac{1}{(-u)_q} = -\frac{1}{(q-1)! u} \sum_{j \geq 0} h_j^{(q-1)} u^j.$$

The family in the numerator is holomorphic at $u = 0$. Comparing the coefficients of $(-1)^n u^n / n!$ proves (8.212). Its Fourier coefficients are $O(|n|^{-q}(1 + \log |n|)^{n_0+1})$ when the fixed Stieltjes index is denoted by n_0. Therefore the Fourier series and its first $q-2$ ordinary derivatives converge absolutely for $q \geq 2$. The derivative formulas for P_k^θ and Q_k^θ follow directly from (8.187) and $\psi(p) = H_{p-1} - \gamma$.

For (8.215), the case $q = 1, n = 0$ is $-\partial_s Z_s^{1/2}|_{s=0}$. Use (8.195), $\zeta(0,a) = \frac{1}{2} - a$, and $\zeta'(0,a) = \log \Gamma(a) - \frac{1}{2} \log(2\pi)$. The difference of the two zeta values at zero is $\frac{1}{2}$, which supplies the term $\frac{1}{2} \log 2$ after differentiation. □

8.29.7 Degeneration to the untwisted finite parts

The zero Fourier coefficient becomes singular as $\theta \downarrow 0$. Deleting it exactly gives a continuous transition to the untwisted normalization, including all the Stieltjes jets.

Theorem 8.95 (Exact zero-mode renormalization)**.** *For every $\rho < 1$,*

$$B_\theta(u) - \Gamma(1-u)(2\pi i \theta)^u \, 1 \longrightarrow B(u) \qquad (\theta \downarrow 0) \tag{8.216}$$

locally uniformly for $|u| < \rho$ after pairing with any smooth periodic test function, and with every fixed number of derivatives in u. Here $B(u) = \Gamma(1-u) R_{-u} = \mathsf{I} - u \sum_{n \geq 0} (-1)^n G_n u^n / n!$ is the already proved untwisted family. The symbol 1 is the constant distribution, which has only the zero Fourier coefficient. Put

$$\mu_n(\theta) = \frac{(-1)^{n+1}}{n+1} \mathcal{B}_{n+1}\big(\gamma + \log(2\pi i \theta), \zeta(2), 2!\zeta(3), \ldots, n!\zeta(n+1)\big). \tag{8.217}$$

Then $\mu_n(\theta) = \widehat{U_n^\theta}(0)$ *and*

$$U_n^\theta - \mu_n(\theta)\,1 \longrightarrow G_n \qquad \text{in } \mathcal{D}'(\mathbb{R}/\mathbb{Z}). \tag{8.218}$$

In particular,

$$\mu_0(\theta) = -\gamma - \log(2\pi i\theta), \qquad \mu_1(\theta) = \frac{(\gamma + \log(2\pi i\theta))^2 + \zeta(2)}{2}.$$

Proof. The subtracted quantity in (8.216) is exactly the $n = 0$ Fourier coefficient. For each $n \neq 0$, the remaining coefficient tends to $\Gamma(1-u)(2\pi in)^u = \widehat{B(u)}(n)$ with the previously fixed branch. Uniformly for $0 < \theta \leq \frac{1}{2}$ and u on a compact subdisk, these coefficients and their fixed order derivatives are bounded by a common polynomial in $|n|$, because $|n+\theta|$ is comparable with $|n|$. Rapid decrease of the test-function Fourier coefficients proves normal convergence and allows termwise order differentiation. Taking the coefficient of $(-1)^{n+1}u^{n+1}/n!$ gives (8.218); the scalar Gamma expansion gives (8.217). □

At $u = 0$, the left side of (8.216) is already $\delta_0 - 1 = \mathsf{I}$. Thus the change of convolution identity is caused exactly by the disappearing zero mode. The same observation explains why the interior alternating identity (8.201) has no added constant, whereas the untwisted digamma convolution has the constant contributed by $-\zeta(2)\mathsf{I}$.

8.30 Unequal twists through beta mixtures and finite resolvent identities

The fixed-twist calculus suggests a next question: how do kernels with different frequency shifts convolve? The Fourier multipliers give an explicit answer. Fractional orders admit a Dirichlet beta mixture of intermediate twists, while positive integer orders reduce to finitely many elementary kernels. These statements use the classical beta integral and partial fractions; they make no literature-priority or arithmetic-independence claim.

Write $K_\alpha^\theta = W_{1-\alpha}^\theta$, so that

$$\widehat{K}_\alpha^\theta(n) = (2\pi i(n+\theta))^{-\alpha}.$$

The twists in the fractional-mixture theorem lie in $(0,1)$, with the logarithms of (8.163). More generally one may put them in a common open interval between consecutive integers. This condition keeps the frequencies at each fixed mode on the same imaginary ray.

Theorem 8.96 (Any finite number of fractional twist kernels)**.** *Let* $d \geq 2$, $\theta_j \in (0,1)$ *and* $\operatorname{Re}\alpha_j > 0$. *Put* $A = \sum_j \alpha_j$ *and let* $\Sigma_{d-1} = \{t_j \geq 0 : \sum_j t_j = 1\}$, *with measure* $dt_1 \cdots dt_{d-1}$. *Then, as periodic distributions,*

$$K_{\alpha_1}^{\theta_1} * \cdots * K_{\alpha_d}^{\theta_d} = \frac{\Gamma(A)}{\prod_j \Gamma(\alpha_j)} \int_{\Sigma_{d-1}} \left(\prod_j t_j^{\alpha_j - 1} \right) K_A^{\sum_j t_j \theta_j}\, dt. \tag{8.219}$$

The mixture includes the zero Fourier coefficient. Its normalized right side, continued as a whole, is entire in all α_j. *For two twists the simplex is the interval* $0 \leq t \leq 1$ *and the intermediate twist is* $t\theta_1 + (1-t)\theta_2$.

Proof. For positive real a_j and $\operatorname{Re}\alpha_j > 0$, change variables $y_j = rt_j$ in the product of Gamma integrals

$$\prod_j a_j^{-\alpha_j} = \frac{1}{\prod_j \Gamma(\alpha_j)} \int_{(0,\infty)^d} \prod_j y_j^{\alpha_j - 1} e^{-\sum_j a_j y_j}\, dy.$$

The Jacobian is r^{d-1}. Integrating r gives the scalar Dirichlet formula with $(\sum_j t_j a_j)^{-A}$. At Fourier mode n, all $n+\theta_j$ have the same sign. Apply the scalar formula to $a_j = |2\pi(n+\theta_j)|$ and restore their common phase $\exp(-i\pi A \operatorname{sgn}(n+\theta_j)/2)$. The resulting coefficient is exactly that of (8.219), including $n = 0$.

The intermediate twists range in a compact subinterval of $(0,1)$. On compact parameter subsets, the kernel coefficients have a common polynomial growth bound, and the simplex weight is integrable. Rapid decrease of the Fourier coefficients of each smooth test function therefore justifies integration and summation. Fourier uniqueness proves the distribution identity. For continuation outside the initial half-planes, subtract finite Taylor polynomials at the simplex faces before integrating; the added monomials have meromorphic beta integrals. Their combined normalized continuation has the same Fourier coefficients as the left side, which are entire in every α_j. Test-function pairing gives normal convergence on compact parameter sets. Thus all apparent poles of the combined continuation are removable. It is this continuation, not a divergent unsubtracted simplex integral, that is asserted to be entire. □

Theorem 8.97 (Finite reduction at every positive integer order)**.** *For distinct real nonintegral θ, ϕ, set $\Delta = 2\pi i(\phi - \theta)$. For positive integers r, s,*

$$K_r^\theta * K_s^\phi = \sum_{j=1}^{r} \frac{(-1)^{r-j}\binom{r+s-j-1}{r-j}}{\Delta^{r+s-j}} K_j^\theta + \sum_{k=1}^{s} \frac{(-1)^{r}\binom{r+s-k-1}{s-k}}{\Delta^{r+s-k}} K_k^\phi. \tag{8.220}$$

Every ordinary representative on $(0,1)$ is elementary, since

$$K_1^\theta(x) = \frac{e^{-2\pi i\theta x}}{1-e^{-2\pi i\theta}}, \qquad K_r^\theta(x) = \frac{(-1)^{r-1}}{(2\pi i)^{r-1}(r-1)!}\partial_\theta^{r-1} K_1^\theta(x). \tag{8.221}$$

If two twists coincide, their orders add: $K_r^\theta * K_s^\theta = K_{r+s}^\theta$.

Proof. At a Fourier mode put $X = 2\pi i(n+\theta)$; the second multiplier has denominator $X + \Delta$. The principal part at $X = 0$ of $X^{-r}(X+\Delta)^{-s}$ is obtained by expanding $(X+\Delta)^{-s}$ through degree $r-1$; this gives the first sum. At $X = -\Delta$, expand X^{-r} in $X + \Delta$ through degree $s-1$; this gives the second sum. Their difference from the rational function has no finite poles and tends to zero at infinity, so it is zero. Apply this identity at every Fourier mode and use uniqueness. The representative for K_1 is (8.167) at $s = 0$. Differentiation of its multiplier gives $\partial_\theta K_r^\theta = -2\pi i r K_{r+1}^\theta$; distributional differentiation in the parameter is legitimate by the same compact Fourier estimates. This proves (8.221). The coincident identity is (8.168). □

For instance, the first unequal-twist convolution is the explicit ordinary integral

$$(K_1^\theta * K_1^\phi)(x) = \frac{K_1^\theta(x) - K_1^\phi(x)}{2\pi i(\phi - \theta)}.$$

Its coincident limit is $-\partial_\theta K_1^\theta/(2\pi i) = K_2^\theta$. Any finite number of distinct twists and positive integer orders is handled by the same principal-part argument; repeated twists are combined first. These integer-order reductions require no common frequency interval. Fractional orders have the precise mixture (8.219). Finite reduction of every mixed Stieltjes order jet to a prescribed list of single Hurwitz values remains a separate question; the mixture does not assert such a reduction.

8.31 Gamma coefficients prove every harmonic-order identity

The continuous Gamma correlations above have a discrete counterpart. Finite harmonic products are coefficients of a Gamma ratio; a normally convergent Gauss sum therefore evaluates an entire family at once. This gives corrected all-order forms of the fifth- and sixth-order conjectures in Choi [113], rather than isolated numerical relations. The harmonic-resonance continuation supplies this route and its broader mixed-jet context [114].

For $n, r \geq 0$, define

$$\prod_{j=1}^{n}\left(1+\frac{u}{j}\right)=\sum_{r=0}^{n}\frac{\mathcal{E}_r(n)}{r!}u^r, \qquad \mathcal{E}_0(n)=1. \tag{8.222}$$

Thus $\mathcal{E}_r(n)$ is $r!$ times the elementary symmetric polynomial in $1, 1/2, \ldots, 1/n$. In particular $\mathcal{E}_1(n) = H_n$ and $\mathcal{E}_2(n) = H_n^2 - H_n^{(2)}$. Equivalently it is the complete exponential Bell polynomial with arguments $H_n, -H_n^{(2)}, 2H_n^{(3)}, \ldots, (-1)^{r-1}(r-1)!H_n^{(r)}$.

Theorem 8.98 (Every pair of harmonic coefficient orders)**.** *For u, v in a neighborhood of zero,*

$$\sum_{n=0}^{\infty}\frac{(1+u)_n(1+v)_n}{(n!)^2(n+1)(n+2)}=\frac{\Gamma(1-u-v)}{\Gamma(2-u)\Gamma(2-v)}. \tag{8.223}$$

Every fixed Taylor coefficient can be taken termwise. Consequently, for all $r, t \geq 0$,

$$\sum_{n=1}^{\infty}\frac{\mathcal{E}_r(n)\mathcal{E}_t(n)}{(n+1)(n+2)}=r!t![u^r v^t]\frac{\Gamma(1-u-v)}{\Gamma(2-u)\Gamma(2-v)}-\frac{1}{2}\mathbf{1}_{r=t=0}. \tag{8.224}$$

These are ordinary absolutely convergent series.

Proof. Since $(3)_n = n!(n+1)(n+2)/2$, the series in (8.223) is $\frac{1}{2}{}_2F_1(1+u, 1+v; 3; 1)$. Gauss summation [112] evaluates it as the displayed Gamma quotient. On a sufficiently small closed polydisc, its summands are uniformly $O(n^{-2+\mathrm{Re}(u+v)})$ by the Gamma-ratio asymptotic. Enlarging that polydisc slightly and applying Cauchy's coefficient formula justifies each derivative or coefficient extraction. Formula (8.222) now proves (8.224). The $n = 0$ summand is $1/2$ and contributes only at $r = t = 0$. □

Corollary 8.99 (Two explicit all-order families)**.** *For every $r \geq 1$ and every $t \geq 0$, respectively,*

$$\sum_{n=1}^{\infty}\frac{\mathcal{E}_r(n)}{(n+1)(n+2)}=r!, \tag{8.225}$$

$$\sum_{n=1}^{\infty}\frac{H_n\mathcal{E}_t(n)}{(n+1)(n+2)}=t!\left(1+\sum_{j=2}^{t+1}\zeta(j)\right). \tag{8.226}$$

The sum on the second right side is empty at $t = 0$.

Proof. Set $v = 0$ in (8.223). Its right side is $1/(1-u)$, proving (8.225). Differentiate first in v and then put $v = 0$. The result is

$$\frac{\psi(2)-\psi(1-u)}{1-u}=\frac{1+\sum_{j\geq 1}\zeta(j+1)u^j}{1-u}, \qquad |u|<1.$$

The normally convergent digamma Taylor expansion at one proves this last equality. Taking the coefficient of u^t proves (8.226). Its $n = 0$ derivative is zero, so no additional constant occurs. □

The cases $r = 5, 6$ prove the intended pure harmonic identities in Choi's Conjecture 4; the cases $t = 5, 6$ prove corrected versions of Conjecture 6. His printed formulas (41)–(46) carry twice the multiplier in (8.226). The finite-product definition also fixes the harmonic orders in every numerator, including the sixth-order term $-120H_n^{(6)}$ and, after multiplication by H_n, the partition term $45H_n^3(H_n^{(2)})^2$. The unambiguous coefficient formula is the corrected statement at all orders. The source comparison and its text-versus-visual evidence boundary are recorded in `verification/ninth-analytic-audit.md`.

For a short independent check of the simplest normalization, summation by parts gives

$$\begin{aligned}\sum_{n\geq 1}\frac{H_n^2}{(n+1)(n+2)} &= \sum_{k\geq 1}\frac{H_k^2 - H_{k-1}^2}{k+1}\\ &= 2\sum_{k\geq 1}\frac{H_{k-1}}{k(k+1)} + \sum_{k\geq 1}\frac{1}{k^2(k+1)}\\ &= 2 + \zeta(2) - 1 = 1 + \zeta(2).\end{aligned}$$

Indeed Tonelli's theorem gives $\sum_{k\geq 1} H_{k-1}/(k(k+1)) = \sum_{j\geq 1} 1/(j(j+1)) = 1$. This isolates the factor of two without any special-function differentiation. Higher mixed orders are finite Gamma Taylor coefficients; no arithmetic-independence assumption is needed.

8.32 Mixed Gamma derivatives and exact harmonic tail identities

The harmonic coefficient calculus has a useful companion at a double zero of a hypergeometric residue. The mixed spectral continuation [123] identifies a normally convergent two-parameter Gamma identity and a weighted square of trigamma differences. We prove the needed first resonance directly from the classical Rogers–Dougall sum [124], rather than assuming the entire pending all-resonance programme. Its mixed derivatives produce ordinary harmonic identities. Differentiating the resulting square identity also suggests an all-orders zeta-tail family; an independent Euler partial-fraction proof of that family closes the loop between experiment and exact algebra.

8.32.1 A normally convergent double-zero germ

Put $\eta = a/2$, $x_n = n + \eta$, and $A = a - 1$. Define

$$Q_n(a;p,q) = \frac{\Gamma(n+a)\Gamma(n+1+p)\Gamma(n+1+q)\Gamma(n+a-p-q)}{\Gamma(n+1)\Gamma(n+a-p)\Gamma(n+a-q)\Gamma(n+1+p+q)}, \tag{8.227}$$

$$B_a(p,q) = \psi(a/2) + \tfrac{1}{2}\{\psi(2) - \psi(1+p) - \psi(1+q) - \psi(a-p-q)\}. \tag{8.228}$$

These are holomorphic for $\Re a > 0$ and p, q sufficiently small on each compact a-set. Both $Q_n(a;p,0)$ and $Q_n(a;0,q)$ are one.

Theorem 8.100 (The first mixed resonance)**.** *For every real $a > 0$, and sufficiently small complex p, q,*

$$\sum_{n\geq 0}\left\{x_n\big(Q_n(a;p,q) - 1\big) + \frac{(A-p-q)pq}{x_n}\right\} = -(A-p-q)pq\,B_a(p,q). \tag{8.229}$$

The bracketed series converges locally normally, including after division by pq at either axis. Every fixed parameter derivative can be taken termwise. The identity extends holomorphically to the corresponding germs over $\Re a > 0$.

Proof. First take $a > 1$, $b = 1 + p$, $c = 1 + q$ and $d_0 = 1 + a - b - c > 0$. Introduce a separate spectral variable u and the centered summand

$$r_n(u) = x_n \frac{\Gamma(n+a)\Gamma(n+b)\Gamma(n+c)\Gamma(n+d_0-u)}{\Gamma(n+1)\Gamma(n+1+a-b)\Gamma(n+1+a-c)\Gamma(n+b+c+u)}.$$

For $0 < \Re u < d_0$, the Rogers–Dougall formula with upper parameters $a, 1 + a/2, b, c, d_0 - u$ and lower parameters $a/2, 1 + a - b, 1 + a - c, b + c + u$ gives

$$\sum_{n\geq 0} r_n(u) = G(u) := \frac{\Gamma(b)\Gamma(c)\Gamma(d_0-u)\Gamma(u)}{2\Gamma(d_0)\Gamma(b+u)\Gamma(c+u)}. \tag{8.230}$$

The classical excess is $2u$; the factor $1/2$ follows from the normalization $x_n = n + a/2$, not the usual Pochhammer summand.

Write

$$(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = (\eta, b - \eta, c - \eta, d_0 - u - \eta).$$

The denominator shifts relative to x_n are $1 - \alpha_i$. The shifted Stirling expansion [49] and Bernoulli reflection therefore give uniformly on compact parameter sets

$$r_n(u) = x_n^{-1-2u}\{1 + C_1(u)x_n^{-2} + O(n^{-4})\}, \qquad C_1(u) = -\tfrac{1}{3}\sum_{i=1}^{4} B_3(\alpha_i).$$

Centering cancels the odd inverse powers. Thus

$$R_2(u) = \sum_{n\geq 0}\{r_n(u) - x_n^{-1-2u} - C_1(u)x_n^{-3-2u}\} = G(u) - \zeta(1+2u,\eta) - C_1(u)\zeta(3+2u,\eta) \tag{8.231}$$

extends holomorphically to $-2 < \Re u < d_0$. The series bound is $O(n^{-5-2\Re u})$ on compact subsets; Cauchy's formula gives normal convergence for each fixed spectral or parameter derivative. The equality first follows by summing the counterterms in the ordinary convergence region, and then by continuation.

At $u = -1$, direct Bernoulli algebra gives

$$\begin{aligned} C_1(-1) &= \rho := -(A - p - q)pq, \\ \tfrac{1}{2}C_1'(-1) + \zeta(-1,\eta) &= \tfrac{1}{2}\{B_2(\eta - p - q) - B_2(\eta)\} = -\tfrac{1}{2}(A - p - q)(p + q). \end{aligned} \tag{8.232}$$

For generic nonzero p, q, G has residue $\rho/2$ and constant term

$$\frac{\rho}{2}\{\psi(2) - \psi(d_0 + 1) - \psi(p) - \psi(q)\}.$$

The resonant Hurwitz term has constant coefficient $C_1'(-1)/2 - \rho\psi(\eta)$. Combining these constants with (8.232), and using $\psi(1 + p) = \psi(p) + 1/p$ and its q counterpart, yields $R_2(-1) = \rho B_a(p, q)$. Since $r_n(-1) = x_n Q_n$, this is (8.229). Holomorphy extends it to the spectral axes.

Finally all Gamma arguments in Q_n and all digamma arguments in B_a stay pole-free for $\Re a > 0$ and small p, q. The centered bound remains $O(n^{-3})$ uniformly on every compact a-set. Each Taylor coefficient is consequently holomorphic in that half-plane and agrees with the asserted coefficient for $\Re a > 1$; the identity theorem extends it throughout. The bracket vanishes identically on both axes. Cauchy's formula on a slightly larger polydisc transfers its $O(n^{-3})$ bound to its holomorphic quotient by pq and to every fixed derivative, proving the divided-series assertion. □

8.32.2 Finite mixed coefficient tables

For $k \geq 2$ set $\Lambda_k(n;a) = (-1)^k \psi^{(k-1)}(n+a) - \psi^{(k-1)}(n+1)$. Logarithmic differentiation of the eight Gamma factors gives

$$\log Q_n = \sum_{r,s \geq 1} \Lambda_{r+s}(n;a) \frac{p^r q^s}{r!s!}. \tag{8.233}$$

Define the finite bivariate Bell coefficient

$$\mathcal{B}_{rs}(n;a) = r!s![p^r q^s] \exp\left(\sum_{i,j \geq 1} \Lambda_{i+j}(n;a) \frac{p^i q^j}{i!j!} \right).$$

Only $i \leq r$, $j \leq s$ and powers at most $\min(r,s)$ are needed. For example

$$\mathcal{B}_{11} = \Lambda_2, \quad \mathcal{B}_{31} = \Lambda_4, \quad \mathcal{B}_{22} = \Lambda_4 + 2\Lambda_2^2, \quad \mathcal{B}_{33} = \Lambda_6 + 9\Lambda_2\Lambda_4 + 9\Lambda_3^2 + 6\Lambda_2^3.$$

Write $B_{rs} = \partial_p^r \partial_q^s B_a(0,0)$. Then

$$B_{00} = \psi(a/2) + \tfrac{1}{2}(1 + \gamma - \psi(a)),$$

and, for $r + s = k \geq 1$,

$$B_{rs} = -\tfrac{1}{2}\{\mathbf{1}_{s=0}\psi^{(r)}(1) + \mathbf{1}_{r=0}\psi^{(s)}(1) + (-1)^k \psi^{(k)}(a)\}. \tag{8.234}$$

Terms with a negative subscript are omitted below.

Corollary 8.101 (All fixed mixed orders)**.** *For $r, s \geq 1$ the absolutely convergent single bracket*

$$S_{rs}(a) = \sum_{n \geq 0} \left\{ x_n \mathcal{B}_{rs}(n;a) + \frac{A\mathbf{1}_{r=s=1} - 2\mathbf{1}_{(r,s)=(2,1)} - 2\mathbf{1}_{(r,s)=(1,2)}}{x_n} \right\} \tag{8.235}$$

equals

$$-rs\{AB_{r-1,s-1} - (r-1)B_{r-2,s-1} - (s-1)B_{r-1,s-2}\}. \tag{8.236}$$

When $r + s \geq 4$ the counterterm is absent.

Proof. Differentiate the normally convergent identity (8.229). Its counterterm is a polynomial of total degree three, while Leibniz's rule on the right gives (8.236). The quotient and normal-convergence statements were proved before differentiation. □

8.32.3 A trigamma square and its arithmetic specializations

Theorem 8.102 (Weighted trigamma square)**.** *For every real $a > 0$,*

$$\sum_{n \geq 0} \left(n + \frac{a}{2}\right) \left(\psi'(n+a) - \psi'(n+1)\right)^2 = \frac{a-1}{4}\psi''(a) + \frac{\psi'(a) - \zeta(2)}{2} + \frac{3(a-1)}{2}\zeta(3). \tag{8.237}$$

The ordinary series is locally normally convergent over $\Re a > 0$, where the square is algebraic.

Proof. The $(2,2)$ and $(3,1)$ rows of (8.236) give

$$\sum_n x_n(\Lambda_4+2\Lambda_2^2)=2A\psi''(a)+4\psi'(a)-4\zeta(2),$$

$$\sum_n x_n\Lambda_4=\tfrac{3A}{2}\{\psi''(1)+\psi''(a)\}+3\{\psi'(a)-\zeta(2)\}.$$

Subtract and divide by two, using $\psi''(1)=-2\zeta(3)$. All series in that calculation converge absolutely. Independently, the polygamma expansion gives $\psi'(n+a)-\psi'(n+1)=O(n^{-2})$ locally uniformly, so the weighted square is $O(n^{-3})$, including at $a=1$. □

The half-integer recurrences reduce the cases $a=3/2$ and $a=1/2$ to

$$\sum_{n\ge0}(4n+3)\{2H^{(2)}_{2n+1}-H^{(2)}_n-\zeta(2)\}^2=\zeta(2)-\zeta(3), \tag{8.238}$$

$$\sum_{n\ge0}(4n+1)\{\zeta(2)-2H^{(2)}_{2n}+H^{(2)}_n\}^2=\zeta(2)+\zeta(3). \tag{8.239}$$

Indeed $\psi'(3/2)=3\zeta(2)-4$, $\psi''(3/2)=-14\zeta(3)+16$, $\psi'(1/2)=3\zeta(2)$ and $\psi''(1/2)=-14\zeta(3)$. The polygamma series express the respective differences as twice the bracketed harmonic tails. At $a=2$, (8.237) reduces to $\sum_{k\ge1}k^{-3}=\zeta(3)$.

8.32.4 An all-orders zeta-tail identity with an independent proof

Put $T_s(n)=\zeta(s)-H^{(s)}_n=\sum_{j>n}j^{-s}$ for $s\ge2$. Differentiating (8.237) at $a=1$ suggests the following finite combination at every order. We prove it independently by Euler partial fractions and positive double sums, so its validity does not depend on the hypergeometric continuation.

Theorem 8.103 (All-order quadratic tail reduction)**.** *For every integer $k\ge2$,*

$$\sum_{n\ge0}\Bigg[\left(n+\tfrac12\right)\sum_{j=1}^{k-1}(j+1)(k-j+1)T_{j+2}(n)T_{k-j+2}(n) - \tfrac12\sum_{j=1}^{k-2}(j+1)(k-j)T_{j+2}(n)T_{k-j+1}(n)\Bigg]=\frac{(k+1)(k+2)}{4}\zeta(k+2). \tag{8.240}$$

Every series displayed on the left is ordinary and absolutely convergent.

Proof. For $a,b\ge3$, summing first over $0\le n<\min(l,m)$ gives

$$\sum_{n\ge0}(n+\tfrac12)T_a(n)T_b(n)=\tfrac12\{\zeta(a+b-2)+\zeta(a,b-2)+\zeta(b,a-2)\}, \tag{8.241}$$

$$\sum_{n\ge0}T_a(n)T_b(n)=\zeta(a+b-1)+\zeta(a,b-1)+\zeta(b,a-1). \tag{8.242}$$

The finite sums of the weights are $\min(l,m)^2/2$ and $\min(l,m)$, respectively. Positivity justifies Tonelli, and the resulting double zeta sums converge. Here $\zeta(r,s)=\sum_{l>m\ge1}l^{-r}m^{-s}$. Substituting these two formulas, using the symmetry in j and $k-j$ or $k-1-j$, reduces (8.240) to

$$\sum_{j=1}^{k-2}(j+1)\zeta(j+2,k-j)+2k\zeta(k+1,1)=\zeta(k+2). \tag{8.243}$$

The diagonal single-zeta coefficient before this last reduction is $(k^2+3k-2)/4$; the proposed value exceeds it by one.

For completeness the classical relation (8.243) has an elementary proof. Iterating

$$\frac{1}{x^r y^s} = \frac{x^{-(r-1)}y^{-s} + x^{-r}y^{-(s-1)}}{x+y}$$

gives, for $x, y > 0$,

$$\frac{1}{x^2y^k} = \sum_{i=0}^{1} \frac{\binom{k+i-1}{i}}{(x+y)^{k+i}x^{2-i}} + \sum_{i=0}^{k-1} \frac{i+1}{(x+y)^{2+i}y^{k-i}}. \tag{8.244}$$

Summing over positive integer x, y is justified by positivity. The first sum becomes $\zeta(k,2) + k\zeta(k+1,1)$; the second becomes $\sum_{i=0}^{k-1}(i+1)\zeta(i+2,k-i)$. On the other hand, separating equal and unequal indices in the product gives the stuffle identity $\zeta(2)\zeta(k) = \zeta(2,k) + \zeta(k,2) + \zeta(k+2)$. Subtracting the two expressions proves (8.243), including $k=2$ where the empty sum is zero. This proves the theorem. □

The first member is the simple ordinary identity

$$\boxed{\sum_{n\geq 0}(2n+1)\{\zeta(3) - H_n^{(3)}\}^2 = \tfrac{3}{2}\zeta(4).} \tag{8.245}$$

The next member is

$$\sum_{n\geq 0}\{12(n+\tfrac{1}{2})T_3(n)T_4(n) - 2T_3(n)^2\} = 5\zeta(5).$$

Corollary 8.104 (Centered tail squares)**.** *For every integer $s \geq 3$,*

$$\sum_{k\geq 1} k\{2T_s(k-1) - k^{-s}\}^2 = 2\zeta(2s-2) + 4\zeta(s,s-2) - \zeta(2s-1). \tag{8.246}$$

At $s=3$ this is the mixed-report evaluation

$$\boxed{\sum_{k\geq 1} k\{2(\zeta(3) - H_{k-1}^{(3)}) - k^{-3}\}^2 = 3\zeta(4) - \zeta(5).} \tag{8.247}$$

Proof. Expand the square and use (8.241), (8.242), and $\sum_{k\geq 1} k^{1-s}T_s(k-1) = \zeta(s,s-1) + \zeta(2s-1)$. The two $\zeta(s,s-1)$ terms cancel. For $s=3$, (8.243) at $k=2$ gives $4\zeta(3,1) = \zeta(4)$, yielding (8.247). All rearranged sums converge absolutely. □

These identities show how a Gamma mixed-derivative experiment can expose an exact Euler relation. The additional all-order tail and centered-tail formulas are collective deductions from classical partial fractions; global priority or independence of their period coordinates is not asserted.

8.33 All centered trigamma-square moments and a polylogarithmic generator

The invariant numerator $(\zeta(2) - H_n^{(2)})^2$ provides an explicit family beyond raw powers of H_n. Define

$$A_n = \zeta(2) - H_n^{(2)} = \psi'(n+1), \qquad \mathcal{T}(s) = \sum_{n=0}^{\infty} \frac{A_n^2}{(n+1/2)^s}. \tag{8.248}$$

The harmonic parity continuation [130] gives an ordinary finite-sum route to these moments, independent of its broader difference-equation classification. The defining series already converges for $\Re s > -1$. Its centered asymptotic expansion contains only even inverse powers. Subtracting finitely many terms continues the Dirichlet series normally to successively larger half-planes, with possible poles only at odd integers $1-2k$. Hence it is regular at every nonpositive even integer. All its values there reduce to a rational number plus a rational multiple of $\zeta(3)$.

Theorem 8.105 (Every nonpositive even trigamma-square value)**.** *For every integer $r \geq 0$,*

$$\begin{aligned}\mathcal{T}(-2r) &= 3B_{2r}(1/2)\zeta(3) \\ &\quad - \frac{1}{2}\sum_{j=1}^{r}\binom{2r}{2j}\frac{2j-1}{2j+1}B_{2r-2j}(1/2)B_{2j-2}.\end{aligned} \tag{8.249}$$

The Bernoulli number convention is $B_1 = -1/2$, and the sum is empty when $r = 0$.

For example,

$$\begin{aligned}\mathcal{T}(0) &= 3\zeta(3), \\ \mathcal{T}(-2) &= -\frac{1}{4}\zeta(3) - \frac{1}{6}, \\ \mathcal{T}(-4) &= \frac{7}{80}\zeta(3) + \frac{1}{30}, \\ \mathcal{T}(-6) &= -\frac{31}{448}\zeta(3) + \frac{1}{672}, \\ \mathcal{T}(-8) &= \frac{127}{1280}\zeta(3) - \frac{17}{540}.\end{aligned} \tag{8.250}$$

The first value is the classical Euler sum. The theorem supplies a single finite formula for all subsequent centered moments.

8.33.1 Equivalent identities of absolutely convergent ordinary sums

Let

$$c_k = \sum_{i=0}^{k-1} B_{2i}(1/2)B_{2k-2-2i}(1/2), \qquad k \geq 1. \tag{8.251}$$

Thus $c_1 = 1$, $c_2 = -1/6$, $c_3 = 47/720$, and $c_4 = -257/5040$. The shifted Stirling expansion gives $A_n \sim \sum_{i\geq 0} B_{2i}(1/2)(n+1/2)^{-2i-1}$. Squaring this expansion gives $A_n^2 \sim \sum_{k\geq 1} c_k(n+1/2)^{-2k}$. For every $r \geq 0$ the exact ordinary convergent identity is

$$\sum_{n=0}^{\infty}\left\{(n+1/2)^{2r}\psi'(n+1)^2 - \sum_{k=1}^{r} c_k(n+1/2)^{2r-2k}\right\} = \mathcal{T}(-2r), \tag{8.252}$$

where the right side is explicitly (8.249). The summands after subtraction are $O(n^{-2})$.

For instance,

$$\sum_{n=0}^{\infty}\left((n+1/2)^2\psi'(n+1)^2 - 1\right) = -\frac{1}{4}\zeta(3) - \frac{1}{6},$$

and

$$\sum_{n=0}^{\infty}\left((n+1/2)^4\psi'(n+1)^2 - (n+1/2)^2 + \frac{1}{6}\right) = \frac{7}{80}\zeta(3) + \frac{1}{30}.$$

These statements need no prescription for summing a divergent series. To prove (8.252), subtract the displayed asymptotic terms in the half-plane of convergence and continue. Their Dirichlet sums at $s = -2r$ are $c_k\zeta(2k - 2r, 1/2) = 0$, since $\zeta(-2m, 1/2) = 0$ for $m \geq 0$.

8.33.2 Proof of the Bernoulli formula by finite summation

For a sequence admitting an expansion in powers of N and $\log N$, write CT_N for its coefficient of $N^0(\log N)^0$. Only finite exact identities followed by known asymptotic expansions will be used here. Put

$$S_r(k) = \sum_{n=0}^{k-1}(n + 1/2)^{2r} = \frac{B_{2r+1}(k + 1/2)}{2r + 1} = \sum_{j=0}^{r} b_{r,j}k^{2j+1}, \tag{8.253}$$

where

$$b_{r,j} = \frac{\binom{2r}{2j}}{2j + 1}B_{2r-2j}(1/2). \tag{8.254}$$

The Bernoulli polynomial $B_{2r+1}(1/2)$ is zero, so this is an odd polynomial in k and has zero constant term.

Proof of Theorem 8.105. First, (8.252) and (8.253) show that

$$\mathcal{T}(-2r) = \mathrm{CT}_N \sum_{n=0}^{N-1}(n + 1/2)^{2r}A_n^2. \tag{8.255}$$

Indeed every subtracted even power has partial sum with zero constant term in N, and the convergent remainder tends to the stated value.

The identity $A_{k-1} = A_k + k^{-2}$ gives finite summation by parts:

$$\sum_{n=0}^{N-1}(n + 1/2)^{2r}A_n^2 = \sum_{k=1}^{N} S_r(k)\left(\frac{2A_k}{k^2} + \frac{1}{k^4}\right) + S_r(N)A_N^2. \tag{8.256}$$

The coefficient $b_{r,0}$ of k contributes

$$\mathrm{CT}_N\left(2\sum_{k=1}^{N}\frac{A_k}{k} + H_N^{(3)} + NA_N^2\right) = 3\zeta(3).$$

Here $\sum_{k\geq 1} A_k/k = \zeta(2, 1) = \zeta(3)$ is the classical Euler identity and $NA_N^2 \to 0$. For completeness, the Euler identity follows from

$$\zeta(2, 1) = \frac{1}{2}\int_0^1 \frac{\log^2(1 - t)}{t}\,dt = \frac{1}{2}\int_0^\infty \frac{u^2}{e^u - 1}\,du = \zeta(3).$$

For $j \geq 1$, it remains to compute the constant of

$$E_j(N) = 2\sum_{k=1}^{N} k^{2j-1}A_k + \sum_{k=1}^{N} k^{2j-3} + N^{2j+1}A_N^2. \tag{8.257}$$

Let

$$F_j(N) = \sum_{k=1}^{N} k^{2j-1} = \frac{B_{2j}(N + 1) - B_{2j}}{2j}.$$

Changing the order in a finite sum gives

$$\sum_{k=1}^{N} k^{2j-1} A_k = A_N F_j(N) + \sum_{\ell=1}^{N} \frac{F_j(\ell-1)}{\ell^2}. \tag{8.258}$$

For $j \geq 2$, the second summand is a sum of polynomial powers of ℓ and has zero constant term as a polynomial in N; $B_{2j-1} = 0$ removes a possible $1/\ell$ term. The other power sum in (8.257) also has zero constant. For $j = 1$ the second sum in (8.258) equals $(N - H_N)/2$; its harmonic term cancels exactly against $\sum k^{-1}$ in (8.257). In all cases,

$$\mathrm{CT}_N\, E_j(N) = 2\,\mathrm{CT}_N(A_N F_j(N)) + \mathrm{CT}_N(N^{2j+1} A_N^2). \tag{8.259}$$

Now use the uncentered tail expansion

$$A_N = \frac{1}{N} - \frac{1}{2N^2} + \sum_{k\geq 1} \frac{B_{2k}}{N^{2k+1}}.$$

The finite coefficient calculation gives

$$\mathrm{CT}_N(A_N F_j(N)) = \frac{3-2j}{4} B_{2j-2}, \qquad \mathrm{CT}_N(N^{2j+1} A_N^2) = -B_{2j-2}. \tag{8.260}$$

To check the first identity, the coefficient of N^ℓ in F_j is $\binom{2j}{\ell} B_{2j-\ell}(1)/(2j)$. Only $\ell = 2$ and $\ell = 2j-1$ can pair with nonzero inverse coefficients of A_N when $j \geq 2$; their contributions are $-(2j-1)B_{2j-2}/4$ and $B_{2j-2}/2$. The case $j = 1$ is the direct product with $F_1(N) = (N^2+N)/2$ and gives $1/4$. For the second identity, an odd inverse exponent in A_N^2 arises only by multiplying $-1/(2N^2)$ by the appropriate odd inverse power; this gives $-B_{2j-2}$, also at $j = 1$.

Combining (8.259) and (8.260) yields

$$\mathrm{CT}_N\, E_j(N) = -\frac{2j-1}{2} B_{2j-2}.$$

Multiplication by $b_{r,j}$, followed by the $j = 0$ contribution, proves (8.249). □

8.33.3 A convergent generator involving classical polylogarithms

The complete sequence has an analytic exponential generating function. Let

$$\mathcal{G}(t) = \sum_{r=0}^{\infty} \mathcal{T}(-2r) \frac{t^{2r}}{(2r)!}.$$

Theorem 8.106 (Polylogarithmic moment generator)**.** *The series $\mathcal{G}$ converges for $|t| < 2\pi$ and equals*

$$\mathcal{G}(t) = \frac{t}{2\sinh(t/2)} \left[3\zeta(3) - \frac{1}{4t} \int_0^t (t-u) u^2 \coth(u/2)\, du \right]. \tag{8.261}$$

The singularities at zero in this expression are removable. For real $0 < t < 2\pi$, an equivalent formula in classical polylogarithms is

$$\mathcal{G}(t) = \frac{t}{2\sinh(t/2)} \left[2\zeta(3) - \frac{t^3}{48} - \frac{t}{2} \mathrm{Li}_2(e^{-t}) - 2\,\mathrm{Li}_3(e^{-t}) + \frac{3\big(\zeta(4) - \mathrm{Li}_4(e^{-t})\big)}{t} \right]. \tag{8.262}$$

Its continuation specified by (8.261) is holomorphic and even throughout $|t| < 2\pi$.

Proof. The Bernoulli generating functions give

$$\frac{t}{2\sinh(t/2)} = \sum_{r\ge 0} B_{2r}(1/2)\frac{t^{2r}}{(2r)!}, \qquad \frac{u}{2}\coth(u/2) = \sum_{k\ge 0} B_{2k}\frac{u^{2k}}{(2k)!}.$$

Both converge in the disk of radius 2π. Integrating the second series twice gives

$$\frac{1}{2t}\int_0^t (t-u)u^2\coth(u/2)\,du = \sum_{j\ge 1}\frac{2j-1}{2j+1}B_{2j-2}\frac{t^{2j}}{(2j)!}.$$

Thus (8.261) is exactly the binomial convolution in (8.249). This proves convergence, analyticity and the integral formula. In particular all apparent odd or logarithmic terms in any branch representation must cancel.

For positive t, expand $(e^u-1)^{-1}$ in its convergent positive exponential series. Direct integration, justified for the integrable factor $(t-u)u^2$, gives

$$\begin{aligned}\int_0^t \frac{(t-u)u^2}{e^u-1}\,du = {} & 2t\zeta(3) - 6\zeta(4) + t^2\operatorname{Li}_2(e^{-t}) \\ & + 4t\operatorname{Li}_3(e^{-t}) + 6\operatorname{Li}_4(e^{-t}).\end{aligned}$$

Finally $\coth(u/2) = 1 + 2/(e^u-1)$ and $\int_0^t (t-u)u^2\,du = t^4/12$ give (8.262). □

The finite-sum proof uses no transcendence assumption and no formal difference Galois theorem. The subtractions in the ordinary sums are part of each identity; without them the higher moments would diverge. The broader reflection-invariant numerator classification, resolvent transforms and independently marked higher-tail moments retain separate audit scopes.

8.34 Digamma powers, harmonic spectral data and exact finite parts

From harmonic Gamma identities to nonlinear finite parts. The three independent fifth-power continuations [131] converge on one all-power mechanism: specify each pole, determine the entire remainder, and only then compare the cutoff constant with spectral Laurent data. We retain the complete polynomial subtraction and every finite Stieltjes correction. This extends the preceding harmonic arithmetic, rather than assuming closure of all nonlinear periods in ordinary constants.

Write $H_n^{(r)} = \sum_{j=1}^n j^{-r}$ and, for a positive integer word $w = (w_1, \dots, w_d)$, put

$$H_n(w) = \sum_{n\ge j_1>\cdots>j_d\ge 1} j_1^{-w_1}\cdots j_d^{-w_d}, \qquad Z(s,w) = \sum_{n\ge 1} H_{n-1}(w)n^{-s}. \tag{8.263}$$

The empty word has value one; a nonempty word has $H_0(w) = 0$. All Z coefficients below mean meromorphic Laurent coefficients, including at a pole. The Stieltjes convention is

$$\zeta(1+u,a) = u^{-1} + \sum_{j\ge 0}(-1)^j\gamma_j(a)u^j/j!, \qquad \gamma = \gamma_0(1). \tag{8.264}$$

If $f(x) = \sum_{j=1}^k a_j x^{-j} + g(x)$ with g continuous on $[0,1]$, the original-coordinate scalar finite part is

$$\operatorname{FP}_x\int_0^1 f(x)\,\mathrm{d}x = \lim_{\epsilon\downarrow 0}\left(\int_\epsilon^1 f(x)\,\mathrm{d}x - \sum_{j=2}^k \frac{a_j\epsilon^{1-j}}{j-1} + a_1\log\epsilon\right). \tag{8.265}$$

Set $Q_k = \mathrm{FP}_x \int_0^1 \psi(x)^k \, \mathrm{d}x$, $k \geq 1$. This scalar problem uses the pole at zero and the upper endpoint one; it is distinct from the distributional shift-contact comparison. For a sequence $F_N = P(\log N) + o(1)$, $\mathrm{CT}_{\log N} F_N = P(0)$. The cutoff comparison below proves its relation with Laurent coefficients.

8.34.1 Pole coefficients and the normalized global expansion

The standard Taylor expansion of $\psi(1+\epsilon)$ and the recurrence $\psi(z+1) = \psi(z) + 1/z$ give [132, §§5.5,5.7]

$$\psi(-n+\epsilon) = -\frac{1}{\epsilon} + b_n + \sum_{r\geq 2} c_{r,n}\epsilon^{r-1}, \tag{8.266}$$

$$b_n = H_n - \gamma, \qquad c_{r,n} = H_n^{(r)} + (-1)^r \zeta(r), \qquad n \geq 0.$$

This is a convergent local expansion for $0 < |\epsilon| < 1$.

For each fixed positive integer k, define the finite polynomial data

$$p_{k,j}(n) = [\epsilon^{k-j}] \left(-1 + b_n\epsilon + \sum_{r=2}^{k} c_{r,n}\epsilon^r \right)^k, \qquad 1 \leq j \leq k, \tag{8.267}$$

$$r_k(n) = p_{k,1}(n), \qquad P_{k,n}(z) = \sum_{j=1}^{k} \frac{p_{k,j}(n)}{(z+n)^j}, \qquad \kappa_k = [\epsilon^0]\psi(\epsilon)^k. \tag{8.268}$$

The symbol κ_k is a local Laurent constant; it will not be confused with the universal contour constant C_k introduced later. The pole coefficients satisfy

$$p_{k,j}(n) = O_k\big((1+\log(n+1))^{k-j}\big). \tag{8.269}$$

Indeed $b_n = O(1+\log(n+1))$ and every $c_{r,n}$ with $r \geq 2$ is bounded. In a term of degree $k-j$ in ϵ, at most $k-j$ factors of b_n can occur.

Theorem 8.107 (Normalized Mittag–Leffler formula for every power)**.** *For $z \notin \{0, -1, -2, \ldots\}$,*

$$\boxed{\psi(z)^k = P_{k,0}(z) + \kappa_k + \sum_{n\geq 1} \big(P_{k,n}(z) - P_{k,n}(0)\big).} \tag{8.270}$$

The series converges normally on compact sets avoiding the poles. In particular the entire remainder is exactly the displayed constant, with no uncomputed entire function.

Proof. Normal convergence of the candidate series follows from (8.269): on a fixed compact set, for sufficiently large n,

$$(z+n)^{-j} - n^{-j} = O(n^{-j-1}),$$

uniformly in z. It remains to determine the possible entire remainder. Put $f(w) = \psi(w)^k - P_{k,0}(w)$, so f is analytic at zero and $f(0) = \kappa_k$. Integrate

$$\frac{z f(w)}{w(w-z)}$$

around $|w| = R_N = N + \frac{1}{2}$, with positive orientation and N large.

We need a uniform estimate on the full circle, including near the negative real axis. On its right half, the usual sectorial digamma asymptotic gives $\psi(w) = O(1 + \log R_N)$. On its left half, use

$$\psi(w) = \psi(1 - w) - \pi \cot(\pi w). \tag{8.271}$$

Here $1 - w$ lies in the right half-plane and has size comparable to R_N. If $|\Im w| \geq 1$, the cotangent is uniformly bounded. If $|\Im w| < 1$, the points of this circle are a fixed positive distance from the integers for all sufficiently large N, because its real intersections are half-integers. The cotangent is again uniformly bounded. Thus $f(w) = O_k((1 + \log R_N)^k)$ on the whole contour, and the integral is $O_k((1 + \log R_N)^k / R_N) = o(1)$.

The residues at $w = z$ and $w = 0$ are $f(z)$ and $-f(0)$. At $w = -n$, the residue is $-P_{k,n}(z) + P_{k,n}(0)$, as follows by expanding $z/[w(w - z)] = 1/(w - z) - 1/w$ there. The residue theorem and passage to the limit give (8.270). This proves the normalization, not just agreement of the principal parts. □

8.34.2 A finite recurrence for the coefficient differences

Write $\Delta p_{k,j}(n) = p_{k,j}(n) - p_{k,j}(n - 1)$ for $n \geq 1$, and $\Delta r_k(n) = \Delta p_{k,1}(n)$.

Proposition 8.108 (Exact difference recurrence)**.** *For $1 \leq j \leq k$,*

$$\Delta p_{k,j}(n) = \sum_{\ell=1}^{k-j} \binom{k}{\ell} \sum_{h=j}^{k-\ell} \binom{\ell + h - j - 1}{h - j} \frac{p_{k-\ell,h}(n-1)}{n^{\ell+h-j}}. \tag{8.272}$$

An empty sum is zero; in particular $\Delta p_{k,k}(n) = 0$.

Proof. The recurrence for ψ gives the local identity

$$\psi(-n + \epsilon) = \psi(-(n-1) + \epsilon) + \frac{1}{n - \epsilon}.$$

Expand its kth power by the binomial theorem. In the term with ℓ copies of $(n - \epsilon)^{-1}$, multiply the pole $p_{k-\ell,h}(n-1)\epsilon^{-h}$ by

$$(n - \epsilon)^{-\ell} = \sum_{q \geq 0} \binom{\ell + q - 1}{q} \frac{\epsilon^q}{n^{\ell+q}}.$$

The coefficient of ϵ^{-j} has $q = h - j$, which gives (8.272). Only finitely many coefficients contribute. □

The recurrence proves, in particular, $\Delta p_{k,j}(n) = O_k((1 + \log n)^k / n)$ for $j < k$. Sharper logarithmic powers are immediate, but are not needed for the exact identities. Formula (8.272) is also a convenient implementation specification: it involves only lower powers and exact rational binomial coefficients.

8.35 The universal contour polynomial

The subtraction constants in (8.270) are individually awkward harmonic expressions. Their combined cutoff asymptotic is much simpler than their separate expansions suggest.

Theorem 8.109 (All coefficients of the large-circle boundary term)**.** *For every fixed integer $k \geq 1$,*

$$\kappa_k - \sum_{n=1}^{N} P_{k,n}(0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} (\log N + i\theta)^k \, \mathrm{d}\theta + o(1)$$
$$= \sum_{m=0}^{\lfloor k/2 \rfloor} \binom{k}{2m} \frac{(-1)^m \pi^{2m}}{2m+1} (\log N)^{k-2m} + o(1). \tag{8.273}$$

Consequently

$$C_k := \mathrm{CT}_{\log N}\left(\kappa_k - \sum_{n=1}^{N} P_{k,n}(0)\right) = \begin{cases} (-1)^m \pi^{2m}/(2m+1), & k = 2m, \\ 0, & k \text{ odd.} \end{cases} \tag{8.274}$$

Proof. The residue at zero of $\psi(w)^k/w$ is κ_k. At $-n$ it is $-P_{k,n}(0)$. Thus, on $R = N + \frac{1}{2}$, the residue theorem gives the exact identity

$$\kappa_k - \sum_{n=1}^{N} P_{k,n}(0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \psi(Re^{i\theta})^k \, \mathrm{d}\theta. \tag{8.275}$$

We compare this with the angular average of the principal logarithm. On the right half-circle, $\psi(w) - \operatorname{Log} w = O(1/R)$ uniformly. On the left half-circle with $|\Im w| \geq \log R$, reflection (8.271) and the exponential expansion of cotangent give

$$\psi(w) - \operatorname{Log} w = O(1/R) + O(e^{-2\pi|\Im w|}).$$

For example, in the upper left quadrant, $\operatorname{Log}(1-w) = \operatorname{Log} w - i\pi + O(1/R)$ and $\pi \cot \pi w = -i\pi + O(e^{-2\pi \Im w})$; the lower quadrant is the conjugate branch calculation.

The remaining arc near the negative real axis has angular length $O(\log R/R)$. On that arc, the bounded-cotangent argument in Theorem 8.107 gives $\psi(w) - \operatorname{Log} w = O(1)$, while both terms are $O(1 + \log R)$. Factoring the difference of the kth powers shows that its contribution to the angular average is $O_k((1+\log R)^k/R)$. The rest of the circle contributes a quantity of the same or smaller order. The logarithm has a jump at the one endpoint of the angular parametrization, which does not affect the integral. Therefore

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} \left(\psi(Re^{i\theta})^k - (\log R + i\theta)^k\right) \mathrm{d}\theta = o(1).$$

Replacing $\log R$ by $\log N$ changes this by $o(1)$. Odd powers of θ integrate to zero and the even angular moments are $(-1)^m \pi^{2m}/(2m+1)$, proving both assertions. □

For orientation, $C_1 = C_3 = C_5 = 0$, $C_2 = -2\zeta(2)$, and $C_4 = 18\zeta(4)$. These are combined boundary constants. They are not the local Laurent constants κ_k.

8.36 A finite-part formula for every positive integer power

Define

$$J_k = \sum_{n \geq 1} \sum_{j=2}^{k} \frac{\Delta p_{k,j}(n)}{(j-1)n^{j-1}}, \tag{8.276}$$

$$D_k(s) = \sum_{n \geq 1} \frac{\Delta r_k(n)}{n^s}, \qquad \Re s > 0. \tag{8.277}$$

The first series is ordinary and absolutely convergent, by Proposition 8.108. The second converges absolutely in its initial half-plane. Its meromorphic continuation to a neighborhood of zero will follow from the proof below.

Theorem 8.110 (The all-power harmonic spectral formula)**.** *For every integer $k \geq 1$,*

$$\boxed{Q_k = C_k + J_k + [s]D_k(s).} \tag{8.278}$$

All data on the right are specified by the finite polynomial calculation (8.267), the exact difference recurrence (8.272), and the coefficient of a meromorphic germ. The identity uses the original-coordinate finite part (8.265).

We first record the exact cutoff comparison that determines the sign and the order of the spectral coefficient.

Lemma 8.111 (Logarithmic cutoffs and Laurent coefficients)**.** *Suppose $a_n = P(\log n) + O(n^{-\delta}(1 + \log n)^M)$, where P is a polynomial and $\delta > 0$. Let $A(s) = \sum_{n\geq 1} a_n n^{-s}$ in its right half-plane of absolute convergence. Then $A(1 + u)$ continues meromorphically near $u = 0$, and*

$$\mathrm{CT}_{\log N} \sum_{n=1}^{N} \frac{a_n}{n} = [u^0]A(1 + u), \tag{8.279}$$

$$\mathrm{CT}_{\log N} \sum_{n=1}^{N} \frac{a_n \log n}{n} = -[u]A(1 + u). \tag{8.280}$$

Proof. For $P(L) = \sum_j c_j L^j$, write

$$A(1 + u) = \sum_j c_j(-1)^j \zeta^{(j)}(1 + u) + \sum_{n\geq 1} \frac{a_n - P(\log n)}{n^{1+u}}.$$

The last series and all its derivatives converge normally for $\Re u > -\delta$. The elementary Stieltjes limit formula gives

$$\sum_{n=1}^{N} \frac{(\log n)^j}{n} = \frac{(\log N)^{j+1}}{j + 1} + \gamma_j + O((1 + \log N)^j/N).$$

The stated error follows by one sum–integral comparison applied to the derivative of $(\log x)^j/x$ on the tail. On the Laurent side, $[u^0](-1)^j\zeta^{(j)}(1 + u) = \gamma_j$ and $[u](-1)^j\zeta^{(j)}(1 + u) = -\gamma_{j+1}$. Termwise comparison proves the first formula and, after inserting one further factor of $\log n$, the second. Absolute convergence handles the subtracted remainder in both cases. □

Proof of Theorem 8.110. The harmonic-number asymptotic expansions, or reflection followed by the positive-axis digamma expansion, show that

$$r_k(n) = R_k(\log n) + O_k((1 + \log n)^M/n), \tag{8.281}$$

$$\Delta r_k(n) = \frac{R_k'(\log n)}{n} + O_k((1 + \log n)^M/n^2), \tag{8.282}$$

$$R_k(L) = \operatorname*{Res}_{\epsilon=0}(L - \pi \cot \pi\epsilon)^k. \tag{8.283}$$

Here and below M may depend on k. To justify the second estimate, use the complete harmonic asymptotic expansions through the $1/n$ term: $r_k(n) = R_k(\log n)+n^{-1}U_k(\log n)+O((1+\log n)^M/n^2)$

with a polynomial U_k. The difference of the explicit $n^{-1}U_k$ terms is $O((1+\log n)^M/n^2)$, and the leading difference is $R_k'(\log n)/n$ to that same accuracy. Thus the second formula does not follow from differencing an unspecified first-order error. It also follows directly from (8.272) and the first two harmonic asymptotic terms.

Apply Lemma 8.111 to $a_n = n\Delta r_k(n)$. It proves the meromorphic continuation of D_k and gives

$$\mathrm{CT}_{\log N} \sum_{n=1}^{N} \Delta r_k(n) \log n = -[s]D_k(s). \tag{8.284}$$

Now integrate the normally convergent series (8.270). After the pole $P_{k,0}$ is removed, the same $O(n^{-j-1})$ estimate proves uniform convergence also on the closed interval $[0,1]$, so integration across its regular endpoint is justified. The finite part of the pole at zero contributes $-\sum_{j=2}^{k} p_{k,j}(0)/(j-1)$; its simple pole has finite part zero. For a pole at $-n$, the integrated simple-pole term is $r_k(n)\log(1+1/n)$, and the higher-pole terms have numerators $p_{k,j}(n)(n^{1-j}-(n+1)^{1-j})/(j-1)$. The constant subtraction integrates to $-P_{k,n}(0)$.

Use the following *finite* summation identities:

$$\begin{aligned}
&\sum_{n=1}^{N} p_{k,j}(n)\big(n^{1-j}-(n+1)^{1-j}\big) \\
&\qquad = p_{k,j}(0) + \sum_{n=1}^{N} \frac{\Delta p_{k,j}(n)}{n^{j-1}} - \frac{p_{k,j}(N)}{(N+1)^{j-1}}, \qquad j \ge 2, \qquad (8.285)
\end{aligned}$$

$$\sum_{n=1}^{N} r_k(n)\log(1+1/n) = r_k(N)\log(N+1) - \sum_{n=1}^{N} \Delta r_k(n)\log n. \tag{8.286}$$

The first terms in (8.285) cancel the finite part of the pole at zero exactly, while its final terms tend to zero. If $J_{k,N}$ denotes the cutoff in (8.276), we obtain

$$\begin{aligned}
Q_k = \lim_{N\to\infty} \Bigg\{ &\kappa_k - \sum_{n=1}^{N} P_{k,n}(0) + J_{k,N} + r_k(N)\log(N+1) \\
&- \sum_{n=1}^{N} \Delta r_k(n)\log n \Bigg\}. \qquad (8.287)
\end{aligned}$$

All terms here arose from finite sums or normally convergent series. By (8.281), the polynomial asymptotic of the boundary $r_k(N)\log(N+1)$ is $(\log N)R_k(\log N)$, which has no constant term. Theorem 8.109 supplies the constant C_k for the first two terms of (8.287), and (8.284) supplies $+[s]D_k$. Finally $J_{k,N} \to J_k$. Taking the constant coefficient of the complete cutoff expression proves (8.278). □

8.36.1 A closed ordinary-series form

The residue polynomial in (8.283) itself has a closed form. This makes every Stieltjes correction explicit, without naming any multiple-zeta Laurent coefficient.

Lemma 8.112 (Closed asymptotic residue polynomial)**.** *For an integer $k \ge 0$, define*

$$R_k(L) = \operatorname*{Res}_{t=0}\big(L - \pi\cot(\pi t)\big)^k.$$

Then

$$\boxed{R_k(L) = -\frac{(L+i\pi)^k - (L-i\pi)^k}{2i\pi}.} \tag{8.288}$$

For $k \geq 1$, if $R_k'(L) = \sum_{j\geq 0} a_{k,j} L^j$, the ordinary Stieltjes contribution is the explicit finite expression

$$\boxed{-\sum_{j\geq 0} a_{k,j}\gamma_{j+1} = k \sum_{m=0}^{\lfloor (k-2)/2 \rfloor} (-1)^m \binom{k-1}{2m+1} \pi^{2m} \gamma_{k-2m-1}.} \tag{8.289}$$

The right side is understood as zero for $k = 1$. Here $\zeta(1+s) = s^{-1} + \sum_{j\geq 0}(-1)^j \gamma_j s^j/j!$.

Proof. Put $f(t) = L - \pi\cot(\pi t)$. The elementary differential identity

$$f'(t) = (f(t) - L)^2 + \pi^2$$

and the zero residue of a meromorphic derivative give, for $k \geq 2$,

$$0 = \frac{1}{k-1} \operatorname*{Res}_{t=0} \frac{\mathrm{d}}{\mathrm{d}t} f(t)^{k-1} = R_k(L) - 2LR_{k-1}(L) + (L^2+\pi^2)R_{k-2}(L).$$

Since $R_0 = 0$ and $R_1 = -1$, solving this recurrence with characteristic roots $L \pm i\pi$ proves (8.288). Expanding its numerator yields

$$R_k(L) = -\sum_{m=0}^{\lfloor (k-1)/2 \rfloor} (-1)^m \binom{k}{2m+1} \pi^{2m} L^{k-2m-1}.$$

On differentiating in L, the possible term with $k - 2m - 1 = 0$ vanishes. Use $(k-2m-1)\binom{k}{2m+1} = k\binom{k-1}{2m+1}$. This gives (8.289). □

To see directly why this finite expression is the Stieltjes term, suppose the residue difference satisfies the previously established asymptotic formula

$$r_k(n) - r_k(n-1) = \frac{R_k'(\log n)}{n} + O\left(\frac{(1+\log n)^M}{n^2}\right).$$

For $D_k(s) = \sum_{n\geq 1}(r_k(n) - r_k(n-1))n^{-s}$, subtraction therefore gives

$$D_k(s) = \sum_{j\geq 0} a_{k,j}(-1)^j \zeta^{(j)}(1+s) + \sum_{n\geq 1}\left(r_k(n) - r_k(n-1) - \frac{R_k'(\log n)}{n}\right) n^{-s}.$$

The second series and its first spectral derivative converge normally near $s = 0$. Since $[s](-1)^j\zeta^{(j)}(1+s) = -\gamma_{j+1}$, it follows that

$$\begin{aligned}[s]D_k(s) = k &\sum_{m=0}^{\lfloor (k-2)/2 \rfloor} (-1)^m \binom{k-1}{2m+1} \pi^{2m}\gamma_{k-2m-1} \\ &- \sum_{n\geq 1}\left(r_k(n) - r_k(n-1) - \frac{R_k'(\log n)}{n}\right)\log n.\end{aligned} \tag{8.290}$$

The final sum is absolutely convergent. For example, the finite Stieltjes terms at powers four and five are respectively $12\gamma_3 - 24\zeta(2)\gamma_1$ and $20\gamma_4 - 120\zeta(2)\gamma_2$.

Corollary 8.113 (Ordinary subtracted series at every power)**.** *With the definitions above,*

$$Q_k = C_k + J_k + k\sum_{m=0}^{\lfloor (k-2)/2\rfloor}(-1)^m\binom{k-1}{2m+1}\pi^{2m}\gamma_{k-2m-1} - \sum_{n\ge 1}\left(\Delta r_k(n) - \frac{R_k'(\log n)}{n}\right)\log n. \tag{8.291}$$

The finite Stieltjes sum is zero at $k = 1$. Both infinite sums are ordinary absolutely convergent sums. The last summand is $O_k((1+\log n)^{M+1}/n^2)$.

This is useful both conceptually and computationally. The right side contains no unspecified regularization and no uncomputed entire remainder. Further terms of the same harmonic asymptotic expansions give convergent Hurwitz-zeta tail accelerations if numerical evaluation is wanted. Such acceleration is optional for the identity itself.

8.37 A finite algorithm in strict harmonic words

The spectral formula becomes finite algebra once its target class is specified. Work over $\mathbb{Q}[\gamma, \zeta(2), \dots, \zeta(k)]$ and put $h_r = H_{n-1}^{(r)}$, $v = 1/n$. Compute the polynomial $p_{k,j}$ from (8.267), and replace $H_n^{(r)}$ by $h_r + v^r$ to obtain its value at n. Subtract its value at $n-1$. This produces every $\Delta p_{k,j}$ as a finite polynomial in $v, h_1, \dots, h_k$.

The finite stuffle recursion, with (a, A) denoting a leading letter followed by a word, is

$$(a, A) * (b, B) = (a, A * (b, B)) + (b, (a, A) * B) + (a + b, A * B). \tag{8.292}$$

It expresses products of the h_r as a finite integer combination of strict words $H_{n-1}(w)$. The coefficients count equal-index possibilities as well as the two strict orderings. Thus

$$h_1^2 = 2H_{n-1}(1,1) + H_{n-1}^{(2)},$$

and a further multiplication gives

$$h_1^3 = 6H_{n-1}(1,1,1) + 3H_{n-1}(2,1) + 3H_{n-1}(1,2) + H_{n-1}^{(3)}.$$

These are finite identities before any infinite sum is taken.

Every positive-integer word has a polynomial harmonic asymptotic

$$H_{n-1}(w) = B_w(\log n) + O((1+\log n)^M/n) \tag{8.293}$$

for some polynomial B_w and integer M. Here is a direct induction that also supplies the continuation needed term by term. For the empty word the assertion is immediate. Given it for w, use $H_N(a, w) = \sum_{m=1}^{N} H_{m-1}(w)m^{-a}$. If $a = 1$, the Stieltjes cutoff formula applied to each power of $\log m$ and the absolutely convergent subtracted remainder give a polynomial in $\log N$ with an $O((1+\log N)^{M'}/N)$ error. If $a \ge 2$, the sum converges and its tail has that same error form, so the polynomial is a constant. Replacing N by $n-1$ preserves the claimed form. Consequently each $Z(1+s, w)$ has a meromorphic germ at zero by Lemma 8.111; for first entry at least two its defining series is already holomorphic near zero.

Theorem 8.114 (Terminating harmonic-word reduction)**.** *At a fixed positive integer power k, the following procedure terminates and yields the exact formula* (8.278)*:*

1. *Compute $p_{k,j}$ and their finite differences by (8.267) or (8.272).*

2. *Expand the summand of J_k and the polynomial $\Delta r_k(n)$ in $v, h_1, \dots, h_k$.*

3. *Apply the finite stuffle recursion to every harmonic monomial.*

4. *Replace $v^q H_{n-1}(w)$ by $Z(q,w)$ for J_k, and by $Z(q+s,w)$ for $D_k(s)$.*

Every first entry in J_k is at least two. Every first entry in $D_k(0)$ is at least one. The resulting word weights are at most $k-1$; coefficient factors in γ and ordinary zeta values account for the remaining formal weight. The prescription introduces only a finite number of specified linear Laurent coefficients.

Proof. Coefficient extraction and finite differences are finite polynomial operations. Each application of the stuffle recursion reduces the combined remaining word length, so it terminates. By (8.272), every monomial in $\Delta p_{k,j}$ contains at least one v, and division by n^{j-1} gives $q \geq j \geq 2$ in J_k. For Δr_k, it gives $q \geq 1$. Therefore all replacements in the initial half-planes are identities of absolutely convergent series. Their continuations at zero agree by the meromorphic subtraction justified by (8.293). Finally assign weight 1 to h_1, γ, v, weight r to $h_r, \zeta(r)$, and use (8.267). Both relevant summand polynomials have weight $k-1$, and stuffle preserves it. □

This is a reduction to a precise finite class of multiple-zeta values and their spectral coefficients. It does not prove that all those values reduce further to products of ordinary zeta values. For higher powers it is appropriate to retain admissible multiple-zeta words unless an exact additional relation is supplied.

8.37.1 Recovery of the lower-power formulas

Define the height-one coefficients

$$\eta = [u]Z(1+u,1), \qquad \theta = [u]Z(1+u,1,1), \qquad \chi = [u]Z(1+u,1,1,1). \tag{8.294}$$

The first instances of Theorem 8.110 are

$$\begin{aligned}
Q_1 &= 0, \\
Q_2 &= 2\gamma_1 - 2\zeta(2), && (8.295)\\
Q_3 &= 3\zeta(2) - 6\eta - 6\gamma\gamma_1, && (8.296)\\
Q_4 &= 12\zeta(3) - 12\gamma\zeta(2) + 18\zeta(4) + 4\zeta'(3) - 24\theta + 24\gamma\eta + 12(\gamma^2 - \zeta(2))\gamma_1. && (8.297)
\end{aligned}$$

The cubic and quartic identities are consistency checks against the incoming reports [131]; they are not presented as new evaluations here. For example,

$$\begin{aligned}
D_2(s) &= -2\zeta(1+s), && J_2 = 0,\\
D_3(s) &= -6Z(1+s,1) + 6\gamma\zeta(1+s), && J_3 = 3\zeta(2),\\
D_4(s) &= -24Z(1+s,1,1) + 24\gamma Z(1+s,1) && \\
&\quad + 12(\zeta(2) - \gamma^2)\zeta(1+s) + 4\zeta(3+s), && J_4 = 12\zeta(3) - 12\gamma\zeta(2).
\end{aligned}$$

At $k=1$ the residue is the constant -1, hence $J_1 = D_1 = 0$; the finite part also follows directly from $\log\Gamma(1) = 0$ and $\log\Gamma(\epsilon) = -\log\epsilon + O(\epsilon)$.

8.38 The fifth power with every finite correction

For compactness write $\zeta_r = \zeta(r)$ in this section. At a fixed pole use $b = b_n$, $c = c_{2,n}$, $d = c_{3,n}$, and $e = c_{4,n}$. Direct coefficient extraction gives

$$\begin{aligned} p_{5,5} &= -1, \qquad p_{5,4} = 5b, \qquad p_{5,3} = 5c - 10b^2, \\ p_{5,2} &= 10b^3 - 20bc + 5d, \\ r_5 &= -5b^4 + 30b^2c - 10c^2 - 20bd + 5e. \end{aligned} \tag{8.298}$$

For reference, the local constant is

$$\kappa_5 = -\gamma^5 + 20\gamma^3\zeta_2 - 55\gamma\zeta_4 - 30\gamma^2\zeta_3 + 20\zeta_2\zeta_3 - 5\zeta_5. \tag{8.299}$$

The universal constant is $C_5 = 0$, a different quantity.

For the finite differences, let b, c, d instead denote the preceding index $n-1$. Then

$$\begin{aligned} \Delta p_{5,4}(n) &= \frac{5}{n}, \\ \Delta p_{5,3}(n) &= -\frac{5}{n^2} - \frac{20b}{n}, \\ \Delta p_{5,2}(n) &= -\frac{5}{n^3} + \frac{10b}{n^2} + \frac{30b^2 - 20c}{n}, \\ \Delta r_5(n) &= \frac{-20b^3 + 60bc - 20d}{n} + \frac{10c}{n^2} + \frac{20b}{n^3}. \end{aligned} \tag{8.300}$$

Equivalently the first numerator in the last line is $5r_4(n-1)$. These identities can be checked simply by replacing (b, c, d, e) by $(b + 1/n, c + 1/n^2, d + 1/n^3, e + 1/n^4)$ in (8.298); Proposition 8.108 gives an independent binomial derivation.

The higher poles therefore contribute

$$\begin{aligned} J_5 &= \sum_{n\geq 1} \left\{ \frac{30(H_{n-1} - \gamma)^2 - 20(H^{(2)}_{n-1} + \zeta_2)}{n^2} - \frac{35}{6n^4} \right\} \\ &= 30\gamma^2\zeta_2 - 60\gamma\zeta_3 + \frac{35}{3}\zeta_4. \end{aligned} \tag{8.301}$$

For completeness, the low-weight evaluations needed in this last step are

$$Z(2,1) = \zeta_3, \quad Z(2,1,1) = \zeta_4, \quad Z(2,2) = \frac{3}{4}\zeta_4, \quad \zeta_2^2 = \frac{5}{2}\zeta_4.$$

They follow from the standard duality, sum, and stuffle identities [132, 133]. In particular,

$$\begin{aligned} \sum_{n\geq 1} \frac{(H_{n-1} - \gamma)^2}{n^2} &= \frac{11}{4}\zeta_4 - 2\gamma\zeta_3 + \gamma^2\zeta_2, \\ \sum_{n\geq 1} \frac{H^{(2)}_{n-1} + \zeta_2}{n^2} &= \frac{13}{4}\zeta_4, \end{aligned}$$

which proves every scalar term in (8.301).

Applying finite stuffle to the last line of (8.300) yields the complete spectral identity

$$\begin{aligned} D_5(s) = &-120Z(1+s,1,1,1)+120\gamma Z(1+s,1,1)\\ &+60(\zeta_2-\gamma^2)Z(1+s,1)+20(\gamma^3-3\gamma\zeta_2+\zeta_3)\zeta(1+s)\\ &+20Z(1+s,3)+10Z(2+s,2)+20Z(3+s,1)\\ &+10\zeta_2\zeta(2+s)-20\gamma\zeta(3+s). \end{aligned} \tag{8.302}$$

All equalities here hold first for $\Re s>0$, and then as meromorphic germs. In the first numerator of (8.300), the words $(2,1)$ and $(1,2)$ from H_{n-1}^3 cancel against the corresponding words in $H_{n-1}H_{n-1}^{(2)}$. The residual (3) word does not cancel. It is responsible for $Z(1+s,3)$.

Set

$$D_{p,q}=[u]Z(p+u,q),\qquad \Omega_4=2D_{1,3}+D_{2,2}+2D_{3,1}. \tag{8.303}$$

For $p\geq 2$ these coefficients are ordinary outer-order derivatives, $D_{p,q}=\partial_1 Z(p,q)$. For $p=1,q=3$, coefficient extraction is required at a simple pole.

Theorem 8.115 (Complete fifth-power finite part)**.** *In the conventions* (8.265)*,* (8.264)*,* (8.294)*, and* (8.303)*,*

$$\begin{aligned} Q_5 = &\,30\gamma^2\zeta_2-60\gamma\zeta_3+\frac{35}{3}\zeta_4-120\chi+120\gamma\theta+60(\zeta_2-\gamma^2)\eta\\ &-20(\gamma^3-3\gamma\zeta_2+\zeta_3)\gamma_1+20D_{1,3}+10D_{2,2}+20D_{3,1}\\ &+10\zeta_2\zeta'(2)-20\gamma\zeta'(3). \end{aligned} \tag{8.304}$$

Equivalently the three mixed terms are $10\Omega_4$*.*

Proof. Use $C_5=0$, (8.301), and the linear coefficient of (8.302) in Theorem 8.110. In particular $[s]\zeta(1+s)=-\gamma_1$, which fixes the sign of its correction. The other coefficients are exactly those named in (8.294) and (8.303). Every step rests on a finite algebraic identity or the normalized analytic results already proved. □

This answers the fifth-power research question in [131]: the fourfold height-one coefficient enters with multiplier -120, and the mixed twofold coefficients and both ordinary zeta derivatives are retained explicitly. The result is not a reduction of χ or Ω_4 to ordinary zeta and Stieltjes constants.

8.38.1 An ordinary convergent formula for the same finite part

The residue polynomial and its derivative are

$$R_5(L)=-5L^4+60\zeta_2L^2-90\zeta_4,\qquad R_5'(L)=-20L^3+120\zeta_2L.$$

Consequently Corollary 8.113 gives the exact alternative

$$\begin{aligned} Q_5 = &\,30\gamma^2\zeta_2-60\gamma\zeta_3+\frac{35}{3}\zeta_4+20\gamma_4-120\zeta_2\gamma_2\\ &-\sum_{n\geq 1}\left\{\Delta r_5(n)+\frac{20\log^3 n-120\zeta_2\log n}{n}\right\}\log n. \end{aligned} \tag{8.305}$$

Here Δr_5 is the explicit harmonic expression (8.300), with b,c,d at $n-1$. It can equally be written in polygamma values at positive integers:

$$b=\psi(n),\qquad c=2\zeta_2-\psi'(n),\qquad d=\tfrac{1}{2}\psi''(n). \tag{8.306}$$

The final series in (8.305) converges absolutely; its summand is $O((1+\log n)^4/n^2)$. This gives a definition-free numerical route to Q_5 using only ordinary Stieltjes constants and ordinary polygamma evaluations, with no multivariate analytic continuation routine.

8.38.2 Explicit kernels for the height-one coefficients

The retained height-one coefficient has a concrete normalized integral representation. We give it at every depth so that the fourfold coefficient is specified independently of the digamma moment. Height-one Laurent constants and related normalized integrals have antecedents in Young's work [134, Section 4]. The proposition below records a direct Mellin subtraction with the coefficient and Stieltjes conventions fixed explicitly for this article. Let

$$\frac{1}{\Gamma(1+u)} = \sum_{j\ge 0} \alpha_j u^j, \qquad K_d(t) = \frac{[-\log(1-e^{-t})]^{d-1}}{(d-1)!(e^t-1)}, \quad d \ge 1. \tag{8.307}$$

Proposition 8.116 (Normalized height-one Mellin coefficient)**.** *For $d \ge 1$,*

$$Z(1+u, 1^{d-1}) = \sum_{j=0}^{d} \alpha_j u^{j-d} + L_d u + O(u^2), \tag{8.308}$$

where

$$\begin{aligned} L_d = \alpha_{d+1} &+ \int_0^1 \log t \left\{ K_d(t) - \frac{(-\log t)^{d-1}}{(d-1)!t} \right\} \, \mathrm{d}t \\ &+ \int_1^\infty \log t \, K_d(t) \, \mathrm{d}t. \end{aligned} \tag{8.309}$$

Both integrals are absolutely convergent. In particular $L_2 = \eta$, $L_3 = \theta$, and $L_4 = \chi$.

Proof. The generating series of $H_{n-1}(1^{d-1})$ gives, for $\Re u > 0$,

$$\Gamma(1+u) Z(1+u, 1^{d-1}) = \int_0^\infty t^u K_d(t) \, \mathrm{d}t.$$

For example it follows by dividing $\mathrm{Li}_{1^{d-1}}(z) = (-\log(1-z))^{d-1}/(d-1)!$ by $1-z$, multiplying by z, and setting $z = e^{-t}$; for $d = 1$ the kernel is simply $1/(e^t - 1)$. Subtract the leading kernel on $(0, 1)$ to write this integral as $u^{-d} + B_d(u)$, with B_d holomorphic near zero. The subtraction is integrable with any fixed power of $\log t$, because its remainder is at worst a polynomial in $|\log t|$ near zero; the kernel decays exponentially at infinity.

The derivative identity

$$K_d(t) = -\frac{1}{d!} \frac{\mathrm{d}}{\mathrm{d}t} [-\log(1-e^{-t})]^d$$

shows that $B_d(0) = 0$: the integrals from ϵ to infinity and of the subtracted kernel from ϵ to 1 have difference

$$\frac{[-\log(1-e^{-\epsilon})]^d - (-\log \epsilon)^d}{d!} \longrightarrow 0.$$

Multiplication by $1/\Gamma(1+u)$ now proves (8.308), with $L_d = \alpha_{d+1} + B_d'(0)$ exactly as in (8.309). □

For the fifth moment the coefficient used in this kernel is explicitly

$$\alpha_5 = \frac{\gamma^5 - 10\gamma^3\zeta_2 + 20\gamma^2\zeta_3 + 15\gamma\zeta_2^2 - 30\gamma\zeta_4 - 20\zeta_2\zeta_3 + 24\zeta\alpha_5}{120}. \tag{8.310}$$

This follows by exponentiating $\log(1/\Gamma(1+u)) = \gamma u + \sum_{r\ge 2} (-1)^{r+1} \zeta(r) u^r / r$. The kernel is an exact analytic representation, not by itself a reduction of the arithmetic content of χ.

8.38.3 Positive Cauchy series and the height-one sign convention

The height-one coefficient formula has a second, ordinary-series form [134]. Put

$$\rho_n = \frac{1}{(n-1)!}\int_0^1 (r)_n \, \mathrm{d}r, \qquad n \geq 1.$$

These are positive rational numbers, with $\rho_n \leq n$.

Proposition 8.117 (An ordinary series for every height-one coefficient)**.** *With $\alpha_j = [u^j]\Gamma(1+u)^{-1}$ and $L_d = [u]Z(1+u, 1^{d-1})$, for every integer $d \geq 1$,*

$$L_d = \alpha_{d+1} + \sum_{n\geq 1} \frac{\rho_n}{n^{d+2}}. \tag{8.311}$$

The sum is ordinary and absolutely convergent. In particular $L_1 = -\gamma_1$; L_2, L_3, L_4 are η, θ, χ in the fifth-power formula.

Proof. Let $\ell(t) = -\log(1-e^{-t})$. It satisfies $\ell(\ell(t)) = t$ and $K_d(t) = -\ell(t)^{d-1}\ell'(t)/(d-1)!$. Substitution into the height-one Mellin integral gives, initially for $\Re u > 0$,

$$\Gamma(1+u)Z(1+u, 1^{d-1}) = \frac{1}{(d-1)!}\int_0^\infty y^{d-1}\ell(y)^u \, \mathrm{d}y.$$

Subtract the elementary integral of $y^{d-1}e^{-uy}/(d-1)!$, which is u^{-d}. The difference is holomorphic near zero: at infinity its integrand is $O(e^{-(1+\Re u)y}y^{d-1})$ locally uniformly in u, and at zero fixed powers of $\log \ell(y)$ are integrable against y^{d-1}. Consequently its derivative at zero is

$$\frac{1}{(d-1)!}\int_0^\infty y^{d-1}\{\log \ell(y) + y\} \, \mathrm{d}y.$$

For $0 < z < 1$, ordinary binomial integration gives

$$\int_0^1 (1-z)^{-r} \, \mathrm{d}r = -\frac{z}{(1-z)\log(1-z)} = 1 + \sum_{n\geq 1}\frac{\rho_n}{n}z^n.$$

Differentiating $\log \ell(y) + y$ and using its zero limit at infinity therefore gives

$$\log \ell(y) + y = \sum_{n\geq 1}\frac{\rho_n}{n^2}e^{-ny}.$$

All terms are positive; Tonelli and the elementary Gamma integral yield $\sum \rho_n/n^{d+2}$. Multiplication by $\Gamma(1+u)^{-1}$ adds α_{d+1}, since the subtracted holomorphic remainder vanishes at zero. This proves the claimed coefficient. Finally $\rho_n \leq n$ follows from $(r)_n \leq n!$ on $0 \leq r \leq 1$, proving convergence. □

This representation is established literature in Cauchy-number notation; the direct involution proof here makes its coefficient and sign explicit. Appendix A.5.2 checks the published description of that sign.

8.38.4 The mixed derivative and its polylogarithmic representations

The stuffle identity in a right half-plane gives

$$Z(1+u,3)=\zeta(1+u)\zeta_3-Z(3,1+u)-\zeta(4+u).$$

Consequently

$$D_{1,3}=-\gamma_1\zeta_3-\partial_2 Z(3,1)-\zeta'(4). \tag{8.312}$$

The inner-order derivative has an absolutely convergent defining series. In particular this transformation replaces a singular outer-order coefficient by a convergent inner-order derivative; it does not remove the nonsymmetric spectral information.

Define the Tornheim function

$$T(a,b,c)=\sum_{m,n\geq 1}\frac{1}{m^a n^b(m+n)^c}$$

in its absolute-convergence domain. A finite partial fraction calculation gives, for u near zero,

$$T(2,2,u)=2Z(2+u,2)+4Z(3+u,1). \tag{8.313}$$

Indeed, for $l=m+n$, summation over $1\leq m<l$ uses

$$\frac{1}{m^2(l-m)^2}=\frac{1}{l^2}\left(\frac{1}{m^2}+\frac{1}{(l-m)^2}+\frac{2}{m(l-m)}\right)$$

and gives $2H^{(2)}_{l-1}/l^2+4H_{l-1}/l^3$. Thus

$$\Omega_4=2D_{1,3}+\tfrac{1}{2}\partial_3 T(2,2,0) \tag{8.314}$$

$$=2(\partial_1-\partial_2)Z(3,1)+\partial_1 Z(2,2)-2\gamma_1\zeta_3-2\zeta'(4). \tag{8.315}$$

Spectral derivatives of Tornheim sums have substantial prior literature, including [32]. These equalities are useful coordinate changes; no arithmetic independence is asserted for the displayed coordinates.

There are also completely convergent one-dimensional representations:

$$D_{1,3}=-\gamma_1\zeta_3+\sum_{n\geq 1}\frac{\zeta(3,n)\log n}{n}, \tag{8.316}$$

$$\partial_3 T(2,2,0)=\int_0^\infty\frac{\mathrm{Li}_2(e^{-t})^2-e^{-t}\zeta_2^2}{t}\,\mathrm{d}t. \tag{8.317}$$

For (8.316), use $H^{(3)}_{n-1}=\zeta_3-\zeta(3,n)$ and extract the linear coefficient. For (8.317), insert the Frullani identity $-\log(m+n)=\int_0^\infty(e^{-(m+n)t}-e^{-t})\,\mathrm{d}t/t$ and sum. Absolute integrability follows either from $\sum_{m,n\geq 1}\log(m+n)/(m^2n^2)<\infty$, or directly from the $O(\log t)$ integrand at zero and exponential decay at infinity. Equations (8.314), (8.316), and (8.317) connect the first mixed correction in a digamma power to a dilogarithm integral and a Hurwitz-zeta tail sum.

8.38.5 What scalar polynomial cancellation proves

Corollary 8.118 (A normalized fifth-degree cancellation)**.** *Use the fixed-coordinate finite parts* $Q_k = \mathrm{FP}_x \int_0^1 \psi(x)^k \,\mathrm{d}x$*, and put*

$$\Omega_4 = 2D_{1,3} + D_{2,2} + 2D_{3,1}, \qquad D_{p,q} = [s]\zeta(p+s,q).$$

Here $[s]$ *denotes the coefficient of* s *in the Laurent expansion, including when a pole is present. Define*

$$\begin{aligned} P_5(T) = T^5 + 5\gamma T^4 + 10\big(\gamma^2 + \zeta(2)\big)T^3 \\ + 10\big(\gamma^3 + 3\gamma\zeta(2) + \zeta(3)\big)T^2. \end{aligned} \tag{8.318}$$

Then

$$\begin{aligned} \mathrm{FP}_x \int_0^1 P_5\big(\psi(x)\big)\,\mathrm{d}x = & -120\chi + 10\Omega_4 + 10\zeta(2)\zeta'(2) \\ & + \frac{260}{3}\zeta(4) - 60\gamma\zeta(4) - 20\gamma^3\zeta(2) - 20\zeta(2)\zeta(3). \end{aligned} \tag{8.319}$$

Among monic polynomials $T^5 + a_4T^4 + a_3T^3 + a_2T^2$ *with coefficients in* $\mathbb{Q}[\gamma, \zeta(2), \zeta(3)]$*, the displayed polynomial is the unique one whose coefficients of* θ, η, γ_1 *vanish in the stated formulas for* Q_2, Q_3, Q_4, Q_5*, when these symbols are treated as formal coordinates.*

Proof. The finite-part operation in one fixed coordinate is linear. In $Q_5 + a_4Q_4 + a_3Q_3 + a_2Q_2$, cancellation of the displayed θ, η, γ_1 coefficients gives, successively,

$$\begin{aligned} 120\gamma - 24a_4 &= 0, \\ 60\big(\zeta(2) - \gamma^2\big) + 24\gamma a_4 - 6a_3 &= 0, \\ -20\big(\gamma^3 - 3\gamma\zeta(2) + \zeta(3)\big) + 12\big(\gamma^2 - \zeta(2)\big)a_4 - 6\gamma a_3 + 2a_2 &= 0. \end{aligned}$$

This triangular system has exactly the coefficients in (8.318). The $\zeta'(3)$ coefficient simultaneously becomes $-20\gamma + 4a_4 = 0$, while the coefficients of χ, Ω_4, and $\zeta(2)\zeta'(2)$ remain respectively $-120, 10, 10$. Before using any even-zeta reduction, the remaining scalar is

$$\frac{35}{3}\zeta(4) + 90\gamma\zeta(4) + 30\zeta(2)^2 - 20\gamma^3\zeta(2) - 60\gamma\zeta(2)^2 - 20\zeta(2)\zeta(3).$$

Substitute $\zeta(2)^2 = \frac{5}{2}\zeta(4)$ to obtain (8.319). □

The uniqueness assertion is deliberately relative to the displayed coefficient presentation and to the normalization excluding constant and linear terms. It does not assert arithmetic independence of $\chi, \Omega_4, \theta, \eta, \gamma_1$, or the impossibility of rewriting Ω_4 in another family of constants. In particular, "cancellation of γ_1" means cancellation of its explicit coefficient while Ω_4 is retained as a coordinate; a different representation of Ω_4 may contain γ_1. Further arithmetic reduction of the mixed-depth term remains a separate question.

8.38.6 Reconciling the regular and singular mixed coordinates

The other two fifth-moment routes [131] package the mixed term as a convergent derivative. Define

$$R(s) = Z(s,3) - \zeta(3)\zeta(s), \qquad J(s) = 2R(s) + 2Z(s+2,1) + Z(s+1,2) - 2\gamma\zeta(s+2) + \zeta(2)\zeta(s+1).$$

Here R is holomorphic for $\Re s > -1$ and J for $\Re s > 0$. Indeed $R(s) = -\sum_{n\ge1} \zeta(3,n)n^{-s}$, whose tail is $O(n^{-\Re s-2})$; the other defining series have the stated normal convergence. Thus $J'(1)$ is an ordinary derivative, and

$$J'(1) = \Omega_4 + 2\zeta(3)\gamma_1 - 2\gamma\zeta'(3) + \zeta(2)\zeta'(2). \tag{8.320}$$

To prove this, first extract $[u]$ in $R(1+u) = Z(1+u,3) - \zeta(3)\zeta(1+u)$; the result is $R'(1) = D_{1,3} + \zeta(3)\gamma_1$. The remaining terms are ordinary derivatives, giving the display. This also proves exact agreement of the three fifth-power formulas. Dropping the $\zeta(3)\gamma_1$ term would break that agreement.

The polynomial P_5 just displayed cancels explicit lower coordinates in the Ω_4 presentation. In the $J'(1)$ presentation, the corresponding nonlinear part is $\widetilde{P}_5(T) = P_5(T) + 10\zeta(3)T^2$. Linearity and $Q_2 = 2\gamma_1 - 2\zeta(2)$ give the exact conversion

$$\operatorname{FP}_x \int_0^1 \big(\widetilde{P}_5 - P_5\big)(\psi(x))\,\mathrm{d}x = 20\zeta(3)\gamma_1 - 20\zeta(2)\zeta(3).$$

The different coefficients therefore reflect different retained coordinates and normalization choices. Neither formal cancellation proves that the remaining mixed period has no alternative arithmetic reduction.

8.39 Normalized primitives and polygamma identities

The same normalization gives a primitive, not only a finite moment. On $\mathbb{C}\setminus(-\infty,0]$ use the principal logarithm.

Theorem 8.119 (An all-power primitive with fixed normalization)**.** *For $k \ge 1$, put*

$$\begin{aligned}\mathcal{A}_k(z) = \kappa_k z + p_{k,1}(0)\operatorname{Log} z + \sum_{j=2}^{k} \frac{p_{k,j}(0)z^{1-j}}{1-j} \\ + \sum_{n\ge1}\Bigg\{ p_{k,1}(n)\log(1+z/n) - zP_{k,n}(0) \\ + \sum_{j=2}^{k} p_{k,j}(n)\frac{(z+n)^{1-j} - n^{1-j}}{1-j}\Bigg\}.\end{aligned} \tag{8.321}$$

The series converges normally on compact subsets of that slit plane, and

$$\mathcal{A}_k'(z) = \psi(z)^k, \qquad \mathcal{A}_k(1) = Q_k. \tag{8.322}$$

Its local Laurent–log expansion at zero has constant coefficient zero, which fixes the otherwise arbitrary additive constant.

Proof. Each brace in (8.321) is the integral from zero to z of $P_{k,n}(w) - P_{k,n}(0)$. For large n and z in a fixed compact set its magnitude is $O_k((1+\log n)^{k-1}/n^2)$, as follows by cancellation of its linear terms or from the bound in the proof of Theorem 8.107. The normally convergent series may therefore be differentiated, and (8.270) gives the first identity. Every term in the braces vanishes at zero and the analytic part begins with $\kappa_k z$. The Laurent and logarithmic terms outside the braces also have no constant coefficient. The definition of the original-coordinate finite part now gives $\mathcal{A}_k(1) = Q_k$. □

For $k = 1$, this is the normalized branch of $\log\Gamma(z)$, whose local expansion begins $-\operatorname{Log} z - \gamma z + \cdots$. For every higher power it gives an explicitly normalized antiderivative through harmonic pole coefficients. Ordinary integrals along paths contained in the slit plane follow by endpoint subtraction of $\mathcal{A}_k$. On other pole-free intervals the same conclusion holds after choosing a consistent analytic continuation.

Corollary 8.120 (Differentiated polygamma identities)**.** *For $m \geq 1$ and z away from the nonpositive integers,*

$$\frac{\mathrm{d}^m}{\mathrm{d}z^m}\psi(z)^k = (-1)^m \sum_{n\geq 0}\sum_{j=1}^{k} \frac{(j)_m p_{k,j}(n)}{(z+n)^{j+m}}. \tag{8.323}$$

The series is normally convergent. Its left side is explicitly

$$\sum_{m_1+\cdots+m_k=m} \frac{m!}{m_1!\cdots m_k!}\prod_{i=1}^{k}\psi^{(m_i)}(z), \tag{8.324}$$

where $\psi^{(0)} = \psi$ and $(j)_m$ is the rising factorial.

Proof. Differentiate (8.270) normally on compact subsets. The constant subtractions disappear after the first derivative and each pole gives the displayed rising factorial. The denominator exponent is at least two, so (8.269) proves normal convergence directly as well. The multinomial identity is the ordinary product rule. □

For example, with $b_n = H_n - \gamma$ the case $k = 2, m = 1$ gives

$$\psi(z)\psi'(z) = -\sum_{n\geq 0}\left\{\frac{1}{(z+n)^3} - \frac{b_n}{(z+n)^2}\right\}.$$

This agrees with the classical polygamma pole series at $k = 1$ and extends it to polynomial products without an undetermined analytic term. The full formulas apply at arbitrary complex z in their specified domains, not only at integer arguments.

The all-power primitive has zero local Laurent–log constant at zero. For a primitive anchored at one, use $\widehat{\mathcal{A}}_k(z) = \mathcal{A}_k(z) - Q_k$; then $\widehat{\mathcal{A}}_k(1) = 0$ and its derivative is unchanged. The numerical value of Q_5 is approximately -8.6903039626245382663, confirmed by separately replayed arithmetic and local-subtraction quadrature routes, and by independent native Wolfram quadrature. These diagnostics carry no interval error certificate. The analytic proofs establish the identities; finite algebra and numerical checks test their implementation.

8.40 Differential polynomials and the disappearance of subtractions

The all-power expansion also supplies a complete calculus for a fixed polynomial in digamma and its positive derivatives. Three further continuations [136] develop this direction. We reconcile here their common analytic core: derivatives remove the frozen residue, so the derivative-containing part has an ordinary, unsubtracted global pole expansion. The integration-by-parts quotient, sixth/seventh coordinate reductions and unequal-tail rank claims remain separate questions; they are not consequences of this convergence theorem alone.

Let $P \in \mathbb{C}[u_0, u_1, \ldots, u_R]$ be a fixed polynomial, and put

$$f_P(z) = P\big(\psi(z), \psi'(z), \ldots, \psi^{(R)}(z)\big), \qquad \phi_P(T) = P(T, 0, \ldots, 0).$$

Write $\mathcal{J} = (u_1, \dots, u_R)$. Thus $P \in \mathcal{J}$ means that every monomial contains at least one positive derivative. The coefficients of P are fixed constants, with no dependence on z. Choose a common pole-order bound M, and extract

$$p_j(n) = [\varepsilon^{-j}] f_P(-n+\varepsilon), \quad r(n) = p_1(n), \quad P_n(z) = \sum_{j=1}^{M} \frac{p_j(n)}{(z+n)^j}.$$

Zero coefficients are understood. The pole data come from the convergent series (8.266), differentiated a finite number of times and multiplied. They are finite harmonic polynomials. For example a monomial $\prod_{a=0}^{R} u_a^{m_a}$ has pole order at most $\sum_a (a+1)m_a$.

Theorem 8.121 (Unsubtracted expansion for the derivative ideal)**.** *For $P \in \mathcal{J}$, the identity*

$$\boxed{f_P(z) = \sum_{n \geq 0} P_n(z)} \tag{8.325}$$

holds with absolute normal convergence away from the nonpositive integers. For some fixed $A \geq 0$, as $n \to \infty$,

$$p_j(n) = O\big((1+\log n)^A\big), \qquad \Delta p_j(n) = O\big(n^{-1}(1+\log n)^A\big), \tag{8.326}$$

$$r(n) = O\big(n^{-1}(1+\log n)^A\big), \qquad \Delta r(n) = O\big(n^{-2}(1+\log n)^A\big). \tag{8.327}$$

Here $\Delta a(n) = a(n) - a(n-1)$.

Proof. Put $K(v) = -\pi \cot(\pi v)$. Its differential equation is $K' = K^2 + \pi^2$. Each positive derivative $K^{(a)}$ is therefore a polynomial in K divisible by $K^2 + \pi^2$: differentiation preserves that divisibility. For any constant L, the polynomial $P(L + K, K', \dots)$ is divisible by $K^2 + \pi^2$. Divide by this factor and integrate the resulting polynomial in K. There is a polynomial T_L with

$$P(L + K(v), K'(v), \dots) = \frac{d}{dv} T_L(K(v)).$$

A meromorphic derivative has zero residue. This is the exact frozen residue cancellation; it uses the derivative in each monomial.

Fix $0 < \rho < 1/2$. Reflection at $-n$ gives

$$\psi(-n+v) = K(v) + \log n + \eta_n(v), \qquad \eta_t(v) = \psi(t+1-v) - \log t \quad (t \geq 1).$$

Uniformly for $|v| \leq \rho$, every fixed finite set of derivatives satisfies

$$\partial_v^a \eta_t(v) = O(t^{-1}), \qquad \partial_t \partial_v^a \eta_t(v) = O(t^{-2}).$$

For $a = 0$ this includes the subtraction of $\log t$ and its derivative; for $a > 0$ it follows directly from the positive-axis polygamma expansions, which may also be obtained by differentiating the normally convergent pole series. Cauchy coefficient extraction on $|v| = \rho$ now bounds $p_j(t)$ and its t derivative by the orders in (8.326). For the residue, first subtract the frozen polynomial with $L = \log t$. Its residue is identically zero, while the difference and its t derivative have bounds $O(t^{-1}(1+\log t)^A)$ and $O(t^{-2}(1+\log t)^A)$. Integrating the t derivative over $[n-1, n]$ proves (8.327); finitely many small indices do not affect these estimates. No derivative of an unspecified remainder is used.

These bounds give absolute normal convergence of the proposed pole series: the simple pole has the extra factor n^{-1}, and every higher pole already has exponent at least two. To determine

the entire remainder, use $|w| = R_N = N + 1/2$. Pole avoidance, reflection and the same sectorial expansions give uniformly on this circle

$$|\psi(w)| \le C(1 + \log R_N), \qquad |\psi^{(a)}(w)| \le C_a\big(R_N^{-1} + e^{-c|\operatorname{Im} w|}\big) \quad (a \ge 1).$$

The cotangent derivatives decay exponentially for large imaginary part and remain bounded for small imaginary part because the circle stays at least $1/2$ from each integer. The right half-plane can be estimated directly. Thus, since every monomial has a positive derivative,

$$|f_P(w)| \le C(1 + \log R_N)^A\big(R_N^{-1} + e^{-c|\operatorname{Im} w|}\big).$$

The angular integral of $e^{-cR_N|\sin\theta|}$ is $O(R_N^{-1})$. Consequently, for fixed z,

$$\oint_{|w|=R_N} \frac{f_P(w)}{w - z}\, dw = O\big((1 + \log R_N)^A / R_N\big) \longrightarrow 0.$$

The residue at z is $f_P(z)$, and the one at $-n$ is $-P_n(z)$. The residue theorem proves (8.325), including its normalization. □

Theorem 8.122 (Ordinary finite-part coordinates for the derivative ideal)**.** *For $P \in \mathcal{J}$, set*

$$J_P = \sum_{n\ge1} \sum_{j=2}^{M} \frac{\Delta p_j(n)}{(j-1)n^{j-1}}, \qquad D_P(s) = \sum_{n\ge1} \frac{\Delta r(n)}{n^s}.$$

Then D_P is holomorphic for $\operatorname{Re} s > -1$, and the original-coordinate finite part is

$$\boxed{\operatorname{FP}_x \int_0^1 f_P(x)\, dx = J_P + D_P'(0) = J_P - \sum_{n\ge1} \Delta r(n) \log n.} \tag{8.328}$$

Every displayed sum is ordinary and absolutely convergent.

Proof. After removing the finitely many pole terms at zero, (8.325) converges uniformly on $[0, 1]$, so it may be integrated there. The simple pole at zero has finite part zero. A higher pole at zero contributes $-p_j(0)/(j-1)$. A pole at $-n$, $n \ge 1$, contributes $r(n)\log(1 + 1/n)$ and $p_j(n)(n^{1-j} - (n+1)^{1-j})/(j-1)$. Use the finite summation identities (8.285) and (8.286). The initial higher-pole terms cancel the finite parts at zero. All boundary terms vanish: $p_j(N)(N+1)^{1-j} \to 0$ for $j \ge 2$, and $r(N)\log(N+1) \to 0$ by (8.327). This gives (8.328). The same bound on Δr proves normal convergence of D_P and all its derivatives on compact subsets of $\operatorname{Re} s > -1$. It justifies the displayed ordinary derivative at zero. □

Corollary 8.123 (An arbitrary fixed differential polynomial)**.** *Write $P = \phi_P(u_0) + P_{\mathrm{mix}}$ with $P_{\mathrm{mix}} \in \mathcal{J}$. The finite part of f_P is the linear combination of the all-power formulas for ϕ_P and (8.328) for P_{mix}. In particular, the large-circle boundary polynomial and the frozen residue polynomial depend only on ϕ_P. If $\phi_P(T) = \sum a_k T^k$, they are respectively*

$$C_P(L) = \sum_k \frac{a_k}{2\pi} \int_{-\pi}^{\pi} (L + i\theta)^k\, d\theta, \qquad R_P(L) = -\frac{\phi_P(L + i\pi) - \phi_P(L - i\pi)}{2i\pi}.$$

The normalized global formula is

$$f_P(z) = P_0(z) + \kappa_P + \sum_{n\ge1} \big(P_n(z) - P_n(0)\big), \qquad \kappa_P = [\varepsilon^0] f_P(\varepsilon).$$

Proof. Apply Theorems 8.107, 8.109 and 8.110 to each scalar monomial, and the two preceding theorems to P_{mix}. The scalar constant term contributes its ordinary integral. For the mixed part, (8.325) implies $\kappa_{P_{\mathrm{mix}}} = \sum_{n\ge 1}(P_{\mathrm{mix}})_n(0)$ by taking the regular constant at zero; this sum converges by (8.326)–(8.327). Hence its unsubtracted expansion is also precisely the displayed normalized expansion. Linearity completes every assertion. □

Corollary 8.124 (Anchored primitives for derivative-containing polynomials)**.** *On $\mathbb{C}\setminus(-\infty,0]$, use the principal logarithm. For z, z_0 in this domain,*

$$F_P(z;z_0) = \sum_{n\ge 0} r(n)\big(\mathrm{Log}(z+n) - \mathrm{Log}(z_0+n)\big) + \sum_{n\ge 0}\sum_{j=2}^{M} \frac{p_j(n)}{j-1}\big((z_0+n)^{1-j} - (z+n)^{1-j}\big)$$

converges normally, has $F_P(z_0;z_0) = 0$ and $\partial_z F_P(z;z_0) = f_P(z)$. For complex anchors retain the difference of logarithms; a principal logarithm of the ratio need not agree.

Proof. On a compact set, the logarithm difference is $O(n^{-1})$ and the inverse-power difference in the jth term is $O(n^{-j})$. The coefficient bounds imply normal convergence. Differentiation and (8.325) prove the derivative and the anchor is immediate. □

As a normalization check, the derivative-containing polynomial $P = u_0u_1$ gives

$$\mathrm{FP}_x \int_0^1 \psi(x)\psi'(x)\,dx = \zeta(2).$$

Indeed its primitive is $\psi^2/2$. Its value at one is $\gamma^2/2$, whereas its local constant at zero is $(\gamma^2 - 2\zeta(2))/2$. More general total derivatives retain exactly this endpoint-minus-local- constant rule. Formula (8.328) applies as well to mixed polynomials which are not total derivatives. It removes singular spectral Laurent coordinates without asserting that their ordinary convergent replacements reduce arithmetically to a smaller constant field.

For the nonlinear example $P = u_1^2$, write the local series as $\psi'(-n+v) = v^{-2} + c_{2,n} + 2c_{3,n}v + \cdots$. The square has $p_4(n) = 1$, $p_3(n) = 0$, $p_2(n) = 2c_{2,n}$ and $r(n) = 4c_{3,n}$. Hence $\Delta p_2(n) = 2n^{-2}$ and $\Delta r(n) = 4n^{-3}$. Formula (8.328) gives the explicit ordinary identity

$$\mathrm{FP}_x \int_0^1 \psi'(x)^2\,dx = 2\zeta(3) + 4\zeta'(3).$$

The logarithmic sum is absolutely convergent. This illustrates why a positive derivative removes the singular spectral germ while retaining an ordinary zeta derivative. Independent local-subtraction quadrature checks this value and the two total-derivative examples; it is a numerical diagnostic, with no interval error claim.

8.41 Canonical integration by parts and Gamma parity

The pole calculus determines a finite part from a differential polynomial; it does not yet identify a preferred polynomial representative. The three continuations in [136] also give compatible answers to that formal question. We retain one triangular basis, its explicit endpoint functional, and the inverse-Gamma parity argument. This is a quotient of vector spaces before evaluation in special values.

Let $\mathcal{A} = \mathbb{Q}[u_0, u_1, \ldots]$, with each polynomial using finitely many variables, and set

$$\mathcal{D} = \sum_{r \geq 0} u_{r+1} \frac{\partial}{\partial u_r}, \qquad \operatorname{wt}(u_r) = r + 1.$$

The factor degree of a monomial is its number of factors, counted with multiplicity. The derivation increases weight by one and preserves factor degree. Under $u_r = \psi^{(r)}(x)$ it becomes d/dx. Write $\mathcal{A}_w$ for weight w and $p(w)$ for the partition number, so $\dim \mathcal{A}_w = p(w)$. For $w \geq 2$, let $\mathcal{B}_w$ be the monomials whose largest variable index occurs at least twice; this includes u_0^w. At weights zero and one use $\mathcal{B}_0 = \{1\}$ and $\mathcal{B}_1 = \{u_0\}$ separately.

Theorem 8.125 (A unique formal normal form at every weight)**.** *For $w \geq 2$,*

$$\mathcal{A}_w = \mathcal{D}\mathcal{A}_{w-1} \oplus \operatorname{span}_{\mathbb{Q}} \mathcal{B}_w. \tag{8.329}$$

Thus every $P \in \mathcal{A}_w$ has a unique decomposition $P = \mathcal{D}Q + N$, with $Q \in \mathcal{A}_{w-1}$ and N in the displayed normal space. A terminating rational algorithm computes both terms. If $p_\ell(w)$ counts partitions of w with exactly ℓ parts, the dimension at factor degree ℓ is $p_\ell(w) - p_\ell(w-1)$. In particular

$$|\mathcal{B}_w| = p(w) - p(w-1), \qquad \sum_{w \geq 0} \dim(\mathcal{A}_w / \mathcal{D}\mathcal{A}_{w-1}) t^w = t + \prod_{j \geq 2} (1 - t^j)^{-1}.$$

Every retained positive derivative has order at most $\lfloor w/2 \rfloor - 1$.

Proof. A monomial outside the normal space has a unique largest index $r \geq 1$. Write it as $u_r u_{r-1}^a V$, where V uses indices at most $r - 2$. The product rule gives

$$u_r u_{r-1}^a V = \mathcal{D}\left(\frac{u_{r-1}^{a+1} V}{a+1} \right) - \frac{u_{r-1}^{a+1} \mathcal{D}V}{a+1}. \tag{8.330}$$

The remainder has strictly smaller largest index, and factor degree is unchanged. Repeated application terminates in the finite set of partitions of w, proving spanning and the algorithm.

For directness, a nonconstant Q with largest index m has in $\mathcal{D}Q$ the nonzero terms $u_{m+1} \partial Q / \partial u_m$. These have largest index $m+1$ exactly once; the other terms have smaller largest index. No element of the normal space has this support. Characteristic zero gives $\partial Q / \partial u_m \neq 0$, so the intersection is zero and $\mathcal{D}$ is injective on positive weights. Equivalently, increasing one largest part of a partition of $w - 1$ gives a distinct leading partition with unique largest part at weight w. Decreasing that unique part is its inverse. The bijection preserves the number of parts, proving both dimension formulas. The constant kernel at weight zero supplies the exceptional t in the generating series. A repeated largest part has size at most $w/2$, giving the derivative order bound. □

For instance the bases at weights four through seven are

$$\begin{array}{c|l} 4 & u_0^4,\ u_1^2 \\ 5 & u_0^5,\ u_0 u_1^2 \\ 6 & u_0^6,\ u_0^2 u_1^2,\ u_1^3,\ u_2^2 \\ 7 & u_0^7,\ u_0^3 u_1^2,\ u_0 u_1^3,\ u_0 u_2^2. \end{array}$$

The image $\mathcal{D}\mathcal{A}$ is not generally an ideal. For example $u_1 = \mathcal{D}u_0$, whereas u_1^2 is a nonzero normal representative at weight four. Multiplication of quotient classes is therefore not being defined. The dimensions count formal representatives, without asserting independence after integration.

8.41.1 The endpoint functional is retained

For a differential polynomial Q, write f_Q for its digamma substitution as in Section 8.40, and define

$$\mathcal{E}(Q) = f_Q(1) - [x^0]f_Q(x). \tag{8.331}$$

The coefficient includes products between pole and regular terms. It is not the product of the individual constant coefficients.

Corollary 8.126 (Exact finite-part integration by parts)**.** *For $P = \mathcal{D}Q + N$ from Theorem 8.125,*

$$\mathrm{FP}_x \int_0^1 f_P(x)\,dx = \mathcal{E}(Q) + \mathrm{FP}_x \int_0^1 f_N(x)\,dx. \tag{8.332}$$

For $\operatorname{Re} a > 0$, the corresponding ordinary shifted identity retains both endpoints:

$$\int_0^1 f_P(a+x)\,dx = f_Q(a+1) - f_Q(a) + \int_0^1 f_N(a+x)\,dx.$$

Proof. Integrate $f_{\mathcal{D}Q} = f_Q'$ over $[\varepsilon, 1]$ before taking the constant term. The answer is $f_Q(1) - f_Q(\varepsilon)$, whose constant is exactly (8.331). Linearity of the fixed-coordinate finite part proves (8.332). With the stated shift the interval avoids all poles, and the fundamental theorem of calculus gives the ordinary identity. □

The functional is explicitly finite. For homogeneous Q of weight $W > 0$, substitute the differentiated truncation

$$-x^{-1} - \gamma + \sum_{j=2}^{W} (-1)^j \zeta(j) x^{j-1}$$

to compute $[x^0]f_Q(x)$. An omitted digamma term starts at x^W; replacing a leading pole by it in a weight-W product raises the total exponent by at least $W+1$, so it cannot contribute a constant. The other endpoint uses $\psi(1) = -\gamma$ and $\psi^{(r)}(1) = (-1)^{r+1} r!\zeta(r+1)$ for $r \geq 1$. Consequently $\mathcal{E}(Q)$ lies in $\mathbb{Q}[\gamma, \zeta(2), \ldots, \zeta(W)]$.

Three useful exact examples are

$$\mathcal{E}(u_0^2) = 2\zeta(2), \qquad \mathcal{E}(u_0 u_1) = -\zeta(3), \qquad \mathcal{E}(u_1 u_2) = 4\zeta(5).$$

Thus dropping a total derivative would lose a nonzero endpoint. Together with the trigamma-square calculation in Section 8.40, the middle identity gives

$$\mathrm{FP}_x \int_0^1 \psi(x)\psi''(x)\,dx = -3\zeta(3) - 4\zeta'(3). \tag{8.333}$$

Also $[x^0]\psi'(x)^2 = \zeta(2)^2 + 6\zeta(4)$, so integrating $(\psi'^2)'/2$ gives

$$\mathrm{FP}_x \int_0^1 \psi'(x)\psi''(x)\,dx = -3\zeta(4). \tag{8.334}$$

The identities are proved by finite pole algebra and endpoint extraction; independent native quadrature is a separate diagnostic.

8.41.2 Inverse-Gamma normal forms have factor parity

Define $A_0 = 1$ and $A_{m+1} = u_0 A_m - \mathcal{D}A_m$. After substitution these are the differential coefficients of the Gamma ratio:

$$A_m(x) = (-1)^m \Gamma(x) \frac{d^m}{dx^m} \Gamma(x)^{-1}, \qquad \frac{\Gamma(x)}{\Gamma(x-t)} = \sum_{m \geq 0} A_m(x) \frac{t^m}{m!}.$$

Let $A_m = N_m + \mathcal{D}T_m$ be the unique normal decomposition, with $N_0 = 1$, $N_1 = u_0$, $T_0 = T_1 = 0$.

Theorem 8.127 (All-degree parity and a recurrence retaining the primitive)**.** *Every monomial in N_m has factor degree congruent to m modulo two. For $m \geq 1$, decompose*

$$u_0 N_m - u_1 T_m = R_m + \mathcal{D}S_m$$

by Theorem 8.125. Then

$$N_{m+1} = R_m, \qquad T_{m+1} = S_m + u_0 T_m - N_m - \mathcal{D}T_m. \tag{8.335}$$

All operations are finite and rational.

Proof. Introduce $p = e^F$, $q = e^{-F}$ with $F' = u_0$ only as a convenient notation: the products below are differential polynomials because the exponential factors cancel. Set

$$L_m = \sum_{j=0}^{m-1} (-1)^j p^{(j)} q^{(m-1-j)}.$$

The product rule telescopes to

$$\mathcal{D}L_m = pq^{(m)} + (-1)^{m-1} p^{(m)} q = (-1)^m A_m(u) - A_m(-u).$$

This is also a purely formal identity: replace $p^{(j)}/p$ by $(-1)^j A_j(-u)$ and $q^{(j)}/q$ by $(-1)^j A_j(u)$. Projection onto the normal space annihilates the derivative and preserves factor degree. Therefore $N_m(-u) = (-1)^m N_m(u)$, proving parity. For the recurrence, substitute $A_m = N_m + \mathcal{D}T_m$ and use $u_0 \mathcal{D}T_m = \mathcal{D}(u_0 T_m) - u_1 T_m$. The result is

$$A_{m+1} = u_0 N_m - u_1 T_m + \mathcal{D}(u_0 T_m - N_m - \mathcal{D}T_m).$$

Uniqueness gives (8.335). □

The first informative remainders are

$$\begin{aligned} N_4 &= u_0^4 - u_1^2, \qquad N_5 = u_0^5 - 5u_0 u_1^2, \\ N_6 &= u_0^6 - 15u_0^2 u_1^2 + u_2^2, \\ N_7 &= u_0^7 - 35u_0^3 u_1^2 + 7u_0 u_2^2, \\ N_8 &= u_0^8 - 70u_0^4 u_1^2 + 28u_0^2 u_2^2 - 7u_1^4 - u_3^2. \end{aligned}$$

In particular u_1^3, although a valid weight-six normal coordinate, is excluded from N_6 by parity. Every identity $A_m = N_m + \mathcal{D}T_m$ yields an exact finite-part identity with $\mathcal{E}(T_m)$ retained. No vanishing endpoint or arithmetic independence is inferred from this formal cancellation.

Independent rational matrix certificates verify the two-sided basis inverses and all monomial decompositions through weight fourteen. Separate exponential Bell and Lagrange-concomitant constructions check the inverse-Gamma coefficients through degree twelve. They check the implementation and the printed examples; the preceding proofs establish the all-weight statements. The Gamma–Appell transport and explicit sixth/seventh spectral-coordinate reductions remain separate audit targets.

Appendix: fully expanded values at the third and quarter points

The reduction recipes of Theorems 8.39 and 8.41, carried out to the end, give the following flattened closed forms. The Γ_2 values live in the atom set announced in Theorem 8.39; the Γ_3 values pick up, exactly as Theorem 8.41 predicts, the three further atoms γ_3, π^4 and $\gamma_4(\frac{1}{q}) - \gamma_4(1-\frac{1}{q})$. The combinations $\gamma + \ln 6\pi$ and $\gamma + \ln 8\pi$ are the constants Λ_3, Λ_4 of Theorem 8.33. Each display has been verified at 90-digit working precision (residuals below 10^{-88}) against $\Gamma_n(p) = \frac{(-1)^{n+1}}{n+1}\big[\zeta^{(n+1)}(0,p) - \zeta^{(n+1)}(0)\big]$ evaluated from Hurwitz-zeta derivative jets.

$$
\begin{aligned}
\Gamma_2\big(\tfrac{1}{3}\big) = {}& \frac{1}{2}\,\gamma_2 + \gamma\,\gamma_1 + \gamma_1\,\ln 2\pi - \frac{1}{2}\,\gamma_1\,\ln 3 \\
& + \frac{4}{9}\,\gamma^3 + \frac{4}{3}\,\gamma^2\,\ln 2\pi - \frac{1}{6}\,\gamma^2\,\ln 3 + \frac{4}{3}\,\gamma\,\ln^2 2\pi \\
& + \frac{7}{6}\,\gamma\,\ln 2\pi\,\ln 3 - \frac{1}{6}\,\gamma\,\ln^2 3 - \frac{4}{9}\,\zeta(3) - \frac{\pi^2}{12}\,\ln 2\pi \\
& + \frac{\pi^2}{24}\,\ln 3 + \frac{4}{9}\,\ln^3 2\pi + \frac{4}{3}\,\ln^2 2\pi\,\ln 3 + \frac{7}{12}\,\ln 2\pi\,\ln^2 3 \\
& - \frac{1}{18}\,\ln^3 3 \\
& - \Big(\big(\gamma + \ln 6\pi\big)^2 - \frac{\pi^2}{12}\Big)\ln\Gamma\big(\tfrac{1}{3}\big) \\
& - \frac{\sqrt{3}}{6\pi}\,\big(\gamma + \ln 6\pi\big)\,\big(\gamma_2(\tfrac{1}{3}) - \gamma_2(\tfrac{2}{3})\big) \\
& - \frac{\sqrt{3}}{18\pi}\,\big(\gamma_3(\tfrac{1}{3}) - \gamma_3(\tfrac{2}{3})\big)
\end{aligned}
$$

$$
\begin{aligned}
\Gamma_2\big(\tfrac{1}{4}\big) = {}& \frac{1}{2}\,\gamma_2 + \gamma\,\gamma_1 + \frac{1}{2}\,\gamma_1\,\ln 2 + \gamma_1\,\ln\pi \\
& + \frac{1}{2}\,\gamma^3 + 2\,\gamma^2\,\ln 2 + \frac{3}{2}\,\gamma^2\,\ln\pi + 6\,\gamma\,\ln^2 2 \\
& + \frac{13}{2}\,\gamma\,\ln 2\,\ln\pi + \frac{3}{2}\,\gamma\,\ln^2\pi - \frac{1}{2}\,\zeta(3) - \frac{\pi^2}{24}\,\ln 2 \\
& - \frac{\pi^2}{12}\,\ln\pi + \frac{49}{6}\,\ln^3 2 + \frac{45}{4}\,\ln^2 2\,\ln\pi + \frac{9}{2}\,\ln 2\,\ln^2\pi \\
& + \frac{1}{2}\,\ln^3\pi \\
& - \Big(\big(\gamma + \ln 8\pi\big)^2 - \frac{\pi^2}{12}\Big)\ln\Gamma\big(\tfrac{1}{4}\big) \\
& - \frac{1}{4\pi}\,\big(\gamma + \ln 8\pi\big)\,\big(\gamma_2(\tfrac{1}{4}) - \gamma_2(\tfrac{3}{4})\big) \\
& - \frac{1}{12\pi}\,\big(\gamma_3(\tfrac{1}{4}) - \gamma_3(\tfrac{3}{4})\big)
\end{aligned}
$$

$$
\begin{aligned}
\Gamma_3\left(\tfrac{1}{3}\right) = &-\frac{1}{2}\,\gamma_3 - \frac{3}{2}\,\gamma\,\gamma_2 - \frac{3}{2}\,\gamma_2\,\ln 2\pi + \frac{3}{4}\,\gamma_2\,\ln 3 \\
&- \frac{3}{2}\,\gamma^2\,\gamma_1 - 3\,\gamma\,\gamma_1\,\ln 2\pi + \frac{3}{2}\,\gamma\,\gamma_1\,\ln 3 - \frac{3}{2}\,\gamma_1\,\ln^2 2\pi \\
&+ \frac{3}{2}\,\gamma_1\,\ln 2\pi\,\ln 3 + \frac{3}{4}\,\gamma_1\,\ln^2 3 - \frac{\pi^2}{8}\,\gamma_1 - \frac{1}{2}\,\gamma^4 \\
&- 2\,\gamma^3\,\ln 2\pi + \frac{1}{4}\,\gamma^3\,\ln 3 - 3\,\gamma^2\,\ln^2 2\pi - \frac{3}{2}\,\gamma^2\,\ln 2\pi\,\ln 3 \\
&+ \frac{3}{8}\,\gamma^2\,\ln^2 3 - \frac{\pi^2}{24}\,\gamma^2 - 2\,\gamma\,\ln^3 2\pi - \frac{15}{4}\,\gamma\,\ln^2 2\pi\,\ln 3 \\
&- \frac{3}{2}\,\gamma\,\ln 2\pi\,\ln^2 3 + \frac{1}{4}\,\gamma\,\ln^3 3 + \frac{\pi^2}{6}\,\gamma\,\ln 2\pi - \frac{\pi^2}{48}\,\gamma\,\ln 3 \\
&+ \frac{5\pi^2}{24}\,\ln^2 2\pi + \frac{\pi^2}{24}\,\ln 2\pi\,\ln 3 - \frac{7\pi^2}{96}\,\ln^2 3 + \frac{\pi^4}{120} \\
&- \frac{1}{2}\,\ln^4 2\pi - 2\,\ln^3 2\pi\,\ln 3 - \frac{15}{8}\,\ln^2 2\pi\,\ln^2 3 - \frac{1}{2}\,\ln 2\pi\,\ln^3 3 \\
&+ \frac{1}{16}\,\ln^4 3 \\
&+ \left(\left(\gamma + \ln 6\pi\right)^3 - \frac{\pi^2}{4}\left(\gamma + \ln 6\pi\right) + 2\,\zeta(3)\right)\ln\Gamma\left(\tfrac{1}{3}\right) \\
&+ \frac{\sqrt{3}}{48\pi}\left(12\left(\gamma + \ln 6\pi\right)^2 - \pi^2\right)\left(\gamma_2(\tfrac{1}{3}) - \gamma_2(\tfrac{2}{3})\right) \\
&+ \frac{\sqrt{3}}{6\pi}\left(\gamma + \ln 6\pi\right)\left(\gamma_3(\tfrac{1}{3}) - \gamma_3(\tfrac{2}{3})\right) \\
&+ \frac{\sqrt{3}}{24\pi}\left(\gamma_4(\tfrac{1}{3}) - \gamma_4(\tfrac{2}{3})\right)
\end{aligned}
$$

$$
\begin{aligned}
\Gamma_3\left(\tfrac{1}{4}\right) = &-\frac{1}{2}\,\gamma_3 - \frac{3}{2}\,\gamma\,\gamma_2 - \frac{3}{4}\,\gamma_2\,\ln 2 - \frac{3}{2}\,\gamma_2\,\ln\pi \\
&- \frac{3}{2}\,\gamma^2\,\gamma_1 - \frac{3}{2}\,\gamma\,\gamma_1\,\ln 2 - 3\,\gamma\,\gamma_1\,\ln\pi + \frac{9}{4}\,\gamma_1\,\ln^2 2 \\
&- \frac{3}{2}\,\gamma_1\,\ln 2\,\ln\pi - \frac{3}{2}\,\gamma_1\,\ln^2\pi - \frac{\pi^2}{8}\,\gamma_1 - \frac{9}{16}\,\gamma^4 \\
&- 3\,\gamma^3\,\ln 2 - \frac{9}{4}\,\gamma^3\,\ln\pi - \frac{45}{4}\,\gamma^2\,\ln^2 2 - \frac{51}{4}\,\gamma^2\,\ln 2\,\ln\pi \\
&- \frac{27}{8}\,\gamma^2\,\ln^2\pi - \frac{\pi^2}{32}\,\gamma^2 - 27\,\gamma\,\ln^3 2 - \frac{153}{4}\,\gamma\,\ln^2 2\,\ln\pi \\
&- \frac{33}{2}\,\gamma\,\ln 2\,\ln^2\pi - \frac{9}{4}\,\gamma\,\ln^3\pi + \frac{\pi^2}{4}\,\gamma\,\ln 2 + \frac{3\pi^2}{16}\,\gamma\,\ln\pi \\
&+ \frac{3\pi^2}{8}\,\ln^2 2 + \frac{11\pi^2}{16}\,\ln 2\,\ln\pi + \frac{7\pi^2}{32}\,\ln^2\pi + \frac{29\pi^4}{3840} \\
&- \frac{207}{8}\,\ln^4 2 - \frac{179}{4}\,\ln^3 2\,\ln\pi - 27\,\ln^2 2\,\ln^2\pi - \frac{27}{4}\,\ln 2\,\ln^3\pi \\
&- \frac{9}{16}\,\ln^4\pi \\
&+ \left(\left(\gamma + \ln 8\pi\right)^3 - \frac{\pi^2}{4}\left(\gamma + \ln 8\pi\right) + 2\,\zeta(3)\right)\ln\Gamma\left(\tfrac{1}{4}\right) \\
&+ \frac{1}{32\pi}\left(12\left(\gamma + \ln 8\pi\right)^2 - \pi^2\right)\left(\gamma_2(\tfrac{1}{4}) - \gamma_2(\tfrac{3}{4})\right) \\
&+ \frac{1}{4\pi}\left(\gamma + \ln 8\pi\right)\left(\gamma_3(\tfrac{1}{4}) - \gamma_3(\tfrac{3}{4})\right) \\
&+ \frac{1}{16\pi}\left(\gamma_4(\tfrac{1}{4}) - \gamma_4(\tfrac{3}{4})\right)
\end{aligned}
$$

Chapter 9

The parameter-derivative tower

Differentiating in the Hurwitz parameter moves the spectral argument upward. The resulting coefficients are elementary symmetric functions of reciprocal integers, so harmonic numbers control every rung. The functional equation then pairs this tower with the integrated tower of the preceding chapter. Explicit character tables show how experiments can be replaced by uniform derivations.

9.1 The uniform master identity

Theorem 9.1 (Harmonic-number master identity)**.** *For all $k \geq 1$ and $n \geq 0$,*

$$\gamma_n^{(k)}(a) = (-1)^{n+k}\, k!\, n! \sum_{i+j=n} e_i(k)\, \frac{\zeta^{(j)}(k{+}1, a)}{j!}, \tag{9.1}$$

where $e_i(k)$ is the i-th elementary symmetric function of $\{1, \frac{1}{2}, \ldots, \frac{1}{k}\}$; in particular $e_0 = 1$, $e_1 = H_k$, $e_2 = \frac{1}{2}(H_k^2 - H_k^{(2)})$. The first two rows are

$$\gamma_1^{(k)}(a) = (-1)^{k+1} k! \left[\zeta'(k{+}1, a) + H_k\, \zeta(k{+}1, a)\right], \tag{9.2}$$

$$\gamma_2^{(k)}(a) = (-1)^k k! \left[\zeta''(k{+}1, a) + 2H_k\, \zeta'(k{+}1, a) + (H_k^2 - H_k^{(2)})\, \zeta(k{+}1, a)\right]. \tag{9.3}$$

Proof. From $\partial_a \zeta(s, a) = -s\, \zeta(s{+}1, a)$, iteration gives $\partial_a^k \zeta(s, a) = (-1)^k (s)_k\, \zeta(s{+}k, a)$ with the rising factorial $(s)_k = s(s{+}1)\cdots(s{+}k{-}1)$. For $k \geq 1$ this annihilates the pole term in (8.40), so

$$\sum_{n \geq 0} \frac{(-1)^n}{n!}\, \gamma_n^{(k)}(a)\, (s-1)^n = (-1)^k (s)_k\, \zeta(s{+}k, a).$$

Set $s = 1 + \varepsilon$. Then $(s)_k = (1+\varepsilon)_k = \prod_{m=1}^{k}(m + \varepsilon) = k! \prod_{m=1}^{k}(1 + \varepsilon/m) = k! \sum_{i \geq 0} e_i(k)\, \varepsilon^i$, the generating identity for the elementary symmetric functions of $\{1/m\}$, while $\zeta(s{+}k, a) = \zeta(k{+}1{+}\varepsilon, a) = \sum_{j \geq 0} \zeta^{(j)}(k{+}1, a)\, \varepsilon^j / j!$ is analytic at $\varepsilon = 0$ because $k{+}1 \geq 2$. Comparing coefficients of ε^n yields (9.1); $n = 1, 2$ give (9.2)–(9.3). □

Theorem 9.1 is the harmonic-number repackaging of Coffey's all-orders Stirling-number ladder [25]: the change of basis from the unsigned Stirling numbers $\left|s(k{+}1, i{+}1)\right|$ to the symmetric functions $e_i(k)$ turns the coefficients into H_k, $H_k^2 - H_k^{(2)}$, ... The specializations $\gamma_1' = \zeta'(2, a) + \zeta(2, a)$, $\gamma_1'' = -2\zeta'(3, a) - 3\zeta(3, a)$, $\gamma_2' = -[\zeta''(2, a) + 2\zeta'(2, a)]$ and $\gamma_2'' = 2\zeta''(3, a) + 6\zeta'(3, a) + 2\zeta(3, a)$ are the $s = 2, 3$ layers used below. Identity (9.1) was verified against numerical k-th parameter-derivatives of γ_1, γ_2 for $k = 1, \ldots, 5$.

9.2 Differentiation and the contact-term correction

For $k \geq 1$, the ordinary function $\gamma_n^{(k)}(x)$ has a power-log singularity of order $k+1$. Its canonical Hadamard finite part on $\mathbb{T}$ is defined by

$$\langle F_{n,k}, \phi \rangle = \mathrm{CT}_{\epsilon \downarrow 0} \int_\epsilon^1 \gamma_n^{(k)}(x)\phi(x)\,\mathrm{d}x. \tag{9.4}$$

Here CT means the coefficient independent of ϵ after the finite expansion in negative powers of ϵ and nonconstant powers of $\log \epsilon$ is removed. The remaining error tends to zero. This convention uses the coordinate $x \in (0,1)$ and no finite counterterm. It agrees with Definition 8.59 when $k = 0$.

Lemma 9.2 (Higher Hurwitz poles as distributions)**.** *For each $k \geq 1$, as $z \to 0$,*

$$Z(1+k-z) = \frac{(-1)^k}{k!}\frac{\delta_0^{(k)}}{z} + \sum_{j \geq 0} \frac{(-1)^j z^j}{j!}\,\mathrm{Fp}_{\mathbb{T}}[\zeta^{[j]}(k+1,x)]. \tag{9.5}$$

The finite parts in this formula have the cutoff convention (9.4).

Proof. Split $\zeta(1+k-z,x) = x^{-k-1+z} + \zeta(1+k-z,1+x)$. Subtract the Taylor polynomial of ϕ at zero through degree k. Its remainder gives an analytic integral near $z=0$. The subtracted monomials give terms $\phi^{(r)}(0)/(r!(z+r-k))$, $0 \leq r \leq k$; only $r=k$ has a pole at zero. Since $\langle \delta_0^{(k)}, \phi \rangle = (-1)^k \phi^{(k)}(0)$, its residue is the one stated. Expanding each regular term and each analytic remainder is exactly the constant-term prescription for the cutoff integrals. This proves the entire Laurent expansion. □

Theorem 9.3 (All-index differentiation anomaly)**.** *For $n \geq 0$ and $k \geq 1$,*

$$D^k G_n = F_{n,k} + (-1)^{n+1} n! e_{n+1}(k)\delta_0^{(k)}. \tag{9.6}$$

In particular, there is no anomaly when $n \geq k$.

Proof. The meromorphic distribution family satisfies

$$D^k Z(s) = (-1)^k (s)_k Z(s+k). \tag{9.7}$$

This follows directly from its Fourier coefficients and the Gamma recurrence; it also follows by repeated integration by parts in a left half-plane and meromorphic continuation. Put $s = 1-z$ and expand

$$(1-z)_k = k! \prod_{r=1}^{k}(1-z/r) = k! \sum_{i=0}^{k} (-1)^i e_i(k) z^i.$$

On the left of (9.7), the pole is $\delta_0^{(k)}/z$ and the regular coefficient of z^n is $D^k G_n/n!$. On the right, multiplying the regular part of Lemma 9.2 by the polynomial gives $F_{n,k}/n!$: this is the finite part of the ordinary pointwise parameter-derivative identity. Multiplying its polar part by that same polynomial gives, in addition, the regular coefficient $(-1)^{n+1} e_{n+1}(k)\delta_0^{(k)}$. Comparison proves the formula. □

The first examples are

$$\begin{aligned} DG_0 &= F_{0,1} - \delta_0', \\ D^2G_0 &= F_{0,2} - \tfrac{3}{2}\delta_0'', \\ D^2G_1 &= F_{1,2} + \tfrac{1}{2}\delta_0''. \end{aligned} \tag{9.8}$$

The point-supported terms disappear upon restriction to $(0,1)$, so ordinary differentiation there cannot detect them. But convolving $\delta_0^{(k)}$ with another distribution differentiates that other factor everywhere. Therefore these terms cannot be discarded before convolution, even when the desired final identity is off the singular point.

The pointwise harmonic master identity (9.1) remains valid. The new contact term governs its canonical periodic extension: point-supported derivatives vanish on the open interval but contribute after convolution.

9.2.1 Convolutions of arbitrary differentiated Stieltjes functions

The same argument gives a closed law for all derivative indices, not only the zeroth Stieltjes function. Put $F_{n,0} = G_n$ and define

$$a_{n,k} = (-1)^{n+1} n! e_{n+1}(k) \quad (k \geq 1), \qquad a_{n,0} = 0.$$

Thus $F_{n,k} = D^k G_n - a_{n,k}\delta_0^{(k)}$ when $k \geq 1$.

Corollary 9.4 (The full differentiated convolution calculus)**.** *Suppose $n, m, k, l \geq 0$, $r = k + l \geq 1$, and the exact rule from Theorem 8.63 is*

$$G_n * G_m = c_{-1}\mathsf{I} + \sum_{j=0}^{n+m+1} c_j G_j.$$

Then

$$\begin{aligned} F_{n,k} * F_{m,l} = {} & \sum_{j=0}^{n+m+1} c_j D^r G_j - a_{n,k} D^r G_m - a_{m,l} D^r G_n \\ & + (c_{-1} + a_{n,k} a_{m,l})\delta_0^{(r)}. \end{aligned} \tag{9.9}$$

Away from the integer point this reduces to

$$\sum_{j=0}^{n+m+1} c_j \gamma_j^{(r)}(x) - a_{n,k}\gamma_m^{(r)}(x) - a_{m,l}\gamma_n^{(r)}(x).$$

In particular, every such convolution has a finite reduction to $\zeta^{[j]}(r+1, x)$ with $0 \leq j \leq n + m + 1$.

Proof. Multiply the two identities $F = DG - a\delta$ at their indicated orders, interpreting the zero-order case through $a_{n,0} = 0$. The product of the first terms is $D^r(G_n * G_m)$, and $D^r\mathsf{I} = \delta_0^{(r)}$. Restricting off zero and applying (9.1) proves the last assertion. □

As an explicit higher-index example, the contact term vanishes for $F_{1,1} = DG_1$. Hence

$$\begin{aligned} (F_{1,1} * F_{1,1})(x) = {} & -2\zeta'''(3,x) - 9\zeta''(3,x) \\ & + [4\zeta(2) - 6]\zeta'(3,x) + [6\zeta(2) + 4\zeta(3)]\zeta(3,x). \end{aligned} \tag{9.10}$$

This follows by twice differentiating (8.81) and using (9.1). It illustrates that the full tower still closes at one spectral integer even when higher Stieltjes indices occur.

9.3 A two-jet collapse for every polygamma convolution

Let $H_0 = 0$ and $H_k = \sum_{j=1}^{k} 1/j$ for $k \geq 1$. Define

$$\Psi_k = \mathrm{Fp}_{\mathbb{T}}\, \psi^{(k)} \qquad (k \geq 0)$$

using the same canonical cutoff convention. Since $\gamma_0 = -\psi$, Theorem 9.3 gives

$$\Psi_0 = -G_0, \qquad \Psi_k = -D^k G_0 - H_k \delta_0^{(k)} \quad (k \geq 1). \tag{9.11}$$

Theorem 9.5 (Polygamma convolution reduction)**.** *Let $k, l \geq 0$ and $r = k + l \geq 1$. Then*

$$\begin{aligned} \Psi_k * \Psi_l = {} & 2D^r G_1 + (H_k + H_l) D^r G_0 \\ & + [H_k H_l - \zeta(2)] \delta_0^{(r)}. \end{aligned} \tag{9.12}$$

Consequently, for $0 < x < 1$,

$$(\Psi_k * \Psi_l)(x) = (-1)^{r+1} r! \left[2\zeta'(r+1, x) + (2H_r - H_k - H_l)\zeta(r+1, x) \right]. \tag{9.13}$$

For $k = l = 0$, the appropriate separate formula is $\Psi_0 * \Psi_0 = 2G_1 - \zeta(2)\mathsf{I}$.

Proof. Convolving (9.11) and using that derivatives may be placed on either factor gives

$$\Psi_k * \Psi_l = D^r (G_0 * G_0) + (H_k + H_l) D^r G_0 + H_k H_l \delta_0^{(r)}.$$

The same expression is valid when one index is zero, since its harmonic number vanishes. Apply (8.78) and $D^r \mathsf{I} = \delta_0^{(r)}$. Restriction to $0 < x < 1$ gives $2\gamma_1^{(r)} + (H_k + H_l)\gamma_0^{(r)}$. Now use (9.1) with $n = 0, 1$:

$$\gamma_1^{(r)} = (-1)^{r+1} r! [\zeta'(r+1, x) + H_r \zeta(r+1, x)], \quad \gamma_0^{(r)} = (-1)^r r! \zeta(r+1, x).$$

This proves (9.13). □

For example,

$$\begin{aligned} (\Psi_0 * \Psi_1)(x) &= 2\zeta'(2, x) + \zeta(2, x), \\ (\Psi_1 * \Psi_1)(x) &= -4\zeta'(3, x) - 2\zeta(3, x), \\ (\Psi_0 * \Psi_2)(x) &= -4\zeta'(3, x) - 3\zeta(3, x). \end{aligned} \tag{9.14}$$

Although the last two convolutions have the same total derivative order, they have different coefficients. Confusing Ψ_k with the kth derivative of Ψ_0 would incorrectly erase this distinction.

9.3.1 A convergent trigamma identity with every subtraction visible

For $0 < x < 1$, set $h = x/2$ and $f(t) = \psi'(t)$. Then

$$\begin{aligned} \mathcal{T}(x) = {} & 2 \int_0^h \left[\frac{f(x-t) - f(x) + t f'(x)}{t^2} + \psi'(1+t) f(x-t) \right] \mathrm{d}t \\ & + \int_x^1 f(t) f(1 + x - t)\, \mathrm{d}t - \frac{2f(x)}{h} - 2f'(x) \log h \end{aligned} \tag{9.15}$$

satisfies

$$\mathcal{T}(x) = -4\zeta'(3, x) - 2\zeta(3, x). \tag{9.16}$$

Indeed, the shift relation $\psi'(t) = t^{-2} + \psi'(1+t)$ and symmetry about $t = x/2$ reduce the first singular convolution integral to twice its left half. Subtracting the constant and linear Taylor terms of $f(x-t)$ leaves a bounded quotient. The constant terms of $\int_\epsilon^h t^{-2}\, \mathrm{d}t$ and $\int_\epsilon^h t^{-1}\, \mathrm{d}t$ are $-1/h$ and $\log h$, respectively. This proves that (9.15) is exactly the canonical finite-part convolution, so Theorem 9.5 applies. Every integral in (9.15) is ordinary and absolutely convergent.

9.4 Contact terms in translated derivative products

For $r \geq 1$ let $h_{r,m} = (-1)^m m! e_{m+1}(r)$, and put $h_{0,m} = 0$. Thus (9.6) reads $F_{m,r} = D^r G_m + h_{r,m}\delta_0^{(r)}$.

Theorem 9.6 (Translated contact law)**.** *For $r, k, m, n \geq 0$, $N = r+k$, and $b = 1-a$ with $0 < a < 1$,*

$$\begin{aligned} &\mathrm{FP}\int_0^1 \gamma_m^{(r)}(x)\gamma_n^{(k)}(\{x+a\})\,dx \\ &\quad = (-1)^r J_{mn}^{(N)}(a) + (-1)^k h_{k,n}\gamma_m^{(N)}(b) + (-1)^r h_{r,m}\gamma_n^{(N)}(a). \end{aligned} \tag{9.17}$$

Proof. The local finite parts use the constant-term convention (9.4). At nonzero shift the singular supports of the two factors are disjoint, so a partition into disjoint endpoint neighborhoods defines the product unambiguously. Substitute the two contact identities. Periodic integration by parts gives the first term. The shifted delta derivative acts at $x = b$, giving $(-1)^k h_{k,n}\gamma_m^{(N)}(b)$, and the unshifted one acts at zero, giving $(-1)^r h_{r,m}\gamma_n^{(N)}(a)$. The product of the two point-supported terms vanishes because their supports are distinct. This proves the formula with the exact scale-one endpoint normalization. □

9.4.1 Every polygamma product in first Hurwitz derivatives

For $r, k \geq 0$ define

$$\Pi_{r,k}(a) = \mathrm{FP}\int_0^1 \psi^{(r)}(x)\psi^{(k)}(\{x+a\})\,\mathrm{d}x.$$

Corollary 9.7 (Polygamma closure)**.** *For $N = r + k \geq 1$,*

$$\begin{aligned} \Pi_{r,k}(a) &= (-1)^r\big[\gamma_1^{(N)}(a) - H_r\psi^{(N)}(a)\big] \\ &\quad + (-1)^k\big[\gamma_1^{(N)}(b) - H_k\psi^{(N)}(b)\big]. \end{aligned} \tag{9.18}$$

Equivalently,

$$\begin{aligned} \Pi_{r,k}(a) = N!\Big\{&(-1)^{k+1}\big[\zeta^{[1]}(N+1,a) + (H_N - H_r)\zeta(N+1,a)\big] \\ &+ (-1)^{r+1}\big[\zeta^{[1]}(N+1,b) + (H_N - H_k)\zeta(N+1,b)\big]\Big\}. \end{aligned} \tag{9.19}$$

The case $N = 0$ is (8.113).

Proof. Take $m = n = 0$ in (9.17), use $\gamma_0^{(r)} = -\psi^{(r)}$, $h_{r,0} = H_r$, and differentiate (8.109). This proves (9.18). The pointwise identities

$$\begin{aligned} \psi^{(N)}(x) &= (-1)^{N+1}N!\zeta(N+1,x), \\ \gamma_1^{(N)}(x) &= (-1)^{N+1}N!\big[\zeta^{[1]}(N+1,x) + H_N\zeta(N+1,x)\big] \end{aligned}$$

then give (9.19). The second follows directly by expanding $\partial_x^N\zeta(1+u,x) = (-1)^N(1+u)_N\zeta(N+1+u,x)$ at $u = 0$; these pointwise ladders are also part of the repository's existing development [109]. □

Two instructive special cases are

$$\Pi_{0,1}(a) = \zeta^{[1]}(2,a) + \zeta(2,a) - \zeta^{[1]}(2,b) = J'_{00}(a) + \psi'(b), \tag{9.20}$$

$$\Pi_{1,1}(a) = 2\zeta^{[1]}(3,a) + 2\zeta^{[1]}(3,b) + \zeta(3,a) + \zeta(3,b). \tag{9.21}$$

The term $+\psi'(b)$ in (9.20) is an explicit counterexample to the unqualified rule "differentiate the finite-part identity under the integral." It is not a counterexample to ordinary differentiation under a genuinely dominated integral.

For reproducibility, let $I_{r,k}(\epsilon)$ be the sum of the two raw cutoff integrals, with arguments split as in (8.91). Then

$$\Pi_{0,1}(a) = \lim_{\epsilon\downarrow 0}\left[I_{0,1}(\epsilon) - \frac{\psi(b)}{\epsilon} - \big(\psi'(a) - \psi'(b)\big)\log\epsilon\right], \tag{9.22}$$

$$\Pi_{1,1}(a) = \lim_{\epsilon\downarrow 0}\left[I_{1,1}(\epsilon) - \frac{\psi'(a) + \psi'(b)}{\epsilon} + \big(\psi''(a) + \psi''(b)\big)\log\epsilon\right]. \tag{9.23}$$

These counterterms can be derived directly from $\psi(x) = -1/x - \gamma + O(x)$ and $\psi'(x) = x^{-2} + O(1)$. They give numerical tests of the contact law that do not implement distributional differentiation.

The full default replays are preserved under `verification/eighth-replay/`. The direct cutoff polygamma calculations check the contact correction without implementing distributional differentiation. The all-index law rests on the Laurent residues and the proof above. The ordinary pointwise master identity has not been retracted or altered.

9.5 Dilation, distribution traces and exact logarithmic contacts

The contact theorem answers an extension question suggested by the correlation reports: classical distribution rows need a normalization correction when interpreted as finite-part identities. Fourier traces make the correction computable at every Stieltjes and derivative order. This also describes how dilation acts on the one-generator convolution algebra, without any arithmetic-independence assumption.

For a positive integer q, define the trace $\mathcal{S}_q$ and covering pullback $\mathcal{P}_q$ on periodic distributions by

$$\widehat{\mathcal{S}_q T}(k) = q\widehat{T}(qk), \qquad \widehat{\mathcal{P}_q T}(k) = \begin{cases} \widehat{T}(k/q), & q \mid k, \\ 0, & q \nmid k. \end{cases}$$

For ordinary periodic functions these are $\mathcal{S}_q f(x) = \sum_{j=0}^{q-1} f((x+j)/q)$ and $\mathcal{P}_q f(x) = f(\{qx\})$. Put $\ell_q = \log q$ and $\mathcal{L}_q = q^{-1}\mathcal{S}_q$.

Theorem 9.8 (The trace translates the convolution generator)**.** *For every $m \geq 0$,*

$$\mathcal{S}_q G_m = q\sum_{j=0}^{m}\binom{m}{j}(-\ell_q)^{m-j}G_j + \frac{q(-\ell_q)^{m+1}}{m+1}\mathsf{I}. \tag{9.24}$$

The normalized trace $\mathcal{L}_q$ is a unital homomorphism of the mean-zero convolution algebra. On its generator $X = G_0$ it acts by $X \mapsto X - \ell_q\mathsf{I}$. These actions compose as $\mathcal{L}_q\mathcal{L}_r = \mathcal{L}_{qr}$.

Proof. The Fourier definition gives $\mathcal{S}_q R_\alpha = q^{1-\alpha}R_\alpha$. Apply this to the Laurent family (8.70):

$$\mathcal{S}_q\left(\frac{\mathsf{I}}{z} + \sum_{m\geq 0} G_m\frac{z^m}{m!}\right) = qe^{-\ell_q z}\left(\frac{\mathsf{I}}{z} + \sum_{m\geq 0} G_m\frac{z^m}{m!}\right).$$

The regular coefficients give (9.24), including the term produced by the pole. On Fourier coefficients, $\mathcal{L}_q(T * S) = (\mathcal{L}_q T) * (\mathcal{L}_q S)$ and $\mathcal{L}_q \mathsf{I} = \mathsf{I}$. The generator formula is the $m = 0$ case, and repeated traces evaluate a Fourier coefficient at qrk. □

Theorem 9.9 (Finite part does not commute with covering pullback)**.** *Let* $\operatorname{Fp}_x \gamma_m(\{qx\})$ *use the constant term in the unscaled local coordinate* $x - j/q$ *at each of the* q *singularities, with right-hand cutoffs tending independently to zero. Then*

$$\mathcal{P}_q G_m = \operatorname{Fp}_x \gamma_m(\{qx\}) + \frac{\ell_q^{m+1}}{q(m+1)} \sum_{j=0}^{q-1} \delta_{j/q}. \tag{9.25}$$

In particular the raw-coordinate finite part has mean $-\ell_q^{m+1}/(m+1)$*, while the covering pullback has mean zero.*

Proof. On the jth segment substitute $y = q(x - j/q)$. Pulling back the cutoff prescription (8.69) gives its local counterterm

$$\frac{\log^{m+1}(q\epsilon)}{q(m+1)} \phi(j/q).$$

The raw-coordinate constant-term prescription keeps only the divergent powers of $\log \epsilon$ in this expression. Their difference is its constant coefficient, $\ell_q^{m+1}\phi(j/q)/(q(m+1))$. All Taylor remainder integrals converge as in the definition of G_m. Summing the segments proves (9.25). The Fourier pullback preserves the zeroth coefficient, and pairing the contact sum with the constant function proves the mean statement. □

For the trace itself, put $c_{m,q} = -q(-\ell_q)^{m+1}/(m+1)$. The classical pointwise multiplication row and its raw-coordinate extension read

$$\operatorname{Fp}_x[\mathcal{S}_q \gamma_m] = q \sum_{j=0}^{m} \binom{m}{j} (-\ell_q)^{m-j} G_j + c_{m,q}\, 1.$$

Comparing with (9.24) gives the separate identity

$$\mathcal{S}_q G_m = \operatorname{Fp}_x[\mathcal{S}_q \gamma_m] - c_{m,q}\delta_0. \tag{9.26}$$

The logarithmic term thus has different signs and support for covering pullback and trace; substituting one convention for the other is invalid.

Theorem 9.10 (Every differentiated trace and its contact)**.** *Use* $F_{m,r}$ *and* $a_{m,r} = (-1)^{m+1} m! e_{m+1}(r)$ *from the contact theorem, and let* $r \geq 1$*. Then*

$$\begin{aligned} \mathcal{S}_q F_{m,r} &= q^{r+1} \sum_{j=0}^{m} \binom{m}{j} (-\ell_q)^{m-j} F_{j,r} + q^{r+1} \kappa_{m,r}(\ell_q) \delta_0^{(r)}, \\ \kappa_{m,r}(\ell) &= \frac{(-\ell)^{m+1}}{m+1} + \sum_{j=0}^{m} \binom{m}{j} (-\ell)^{m-j} a_{j,r} - a_{m,r}. \end{aligned} \tag{9.27}$$

The first sum is exactly $\operatorname{Fp}_x[\mathcal{S}_q \gamma_m^{(r)}]$*. For the canonical polygamma finite parts, the particularly simple all-order specialization is*

$$\mathcal{S}_q \Psi_r = q^{r+1}(\Psi_r + \ell_q \delta_0^{(r)}), \qquad r \geq 1. \tag{9.28}$$

Proof. Fourier coefficients give $\mathcal{S}_q D^r = q^r D^r \mathcal{S}_q$ and $\mathcal{S}_q \delta_0^{(r)} = q^{r+1}\delta_0^{(r)}$. Apply these identities to $F_{m,r} = D^r G_m - a_{m,r}\delta_0^{(r)}$, use (9.24), and then replace each $D^r G_j$ by $F_{j,r} + a_{j,r}\delta_0^{(r)}$. Because $D^r \mathsf{I} = \delta_0^{(r)}$, the contact coefficient is precisely the displayed $\kappa_{m,r}$. Differentiating the classical pointwise multiplication row r times proves the raw-coordinate interpretation of the first sum. Finally $\Psi_r = -F_{0,r}$ and $\kappa_{0,r} = -\ell_q$, proving (9.28). □

These identities answer the normalization question for integer covering maps and their distribution traces at every index. They also show that the polynomial convolution algebra is preserved by the normalized trace. Twisted character projections and products at several independent shifted singularities still require separate calculations. Exact Fourier-polynomial and local logarithmic-counterterm controls, with a corruption check, are recorded in `verification/certify_stieltjes_dilation.py`.

9.6 The uniform rational-argument landscape

Three structural facts hold uniformly in k; all follow from $\sum_p \chi(p)\,\zeta(s,p/q) = q^s L(s,\chi)$ (and its multiplicative cousin) differentiated in s, fed through Theorem 9.1.

Proposition 9.11 (Character coordinate)**.** *For a primitive nonprincipal Dirichlet character χ modulo $q > 1$ and all $k \geq 1$,*

$$\sum_{p=1}^{q-1} \chi(p)\,\gamma_1^{(k)}(p/q) = (-1)^{k+1}k!\,q^{k+1}\big[L'(k{+}1,\chi) + (H_k + \ln q)\,L(k{+}1,\chi)\big]. \tag{9.29}$$

Proposition 9.12 (Distribution row)**.** *For $d \geq 2$ and all $k \geq 1$, with $\psi^{(k)}(a) = (-1)^{k+1}k!\,\zeta(k{+}1,a)$,*

$$\sum_{j=0}^{d-1} \gamma_1^{(k)}\Big(\tfrac{a+j}{d}\Big) = d^{\,k+1}\gamma_1^{(k)}(a) + d^{\,k+1}\ln d\;\psi^{(k)}(a). \tag{9.30}$$

Proof of (9.29) *and* (9.30). Apply the character sum to (9.2) for (9.29), and the divisor sum for (9.30). For the character identity, use $\sum_p \chi(p)\zeta(s,p/q) = q^s L(s,\chi)$ and its s-derivative $q^s \ln q\, L(s,\chi) + q^s L'(s,\chi)$ at $s = k{+}1$; collect. For (9.30), use $\sum_{j<d}\zeta(s,(a{+}j)/d) = d^s\zeta(s,a)$ and its s-derivative, and recognize $(-1)^{k+1}k!\,\zeta(k{+}1,a) = \psi^{(k)}(a)$. □

Corollary 9.13 (Uniform formal distribution ranks)**.** *For every $q \geq 2$ and every parameter derivative order $k \geq 1$, the specified distribution presentation, with the endpoint and lower-index data treated as background, has residual dimension $\varphi(q) - 1$. This is a formal quotient statement, not arithmetic independence of the evaluated constants.*

The original finite rank experiments suggested this law. The following polynomial character normal form proves it without extrapolation, also at arbitrary weights and in finite jet algebras.

9.7 Weighted distribution modules and exact rational-grid ranks

The distribution relations in the Stieltjes drafts have a simple representation-theoretic structure. The relevant dimension is the dimension of a formal quotient by specified relations. The classical theory of universal distributions provides the broader setting [52, 53]. Here we give an explicit character normal form, including its behavior at exceptional weights and in finite Taylor jets. This proves the two all-denominator rank statements suggested by the source computations, without making an independence assertion about evaluated special values.

9.7.1 The relation module and its complete character decomposition

Fix $q \geq 2$, and put

$$G_q = \frac{1}{q}\mathbb{Z}/\mathbb{Z}, \qquad U(q) = (\mathbb{Z}/q\mathbb{Z})^\times.$$

Let $\mathcal{A}$ be a commutative unital $\mathbb{C}$-algebra and choose arbitrary $t_p \in \mathcal{A}$, one for each prime $p \mid q$. The choices may vanish or be nilpotent. Let

$$\mathcal{V}_q = \bigoplus_{x \in G_q} \mathcal{A}[x].$$

For $p \mid q$ and $x \in pG_q$, define the prime distribution row

$$D_p(x) = \sum_{\substack{y \in G_q \\ py = x}} [y] - t_p[x]. \tag{9.31}$$

Write $\mathcal{R}_q$ for the $\mathcal{A}$-span of these rows and $\mathcal{U}_q = \mathcal{V}_q/\mathcal{R}_q$. The unit group acts by $u[x] = [ux]$, and each relation map is equivariant.

For $d \mid q$, set $t_d = \prod_{p \mid d} t_p^{v_p(d)}$. Rows for composite divisors introduce no additional relations. Indeed, if $ab \mid q$ and $x \in abG_q$, then

$$D_{ab}(x) = \sum_{\substack{z \in G_q \\ az = x}} D_b(z) + t_b D_a(x). \tag{9.32}$$

Every z in this sum belongs to bG_q, so the rows on the right are legitimate grid rows. Expanding the two sums proves the identity. Induction on the number of prime factors of d, counted with multiplicity, proves the assertion for every divisor.

Let χ run through all characters of $U(q)$, and let $f = f_\chi$ be the conductor of its primitive inducing character χ^*. For $f \mid d \mid q$, let χ_d denote the induced character modulo d, and define

$$Y_{d,\chi} = \sum_{a \in U(d)} \chi_d(a)[a/d]. \tag{9.33}$$

For the principal character at $d = 1$, this means $Y_{1,\mathbf{1}} = [0]$. These are formal vectors; no numerical L-value is being declared independent.

Lemma 9.14 (All character components of the rows)**.** *The χ^{-1}-eigenspace of $\mathcal{V}_q$ is free over $\mathcal{A}$, with basis*

$$\{Y_{d,\chi} : f_\chi \mid d \mid q\}.$$

Its relation submodule is generated exactly by the edges

$$\begin{aligned} Y_{pd,\chi} - t_p Y_{d,\chi}, &\qquad p \mid d, \\ Y_{pd,\chi} - (t_p - \chi^*(p)) Y_{d,\chi}, &\qquad p \nmid d, \end{aligned} \qquad f_\chi \mid d, \quad pd \mid q. \tag{9.34}$$

In particular, there are no additional character relations arising from denominators smaller than, or not divisible by, f_χ.

Proof. The elements of G_q of exact denominator d form a single $U(q)$-orbit, naturally identified with $U(d)$. Reduction $U(q) \to U(d)$ is surjective. The character χ factors through this map exactly when $f_\chi \mid d$; otherwise its projection on that orbit is zero. More explicitly, the idempotent

$$E_\chi = \frac{1}{\varphi(q)} \sum_{u \in U(q)} \chi(u) u$$

satisfies, for $a \in U(d)$,

$$E_\chi[a/d] = \begin{cases} \dfrac{\chi_d(a)^{-1}}{\varphi(d)} Y_{d,\chi}, & f_\chi \mid d, \\ 0, & f_\chi \nmid d. \end{cases} \tag{9.35}$$

This follows by grouping u according to its residue modulo d and using character orthogonality on the kernel of reduction. The idempotents are mutually orthogonal and sum to the identity. All their scalar denominators are units in $\mathcal{A}$, so the argument applies equally to a nonreduced algebra.

For a fixed $p \mid q$, consider the equivariant map

$$D_p : \mathcal{A}[pG_q] \longrightarrow \mathcal{V}_q, \qquad [x] \longmapsto D_p(x).$$

The exact-denominator orbits in its domain have $pd \mid q$. By (9.35), their only nonzero χ-components have $f_\chi \mid d$, and each such component has the single generator $Y_{d,\chi}$. Consequently, computing $D_p(Y_{d,\chi})$ computes *every* projected relation for that p; an orbit on which the projector vanishes cannot create a hidden relation in the target.

If $p \mid d$, all p preimages of each a/d, $a \in U(d)$, have exact denominator pd. Summing with weights $\chi_d(a)$ therefore gives $Y_{pd,\chi} - t_p Y_{d,\chi}$. If $p \nmid d$, one preimage has denominator d, while the other $p-1$ have denominator pd. In the former preimage, $pc \equiv a \pmod{d}$, so its character coefficient is $\chi_d(a) = \chi^*(p)\chi_d(c)$. Thus the sum is

$$Y_{pd,\chi} + \chi^*(p) Y_{d,\chi} - t_p Y_{d,\chi}.$$

Here $p \nmid f_\chi$, as $f_\chi \mid d$, so $\chi^*(p)$ is a nonzero root of unity. This proves both the edge formulas and their completeness. □

Theorem 9.15 (Polynomial normal form)**.** *For $f = f_\chi \mid d \mid q$, define*

$$A_{d,\chi}(\mathbf{t}) = \prod_{p \mid f} t_p^{v_p(d) - v_p(f)} \prod_{\substack{p \mid d \\ p \nmid f}} t_p^{v_p(d)-1}(t_p - \chi^*(p)). \tag{9.36}$$

Empty products are 1, so $A_{f,\chi} = 1$. Then the full χ-relation submodule is

$$\mathcal{R}_{q,\chi} = \bigoplus_{\substack{f \mid d \mid q \\ d \neq f}} \mathcal{A}\big(Y_{d,\chi} - A_{d,\chi} Y_{f,\chi}\big). \tag{9.37}$$

In particular,

$$\mathcal{U}_q \cong \bigoplus_{\chi \in \widehat{U(q)}} \mathcal{A}\, Y_{f_\chi,\chi} \cong \mathcal{A}^{\varphi(q)}. \tag{9.38}$$

The character normal form of every vector is obtained by replacing $Y_{d,\chi}$ with $A_{d,\chi} Y_{f_\chi,\chi}$.

Proof. The polynomials $A_{d,\chi}$ satisfy precisely the edge recursions (9.34): a prime already dividing d contributes t_p, and a prime introduced for the first time outside f contributes $t_p - \chi^*(p)$. The factors for distinct primes commute.

Follow any chain of prime extensions from f to d. Successively eliminating the intermediate Y's expresses $Y_{d,\chi} - A_{d,\chi}Y_{f,\chi}$ as a combination of the edge relations. Conversely, write an edge as $Y_{pd,\chi} - cY_{d,\chi}$, where $A_{pd,\chi} = cA_{d,\chi}$. It equals

$$\big(Y_{pd,\chi} - A_{pd,\chi}Y_{f,\chi}\big) - c\big(Y_{d,\chi} - A_{d,\chi}Y_{f,\chi}\big).$$

Lemma 9.14 therefore gives equality of the two relation spans.

The displayed generators in (9.37) are independent: the coefficient of $Y_{d,\chi}$, for $d \neq f$, in a linear combination is exactly the coefficient of its own generator. Together with $Y_{f,\chi}$, they form a triangular basis of the whole character block. Its quotient is consequently free of rank one. Summing over all $\varphi(q)$ characters proves (9.38). □

Remark 9.16. The proof divides by neither t_p nor $t_p - \chi^*(p)$. It proves freeness first over $\mathbb{C}[t_p : p \mid q]$ and then over every $\mathbb{C}$-algebra obtained by specialization. Exceptional weights can therefore make a preferred coordinate vanish, but cannot change the rank of the full quotient. This distinction is essential when Euler factors are used to solve a relation system.

9.7.2 Anchors, reflection, and the ranks in the drafts

In this subsection take $\mathcal{A} = \mathbb{C}$. In a Hurwitz-zeta grid, use the representatives $0 < a \leq 1$: the group element 0 represents $a = 1$, not the singular Hurwitz parameter $a = 0$. Treating the endpoint as known means imposing the homogeneous anchor $[0] = 0$. It removes exactly the principal generator $Y_{1,\mathbf{1}}$. Hence

$$\dim_{\mathbb{C}} \mathcal{V}_q/(\mathcal{R}_q + \mathbb{C}[0]) = \varphi(q) - 1. \tag{9.39}$$

There are $q - 1$ remaining grid variables, so their distribution coefficient matrix has rank $q - \varphi(q)$. At rational weights t_p, the matrix has rational entries and this is also its rank over $\mathbb{Q}$.

Let $J[x] = [-x]$. On the χ-block, J acts as $\chi(-1)$. Thus the additional homogeneous reflection rows

$$[x] + (-1)^k[-x] = 0, \qquad x \neq 0, \tag{9.40}$$

retain exactly the characters with $\chi(-1) = (-1)^{k+1}$. After anchoring, the omission of $x = 0$ in (9.40) has no effect.

Corollary 9.17 (All-denominator formal ranks)**.** *For $q \geq 3$ and every integer $k \geq 0$, the formal quotient by distribution, endpoint, and the reflection rows (9.40) has dimension*

$$r_{q,k} = \begin{cases} \varphi(q)/2 - 1, & k \text{ odd}, \\ \varphi(q)/2, & k \text{ even}. \end{cases} \tag{9.41}$$

This conclusion holds for arbitrary prime weights. In particular it proves the all-q, all-k formal rank law for the negative Hurwitz-zeta derivative ladder. The formal distribution-only rank in the Stieltjes parameter derivative tower is $q - \varphi(q)$ for every $q \geq 2$, every parameter derivative order $k \geq 1$, and every Stieltjes index $n \geq 1$, relative to the endpoint and lower-index data.

Proof. For $q \geq 3$, the element -1 is a nonidentity involution in $U(q)$. Character orthogonality gives $\sum_\chi \chi(-1) = 0$, and each summand is 1 or -1. There are consequently $\varphi(q)/2$ characters of each

parity. Reflection retains the parity stated above. The principal character is even, so the endpoint removes one additional dimension exactly when k is odd. This proves (9.41).

For the negative ladder, differentiation of the Hurwitz multiplication formula at $s = -k$ gives

$$\sum_{j=0}^{d-1} \zeta'\left(-k, \frac{a+j}{d}\right) = d^{-k}\big(\zeta'(-k,a) + \log d\,\zeta(-k,a)\big). \tag{9.42}$$

The Bernoulli-polynomial values $\zeta(-k,a)$ are lower data. Thus the homogeneous distribution weight is $t_p = p^{-k}$. The homogeneous reflection is (9.40); an explicit right-hand side is given below. This applies the first part to the ladder in the source article.

For the parameter derivative tower, the Laurent expansion is

$$\zeta(s,a) = \frac{1}{s-1} + \sum_{n\ge 0} \frac{(-1)^n}{n!}\gamma_n(a)(s-1)^n. \tag{9.43}$$

For $k \ge 1$, its pole disappears after differentiation in a:

$$F_k(\epsilon,a) := \partial_a^k \zeta(1+\epsilon,a) = (-1)^k(1+\epsilon)_k\zeta(k+1+\epsilon,a) = \sum_{n\ge 0}\frac{(-1)^n}{n!}\gamma_n^{(k)}(a)\epsilon^n. \tag{9.44}$$

This standard identity is the starting point of the all-order parameter derivative formula of Coffey [25]. Differentiating the Hurwitz multiplication formula k times in a, including its chain-rule factors, gives

$$\sum_{j=0}^{d-1} F_k\left(\epsilon, \frac{a+j}{d}\right) = d^{k+1+\epsilon}F_k(\epsilon,a).$$

Comparing coefficients yields the exact triangular identity

$$\sum_{j=0}^{d-1}\gamma_n^{(k)}\left(\frac{a+j}{d}\right) = d^{k+1}\sum_{r=0}^{n}\binom{n}{r}(-\log d)^{n-r}\gamma_r^{(k)}(a). \tag{9.45}$$

With lower indices $r < n$ treated as specified data, the homogeneous weight is $t_p = p^{k+1}$, independently of n. Equation (9.39) now gives rank $q - \varphi(q)$. □

For completeness, the right-hand side of the negative reflection can be written without introducing another Hurwitz derivative. For $0 < a < 1$, set

$$C_j(a) = \sum_{m\ge 1}\frac{\cos(2\pi m a)}{m^j}, \qquad S_j(a) = \sum_{m\ge 1}\frac{\sin(2\pi m a)}{m^j}.$$

For $k \ge 1$, these series at $j = k+1$ converge absolutely, and the Fourier form of the Hurwitz functional equation [49] gives

$$\zeta'(-k,a) + (-1)^k\zeta'(-k,1-a) = \frac{(-1)^{\lfloor k/2\rfloor}k!}{(2\pi)^k}\begin{cases} C_{k+1}(a), & k \text{ even},\\ S_{k+1}(a), & k \text{ odd}.\end{cases} \tag{9.46}$$

Indeed, the Fourier expansion of $\zeta(s,a)$ has the factor

$$\frac{2\Gamma(1-s)}{(2\pi)^{1-s}}\left(\sin\frac{\pi s}{2}\,C_{1-s}(a) + \cos\frac{\pi s}{2}\,S_{1-s}(a)\right).$$

The indicated reflection selects a sine or cosine prefactor that vanishes at $s = -k$. Only the derivative of that prefactor survives, giving (9.46). This also explains why reflection is an inhomogeneous relation with depth-one data in the inventory.

Corollary 9.18 (The formal gamma grid)**.** *For $q \geq 3$, the $q-1$ formal coordinates $\log\Gamma(a/q)$, $1 \leq a < q$, modulo multiplication and reflection and relative to their stated elementary right-hand sides, have quotient dimension $\varphi(q)/2$.*

Proof. Lerch's identity $\zeta'(0,a) = \log\Gamma(a) - \frac{1}{2}\log(2\pi)$ translates gamma multiplication into (9.42) with $k=0$, hence $t_p = 1$. The reflection formula

$$\zeta'(0,a) + \zeta'(0,1-a) = -\log(2\sin\pi a)$$

has homogeneous part $[x] + [-x] = 0$. Apply (9.41) with $k=0$. □

Remark 9.19 (What these rank theorems settle). The first Stieltjes source states a finite verification followed by an all-denominator conjecture; the parameter-derivative source states a general rank theorem while reporting only a finite calculation. Corollary 9.17 supplies a proof for the specified relation modules in both cases. The README had already identified the second proof gap. Corollary 9.18 likewise determines the rank of the stated gamma relation matrix. None of these results says that the evaluated numbers admit no further relations. In particular, the claim that multiplication and reflection generate every algebraic multiplicative relation among rational gamma values remains part of the conjectural completeness problem, not a consequence of the matrix rank.

9.7.3 Analytic weights, exceptional coordinates, and finite jets

Specialize $t_p = p^s = \exp(s\log p)$, using the real logarithm of the prime. Formula (9.36) becomes

$$A_{d,\chi}(s) = \left(\frac{d}{f_\chi}\right)^s \prod_{\substack{p\mid d \\ p\nmid f_\chi}} \left(1 - \chi^*(p)p^{-s}\right). \tag{9.47}$$

This is exactly the familiar imprimitive Euler-factor pattern for Dirichlet L-functions [49]. The normal-form proof shows why it appears in the formal relations before any evaluation is made.

The symbols of exact denominator q have character coordinates $Y_{q,\chi}$. In the quotient, their χ-coordinate is

$$Y_{q,\chi} = A_{q,\chi}(s) Y_{f_\chi,\chi}. \tag{9.48}$$

They form a basis of the full quotient precisely when every $A_{q,\chi}(s)$ is nonzero. In particular this holds when $\operatorname{Re} s \neq 0$, since $\chi^*(p)$ has modulus one. All nonzero integer weights in the two Stieltjes articles are therefore free of this coordinate obstruction. On the imaginary axis the top-denominator coordinates can fail, while the conductor coordinates of Theorem 9.15 remain a basis.

Theorem 9.20 (Exact defect of top-denominator jets)**.** *Fix $s_0 \in \mathbb{C}$ and $N \geq 0$, and work over $\mathcal{A}_N = \mathbb{C}[\epsilon]/(\epsilon^{N+1})$ with $t_p = p^{s_0+\epsilon}$, truncated in this ring. The full distribution module is free of rank $\varphi(q)$ over $\mathcal{A}_N$, hence has complex dimension $(N+1)\varphi(q)$. Set*

$$r_\chi(s_0) = \#\{p \mid q : p \nmid f_\chi,\ p^{s_0} = \chi^*(p)\}. \tag{9.49}$$

The image of the top-denominator submodule in its χ-block has complex dimension

$$\max\{N+1-r_\chi(s_0), 0\}, \tag{9.50}$$

and its cokernel has dimension $\min\{r_\chi(s_0), N+1\}$. An endpoint anchor or a fixed-parity reflection restricts these formulas to the retained characters. The resulting full quotient remains free over $\mathcal{A}_N$ with the ranks already computed.

Proof. Freeness is Theorem 9.15, applied to $\mathcal{A}_N$. A resonant factor in (9.47) has a simple zero at s_0:

$$\frac{d}{ds}(1-\chi^*(p)p^{-s})\Big|_{s=s_0} = \log p \neq 0.$$

All other factors, including $(q/f_\chi)^s$, are nonzero there. In the ring of analytic germs,

$$A_{q,\chi}(s_0+\epsilon) = \epsilon^{r_\chi(s_0)}u_\chi(\epsilon), \qquad u_\chi(0)\neq 0.$$

After truncation, u_χ is a unit. The top-denominator image in the free rank-one block is therefore the ideal $\epsilon^{r_\chi(s_0)}\mathcal{A}_N$. Its basis consists of the powers $\epsilon^{r_\chi},\dots,\epsilon^N$ when $r_\chi \le N$, and it is zero otherwise. This proves both dimensions. Anchoring and reflection remove entire character blocks, so the same calculation applies to each retained block. □

Comparing Taylor coefficients of a weighted distribution identity gives the corresponding stacked finite-jet system. If one identifies the coefficient at order j with $\epsilon^{N-j}[x]$, its relation space is the $\mathcal{A}_N$-span above; factorial normalizations merely rescale coordinates. Thus freeness also gives the rank of the complete simultaneous jet system, rather than just its successive highest layers.

Proposition 9.21 (Multiplicity of nonzero resonances)**.** *Resonances occur only on* $\operatorname{Re} s_0 = 0$*. For* $s_0 \neq 0$*, each* $r_\chi(s_0)$ *is at most one. Multiple resonant primes in a single character block can occur only at* $s_0 = 0$*.*

Proof. Taking absolute values in $p^{s_0} = \chi^*(p)$ gives $\operatorname{Re} s_0 = 0$. Write $s_0 = it$, with $t \neq 0$. Because every character value is a root of unity, resonance at p implies $t\log p/(2\pi) \in \mathbb{Q}$. This rational number is nonzero. If distinct primes p and ℓ both resonate, then $\log p/\log \ell \in \mathbb{Q}$. Writing this positive rational as a/b would give $p^b = \ell^a$ for positive integers a, b, contradicting unique factorization. □

For a concrete origin resonance, take $q = 14$ and let χ^* be the quadratic character modulo 7. Its conductor is 7, its parity is odd, and $\chi^*(2) = 1$. Then

$$Y_{14,\chi}(s) = (2^s - 1)Y_{7,\chi}(s). \tag{9.51}$$

At $s = 0$, the top-denominator coordinate vanishes although the conductor direction persists. For any analytic realization of this relation,

$$\begin{aligned} Y'_{14,\chi}(0) &= \log 2\, Y_{7,\chi}(0), \\ Y''_{14,\chi}(0) &= 2\log 2\, Y'_{7,\chi}(0) + (\log 2)^2 Y_{7,\chi}(0). \end{aligned} \tag{9.52}$$

Thus recovering the first conductor derivative from these top-denominator data requires the second top-denominator jet. The actual realization

$$Y_{d,\chi}(s) = \sum_{a\in U(d)} \chi_d(a)\zeta(s, a/d) = d^s L(s,\chi_d)$$

obeys the same identity. This example concerns an imprimitive Euler factor; it is distinct from the trivial-zero regularization of the primitive L-function in the next section.

9.8 Integral distribution, reflection and modular specialization

The character basis above gives efficient coordinates over a splitting field. To settle the integral question, retain the original grid symbols: Fourier transport is exact over its cyclotomic coefficient field, but inverse transforms introduce group-order denominators and need not be unimodular over integral rings. The following division-free construction proves an all-ring basis before imposing reflection. The resulting reflection quotients can have two-torsion even though the unreflected module is free. This develops the two overlapping integral continuations through a single proof chain: point basis, resolution, fixed-point homology, scalar torsion, jets and analytic identities.

The ordinary distribution and its sign cohomology are classical [52, 92]; Ouyang's universal norm distribution and Anderson-type resolution supply broader antecedents [53]. We give the complete proofs for the specified weighted presentation. Neither formal freeness nor a formal torsion calculation asserts independence of evaluated periods.

For a positive integer q, let

$$X_q = (q^{-1}\mathbb{Z})/\mathbb{Z}, \qquad P(q) = \{p : p \text{ is prime and } p \mid q\}, \qquad R_q = \mathbb{Z}[A_p : p \in P(q)].$$

We write $\mathrm{den}(x)$ for the exact order of x in $\mathbb{Q}/\mathbb{Z}$, and set $\mathrm{den}(0) = 1$. A symbol e_x is not itself a special value. It is a generator in a free module. Under polylogarithm evaluation, e_0 will become $\mathrm{Li}_s(1) = \zeta(s)$. Under Hurwitz evaluation, it becomes $\zeta(s, 1)$, not the undefined expression $\zeta(s, 0)$.

For $d \mid q$, put $A_d = \prod_{p|d} A_p^{v_p(d)}$. Weights on composite divisors are thus multiplicative; an unrelated variable for every divisor would define a different module. The universal module and its relation submodule are

$$V_q = \bigoplus_{x \in X_q} R_q e_x, \tag{9.53}$$

$$r_{p,x} = \sum_{\substack{y \in X_q \\ py = x}} e_y - A_p e_x \qquad (p \mid q,\ x \in X_{q/p}), \tag{9.54}$$

$$\mathcal{R}_q = \langle r_{p,x} \rangle_{R_q}, \qquad \mathcal{D}_q = V_q/\mathcal{R}_q. \tag{9.55}$$

Every sum in (9.54) has exactly p terms. For a commutative ring S and weights $a_p \in S$, write $\mathcal{D}_q(S, a)$ for the same presentation after $A_p \mapsto a_p$.

Lemma 9.22 (Prime rows suffice)**.** *The relation $r_{d,x} = \sum_{dy=x} e_y - A_d e_x$, for $d \mid q$ and $x \in X_{q/d}$, follows from the prime rows without division.*

Proof. For $ab \mid q$ and $x \in X_{q/(ab)}$, direct expansion gives

$$r_{ab,x} = \sum_{ay=x} r_{b,y} + A_b r_{a,x}.$$

Here all y belong to $X_{q/b}$, so every displayed prime or divisor row is legitimate. The intermediate symbols cancel. Induction on the number of prime factors of d, counted with multiplicity, proves the claim. □

This elementary observation already prevents a common ambiguity: "distribution rank" is meaningful only after the row vocabulary, endpoint convention, coefficient ring, and divisor-weight convention have all been fixed.

9.8.1 A division-free integral basis and a unit relation minor

Choosing the surviving grid points

Write $U_m = (\mathbb{Z}/m\mathbb{Z})^\times$ and use a one-element set U_1. For every prime p and exponent $a \geq 1$, choose a distinguished lift in each fiber of $U_{p^a} \to U_{p^{a-1}}$. At $a = 1$, this means choosing one member of U_p. Let $E_{p,a}$ be the remaining members. Thus

$$|E_{p,a}| = \varphi(p^a) - \varphi(p^{a-1}), \qquad \varphi(1) = 1. \tag{9.56}$$

The following fixed choice makes the construction executable:

$$\begin{aligned} E_{p,1} &= \{2, 3, \ldots, p-1\}, \\ E_{p,a} &= \{b + jp^{a-1} : 1 \leq b < p^{a-1},\ p \nmid b,\ 1 \leq j \leq p-1\} \quad (a \geq 2). \end{aligned} \tag{9.57}$$

For $p = 2$, the first set is empty. This is intentional.

Every point x of exact denominator m has a unique additive primary expansion

$$x = \sum_{p^a \| m} \frac{b_p}{p^a} \pmod{\mathbb{Z}}, \qquad b_p \in U_{p^a}. \tag{9.58}$$

For $x = n/m$ in lowest terms, $b_p \equiv n(m/p^a)^{-1} \pmod{p^a}$. Define $\mathcal{B}_q$ to contain 0 and all points of denominator $m \mid q$ for which $b_p \in E_{p,a}$ at every nonzero primary component. We call a primary component *bad* when its residue is the distinguished, discarded lift. This adjective refers only to a basis choice.

The cardinality telescopes prime by prime:

$$|\mathcal{B}_q| = \prod_{p^e \| q} \left(1 + \sum_{a=1}^{e} \big(\varphi(p^a) - \varphi(p^{a-1})\big)\right) = \varphi(q). \tag{9.59}$$

The point 0 must be retained for this count and for the normal forms.

Theorem 9.23 (Integral normal form)**.** *The images of e_b, $b \in \mathcal{B}_q$, form an R_q-basis of $\mathcal{D}_q$. Consequently, for every commutative ring S and every choice of weights $a_p \in S$, the same points form an S-basis of $\mathcal{D}_q(S, a)$. In particular this module is free of rank $\varphi(q)$ even in positive characteristic, at zero weights, and over nonreduced rings.*

The spanning algorithm is immediate and is proved next. Its independence is established in Section 9.8.2, without dividing by any group order or weight.

The terminating straightening rule

Let $y \notin \mathcal{B}_q$, let $m = \operatorname{den}(y)$, and choose any bad prime p for its primary coordinates. The row with target py gives

$$e_y = A_p e_{py} - \sum_{\substack{pz = py \\ z \neq y}} e_z \quad \text{in } \mathcal{D}_q. \tag{9.60}$$

All terms on the right have denominator dividing m. At exact denominator m, the p-primary components of the other solutions are the allowed lifts in the same fiber; all other primary components are unchanged. When the p-component has order p, there is also one solution with zero p-component, hence lower denominator. The point py itself has denominator m/p.

Order points first by their positive denominator and then by the number of bad primary components. Every right-hand term of (9.60) is strictly earlier in this order. Recursion therefore terminates, and proves that $\mathcal{B}_q$ spans. The implementation always chooses the smallest bad prime, but the theorem permits any choice. Independence will imply that all choices yield the same normal form.

Proposition 9.24 (Support and polynomial degree)**.** *Write the normal form of e_y, with $m = \mathrm{den}(y)$, as*

$$\mathrm{NF}(e_y) = \sum_{b \in \mathcal{B}_q} c_{y,b}(\mathbf{A}) e_b.$$

A nonzero coefficient has $\mathrm{den}(b) = d \mid m$, and

$$\deg_{A_p} c_{y,b} \le v_p(m/d) \qquad (p \mid q). \tag{9.61}$$

All coefficients belong to $\mathbb{Z}[\mathbf{A}]$.

Proof. The recursion never increases a primary denominator. Multiplication by A_p occurs only in the term e_{py}, where that denominator has lost one power of p. Every path from denominator m to denominator d can therefore contain at most $v_p(m/d)$ such multiplications. The other coefficients in the recursion are -1. Summing the finitely many paths proves the assertions, including possible cancellations. □

There is no assertion here that the coefficient of e_0 factors as a product of $A_p - 1$. At level 15 the coefficient in one normal form is $1 - A_3A_5$; this distinction will matter at the spectral pole.

An explicit maximal minor equal to one

For each nonbasis point y, select the single row $r_{p,py}$ used in its first straightening step. Select the nonbasis points as columns, and order rows by their selected pivot. The restricted matrix is triangular, with every diagonal entry equal to 1: all other nonbasis columns are earlier in the terminating order.

The selected rows are distinct. Indeed a selected row has its chosen point as its unique latest nonbasis point, with coefficient 1. Two identical rows could not have different such latest points. We have proved the following stronger integrality statement.

Theorem 9.25 (Universal unit minor and relation basis)**.** *The prime-relation matrix contains an explicitly selected $(q - \varphi(q)) \times (q - \varphi(q))$ minor equal to 1 in R_q. The selected rows form an R_q-basis of the full relation module $\mathcal{R}_q$. After specializing the weights to integers, all nonzero Smith invariants of the complete prime-relation matrix are 1; there are exactly $q - \varphi(q)$ of them.*

Proof. Triangularity proves the minor statement. Selected-row elimination reduces every vector to a combination of basis-point symbols. By Theorem 9.23, a vector in $\mathcal{R}_q$ has zero such remainder, so every raw row is a combination of the selected ones. The minor proves their independence. The same argument specializes without division. Equivalently, after an integer specialization the image lattice is saturated and has the indicated rank; the unit minor also proves directly that all Smith invariants are units. □

The proof order is not circular: the resolution below proves Theorem 9.23 independently of the assertion that the selected rows span all relations.

Certificates rather than opaque elimination

The straightener can retain the exact row combination at every step. If C_y denotes a certificate for $e_y - \mathrm{NF}(e_y)$, then

$$C_y = r_{p,py} + A_p C_{py} - \sum_{\substack{pz=py \\ z\neq y}} C_z, \qquad C_b = 0 \ (b \in \mathcal{B}_q). \tag{9.62}$$

A verifier need not trust the recursion. It can expand the listed raw rows in the original q-generator free module, multiply their explicit integer polynomials, and compare coefficients. The accompanying verifier does exactly this for the three frozen identities.

Example 9.26 (Small levels). The specified basis choices give

$$\mathcal{B}_4 = \{0, 3/4\}, \qquad \mathcal{B}_{12} = \{0, 5/12, 2/3, 3/4\},$$

$$\mathcal{B}_{15} = \{0, 1/15, 4/15, 2/5, 7/15, 3/5, 2/3, 4/5\}.$$

Also $\mathcal{B}_{30} = \mathcal{B}_{15}$ as sets of rational points. An exact primary component of order two has no allowed residue, which explains this last equality. It does not make the weight A_2 irrelevant to all normal-form coefficients.

The procedure terminates with polynomial coefficients, but we do not assert a polynomial-time bound in the bit length of q. There are q raw grid points to begin with, and certificate coefficients can grow. The finite support and degree bounds are rigorous; sharper complexity and coefficient-height estimates remain separate questions.

A uniform coefficient-height bound

For $q > 1$, let $s = |P(q)|$, let $P_{\max}$ be its largest prime divisor, and put $H(x) = \sum_p v_p(\mathrm{den}(x))$, $b(x) = \#\{\text{bad primary components of } x\}$ and $W(x) = (s+1)H(x) + b(x)$. Summing absolute values over all normal-form coordinates and all polynomial monomials gives

$$\| \mathrm{NF}(e_x)\|_1 \leq P_{\max}^{W(x)}.$$

Indeed every straightening edge lowers W by at least one: at equal denominator a bad component is removed, while a decrease in H absorbs any possible increase of at most s in b. There are at most $P_{\max}$ terms in a step, and multiplication by A_p preserves the coefficient norm. Induction proves the bound. The sharper coordinate multidegrees remain those in (9.61). The estimate is a uniform height bound, not an optimal complexity assertion.

The complete level-twelve normal-form table

Let $E_a = [e_{a/12}]$, $\alpha = A_2$, $\beta = A_3$. The retained basis is E_0, E_5, E_8, E_9. Each row gives the four coefficients of E_a in that basis.

a	E_0	E_5	E_8	E_9
0	1	0	0	0
1	$\alpha\beta(\alpha-1)$	-1	0	$-\beta-1$
2	$\alpha(\beta-1)$	0	$-\alpha-1$	0
3	$\alpha(\alpha-1)$	0	0	-1
4	$\beta-1$	0	-1	0
5	0	1	0	0
6	$\alpha-1$	0	0	0
7	$\alpha(\beta-\alpha)$	1	$-\alpha(\alpha+1)$	$\beta+1$
8	0	0	1	0
9	0	0	0	1
10	$1-\beta$	0	$\alpha+1$	0
11	$\alpha(1-\beta)$	-1	$\alpha(\alpha+1)$	0

The companion file `normal_forms_q12.json` contains the unfactored integer monomial data. At arbitrary complex order, substitute $\alpha = 2^{1-s}$, $\beta = 3^{1-s}$ and $E_a = \mathrm{Li}_s(e^{2\pi i a/12})$. At formal integral jet weights, substitute polynomials modulo u^L instead. The same table supports both uses.

9.8.2 A finite weighted resolution and a proof of independence

The complex

Order the primes dividing q. For $S \subseteq P(q)$ put $p_S = \prod_{p\in S} p$, and define

$$C_j = \bigoplus_{\substack{S\subseteq P(q)\\ |S|=j}} R_q[X_{q/p_S}]. \tag{9.63}$$

A generator is written $[S, x]$. Define

$$d[S,x] = \sum_{p\in S} (-1)^{\#\{\ell\in S:\ell<p\}} \left(\sum_{py=x} [S\setminus\{p\}, y] - A_p[S\setminus\{p\}, x] \right). \tag{9.64}$$

The roots in each inner sum belong to $X_{q/p_{S\setminus\{p\}}}$. The augmentation $C_0 \to \mathcal{D}_q$ is the defining quotient.

Lemma 9.27. *The operators in distinct prime directions commute before the exterior signs are applied. Hence $d^2 = 0$, and the image of $C_1 \to C_0$ is exactly $\mathcal{R}_q$.*

Proof. Both successive root sums in prime directions p and ℓ enumerate once all solutions of $p\ell z = x$. The two terms consisting of one root sum and one multiplication by a weight agree in either order, and so do the terms with coefficient A_pA_ℓ. All intermediate points lie in the required grids. The exterior signs make each pair cancel in d^2. The assertion about C_1 is its definition. □

A filtration that removes the weights

Associate to a generator the integer

$$\nu(S,x) = p_S \operatorname{den}(x), \tag{9.65}$$

which divides q. Filter the complex by increasing ν. If $p \in S$, a root of $py = x$ has denominator either $\operatorname{den}(x)$ or $p\operatorname{den}(x)$. Thus the differential does not increase ν. Its top-filtration part retains exactly the roots of denominator $p\operatorname{den}(x)$; the term with coefficient A_p lies strictly lower. The associated graded differential is consequently independent of all weights.

Fix a divisor $m \mid q$. In its associated graded summand, a generator with subset S has exact denominator m/p_S, so necessarily $S \subseteq P(m)$. For $p^a \parallel m$, the local coordinate belongs to $U_{p^{a-1}}$ when $p \in S$, and to U_{p^a} otherwise. At exponent zero the unique local coordinate is zero, represented by the one-element set U_1.

There is a harmless CRT twist: division by a prime changes the other primary coordinates by a unit. We spell out its removal because omitting it would leave a gap in a claimed tensor-product argument. If x_ℓ is the ℓ-primary component of x, replace it in the S-summand by

$$z_\ell = \left(\prod_{p \in S \setminus \{\ell\}} p \right)^{-1} x_\ell. \tag{9.66}$$

The inverse exists on the ℓ-primary group. Under a differential edge removing p, the coordinates z_ℓ for $\ell \neq p$ are unchanged: the factor p^{-1} in a root coordinate cancels the removal of p from the product. The p-coordinate is carried to all its lifts under multiplication by p.

It follows that the associated graded summand at m is the tensor product, with its usual exterior signs, of the two-term complexes

$$R_q[U_{p^{a-1}}] \xrightarrow{N_{p,a}} R_q[U_{p^a}], \qquad N_{p,a}(b) = \sum_{\substack{u \in U_{p^a} \\ u \mapsto b}} u, \quad p^a \parallel m. \tag{9.67}$$

For $a = 1$, the image of the single generator is the sum of all $p-1$ nonzero residues. For $a \geq 2$, each fiber has p elements.

Lemma 9.28 (Local splitting)**.** *Each $N_{p,a}$ is a split injection. Its cokernel is free with basis the images of $E_{p,a}$.*

Proof. Different fibers have disjoint supports. In one fiber, the sum of its basis vectors has a coefficient equal to 1 on the distinguished lift. Replace that one vector by the fiber sum; this is a unimodular basis change. The remaining vectors are exactly the allowed lifts. Doing this in all fibers proves both claims over any coefficient ring. □

Theorem 9.29 (Exact weighted resolution)**.** *The augmented complex*

$$0 \longrightarrow C_{|P(q)|} \longrightarrow \cdots \longrightarrow C_1 \longrightarrow C_0 \longrightarrow \mathcal{D}_q \longrightarrow 0$$

is exact. The images of $\mathcal{B}_q$ are a basis of $\mathcal{D}_q$, proving Theorem 9.23.

Proof. By Lemma 9.28, each local two-term complex is a direct sum of a contractible identity complex and its free cokernel in degree zero. Its finite tensor product is therefore acyclic in positive

degrees. Its degree-zero homology has the tensor-product basis of the allowed local residues. In the graded summand at m, these are exactly the basis points of denominator m.

Now lift through the finite filtration. If a positive-degree cycle has highest nonzero filtration m, its highest component is a boundary in the associated graded complex. Subtract a lift of that boundary to obtain a cycle of lower filtration. Repetition makes the cycle a boundary in the original complex. This proves positive-degree exactness without any convergence issue.

For degree zero, suppose a linear combination of allowed points is dw with $w \in C_1$, and let M be the highest filtration present in that linear combination. If w has higher filtration, its leading component is a cycle in degree one of the associated graded complex. Positive-degree exactness there allows us to subtract a lifted boundary dv, $v \in C_2$, from w and lower its filtration without changing dw. Repetition gives a preimage of filtration at most M. Its leading boundary would now be a relation among the allowed points in the graded summand at M. Those points form a free basis of its zero-degree homology, so all their coefficients must vanish. Descending M proves independence. Spanning was proved by (9.60). Thus the claimed points form a basis of $H_0(C_\bullet) = \mathcal{D}_q$. □

Base change and the rank identity

The proof works verbatim over an arbitrary commutative ring with arbitrary weights. Alternatively, the augmented complex over R_q is split exact as an underlying R_q-module complex: its degree-zero homology is free, and successive surjections onto projective kernels split. Tensoring with any R_q-algebra therefore preserves exactness. No flatness hypothesis on the target algebra is needed.

Taking the alternating ranks of the chain groups gives a useful check:

$$\operatorname{rank} \mathcal{D}_q = \sum_{S \subseteq P(q)} (-1)^{|S|} \frac{q}{p_S} = q \prod_{p \mid q} (1 - p^{-1}) = \varphi(q). \tag{9.68}$$

This rank computation alone would not prove integral freeness. The local unimodular splittings and the lifted basis are the missing integral information.

Remark 9.30 (Relation to character normal forms). Over a characteristic-zero splitting field, both the conductor-character basis in the source and the present raw-point basis describe the same free module. Their transition matrix is a legitimate field-valued change of coordinates. Its construction can involve divisions by character-group orders. The raw-point basis is designed to avoid those divisions, not to replace the arithmetic convenience of character coordinates where those coordinates are available.

9.8.3 Reflection and the exact integral torsion formula

Reflection is a separate quotient operation

The map $c(e_x) = e_{-x}$ preserves the relations and satisfies $c^2 = 1$. It acts on the entire resolution by $c[S, x] = [S, -x]$. For integer weights a_p, let $D = \mathcal{D}_q(\mathbb{Z}, a)$ and define

$$D^+ = D/(c-1)D, \qquad D^- = D/(c+1)D.$$

These are *coinvariants with a prescribed sign*, not the submodules of invariant and anti-invariant elements. Over a field of characteristic not two, the corresponding eigenspaces and coinvariants agree. Over $\mathbb{Z}$ this distinction is essential.

For any q, put

$$r(q) = \#\{p > 2 : p \mid q\} + \mathbf{1}_{4|q}. \tag{9.69}$$

For $q > 2$, we have $r(q) \geq 1$ and set $n(q) = \varphi(q)/2$, $h(q) = 2^{r(q)-1}$.

Theorem 9.31 (Weighted integral reflection theorem)**.** *Let $q > 2$ and specialize every A_p to an integer a_p. For each $\varepsilon \in \{1, -1\}$,*

$$D/(c-\varepsilon)D \simeq \mathbb{Z}^{n(q)} \oplus (\mathbb{Z}/2\mathbb{Z})^{t(q,a)}, \tag{9.70}$$

where

$$t(q,a) = \begin{cases} h(q), & a_p \text{ is odd for every odd prime } p \mid q, \\ 0, & \text{some odd prime } p \mid q \text{ has } a_p \text{ even.} \end{cases} \tag{9.71}$$

In particular the answer is independent of the integer a_2. At $q = 1$ or $q = 2$, the separate answers are $D^+ = \mathbb{Z}$ and $D^- = \mathbb{Z}/2\mathbb{Z}$.

The absence of odd torsion is an elementary property of an involution on a free lattice. The exact multiplicity is the substantive calculation. We prove it through the fixed points of the resolution.

Two elementary facts about involutions

For a module M with an involution c, define its two periodic cohomology groups by

$$\widehat{H}^0(M) = \ker(c-1)/(1+c)M, \qquad \widehat{H}^1(M) = \ker(1+c)/(c-1)M, \tag{9.72}$$

and repeat with period two. These are the cohomology groups of the complex whose differentials alternate between $c-1$ and $1+c$, with $c-1$ leaving even degree. This definition, rather than any unmentioned cohomology convention, fixes all signs and parity labels.

Lemma 9.32 (Torsion in sign coinvariants)**.** *If M is a finite-rank free abelian group, every torsion class in $M/(c-\varepsilon)M$ is killed by 2. Its torsion subgroup is $\widehat{H}^1(M)$ for $\varepsilon = 1$, and $\widehat{H}^0(M)$ for $\varepsilon = -1$.*

Proof. Suppose $kv = (c-\varepsilon)w$ with $k \neq 0$. Applying $c+\varepsilon$ gives $k(c+\varepsilon)v = 0$, hence $(c+\varepsilon)v = 0$ because M is torsion-free. Then $(c-\varepsilon)v = -2\varepsilon v$, so the class of v is killed by 2. Conversely a vector in $\ker(c+\varepsilon)$ represents a class killed by 2. Taking its quotient by $(c-\varepsilon)M$ gives exactly the two groups in (9.72). □

Lemma 9.33 (Balanced free ranks)**.** *For every field of characteristic different from two, and arbitrary weights in that field, the two reflection coinvariants of $\mathcal{D}_q$ have dimension $\varphi(q)/2$ when $q > 2$.*

Proof. On a permutation module with u basis points and f fixed points, the two eigenspace dimensions are the integers $(u+f)/2$ and $(u-f)/2$. The sign-eigenspace functors are exact when 2 is invertible, so apply this count to the resolution. The alternating sum of the u's is $\varphi(q)$ by (9.68). The fixed-point count for X_m is one if m is odd and two if m is even. Thus its alternating sum is

$$\sum_{S \subseteq P(q)} (-1)^{|S|} \gcd(2, q/p_S) = 0 \qquad (q > 2).$$

For odd q, this is $(1-1)^{|P(q)|}$. If $v_2(q) = 1$, summing first over whether $2 \in S$ leaves $(2-1)(1-1)^k$, where k is the number of odd prime divisors and is positive. If $v_2(q) \geq 2$, that first sum is $2-2$. Hence both dimensions are $\varphi(q)/2$.

This uses integer dimension counts in the chain modules, not merely a trace equality in the coefficient field. The latter alone would not determine dimensions in positive characteristic. □

Removing every nonfixed orbit from periodic cohomology

This step explains why the calculation is small despite the q original grid points. Let C_j^{mov} be the span of the nonfixed points in each summand of C_j. It is a subcomplex: if $x \neq -x$ and $py = x$, then y cannot be fixed; the term $-a_p[S \setminus \{p\}, x]$ is also nonfixed. Each term of this subcomplex is a direct sum of copies of the regular permutation module $S[\mathbb{Z}/2\mathbb{Z}]$ over the chosen coefficient ring S.

The periodic complex of a regular two-point orbit is exact over *every* commutative ring. Indeed the kernel of the difference map is the diagonal, which is the image of the sum map; the kernel of the sum map consists of $(u, -u)$, which is the image of the difference map. This assertion requires neither division by 2 nor absence of 2-torsion.

Let $K_\bullet = C_\bullet / C_\bullet^{\mathrm{mov}}$. Its generators are only the fixed symbols $[S, 0]$ and, when available, $[S, 1/2]$. The action on $K_\bullet$ is trivial. Forming the double complex of the periodic cohomology complexes gives

$$\widehat{H}^\bullet(\mathcal{D}_q(S, a)) \simeq H^\bullet \operatorname{Tot}\big(T(K_\bullet)\big). \tag{9.73}$$

Here T denotes the alternating difference/sum complex just defined, and the chain degree of K_j contributes $-j$ to total degree.

For completeness, this comparison can be justified without assuming a spectral sequence degenerates. The augmented resolution is exact after base change, so taking horizontal homology in the double complex replaces $C_\bullet$ by D in degree zero. Removing the moving subcomplex does not change total cohomology because each of its vertical columns is exact. In both statements the horizontal width is bounded by $|P(q)|$, so in each total degree only finitely many columns occur; successive elimination along these columns justifies the two comparisons. In particular, no convergence of an infinite horizontal filtration is being used.

The fixed-point complex is a Koszul complex

For a sequence $b_1, \dots, b_r$ in a ring S, let $K_S(b_1, \dots, b_r)$ denote the tensor product of the two-term complexes $S \xrightarrow{b_i} S$, with source in chain degree one. This is the Koszul complex; its definition is all that is needed below.

For an odd prime p, the fixed-point part of its distribution map is multiplication by $1 - a_p$ on each available fixed symbol. Of the p roots of a fixed point, exactly one is fixed, and is the original point. The other roots disappear in the quotient by moving orbits. The factor at 2 has two different forms.

When $v_2(q) = 1$, its source has only 0 and its target has $0, 1/2$. Its column is

$$\begin{pmatrix} 1 - a_2 \\ 1 \end{pmatrix}. \tag{9.74}$$

The unit entry splits off an identity complex, leaving one free copy of S in degree zero. Therefore this prime contributes no Koszul generator to the fixed-point complex.

When $v_2(q) \geq 2$, both grids have two fixed points. In the order $(0, 1/2)$, the matrix is

$$B(a_2) = \begin{pmatrix} 1 - a_2 & 0 \\ 1 & -a_2 \end{pmatrix}. \tag{9.75}$$

The two roots of $1/2$ are $1/4, 3/4$, which are moving points; this explains the second column. Unimodular row and column operations transform this matrix to $\operatorname{diag}(1, a_2(a_2 - 1))$, up to a unit sign. These operations are valid over $\mathbb{Z}[a_2]$ and over every specialization, because they pivot on the

displayed 1. Odd-prime maps are scalar, so the operations commute with all other directions of the complex.

We have proved the following ring-uniform reduction.

Proposition 9.34 (Fixed-point reduction)**.** *Over any commutative ring, $K_\bullet$ is chain-homotopy equivalent to*

$$K_S(\boldsymbol{\delta}), \qquad \boldsymbol{\delta} = \big(a_p - 1 : \ p > 2, \ p \mid q\big) \text{ together with } a_2(a_2-1) \text{ if } 4 \mid q. \tag{9.76}$$

It has $r(q)$ active scalar entries. At $q = 1, 2$, this is the empty Koszul complex S in degree zero.

Completing the integer calculation

For $S = \mathbb{Z}$, the periodic complex on a trivial module alternates between 0 and multiplication by 2. It separates into adjacent two-term blocks. Tensoring one such block with the free complex $K_{\mathbb{Z}}(\boldsymbol{\delta})$ computes $K_{\mathbb{Z}}(\boldsymbol{\delta}) \otimes \mathbb{F}_2$: multiplication by 2 is a free resolution of reduction modulo two. Consequently

$$\widehat{H}^{\varepsilon}(D) \simeq \bigoplus_{\substack{0 \le j \le r(q) \\ j \equiv \varepsilon \ (2)}} H_j\big(K_{\mathbb{F}_2}(\overline{\boldsymbol{\delta}})\big), \qquad \varepsilon = 0, 1. \tag{9.77}$$

Only an isomorphism of abelian groups is asserted here; no extra multiplicative structure is required.

If some odd a_p is even, its active scalar reduces to 1, so its two-term complex is contractible and all the homology vanishes. Otherwise every odd-prime scalar is zero modulo two. The remaining scalar $a_2(a_2 - 1)$, when present, is also zero modulo two for every integer a_2. All differentials in the reduced active complex therefore vanish, and

$$\dim_{\mathbb{F}_2} H_j = \binom{r(q)}{j}.$$

For $r(q) \ge 1$, the sums over even and odd j are both $2^{r(q)-1}$. Lemmas 9.32 and 9.33 now prove Theorem 9.31. When $r(q) = 0$, the only homology is in degree zero, giving the stated distinct answers for $q = 1, 2$.

Examples and the classical specialization

At level four, put $a = a_2$. The basis is $(e_0, e_{3/4})$, and

$$e_{1/2} = (a-1)e_0, \qquad e_{1/4} = a(a-1)e_0 - e_{3/4}, \qquad c = \begin{pmatrix} 1 & a(a-1) \\ 0 & -1 \end{pmatrix}.$$

For integer a, the nonzero Smith factor of either $c - 1$ or $c + 1$ is 2, because $a(a-1)$ is even. This two-dimensional example exhibits both the unavoidable parity and the irrelevance of the integer value of a_2.

For all odd-prime weights odd, the following values result:

q	4	12	15	30	60	105
$n(q)$	1	2	4	4	8	24
$r(q)$	1	2	2	2	3	3
$h(q)$	1	2	2	2	4	4

For example, either sign at level 60 gives $\mathbb{Z}^8 \oplus (\mathbb{Z}/2)^4$ in the exceptional parity case, but only $\mathbb{Z}^8$ as soon as a_3 or a_5 is even.

When all $a_p = 1$, the count agrees with the classical ordinary sign-cohomology theorem of Kubert [92]; twice an odd level has the same effective prime count as that odd level. The weighted parity criterion, the fixed-point matrices, and the later jet calculation are the present explicit continuation of the manuscript's integral question.

9.8.4 Characteristic two, the universal support, and all jet lengths

A characteristic-two homology theorem over arbitrary algebras

Assume now that the coefficient ring S has characteristic two. Write $N = c - 1 = c + 1$. Then $N^2 = 0$, and both periodic cohomology groups are the same module $\ker N / \operatorname{im} N$. On a fixed-point module both periodic differentials vanish. The comparison (9.73) therefore has a particularly simple consequence.

Theorem 9.35 (Koszul model in characteristic two)**.** *For every commutative characteristic-two ring S and every choice of its weights,*

$$\frac{\ker(N : \mathcal{D}_q(S,a) \to \mathcal{D}_q(S,a))}{\operatorname{im} N} \simeq \bigoplus_{j=0}^{r(q)} H_j(K_S(\boldsymbol{\delta})), \tag{9.78}$$

where the active sequence is (9.76)*.*

Proof. Remove the moving-orbit subcomplex as in Section 9.8.3. On its fixed quotient the vertical periodic differential is zero. Total cohomology is therefore the direct sum of the horizontal homology groups, one from each of the finitely many chain degrees. Apply the chain-homotopy reduction of Proposition 9.34. □

The universal obstruction is a reduced complete intersection

Let $R_2 = \mathbb{F}_2[A_p : p \mid q]$ and let $I_q \subseteq R_2$ be the ideal

$$I_q = (A_p - 1 : \ p > 2,\ p \mid q) + (A_2(A_2 - 1) : \ 4 \mid q), \tag{9.79}$$

where the second summand is omitted unless indicated. Its generators form a regular sequence. Indeed the odd-prime entries are independent linear variables shifted by one; after removing them, the possible last entry is a nonzero monic polynomial in the still-free variable A_2. At each step multiplication by that entry is injective in the quotient by the earlier ones.

The Koszul complex on a regular sequence resolves its quotient. This can be seen inductively here without a general regular-sequence criterion: tensor a free resolution of the previous quotient with $[R_2 \xrightarrow{\delta} R_2]$. Its homology consists of the kernel and cokernel of multiplication by δ on that quotient. The kernel is zero, and the cokernel is the next quotient.

Corollary 9.36 (Universal characteristic-two support)**.** *Over R_2, the reflection homology is a cyclic module:*

$$\ker N / \operatorname{im} N \simeq R_2 / I_q. \tag{9.80}$$

Its zeroth Fitting ideal is exactly I_q. The closed locus defined by I_q is reduced. For odd $q > 1$ it is the single point with every weight equal to one. For $4 \mid q$ it has two points, with all odd-prime weights equal to one and $A_2 = 0$ or 1. For q twice an odd number, A_2 is free and the locus is an affine line.

Proof. The positive Koszul homology vanishes by the preceding argument; apply Theorem 9.35. A cyclic module R_2/I_q has zeroth Fitting ideal I_q by its one-generator presentation. After the odd linear equations, the only possible further equation is $A_2(A_2-1)$, whose two factors are relatively prime. Its quotient is a product of two fields, or of two copies of the remaining coefficient ring if one extends scalars. The stated descriptions and reducedness follow. □

The cyclic universal answer does *not* imply that every fiber of the reflection homology is one-dimensional. Homology need not commute with a nonflat specialization. At an exceptional field-valued point the differentials of the specialized Koszul complex all vanish, so its homology has total dimension $2^{r(q)}$. This is the precise source of the larger fiber defect.

There is also an integral universal formulation. With the periodic convention (9.72), the same fixed-complex calculation, followed by reduction modulo two, gives

$$\widehat{H}^{2j}(\mathcal{D}_q) \simeq R_q/(2,\boldsymbol{\delta}), \qquad \widehat{H}^{2j+1}(\mathcal{D}_q) = 0. \tag{9.81}$$

Here $\mathcal{D}_q$ is still over the polynomial ring, not an integer weight fiber. The regularity argument is in $R_q/2R_q$. Odd periodic cohomology appearing after integer specialization is another nonflat-base-change effect, not a contradiction to (9.81).

The exact rank locus over every field

For a field k of characteristic two, a nonzero active scalar is a unit and makes the Koszul complex contractible. If all active scalars vanish, its differentials vanish and its total homology dimension is $2^{r(q)}$. To pass from homology to the desired coinvariants, note that an endomorphism N with $N^2=0$ on a d-dimensional vector space satisfies

$$\dim(\ker N/\operatorname{im} N) = d - 2\operatorname{rank} N, \qquad \dim\operatorname{coker} N = \frac{d+\dim(\ker N/\operatorname{im} N)}{2}. \tag{9.82}$$

Theorem 9.37 (All-field reflection dimension)**.** *Let $q>2$, let k be a field, and use arbitrary weights $a_p \in k$. If $\operatorname{char} k \neq 2$, both sign coinvariants have dimension $n(q)$. If $\operatorname{char} k = 2$, the signs coincide and*

$$\dim_k \mathcal{D}_q(k,a)/(c-1)\mathcal{D}_q(k,a) = n(q) + h(q)\mathbf{1}_{\mathcal{E}_q(a)}, \tag{9.83}$$

where $\mathcal{E}_q(a)$ means

$$a_p = 1 \quad \textit{for every odd } p \mid q, \qquad a_2 \in \{0,1\} \quad \textit{if } 4 \mid q.$$

There is no condition on a_2 when $v_2(q)=1$.

Proof. For characteristic different from two use Lemma 9.33. In characteristic two use the Koszul calculation and (9.82) with $d=\varphi(q)$. Over a field, $a_2(a_2-1)=0$ is equivalent to $a_2=0$ or 1. □

Over $\mathbb{F}_2$, the condition at 2 is automatic. Over $\mathbb{F}_4$ it is not: a weight satisfying $a_2^2+a_2+1=0$ is neither zero nor one, and the reflection defect disappears when $4 \mid q$. Thus an experiment restricted to binary weights would miss part of the geometry. The exact regression suite includes all four field weights at selected levels to test this distinction.

A complete one-parameter Smith law

Let k have characteristic two, let $O = k[[t]]$, and choose power series weights $a_p(t)$. The module $\mathcal{D}_q(O, a)$ is free of rank $2n = \varphi(q)$. Form the active sequence $\boldsymbol{\delta}(t)$ as before, and define its contact order by

$$d = \min_i v_t(\delta_i(t)), \qquad v_t(0) = \infty. \tag{9.84}$$

The word "contact" refers to the pullback of the explicit ideal I_q along the chosen formal curve. It is not an approximation to a root of a transcendental function.

Each active prime contributes a factor at least two to $\varphi(q)$, so $\varphi(q) \geq 2^{r(q)}$ and $n \geq h$. Thus all block sizes in the following statement are nonnegative.

Theorem 9.38 (Contact-order Smith law)**.** *Let $q > 2$, $n = n(q)$, $h = h(q)$, and use the above characteristic-two power series weights. If $d < \infty$, the Smith diagonal of $N = c - 1$ over O is, up to units,*

$$\operatorname{diag}\big(\underbrace{1, \dots, 1}_{n-h}, \underbrace{t^d, \dots, t^d}_{h}, \underbrace{0, \dots, 0}_{n}\big). \tag{9.85}$$

When $d = 0$, the first two blocks together are n unit entries. When $d = \infty$, replace the h entries t^d by zero; there are then $n - h$ unit entries and $n + h$ zeros.

For every integer $L \geq 1$, put $O_L = k[t]/(t^L)$. The complete jet cokernel formula is

$$\dim_k \frac{\mathcal{D}_q(O_L, a)}{(c-1)\mathcal{D}_q(O_L, a)} = nL + h \min(d, L), \tag{9.86}$$

with $\min(\infty, L) = L$.

Proof. First work over O_L and put $d_L = \min(d, L)$. Elementary invertible operations on the active sequence identify its Koszul complex with that on $(t^{d_L}, 0, \dots, 0)$, with the convention that $t^L = 0$. To see this explicitly when $d_L < L$, choose an entry with least valuation, multiply it by a unit to make it t^{d_L}, and subtract its multiples from all other entries. Every other entry is divisible by it in O_L. If $d_L = L$, all entries already vanish. Invertible changes of the sequence induce invertible changes on its exterior algebra, hence isomorphisms of Koszul complexes.

The kernel and cokernel of multiplication by t^{d_L} on O_L each have k-dimension d_L. Tensoring with $r - 1$ zero-differential two-term complexes multiplies total homology dimension by 2^{r-1}. The right-hand side of (9.78) therefore has k-dimension $2^r d_L$. The underlying vector space of $\mathcal{D}_q(O_L, a)$ has dimension $2nL$, by the integral basis theorem. Applying (9.82) proves (9.86).

It remains to show that these length formulas determine all Smith exponents, rather than just their sum. A matrix over the discrete valuation ring O has nonzero invariant factors t^{e_i} up to units, together with some zero factors. If there are z zero factors, its cokernel modulo t^L has length

$$zL + \sum_i \min(e_i, L).$$

The first difference in L is $z + \#\{i : e_i \geq L\}$ for $L \geq 1$. For finite d, the formula already proved has first difference $n + h$ for $1 \leq L \leq d$, and n for $L > d$. Thus exactly n factors are zero and exactly h nonunit factors have exponent d; the remaining $n - h$ factors are units. For $d = 0$ there are no positive exponents. For $d = \infty$, the first difference is always $n + h$, so there are $n + h$ zero factors and no positive finite exponents. This proves every case of (9.85). □

Examples of the jet law

At level 15, take

$$a_3(t) = 1 + t^2, \qquad a_5(t) = 1 + t^5.$$

Here $(n, r, h) = (4, 2, 2)$ and $d = 2$. The reflection matrix has Smith entries $1, 1, t^2, t^2, 0, 0, 0, 0$, and its jet cokernel dimensions at $L = 1, 2, 3, 4$ are $6, 12, 16, 20$. A rank calculation at $t = 0$ alone would detect the first number, but not the exponent two.

At level 12, take $a_2(t) = t^3$ and $a_3(t) = 1 + t^5$. Then $a_2(a_2 - 1) = t^3(1 + t^3)$, so $d = 3$. Since $(n, h) = (2, 2)$, there are no unit Smith factors and the diagonal is $t^3, t^3, 0, 0$. Every jet of length at most three has trivial reflection matrix up to equivalence; later lengths detect the nonzero factors.

If the entire curve is contained in the exceptional locus, then $d = \infty$ and the dimension is $(n + h)L$. If any active scalar has a nonzero constant term, $d = 0$ and the dimension is nL. These are exact boundary cases of the same formula. In contrast, the *unreflected* distribution module always has dimension $\varphi(q)L$ at length L and never exhibits this jump.

9.8.5 Integral jets and why abelian freeness is insufficient

For a finite free $\mathbb{Z}$-algebra B, put $C_\pm(\mathcal{D}_B) = \mathcal{D}_q(B, a)/(c \mp 1)\mathcal{D}_q(B, a)$. The scalar Tate calculation extends to this coefficient ring.

Proposition 9.39. *With $\overline{B} = B/2B$ and the reduced active sequence $\overline{\boldsymbol{\delta}}$, the two integer-torsion submodules are*

$$\operatorname{Tor}_{\mathbb{Z}} C_+(\mathcal{D}_B) \simeq \bigoplus_{j \text{ odd}} H_j(K_{\overline{B}}(\overline{\boldsymbol{\delta}})), \qquad \operatorname{Tor}_{\mathbb{Z}} C_-(\mathcal{D}_B) \simeq \bigoplus_{j \text{ even}} H_j(K_{\overline{B}}(\overline{\boldsymbol{\delta}})).$$

These are isomorphisms of B-modules. Every torsion class is killed by two. For $q > 2$ and $T = \operatorname{rank}_{\mathbb{Z}} B$, the free abelian rank of both quotients is $T\varphi(q)/2$.

Proof. The underlying distribution lattice is free over $\mathbb{Z}$ by the all-ring basis theorem. Thus the torsion lemma still applies. In the fixed-point periodic complex, multiplication by two is injective on B and its free chain modules. Its adjacent two-term blocks resolve reduction modulo two, exactly as in (9.77); the bounded chain width gives the same parity sums as B-modules. Over $B \otimes \mathbb{Q}$, reflection eigenspaces are exact direct summands. The fixed-point Euler count in the balanced-rank proof is multiplied by T and remains zero, giving the asserted free ranks. This argument does not assume that the reflected torsion-free part is a free B-module. □

9.8.6 An exact Smith formula for arbitrary finite integral jets

Let

$$B_L = \mathbb{Z}[u]/(u^L), \quad L \geq 1, \qquad a_p(u) \in B_L,$$

and reduce coefficients modulo two. In $\overline{B}_L = \mathbb{F}_2[u]/(u^L)$, form the parameters (9.76). For an element f of this ring, define

$$\nu(f) = \min\{j : [u^j]f \neq 0\}, \qquad \nu(0) = L, \quad v = \min_{f \in \boldsymbol{f}} \nu(f).$$

Here $\boldsymbol{f} = \overline{\boldsymbol{\delta}}$ and $r(q)$ is the active-prime count in (9.69). For $q > 2$, the list is nonempty.

Theorem 9.40 (Minimum-valuation jet formula)**.** *For $q > 2$, the two reflected jet quotients have the following Smith decomposition as abelian groups:*

$$C_{\pm}(\mathcal{D}_{B_L}) \simeq \mathbb{Z}^{L\varphi(q)/2} \oplus (\mathbb{Z}/2\mathbb{Z})^{v\,2^{r(q)-1}}. \tag{9.87}$$

More precisely, their integer-torsion submodules are each isomorphic, as B_L-modules, to

$$\big(B_L/(2,u^v)\big)^{2^{r(q)-1}}. \tag{9.88}$$

The direct-sum decomposition in (9.87) is a statement over $\mathbb{Z}$, not a claim that the torsion-free part is free over B_L.

Proof. In the principal ideal ring $\overline{B}_L$, choose a parameter of least valuation. It is a unit times u^v unless every parameter is zero, which is the same argument with $v = L$. A change of basis in the free module defining the Koszul complex transforms the parameter list into $(u^v, 0, \dots, 0)$: first divide by its unit factor, then subtract its multiples from the other entries. All these are invertible operations over $\overline{B}_L$.

The complex is now the tensor product of the two-term complex given by multiplication by u^v and $r(q) - 1$ zero-differential exterior factors. Both the kernel and the cokernel of multiplication by u^v are isomorphic to $\overline{B}_L/(u^v)$; the kernel is the ideal $u^{L-v}\overline{B}_L$. Consequently

$$H_k\big(K_{\overline{B}_L}(\boldsymbol{f})\big) \simeq \big(\overline{B}_L/(u^v)\big)^{\binom{r(q)}{k}}.$$

Summing either degree parity in Theorem 9.39 proves (9.88). Each copy has $\mathbb{F}_2$-dimension v, so Lemma 9.32 gives the torsion multiplicity in (9.87).

The unreflected underlying lattice has rank $L\varphi(q)$. The alternating fixed-point trace used in Theorem 9.31 is multiplied by L, and still vanishes. Thus each rational eigenspace has dimension $L\varphi(q)/2$. This proves the free rank and completes the Smith calculation. □

Example 9.41 (Five jet coefficients at level fifteen). Take $L = 5$, $a_3(u) = 1+u^2$, and $a_5(u) = 1+2u+u^3$. The binary parameters are u^2, u^3, so $v = 2$ and $r(q) = 2$. Therefore

$$C_{\pm}(\mathcal{D}_{B_5}) \simeq \mathbb{Z}^{20} \oplus (\mathbb{Z}/2\mathbb{Z})^4.$$

The constant jets $a_3 = a_5 = 1$ would instead give ten two-torsion factors. The constant terms of the weights agree; their higher binary coefficients change the integral reflection structure.

Example 9.42 (The two-adic jet parameter matters). At $q = 12$, $L = 4$, let $a_2(u) = 1 + u$ and $a_3(u) = 1 + u^3$. The two parameters are u^3 and $u + u^2$, not simply $1 - a_3$ and $1 - a_2$. Thus $v = 1$, $r(q) = 2$, and

$$C_{\pm}(\mathcal{D}_{B_4}) \simeq \mathbb{Z}^8 \oplus (\mathbb{Z}/2\mathbb{Z})^2.$$

Although the parity of a *scalar* two-prime weight does not affect Theorem 9.31, a nonconstant two-prime jet can affect Theorem 9.40.

Why the free abelian part need not be jet-free

Take $q = 3$, $L = 2$, and $a_3 = 1 + u$. In the unreflected basis $e = e_0$, $f = e_{2/3}$, reflection satisfies

$$ce = e, \qquad cf = ue - f.$$

Hence

$$C_{+}(\mathcal{D}_{B_2}) = B_2 e \oplus B_2 f \big/ \langle ue - 2f \rangle. \tag{9.89}$$

As an abelian group this has free generators e, f and a torsion generator uf of order two. In its torsion-free quotient, multiplication by u sends e to $2f$ and f to zero. This is not a free rank-one B_2-module: the image of u in a free rank-one module is a primitive rank-one sublattice, whereas here the image is $2\mathbb{Z}f$. Thus a claim of B_L-freeness for the reflected torsion-free part would be false. By contrast, $C_- = B_2 f \oplus (B_2/(2,u))e$ in this example. Both signs have the same abelian Smith factors, but different module extensions over the jet ring.

Multivariate jets and arithmetic weights

For a finite free $\mathbb{Z}$-algebra B of rank L, the Koszul theorem remains an exact answer. If $q > 2$ and $h_k = \dim_{\mathbb{F}_2} H_k(K_{B/2B}(\boldsymbol{f}))$, then

$$\operatorname{Tor}_{\mathbb{Z}} C_+ \simeq (\mathbb{Z}/2\mathbb{Z})^{\sum_{k \text{ odd}} h_k}, \qquad \operatorname{Tor}_{\mathbb{Z}} C_- \simeq (\mathbb{Z}/2\mathbb{Z})^{\sum_{k \text{ even}} h_k}.$$

The two sums are equal: the Euler characteristic of the Koszul complex is $L(1-1)^{r(q)} = 0$. A closed minimum-valuation description is special to the one-variable principal ideal ring; it must not be transferred unchanged to a multivariate Artinian ring.

For actual positive integral polylogarithm orders, $A_p = p^{1-n}$ usually belongs to $\mathbb{Z}[1/q]$, rather than $\mathbb{Z}$. The corresponding coefficient ring must be stated. If q is even, inverting q also inverts two and removes the reflected two-torsion. For odd q, every p^{1-n} is one modulo two, and the ordinary maximal multiplicity persists over $\mathbb{Z}[1/q]$. These are statements about the chosen formal coefficient lattice, not about independence of the evaluated periods.

9.8.7 Separable multivariable jets: an exact product law

The one-variable minimum-valuation formula has a useful multivariable extension when the active parameters separate into coordinate powers. The general Koszul answer remains necessary for parameters that do not have this form. In particular, replacing several coordinate valuations by their minimum would give the wrong answer here.

Let $q > 2$, put $r = r(q)$, $n = \varphi(q)/2$, $h = 2^{r-1}$, and set

$$S = k[t_1, \dots, t_r]/(t_1^{L_1}, \dots, t_r^{L_r}), \qquad L_i \geq 1, \qquad \operatorname{char} k = 2.$$

Suppose the active sequence, after permutation and multiplication of individual entries by units, is $(t_1^{d_1}, \dots, t_r^{d_r})$, where $d_i \geq 1$; a zero entry is allowed by taking $d_i = \infty$. Write $v_i = \min(d_i, L_i)$ and $T = \prod_i L_i$.

Theorem 9.43 (Multivariable reflection product law)**.** *Under these hypotheses,*

$$\dim_k H_j(K_S(\boldsymbol{\delta})) = \binom{r}{j} \prod_{i=1}^{r} v_i, \qquad 0 \leq j \leq r, \tag{9.90}$$

$$\dim_k \mathcal{D}_q(S,a)/(c-1)\mathcal{D}_q(S,a) = nT + h \prod_{i=1}^{r} v_i. \tag{9.91}$$

The unreflected dimension is $\varphi(q)T$. For integral weights in $B = \mathbb{Z}[t_1, \dots, t_r]/(t_1^{L_1}, \dots, t_r^{L_r})$ whose reduced active sequence has the above form, both reflected coinvariants have the abelian-group decomposition

$$C_\pm(\mathcal{D}_B) \simeq \mathbb{Z}^{nT} \oplus (\mathbb{Z}/2\mathbb{Z})^{h \prod_i v_i}.$$

Their integer-torsion submodules are, more precisely, isomorphic as B-modules to

$$(B/(2, t_1^{v_1}, \dots, t_r^{v_r}))^h .$$

The integral assertion uses $k = \mathbb{F}_2$.

Proof. Multiplying an active entry by a unit and permuting entries induces an invertible change of the exterior generators of its Koszul complex. After that change, the complex is the tensor product over k of the two-term complexes

$$k[t_i]/(t_i^{L_i}) \xrightarrow{t_i^{d_i}} k[t_i]/(t_i^{L_i}).$$

Each has kernel and cokernel of dimension v_i, in degrees one and zero respectively. As a module over its truncated polynomial ring, each homology group is isomorphic to $k[t_i]/(t_i^{v_i})$; multiplication by $t_i^{L_i - v_i}$ gives the kernel isomorphism. This includes the zero-map case $v_i = L_i$.

The Künneth formula over the field k has no Tor corrections. In total degree j there is one tensor summand for each j-element subset of the r factors. Each summand has dimension $\prod_i v_i$ and is isomorphic, as an S-module, to $S/(t_1^{v_1}, \dots, t_r^{v_r})$. This proves (9.90) and its module refinement. The characteristic-two fixed-point theorem gives total reflection homology of dimension $2^r \prod_i v_i$. Since $\mathcal{D}_q(S, a)$ has dimension $2nT$, the square-zero rank identity (9.82) gives (9.91). For the integral statement apply the finite-free-algebra Tate formula of Proposition 9.39. The even and odd binomial sums are both 2^{r-1}; the balanced-rank calculation gives free rank nT. Finally a finitely generated abelian group splits into its free part and its torsion subgroup. No splitting as B-modules or freeness of its torsion-free part is claimed. □

Example 9.44 (Two independent deformation directions at level fifteen). Take $B = \mathbb{Z}[t, u]/(t^3, u^4)$, $a_3 = 1 + t^2$ and $a_5 = 1 + u^3$. Then $n = 4$, $r = 2$, $h = 2$, $T = 12$ and $(v_1, v_2) = (2, 3)$. Both integral signs give

$$C_{\pm}(\mathcal{D}_B) \simeq \mathbb{Z}^{48} \oplus (\mathbb{Z}/2\mathbb{Z})^{12},$$

and the characteristic-two reflected dimension is 60. The defect depends on the product $2 \cdot 3$, not on a single minimum. At an active two-prime coordinate one may take $a_2 = t_i^{d_i}$ or $a_2 = 1 + t_i^{d_i}$: in either case $a_2(a_2 - 1) = t_i^{d_i}(1 + t_i^{d_i})$, and the second factor is a unit.

The independent raw-matrix verifier expands the original prime rows and reflection rows in point/monomial coordinates over $\mathbb{F}_2$. It imports neither the normal-form generator nor the Koszul calculation. It checks 210 finite cases, including both two-prime branches, vanished coordinate powers, unequal jet lengths, and a three-prime level. A corruption control rejects omission of reflection rows. These computations corroborate the formula; the proof above establishes its full scope.

9.8.8 Exact cyclotomic identities and pole-cancelled spectral jets

Evaluating an integral certificate

For $x \in \mathbb{Q}/\mathbb{Z}$, write

$$\mathcal{L}_s(x) = \mathrm{Li}_s(e^{2\pi i x}).$$

At $x = 0$, this means $\zeta(s)$. For $x \neq 0$, the spectral function is entire. One way to see this, and to fix its continuation, is the finite residue-class formula

$$\mathcal{L}_s(a/q) = q^{-s} \sum_{u=1}^{q} e^{2\pi i a u/q} \zeta(s, u/q). \tag{9.92}$$

For $a \not\equiv 0 \pmod{q}$, the residues of the simple poles at $s = 1$ sum to zero. The standard meromorphic continuation of Hurwitz zeta has no other spectral poles [93].

For $\Re s > 1$, absolute convergence gives

$$\sum_{py=x} \mathcal{L}_s(y) = p^{1-s}\mathcal{L}_s(x). \tag{9.93}$$

Indeed the root sum cancels every series index not divisible by p; for index pk it contributes $pe^{2\pi ikx}$. Continuation then proves the meromorphic identity for every complex s. Thus an integral polynomial certificate becomes an exact analytic identity under

$$e_x \longmapsto \mathcal{L}_s(x), \qquad A_p \longmapsto p^{1-s}, \tag{9.94}$$

where real positive p fixes the exponential unambiguously. No integer relation search or floating-point acceptance threshold occurs in this step.

Proposition 9.45 (A removable endpoint coefficient)**.** *For $y \neq 0$, the coefficient $c_{y,0}(\mathbf{A})$ in its integral normal form vanishes at $A_p = 1$ for all p. Hence its polylogarithmic specialization cancels the simple pole of $\zeta(s)$ at $s = 1$.*

Proof. At ordinary weights, the functional taking e_0 to 1 and all other grid symbols to 0 annihilates every distribution row. For $x = 0$, both the root sum and e_x contain exactly one copy of e_0; for $x \neq 0$, neither does. Apply the functional to the normal form of e_y. The second assertion follows because $p^{1-s} = 1$ at $s = 1$ and the coefficient is analytic in s. □

The vanishing need only be first order. It is not legitimate to infer one vanishing factor for each prime merely from the number of prime variables in a raw-point normal form.

A three-row level-twelve identity

In the level-12 free module, define

$$F_{12} = e_{1/12} + e_{5/12} + (A_3 + 1)e_{3/4} - A_2A_3(A_2 - 1)e_0.$$

The following is an equality before taking the quotient:

$$F_{12} = r_{3,1/4} + A_3r_{2,1/2} + A_2A_3r_{2,0}. \tag{9.95}$$

For example, the first row contains $e_{1/12} + e_{5/12} + e_{3/4} - A_3e_{1/4}$; the next two rows cancel $e_{1/4}$ and then $e_{1/2}$. This verifies the certificate by direct inspection.

Corollary 9.46. *For every complex s, with the removable value used at $s = 1$,*

$$\mathcal{L}_s(1/12) + \mathcal{L}_s(5/12) + (3^{1-s} + 1)\mathcal{L}_s(3/4) = 6^{1-s}(2^{1-s} - 1)\zeta(s). \tag{9.96}$$

At $s = 1$, the right-hand side is $-\log 2$.

Proof. Evaluate (9.95) using (9.94). The value at one follows from $2^{1-s} - 1 = -(s-1)\log 2 + O((s-1)^2)$. □

A four-row level-fifteen identity

Set

$$U = \{1/15, 4/15, 7/15, 2/5, 4/5\}. \tag{9.97}$$

In the level-15 free module define

$$F_{15} = e_{8/15} - \sum_{u\in U} e_u - (A_3+1)e_{3/5} - (A_5+1)e_{2/3} - (1 - A_3A_5)e_0. \tag{9.98}$$

A short exact certificate is

$$F_{15} = r_{3,3/5} - r_{5,1/3} - A_5 r_{3,0} - r_{5,0}. \tag{9.99}$$

The roots in the first two rows are respectively

$$\{1/5, 8/15, 13/15\}, \qquad \{1/15, 4/15, 7/15, 2/3, 13/15\}.$$

The 13/15 terms cancel. The last two rows remove 1/3 and 1/5, and leave precisely (9.98). In particular the coefficient of e_0 in the normal form of $e_{8/15}$ is $1 - A_3A_5$, not $(A_3 - 1)(A_5 - 1)$.

Theorem 9.47 (All-order level-fifteen identity)**.** *For every complex spectral order s,*

$$\begin{aligned}\mathcal{L}_s(8/15) - \sum_{u\in U} \mathcal{L}_s(u) - (3^{1-s}+1)\mathcal{L}_s(3/5) - (5^{1-s}+1)\mathcal{L}_s(2/3)& \\ = (1 - 15^{1-s})\zeta(s).&\end{aligned} \tag{9.100}$$

Both sides extend holomorphically through $s = 1$, where the right-hand side is $\log 15$.

Proof. Apply (9.94) to the four-row identity. The endpoint value follows from $1 - 15^{1-s} = (s - 1)\log 15 + O((s-1)^2)$ and the residue one of $\zeta(s)$. □

The theorem supplies infinitely many exact identities, including ordinary positive integer weights, negative orders, and derivatives in spectral order. It is generated by classical distribution rather than by a new numerical relation conjecture.

A compatible level-30 normal form follows from the extra row $r_{2,1/15}$:

$$\begin{aligned}e_{1/30} = (A_2 - 1)e_{1/15} - (A_3+1)e_{3/5} - (A_5+1)e_{2/3}& \\ - e_{4/15} - e_{7/15} - e_{2/5} - e_{4/5} + (A_3A_5 - 1)e_0.&\end{aligned} \tag{9.101}$$

Its residual has the five-row certificate

$$r_{2,1/15} - r_{3,3/5} + r_{5,1/3} + A_5 r_{3,0} + r_{5,0}.$$

All rows can be read in the level-30 grid. This illustrates compatibility of the straightener with an enlarged level and a weight whose order-one specialization vanishes in a particular coefficient.

Every spectral derivative, with the pole term retained

Let

$$\mathcal{L}^{[j]}(x) = \frac{\partial^j}{\partial s^j}\mathcal{L}_s(x)\Big|_{s=1}, \qquad x \neq 0,$$

and write the Laurent expansion

$$\zeta(1+t)=\frac{1}{t}+\sum_{m\ge 0}\frac{(-1)^m\gamma_m}{m!}t^m. \tag{9.102}$$

For a positive prime p and $N\ge 0$, define the explicit finite operator

$$\mathcal{T}_{p,N}(x)=\mathcal{L}^{[N]}(x)+\sum_{j=0}^{N}\binom{N}{j}(-\log p)^j\mathcal{L}^{[N-j]}(x). \tag{9.103}$$

It is the Nth derivative of $(p^{1-s}+1)\mathcal{L}_s(x)$ at $s=1$.

Theorem 9.48 (A level-fifteen Stieltjes–polylogarithm jet family)**.** *For every integer $N\ge 0$, put $\ell=\log 15$. Then*

$$\begin{aligned}\mathcal{L}^{[N]}(8/15)-\sum_{u\in U}\mathcal{L}^{[N]}(u)-\mathcal{T}_{3,N}(3/5)-\mathcal{T}_{5,N}(2/3)\\ =(-1)^N\left\{\frac{\ell^{N+1}}{N+1}-\sum_{j=1}^{N}\binom{N}{j}\ell^j\gamma_{N-j}\right\}.\end{aligned} \tag{9.104}$$

The sum on the right is empty at $N=0$.

Proof. All nonzero-root terms of (9.100) are entire in s, so differentiate the holomorphic continuation N times. The weighted terms give (9.103) by Leibniz's rule. For the right side use

$$1-e^{-\ell t}=\sum_{j\ge 1}\frac{(-1)^{j+1}\ell^j}{j!}t^j$$

and multiply by (9.102). The pole term contributes $(-1)^N\ell^{N+1}/(N+1)$ to the Nth derivative. The regular Laurent coefficients contribute $(-1)^{N+1}\sum_{j=1}^{N}\binom{N}{j}\ell^j\gamma_{N-j}$. This is the stated formula. □

The first derivative gives the especially compact identity

$$\begin{aligned}\mathcal{L}^{[1]}(8/15)-\sum_{u\in U}\mathcal{L}^{[1]}(u)-2\mathcal{L}^{[1]}(3/5)-2\mathcal{L}^{[1]}(2/3)+\log 3\,\mathcal{L}^{[0]}(3/5)+\log 5\,\mathcal{L}^{[0]}(2/3)\\ =\gamma_0\log 15-\tfrac{1}{2}\log^2 15.\end{aligned} \tag{9.105}$$

For $N=2$, the scalar right-hand side is

$$\frac{\log^3 15}{3}-\gamma_0\log^2 15-2\gamma_1\log 15.$$

These terms would be wrong if one substituted $s=1$ into the coefficient of $\zeta(s)$ before cancelling its pole.

More generally, if an analytic coefficient $c(t)$ satisfies $c(0)=0$, then

$$\frac{d^N}{dt^N}\{c(t)\zeta(1+t)\}\Big|_{t=0}=\frac{c^{(N+1)}(0)}{N+1}+\sum_{j=0}^{N}\binom{N}{j}c^{(j)}(0)(-1)^{N-j}\gamma_{N-j}. \tag{9.106}$$

This coefficient identity gives a general implementation rule for every raw-point certificate, not just the displayed level.

An elementary endpoint and branch check

For $0 < x < 1$, the chosen branch gives

$$\mathcal{L}^{[0]}(x) = -\log(2\sin \pi x) + i\pi(1/2 - x).$$

Taking real parts of (9.100) at $s = 1$ gives

$$\frac{\prod_{u\in U}(2\sin \pi u)(2\sin(3\pi/5))^2(2\sin(2\pi/3))^2}{2\sin(8\pi/15)} = 15.$$

Since $\sin(8\pi/15) = \sin(7\pi/15)$, it simplifies to

$$(2\sin(\pi/15))(2\sin(4\pi/15))(2\sin(2\pi/5))^3(2\sin(\pi/5)) = 5. \tag{9.107}$$

All factors are positive, so exponentiating the real logarithmic identity introduces no ambiguity. The imaginary parts cancel by the displayed linear argument formula. This is an elementary specialization and branch check, not a claim of novelty for the trigonometric product itself.

A division-free Hurwitz–Stieltjes translation

The same polynomial certificates evaluate under

$$A_p = p^s, \qquad e_x \mapsto Z_x(s), \qquad Z_x(s) = \begin{cases} \zeta(s,x), & 0 < x < 1, \\ \zeta(s,1), & x = 0. \end{cases}$$

The distribution identity here is $\sum_{py=x} Z_y(s) = p^s Z_x(s)$; it follows by splitting the Hurwitz series into residue classes, including the endpoint convention. Let $c_b(t)$ be the normal-form coefficient of e_b for e_x after $A_p = p^{1+t}$, and put $C(t) = \sum_b c_b(t)$. Residue comparison gives $C(0) = 1$. If $\gamma_j(b^*)$ denotes the generalized Stieltjes coefficient with $b^* = b$ for $b \neq 0$ and $b^* = 1$ for $b = 0$, expansion of the exact normal form gives, for all $N \geq 0$,

$$\gamma_N(x^*) = \frac{(-1)^N C^{(N+1)}(0)}{N+1} + \sum_{b\in\mathcal{B}_q}\sum_{j=0}^{N}(-1)^j\binom{N}{j}c_b^{(j)}(0)\gamma_{N-j}(b^*). \tag{9.108}$$

To verify the sign, multiply each Laurent series by $c_b(t)$, compare t^N, and multiply by $(-1)^N N!$. The pole part is $C(t)/t$ and contributes the first term. Thus even the underived constants require a coefficient derivative. This is the raw-basis version of the manuscript's pole-corrected jet philosophy, now with division-free integral certificates for the coefficients.

Every coefficient derivative in (9.108) is a finite integer combination of products of logarithms of primes and integer prime powers. There is no need to differentiate a solved matrix inverse or to redo a relation search separately at each derivative order.

Level-twelve derivatives and cyclotomic products

For the coefficient $c(t) = 12^{-t} - 6^{-t}$ in (9.96), set $D_j = (-\log 12)^j - (-\log 6)^j$. The finite-part rule (9.106) yields, for every $N \geq 0$,

$$\mathcal{L}^{[N]}(1/12) + \mathcal{L}^{[N]}(5/12) + \mathcal{T}_{3,N}(3/4) = \frac{D_{N+1}}{N+1} + \sum_{j=1}^{N}\binom{N}{j}D_j(-1)^{N-j}\gamma_{N-j}. \tag{9.109}$$

For example its first derivative has right side $(\log^2 12 - \log^2 6)/2 - \gamma_0 \log 2$. At order one of the polylogarithm, exponentiating the exact logarithmic identities gives, with $\zeta_q = e^{2\pi i/q}$,

$$(1-\zeta_{12})(1-\zeta_{12}^5)(1-\zeta_{12}^9)^2 = 2, \tag{9.110}$$

$$(1-\zeta_{30})(1-\zeta_{30}^8)(1-\zeta_{30}^{12})(1-\zeta_{30}^{14})(1-\zeta_{30}^{18})^2(1-\zeta_{30}^{20})^2(1-\zeta_{30}^{24}) = 15. \tag{9.111}$$

The level-thirty identity is the analytic specialization of (9.101); its endpoint coefficient has removable value $-\log 15$. Exponentiation proves the product without a branch ambiguity, while the converse would lose an additive multiple of $2\pi i$. The preserved independent polynomial verifier also proves that each product polynomial minus its displayed value is divisible by the corresponding cyclotomic polynomial.

9.9 Primitive-grid determinants, local defects and finite descent

The polynomial character normal form above remains valid over arbitrary complex coefficient algebras. Here identify the prime weights A_p with t_p, write $\mathcal{D}_q$ for that quotient, and put $v_{m,\chi} = Y_{m,\chi}$, $b_\chi = Y_{f,\chi}$ for $f = \mathrm{cond}(\chi)$. The coefficient $E_{m,\chi}$ is $A_{m,\chi}$ of Theorem 9.15. Thus

$$[v_{m,\chi}] = E_{m,\chi} b_\chi. \tag{9.112}$$

No Euler-factor division is used. Let $U_q = \{u \bmod q : (u,q) = 1\}$, and use $r = 0$ for the Hurwitz endpoint $x_r = 1$. Distinguish the rank of the full quotient from the image of its primitive-grid coordinate map.

Conductor coordinates are polynomial and globally valid, but it is often more convenient to use the $\varphi(q)$ primitive-grid points $e_{u/q}$, $u \in U_q$. Fourier-transforming those points gives $v_{q,\chi}$. By Theorem 9.15, the map from their formal span into $\mathcal{D}_q$ is diagonal in character coordinates:

$$v_{q,\chi} \longmapsto E_{q,\chi} b_\chi. \tag{9.113}$$

Its determinant in these specified Fourier and conductor bases is $\Delta_q = \prod_{\chi \bmod q} E_{q,\chi}$.

For $p^e \| q$, write

$$M_p = q/p^e, \qquad h_p = \mathrm{ord}_{M_p}(p),$$

with $h_p = 1$ when $M_p = 1$.

Theorem 9.49 (Determinant factorization)**.** *One has the polynomial identity*

$$\boxed{\Delta_q(\mathbf{A}) = \prod_{p^e \| q} A_p^{\varphi(M_p)(p^{e-1}-1)} \big(A_p^{h_p} - 1\big)^{\varphi(M_p)/h_p}.} \tag{9.114}$$

Thus primitive-grid coordinates form a basis precisely when all the factors in (9.114) are invertible in a field specialization. Their failure does not change the rank of $\mathcal{D}_q$.

Proof. Fix $p^e \| q$ and inspect the p factor of $\prod_\chi E_{q,\chi}$. Characters whose conductor is not divisible by p are exactly the characters of U_{M_p}. Their additional factors are $A_p - \chi(p)$. In the character group of U_{M_p}, the values $\chi(p)$ run through every h_pth root of unity with multiplicity $\varphi(M_p)/h_p$. This follows by duality from the fact that p has order h_p in U_{M_p}. Their product is therefore $(A_p^{h_p} - 1)^{\varphi(M_p)/h_p}$.

It remains to count the pure powers of A_p. If $a = v_p(f)$, the exponent contributed by one character is $e - \max(1, a)$. Writing this as a sum of indicators gives

$$\sum_{\chi \bmod q} \big(e - \max(1, v_p(f_\chi))\big) = \sum_{j=1}^{e-1} \#\{\chi : v_p(f_\chi) \le j\}.$$

The count in the last sum is $\varphi(p^j)\varphi(M_p)$, since these are exactly the characters induced from level $p^j M_p$. Hence the exponent is $\varphi(M_p)\sum_{j=1}^{e-1}\varphi(p^j) = \varphi(M_p)(p^{e-1}-1)$. Multiply over the primes. □

Example 9.50 (Level twelve). Let χ_3 and χ_4 be the nonprincipal quadratic characters of conductors three and four, and let $\chi_{12} = \chi_3\chi_4$. Their conductor generators are

$$b_3 = e_{1/3} - e_{2/3}, \qquad b_4 = e_{1/4} - e_{3/4},$$
$$b_{12} = e_{1/12} - e_{5/12} - e_{7/12} + e_{11/12}.$$

For $r \in U_{12}$, the normal form reads

$$[e_{r/12}] = \frac{1}{4}\big[A_2(A_2-1)(A_3-1)e_0 + \chi_3(r)A_2(A_2+1)b_3 + \chi_4(r)(A_3+1)b_4 + \chi_{12}(r)b_{12}\big]. \tag{9.115}$$

Consequently

$$\Delta_{12} = A_2^2(A_2^2-1)(A_3^2-1).$$

At $A_2 = A_3 = 1$, the primitive-grid span misses the principal conductor coordinate, even though the full quotient still has rank four.

9.9.1 Local jet defects of the primitive-grid map

Specialize to $A_p(s) = p^s$, and let $s = s_0 + t$. All exponential factors are nonzero. A factor $p^s - \chi(p)$ vanishes precisely when $p^{s_0} = \chi(p)$, and any such zero is simple because its derivative is $p^{s_0}\log p \neq 0$.

Theorem 9.51 (Local Smith factors and primitive-jet ranks)**.** *Define*

$$\rho_\chi(s_0) = \#\{p \mid q : p \nmid f_\chi,\ p^{s_0} = \chi(p)\}. \tag{9.116}$$

Over $\mathbb{C}[[t]]$, the diagonal factors of the primitive-grid map are units times $t^{\rho_\chi(s_0)}$. Consequently its rank on order-N truncated jets, as a complex linear map, is

$$\sum_{\chi \bmod q} \max\{N+1-\rho_\chi(s_0), 0\}. \tag{9.117}$$

At $s_0 = 0$, this is determined by $\rho_\chi(0) = \#\{p \mid q : p \nmid f_\chi,\ \chi(p) = 1\}$.

Proof. Factor $E_{q,\chi}(s_0+t)$ into its simple vanishing factors and a unit. The primitive Fourier matrix is constant and invertible, so these are also the local invariant factors, up to permutation and units. Multiplication by t^r on $\mathbb{C}[t]/(t^{N+1})$ has rank $\max(N+1-r, 0)$. Summing over the diagonal coordinates proves the formula. This is the rank of a *coordinate inclusion map*, not the rank of the distribution relation matrix. □

Corollary 9.52 (Multiple-prime resonances are central)**.** *If $s_0 \neq 0$, at most one prime can satisfy $p^{s_0 h_p} = 1$. At a nonzero resonant point for that prime p, the primitive-grid rank defect is $\varphi(M_p)/h_p$, and every affected local factor has order one. The order-N inclusion rank is then $(N+1)\varphi(q) - \varphi(M_p)/h_p$ for all $N \geq 0$.*

Proof. A resonance forces s_0 to be purely imaginary. Resonances at distinct primes would make $\log p/\log \ell$ rational, because each phase is a rational multiple of 2π. Unique prime factorization rules this out unless $s_0 = 0$. For the one resonant prime, the multiplicity count in Theorem 9.49 gives the number of affected characters. Apply Theorem 9.51. □

Example 9.53 (Two exact jet-rank profiles). At level 12 and $s_0 = 0$, the principal character has $\rho = 2$ and the other three characters have $\rho = 0$. For $N = 0, 1, 2$, the primitive-jet ranks are $3, 6, 10$, while the full quotient dimensions are $4, 8, 12$. At level 21, the principal character has $\rho = 2$, the character induced from conductor three has $\rho = 1$, and the other ten characters have $\rho = 0$. The inclusion ranks are $10, 21, 33$, compared with full quotient dimensions $12, 24, 36$. Indeed,

$$\Delta_{21} = (A_3^6 - 1)(A_7 - 1)^2.$$

Exact character-phase certificates for these examples are supplied.

9.10 Explicit primitive-grid reduction coefficients

Away from the exceptional locus, the conductor normal form can be inverted to express all grid points through the primitive ones. For Hurwitz weights there is a particularly concrete answer that avoids character-field arithmetic.

Let $\mathcal{S}_q$ be the multiplicative semigroup of positive integers whose prime factors divide q, including 1. For $\operatorname{Re} s > 0$, define

$$C_{r,u}(s) = \sum_{\substack{d \in \mathcal{S}_q \\ du \equiv r \ (q)}} d^{-s}, \qquad 0 \le r < q, \quad u \in U_q. \tag{9.118}$$

The series is absolutely convergent, since $\sum_{d \in \mathcal{S}_q} d^{-\sigma} = \prod_{p|q} (1 - p^{-\sigma})^{-1}$ for $\sigma > 0$.

Theorem 9.54 (Primitive-grid extension)**.** *For $\operatorname{Re} s > 0$, the $q \times \varphi(q)$ matrix $C(s)$ satisfies every prime and divisor distribution relation with weights p^s, and its primitive rows form the identity matrix. It is the unique extension matrix from primitive-grid data. In particular,*

$$\zeta(s, r/q) = \sum_{u \in U_q} C_{r,u}(s) \zeta(s, u/q) \tag{9.119}$$

with the endpoint convention for $r = 0$, initially for $\operatorname{Re} s > 1$ and then meromorphically wherever the coefficients are defined.

Proof. For a primitive row r, the congruence $du \equiv r \pmod q$ forces d to be prime to q, hence $d = 1$. This gives the identity submatrix. For a prime distribution row indexed by $a \equiv 0 \pmod p$,

$$\sum_{pr \equiv a \ (q)} C_{r,u}(s) = \sum_{\substack{d \in \mathcal{S}_q \\ pdu \equiv a \ (q)}} d^{-s} = p^s \sum_{\substack{d' \in \mathcal{S}_q \\ d'u \equiv a \ (q)}} (d')^{-s}.$$

In the last step $d' = pd$; conversely every d' in the last sum is divisible by p, because u is a unit and a is divisible by p. The last expression is $p^s C_{a,u}(s)$. Prime sufficiency gives all divisor rows. Theorem 9.15 and the identity submatrix show that these $\varphi(q)$ independent columns exhaust the solution space. Hurwitz zeta satisfies the same distribution equations by regrouping its absolutely convergent defining series, proving (9.119). □

9.10.1 A finite formula, including negative spectral parameters

For $0 \le r < q$, let

$$g = \gcd(r, q), \quad m = q/g, \quad a = r/g, \quad P_r = \{p \mid q : p \nmid m\}.$$

When $r = 0$, this means $g = q$, $m = 1$, $a = 0$. For $p \in P_r$, put $h_{p,m} = \operatorname{ord}_m(p)$, with order one modulo one.

Theorem 9.55 (Finite residue formula)**.** *The meromorphic continuation of* (9.118) *is*

$$C_{r,u}(s) = g^{-s} \frac{\displaystyle\sum_{0\le b_p<h_{p,m}\ (p\in P_r)} \ \prod_{p\in P_r} p^{-sb_p}}{\displaystyle\prod_{p\in P_r}(1-p^{-sh_{p,m}})}, \qquad u\prod_{p\in P_r} p^{b_p}\equiv a\ (m) \tag{9.120}$$

Empty products have their usual value one. In particular,

$$C_{0,u}(s) = q^{-s}\prod_{p\mid q}(1-p^{-s})^{-1}, \tag{9.121}$$

independently of u.

Proof. The congruence in (9.118) forces the valuation of d at an unsaturated prime, one dividing m, to be exactly $v_p(g)$. At a saturated prime, one not dividing m, it is at least $v_p(g)$. Thus $d = g\prod_{p\in P_r} p^{c_p}$ with $c_p \ge 0$, and the remaining congruence is $u\prod p^{c_p} \equiv a \pmod m$. Write $c_p = b_p + h_{p,m}j_p$ and sum the independent geometric series in j_p. This proves the formula in $\operatorname{Re} s > 0$ and hence its meromorphic continuation. The endpoint has trivial congruence modulo one. □

Formula (9.120), *not* the divergent smooth-number series, is the definition used at negative real s. It has no pole at any nonzero real s, and the continued distribution identities remain valid there. At a zero of a displayed denominator, one should use the polynomial conductor normal form rather than cancel poles numerically. Some individual coefficients may have removable singularities, but Theorem 9.49 gives the criterion for the entire primitive-grid coordinate system.

Corollary 9.56 (Exact support and signs)**.** *Let H_r be the subgroup of U_m generated by the classes of the primes in P_r. For every nonzero real s,*

$$C_{r,u}(s) \ne 0 \quad\Longleftrightarrow\quad au^{-1} \in H_r.$$

The number of supported primitive coordinates is

$$\frac{\varphi(q)}{\varphi(m)}|H_r|. \tag{9.122}$$

All supported coefficients are positive for $s > 0$. For $s < 0$, they all have sign $(-1)^{|P_r|}$.

Proof. The numerator of (9.120) is a sum of positive real terms and is nonempty exactly under the subgroup condition. The reduction map $U_q \to U_m$ has fibers of cardinality $\varphi(q)/\varphi(m)$, giving (9.122). For $s > 0$ every denominator factor is positive. For $s < 0$ each of the $|P_r|$ denominator factors is negative, while the numerator and g^{-s} are positive. □

Corollary 9.57 (Complete monotonicity and normalization)**.** *For $s > 0$ and $j \ge 0$, put*

$$W^{[j]}_{r,u}(s) = (-1)^j \frac{d^j}{ds^j} C_{r,u}(s) = \sum_{\substack{d\in\mathcal{S}_q \\ du\equiv r\ (q)}} \frac{(\log d)^j}{d^s}. \tag{9.123}$$

These quantities are nonnegative. Moreover, $\sum_u C_{r,u}(1) = 1$ for every r, and $0 \le C_{r,u}(k) \le 1$ for integer $k \ge 1$.

Proof. The logarithmically weighted semigroup series converges locally uniformly in $\operatorname{Re} s > 0$, so termwise differentiation is valid. The constant grid vector satisfies the distribution equations at $s = 1$, and its primitive entries are all one. Uniqueness of extension proves the row-sum identity. Monotonicity gives $C_{r,u}(k) \le C_{r,u}(1) \le 1$. □

Example 9.58 (Two reductions at denominator six). The nontrivial coefficients for $r = 2$ are

$$C_{2,1}(s) = \frac{2^s}{2^{2s} - 1}, \qquad C_{2,5}(s) = \frac{1}{2^{2s} - 1}.$$

Thus, meromorphically,

$$\zeta(s, 1/3) = \frac{2^s \zeta(s, 1/6) + \zeta(s, 5/6)}{2^{2s} - 1}, \tag{9.124}$$

$$\zeta(s, 1/2) = \frac{\zeta(s, 1/6) + \zeta(s, 5/6)}{3^s - 1}. \tag{9.125}$$

At $s = 1$, the coefficients in (9.124) are $2/3$ and $1/3$. At $s = -1$ they are $-2/3$ and $-4/3$, illustrating the sign theorem without using a divergent series. The poles at $s = 0$ reflect a singular primitive basis, not a failure of the underlying distribution relations.

9.10.2 Denominator certificates and construction cost

Let M_p also denote, in this paragraph only, the pushforward matrix on grid vectors defined by

$$(M_p f)_r = \sum_{pu \equiv r \ (q)} f_u.$$

To avoid confusion with the integer complement in Section 9.9, write that complement as $q_p = q/p^e$ here and put $h_p = \operatorname{ord}_{q_p}(p)$. Multiplication of residues gives $M_p^{e+h_p} = M_p^e$. For $z = p^{-s}$,

$$(I - zM_p)^{-1} = \sum_{j=0}^{e-1} z^j M_p^j + \frac{z^e}{1 - z^{h_p}} \sum_{j=0}^{h_p - 1} z^j M_p^{e+j}. \tag{9.126}$$

This follows first from a geometric series for $|z| < 1$, then as a rational matrix identity. The commuting product of these inverses, applied to vectors supported on U_q, is the extension $C(s)$.

Proposition 9.59 (A uniform rational denominator)**.** *For integer $k \ge 1$, every reduced denominator of $C_{r,u}(k)$ divides*

$$D_q(k) = \prod_{p^e \| q} p^{k(e-1)} \bigl(p^{kh_p} - 1\bigr). \tag{9.127}$$

Its bit length is $O(kq \log q)$. The coefficients of the raw derivative $C_{r,u}^{(j)}(k)$, viewed as a polynomial in the symbols $\log p$, have denominators dividing $D_q(k)^{j+1}$.

Proof. Every scalar coefficient on the right of (9.126) has denominator dividing $p^{k(e-1)}(p^{kh_p} - 1)$. Multiplication over primes proves the first claim. Since $h_p \le q$ and $\sum_{p|q} \log p \le \log q$, its logarithm is at most a constant times $kq \log q$. The matrix product can be written as a ratio of integer polynomials in p^s, with denominator obtained from (9.127) by replacing k with s. Repeated differentiation introduces polynomial factors in $\log p$ and raises the denominator power by at most one each time. For normalized Taylor coefficients, an additional factor $j!$ may be needed. □

The implementation computes (9.120) by convolution on residue classes modulo m, separately for each $m \mid q$. At a fixed m, each prime update takes at most $\varphi(m)^2$ rational arithmetic operations. A complete matrix can therefore be constructed in

$$O\left(\omega(q)\sum_{m|q}\varphi(m)^2 + q\varphi(q)\right) = O\big(\omega(q)q^2\big)$$

rational arithmetic operations, using $\sum_{m|q}\varphi(m) = q$. This is an arithmetic-operation bound, not a constant-cost bit model or a claim of optimality. The denominator estimate supplies a conservative size control at positive integer orders.

9.11 All-orders Hurwitz–Stieltjes identities

Write the Laurent expansion as

$$\zeta(1+t,x) = \frac{1}{t} + \sum_{n\ge 0}\frac{(-1)^n}{n!}\gamma_n(x)t^n, \qquad x > 0. \tag{9.128}$$

Here $\gamma_n^{(k)}$ always means k derivatives in x; a derivative on $C(s)$ is in the spectral variable. The coefficient identities in this section are analytic theorems, distinct from formal rank claims.

Theorem 9.60 (Derived Stieltjes reduction)**.** *For $k \ge 1$, $n \ge 0$, and $x_r = r/q$ with the endpoint convention,*

$$\boxed{\gamma_n^{(k)}(x_r) = \sum_{u\in U_q}\sum_{j=0}^{n}\binom{n}{j}W_{r,u}^{[j]}(k+1)\,\gamma_{n-j}^{(k)}(u/q).} \tag{9.129}$$

All the coefficient functions are given by the finite formula (9.120) and its derivatives.

Proof. Differentiating Hurwitz zeta in its parameter yields

$$\partial_x^k\zeta(1+t,x) = (-1)^k(1+t)_k\zeta(k+1+t,x).$$

The pole in (9.128) is independent of x, so for $k \ge 1$ the left side is $\sum_n(-1)^n\gamma_n^{(k)}(x)t^n/n!$. Apply (9.119) at spectral parameter $k+1+t$ and multiply by the common scalar $(-1)^k(1+t)_k$. Taylor multiplication gives (9.129), because $(-1)^jC_{r,u}^{(j)} = W_{r,u}^{[j]}$. □

This proof also explains why the harmonic-number master identity in Chapter 8 is compatible with the same reduction matrix at every Stieltjes order: the rising factorial is common to every rational argument. It does not create new independent distribution coordinates.

Theorem 9.61 (Underived Stieltjes reduction, with the pole term)**.** *For $n \ge 0$,*

$$\boxed{\gamma_n(x_r) = \sum_{u\in U_q}\sum_{j=0}^{n}\binom{n}{j}W_{r,u}^{[j]}(1)\gamma_{n-j}(u/q) - \frac{1}{n+1}\sum_{u\in U_q}W_{r,u}^{[n+1]}(1).} \tag{9.130}$$

Proof. Apply (9.119) at $s = 1+t$. The coefficients are analytic there. Multiplying their Taylor expansions by (9.128), the coefficient of t^n receives the usual product terms and also $\sum_u C_{r,u}^{(n+1)}(1)/(n+1)!$ from the pole $1/t$. Multiplication by $(-1)^n n!$ converts that additional term to the negative last term of (9.130). The coefficient of $1/t$ agrees because $\sum_u C_{r,u}(1) = 1$. □

Omitting the last term would already give an incorrect formula for γ_0. The point is not a rank correction: a known scalar right side does not change a homogeneous rank. It is a correction to the actual analytic value identity.

Example 9.62 (An explicit first-order identity). From (9.124), direct differentiation at $s = 1$ gives

$$W_{2,1}^{[1]}(1) = \frac{10}{9}\log 2, \qquad W_{2,5}^{[1]}(1) = \frac{8}{9}\log 2.$$

Consequently

$$\gamma_0(1/3) = \frac{2}{3}\gamma_0(1/6) + \frac{1}{3}\gamma_0(5/6) - 2\log 2. \tag{9.131}$$

At $s = 2$, the analogous coefficients give

$$\begin{aligned}\gamma_1'(1/3) = \frac{4}{15}\gamma_1'(1/6) + \frac{1}{15}\gamma_1'(5/6) \\ + \frac{68\log 2}{225}\gamma_0'(1/6) + \frac{32\log 2}{225}\gamma_0'(5/6).\end{aligned} \tag{9.132}$$

Since $\gamma_0' = -\psi'$, the signs can also be checked against the parameter-differentiated digamma multiplication identity. The verification program checks (9.130) through $n = 2$ and (9.132) by numerical parameter differentiation of Stieltjes functions.

9.12 Finite conductor descent for all Stieltjes layers

For a primitive character χ of conductor $f \mid d$, define

$$G_{d,\chi}^{(n,k)} = \sum_{a \in \mathcal{U}_d} \chi(a)\gamma_n^{(k)}(a/d). \tag{9.133}$$

Set

$$R(d,f) = \prod_{\substack{p \mid d \\ p \nmid f}} p, \qquad m_r = \frac{d}{fr} \quad (r \mid R(d,f)). \tag{9.134}$$

Each m_r is a positive integer. Upon substituting $t_p = p^s$, (9.36) becomes the entire exponential polynomial

$$\begin{aligned}A_{d,f,\chi}(s) &= \left(\frac{d}{f}\right)^s \prod_{\substack{p \mid d \\ p \nmid f}} (1 - \chi(p)p^{-s}) \\ &= \sum_{r \mid R(d,f)} \mu(r)\chi(r)m_r^s.\end{aligned} \tag{9.135}$$

In particular,

$$A_{d,f,\chi}^{(j)}(s) = \sum_{r \mid R(d,f)} \mu(r)\chi(r)m_r^s \log^j m_r. \tag{9.136}$$

Theorem 9.63 (All-index, all-parameter-order descent)**.** *For $n \geq 0$, $k \geq 1$ and $f \mid d$,*

$$\begin{aligned}G_{d,\chi}^{(n,k)} = \sum_{r \mid R(d,f)} \mu(r)\chi(r)m_r^{k+1} \\ \cdot \sum_{j=0}^{n} (-1)^j \binom{n}{j} \log^j m_r \, G_{f,\chi}^{(n-j,k)}.\end{aligned} \tag{9.137}$$

No Euler-factor denominator occurs. Equivalently,

$$G_{d,\chi}^{(n,k)} = \sum_{j=0}^{n} (-1)^j \binom{n}{j} A_{d,f,\chi}^{(j)}(k+1) G_{f,\chi}^{(n-j,k)}. \tag{9.138}$$

Proof. For $\operatorname{Re} s > 1$, sorting a Dirichlet series by residue classes gives

$$\sum_{a \in \mathcal{U}_d} \chi(a)\zeta(s, a/d) = d^s L(s, \chi_d),$$

where χ_d is the induced character modulo d. Removing the extra Euler factors from its primitive L-series yields

$$L(s, \chi_d) = L(s, \chi) \prod_{\substack{p \mid d \\ p \nmid f}} (1 - \chi(p)p^{-s}).$$

Thus the character sum at d is $A_{d,f,\chi}(s)$ times the character sum at f. This is also the analytic realization of Theorem 9.15.

Insert $s = k + 1 + \varepsilon$, multiply by the common factor $(-1)^k(1+\varepsilon)_k$, and use (9.1). Comparison of $(-1)^n \varepsilon^n / n!$ gives (9.138); the sign $(-1)^j$ comes from the Stieltjes Laurent convention. Finally apply (9.136). □

9.12.1 Concrete second-index identities at level twelve

The multipliers are

$$A_{12,3,\chi_3}(s) = 4^s + 2^s, \qquad A_{12,4,\chi_4}(s) = 3^s + 1.$$

Therefore, at $n = 2$, $k = 1$,

$$G_{12,\chi_3}^{(2,1)} = 20 G_{3,\chi_3}^{(2,1)} - 72 \log 2 \, G_{3,\chi_3}^{(1,1)} + 68 \log^2 2 \, G_{3,\chi_3}^{(0,1)}, \tag{9.139}$$

$$G_{12,\chi_4}^{(2,1)} = 10 G_{4,\chi_4}^{(2,1)} - 18 \log 3 \, G_{4,\chi_4}^{(1,1)} + 9 \log^2 3 \, G_{4,\chi_4}^{(0,1)}. \tag{9.140}$$

For instance,

$$G_{12,\chi_3}^{(2,1)} = \gamma_2'(1/12) - \gamma_2'(5/12) + \gamma_2'(7/12) - \gamma_2'(11/12),$$

and $G_{3,\chi_3}^{(2,1)} = \gamma_2'(1/3) - \gamma_2'(2/3)$. These are exact identities between explicitly specified signed sums; they require no integer-relation search. The two identities were also checked numerically using independent finite Hurwitz character sums.

9.12.2 The undifferentiated layer and its pole correction

For $k = 0$, a principal character sum still contains the Hurwitz pole. The same formula without a correction would be wrong.

Proposition 9.64 (The $k = 0$ correction)**.** *For nonprincipal primitive χ the formula* (9.138) *remains valid with $k = 0$. For $f = 1$, put $A_d(s) = A_{d,1,1}(s)$. Then*

$$G_{d,1}^{(n,0)} = \sum_{j=0}^{n} (-1)^j \binom{n}{j} A_d^{(j)}(1) \gamma_{n-j}(1) + \frac{(-1)^n}{n+1} A_d^{(n+1)}(1). \tag{9.141}$$

Proof. For a nonprincipal character, the sum of its values over units is zero, so the $1/\varepsilon$ residues cancel. Coefficient comparison is unchanged. For the principal conductor, multiply

$$A_d(1+\varepsilon)\left(\frac{1}{\varepsilon}+\sum_{n\ge 0}\frac{(-1)^n\gamma_n(1)}{n!}\varepsilon^n\right).$$

The pole term contributes $A_d^{(n+1)}(1)/(n+1)!$ to the coefficient of ε^n. Multiplying by $(-1)^n n!$ gives the correction in (9.141). The residue itself is $A_d(1) = d\prod_{p\mid d}(1-p^{-1}) = \varphi(d)$, as required. □

For the non-character multiplication formula this specializes to

$$\sum_{j=0}^{d-1}\gamma_n\left(\frac{a+j}{d}\right) = d\sum_{r=0}^{n}\binom{n}{r}(-\log d)^r\gamma_{n-r}(a) + \frac{(-1)^n d\log^{n+1} d}{n+1}. \tag{9.142}$$

The inhomogeneous term is the residue contribution, not an optional normalization.

9.13 Cyclotomic polylogarithm traces

The same finite module applies to a second evaluation. Put $\zeta_m = \exp(2\pi\mathrm{i}/m)$ and evaluate $e_x \mapsto \mathrm{Li}_s(\mathrm{e}^{2\pi\mathrm{i}x})$. For $\mathrm{Re}\, s > 1$, grouping the defining series by multiples of p gives

$$\sum_{py=x}\mathrm{Li}_s(\mathrm{e}^{2\pi\mathrm{i}y}) = p^{1-s}\,\mathrm{Li}_s(\mathrm{e}^{2\pi\mathrm{i}x}).$$

Thus the weights are now $A_p = p^{1-s}$, and analytic continuation extends the identity wherever the individual expressions are defined.

For a primitive character χ of conductor $f \mid q$, define

$$\mathcal{P}_{q,\chi}(s) = \sum_{u\in U_q}\chi(u)\,\mathrm{Li}_s(\zeta_q^u), \qquad \tau(\chi) = \sum_{a\in U_f}\chi(a)\zeta_f^a. \tag{9.143}$$

At conductor one take $\tau(\mathbf{1}) = 1$ and $L(s,\mathbf{1}) = \zeta(s)$.

Theorem 9.65 (Conductor-lifted trace identity)**.** *As a meromorphic identity, and with removable singularities interpreted by continuation,*

$$\boxed{\mathcal{P}_{q,\chi}(s) = (q/f)^{1-s}\prod_{\substack{p\mid q\\ p\nmid f}}\bigl(1-\chi(p)p^{s-1}\bigr)\tau(\chi)L(s,\overline{\chi}).} \tag{9.144}$$

For every $q > 1$, the left side is an entire function of the spectral variable s.

Proof. At primitive conductor $f > 1$, absolute convergence and the primitive Gauss-sum identity give

$$\sum_{a\in U_f}\chi(a)\,\mathrm{Li}_s(\zeta_f^a) = \sum_{n\ge 1}\frac{\sum_a \chi(a)\zeta_f^{an}}{n^s} = \tau(\chi)L(s,\overline{\chi}).$$

The Gauss-sum convention, including the complex conjugate on χ, is the one in [80]. Applying (9.112) with $A_p = p^{1-s}$ multiplies this primitive trace by $E_{q,\chi}$. Extracting p^{1-s} from each first-appearance factor gives precisely the factor in (9.144). The conductor-one case follows directly from $\mathrm{Li}_s(1) = \zeta(s)$ and the same polynomial identity.

For completeness, when $z = \zeta_q^u \neq 1$, regrouping by residues gives

$$\mathrm{Li}_s(z) = q^{-s} \sum_{a=1}^{q} z^a \zeta(s, a/q). \tag{9.145}$$

Hurwitz zeta has only its simple pole at $s = 1$, whose coefficient in this finite sum is $\sum_a z^a = 0$. Hence $\mathrm{Li}_s(z)$ is entire in s. All arguments in the primitive level-q trace differ from one. □

Remark 9.66 (The character phase is consequential). For the order-four character modulo five with $\chi(2) = \mathrm{i}$, the level-ten lifting factor is $2^{1-s} - \mathrm{i}$, not $2^{1-s} + \mathrm{i}$. A complex-character numerical check is included to catch an accidental interchange of χ and $\overline{\chi}$. Real quadratic examples alone would not detect that error.

9.13.1 Exact nonprincipal vanishing orders

For a nonprincipal primitive χ, set

$$\rho_\chi = \#\{p \mid q : p \nmid f,\ \chi(p) = 1\}.$$

These are the same conductor statistics that controlled the primitive-grid jet defects at Hurwitz order zero.

Theorem 9.67 (Character trace zeros at order one)**.** *The entire function $\mathcal{P}_{q,\chi}(s)$ has a zero of exact order ρ_χ at $s = 1$, where order zero means a nonzero value. All lower derivatives vanish, and*

$$\begin{aligned}\mathcal{P}_{q,\chi}^{(\rho_\chi)}(1) = (-1)^{\rho_\chi} \rho_\chi!\, \tau(\chi) L(1, \overline{\chi}) \\ \times \prod_{\substack{p \mid q,\ p \nmid f \\ \chi(p) = 1}} \log p \prod_{\substack{p \mid q,\ p \nmid f \\ \chi(p) \neq 1}} (1 - \chi(p)).\end{aligned} \tag{9.146}$$

Proof. Each factor with $\chi(p) = 1$ in (9.144) is $1 - p^{s-1} = -(s-1)\log p + O((s-1)^2)$. All other displayed factors are nonzero at $s = 1$. A primitive Gauss sum is nonzero, and Dirichlet's nonvanishing theorem states $L(1, \overline{\chi}) \neq 0$ for nonprincipal characters [80, Eq. 25.15.9]. Thus there is no additional zero. Taking the first nonzero Taylor coefficient proves the formula. □

Increasing the exponents of primes already present in q changes higher Taylor coefficients but not this first nonzero derivative: $(q/f)^{1-s}$ equals one at $s = 1$. This invariance provides an additional family of identities between traces at different levels.

9.13.2 The principal trace and its pole cancellation

Write $\mathcal{P}_q = \mathcal{P}_{q,\mathbf{1}}$ and define

$$r = \omega(q), \quad P = \prod_{p \mid q} \log p, \qquad B = \log q - \tfrac{1}{2} \log \mathrm{rad}(q).$$

The conductor-one factor contains a pole, so the vanishing order is one less than the number of primes, not equal to that number.

Theorem 9.68 (Principal trace jet generator)**.** *In a neighborhood of $t = 0$,*

$$\boxed{\mathcal{P}_q(1+t) = (-1)^r P\, t^{r-1}\, \mathrm{e}^{-Bt}\, [t\zeta(1+t)] \prod_{p\mid q} \frac{\sinh(t\log p/2)}{t\log p/2}.} \tag{9.147}$$

In particular its zero at $s = 1$ has exact order $r - 1$, and

$$\mathcal{P}_q^{(j)}(1) = 0 \qquad (0 \le j < r-1), \tag{9.148}$$

$$\mathcal{P}_q^{(r-1)}(1) = (-1)^r (r-1)! P, \tag{9.149}$$

$$\mathcal{P}_q^{(r)}(1) = (-1)^r r! P(\gamma - B), \tag{9.150}$$

$$\mathcal{P}_q^{(r+1)}(1) = (-1)^r (r+1)! P \left[-\gamma_1 - B\gamma + \frac{B^2}{2} + \frac{1}{24} \sum_{p\mid q} (\log p)^2 \right]. \tag{9.151}$$

Here γ_1 is the ordinary first Stieltjes constant in (9.128)*.*

Proof. The principal case of (9.144) is

$$\mathcal{P}_q(1+t) = q^{-t} \prod_{p\mid q} (1 - p^t)\zeta(1+t).$$

Use $1 - \mathrm{e}^x = -x\mathrm{e}^{x/2}\sinh(x/2)/(x/2)$ in every factor. This proves (9.147). The factors in square brackets and in the hyperbolic-sine product are analytic and equal to one at zero. Expanding them gives

$$t\zeta(1+t) = 1 + \gamma t - \gamma_1 t^2 + O(t^3), \qquad \prod_p \frac{\sinh(t\log p/2)}{t\log p/2} = 1 + \frac{t^2}{24} \sum_p (\log p)^2 + O(t^4).$$

Multiplication by e^{-Bt} gives all the displayed derivatives. □

9.13.3 Concrete exact identities

The following examples are proved consequences of the preceding theorems, not integer-relation conjectures. The derivatives are in the *order* of the polylogarithm, not its argument.

At level thirty,

$$\sum_{u\in U_{30}} \mathrm{Li}_1(\zeta_{30}^u) = 0, \tag{9.152}$$

$$\sum_{u\in U_{30}} \partial_s^1 \mathrm{Li}_s(\zeta_{30}^u)\big|_{s=1} = 0, \tag{9.153}$$

$$\sum_{u\in U_{30}} \partial_s^2 \mathrm{Li}_s(\zeta_{30}^u)\big|_{s=1} = \boxed{-2\log 2\log 3\log 5}. \tag{9.154}$$

The third derivative of this trace is $-6\log 2\log 3\log 5\,[\gamma - \frac{1}{2}\log 30]$. At level 210, derivatives of orders zero, one, and two vanish, while

$$\sum_{u\in U_{210}} \partial_s^3 \mathrm{Li}_s(\zeta_{210}^u)\big|_{s=1} = 6\log 2\log 3\log 5\log 7. \tag{9.155}$$

For the quadratic character χ_3 with $\chi_3(1) = 1$ and $\chi_3(2) = -1$, the primitive trace at order one is $\tau(\chi_3)L(1, \chi_3) = \mathrm{i}\pi/3$. Since $\chi_3(7) = \chi_3(13) = 1$, one obtains

$$\sum_{u \in U_{21}} \chi_3(u)\, \partial_s^1 \operatorname{Li}_s(\zeta_{21}^u)\big|_{s=1} = -\frac{\mathrm{i}\pi}{3} \log 7, \tag{9.156}$$

$$\sum_{u \in U_{273}} \chi_3(u)\, \partial_s^2 \operatorname{Li}_s(\zeta_{273}^u)\big|_{s=1} = \frac{2\mathrm{i}\pi}{3} \log 7 \log 13. \tag{9.157}$$

In the first equation the order-zero trace vanishes; in the second, both order-zero and first-order traces vanish. Likewise the character χ_4 gives

$$\sum_{u \in U_{20}} \chi_4(u)\, \partial_s^1 \operatorname{Li}_s(\zeta_{20}^u)\big|_{s=1} = -\frac{\mathrm{i}\pi}{2} \log 5. \tag{9.158}$$

These primitive trace constants follow directly from $\operatorname{Li}_1(z) = -\log(1 - z)$ with the values continued from inside the unit disk, so the signs do not depend on an unspecified logarithm branch.

For the real quadratic character χ_5, positive on residues $1, 4$ and negative on $2, 3$, let $\phi = (1 + \sqrt{5})/2$. Pairing conjugates in the primitive trace gives

$$\mathcal{P}_{5,\chi_5}(1) = 2 \log \frac{\sin(2\pi/5)}{\sin(\pi/5)} = 2 \log \phi.$$

As $\chi_5(11) = 1$,

$$\sum_{u \in U_{55}} \chi_5(u)\, \partial_s^1 \operatorname{Li}_s(\zeta_{55}^u)\big|_{s=1} = -2 \log \phi \log 11. \tag{9.159}$$

This derivation needs no numerical integer-relation search or unproved independence statement about logarithms.

9.13.4 A stable independent formula for numerical checks

Differentiation of (9.145) at its cancelled pole gives, for $z = \zeta_q^u \neq 1$,

$$\partial_s^k \operatorname{Li}_s(z)\Big|_{s=1} = \frac{(-1)^k}{q} \sum_{a=1}^{q} z^a \sum_{j=0}^{k} \binom{k}{j} (\log q)^{k-j} \gamma_j(a/q). \tag{9.160}$$

There is no pole term, because its complete Fourier coefficient vanishes identically. This supplies a finite Stieltjes-coordinate evaluator for the trace identities, independently of their Euler-factor right sides. In development, direct tiny-step numerical differentiation of a polylogarithm routine near integral order gave an unreliable trace derivative; the pole-cancelled formula avoids that cancellation problem. This observation is an implementation diagnostic, not a claimed theorem about all versions of a library.

9.13.5 A level-260 specialization and convention reconciliation

We use the unbarred weight $\chi(a)$ in $\mathcal{P}_{q,\chi}$. An alternative convention weights the trace by $\overline{\chi}(a)$; it conjugates the character factors and interchanges the primitive L-character. All formulas here use the first convention. The following exact specialization adds two primes with character value one.

Let χ be primitive of conductor $f > 1$, and $f \mid q$. Among the added primes, define

$$\mathcal{S} = \{p \mid q : p \nmid f,\ \chi(p) = 1\}, \qquad r_\chi = |\mathcal{S}|.$$

The remaining added primes form $\mathcal{N}$.

Theorem 9.69 (Twisted leading jet)**.** *The trace $\mathcal{P}_{q,\chi}$ vanishes to order at least r_χ at one, and*

$$\mathcal{P}_{q,\chi}^{(r_\chi)}(1) = (-1)^{r_\chi} r_\chi!\, \mathcal{P}_{f,\chi}(1) \prod_{p\in\mathcal{S}} \log p \prod_{p\in\mathcal{N}} (1-\chi(p)). \tag{9.161}$$

Its order is exactly r_χ whenever $\mathcal{P}_{f,\chi}(1) \neq 0$. More generally its exact order is $r_\chi + \operatorname{ord}_{s=1} \mathcal{P}_{f,\chi}(s)$.

Proof. The multiplier is

$$B_{q,f,\chi}(1+\varepsilon) = (q/f)^{-\varepsilon} \prod_{\substack{p\mid q\\ p\nmid f}} (1-\chi(p)p^{\varepsilon}).$$

The factors indexed by $\mathcal{S}$ have simple zeros with leading coefficient $-\log p$. All other factors are nonzero at zero and have value $1-\chi(p)$. Multiply their leading terms by the Taylor series of $\mathcal{P}_{f,\chi}(1+\varepsilon)$. □

For primitive nonprincipal characters, the classical nonvanishing $L(1,\overline{\chi}) \neq 0$, together with the nonzero primitive Gauss sum in (9.144), supplies the stated nonvanishing hypothesis [80]. The explicit example below does not need that theorem; its primitive trace is computed directly from logarithms.

For $\chi_4(a) = (-1)^{(a-1)/2}$ on odd integers,

$$\mathcal{P}_{4,\chi_4}(1) = \operatorname{Li}_1(\mathrm{i}) - \operatorname{Li}_1(-\mathrm{i}) = \frac{\mathrm{i}\pi}{2}.$$

Take $q = 260 = 4\cdot 5\cdot 13$. Both added primes have $\chi_4(p) = 1$. Consequently

$$\sum_{a\in\mathcal{U}_{260}} \chi_4(a) \operatorname{Li}_1(\zeta_{260}^a) = 0, \tag{9.162}$$

$$\sum_{a\in\mathcal{U}_{260}} \chi_4(a)\, \partial_s^1 \operatorname{Li}_s(\zeta_{260}^a)\big|_{s=1} = 0, \tag{9.163}$$

$$\sum_{a\in\mathcal{U}_{260}} \chi_4(a)\, \partial_s^2 \operatorname{Li}_s(\zeta_{260}^a)\big|_{s=1} = \mathrm{i}\pi \log 5 \log 13. \tag{9.164}$$

This is a proved identity, not an integer-relation conjecture. More generally, for distinct primes $p_1,\dots,p_r \equiv 1 \pmod 4$ and $q = 4p_1\cdots p_r$,

$$\sum_{a\in\mathcal{U}_q} \chi_4(a)\, \partial_s^r \operatorname{Li}_s(\zeta_q^a)\big|_{s=1} = (-1)^r r! \frac{\mathrm{i}\pi}{2} \prod_{j=1}^{r} \log p_j, \tag{9.165}$$

with all lower order traces vanishing. The same leading identity persists if the added prime factors occur with higher exponents.

9.14 Explicit tables for the second constant

We record the closed forms of $\gamma_2' = -[\zeta''(2,a) + 2\zeta'(2,a)]$ (the $s=2$ layer) and $\gamma_2'' = 2\zeta''(3,a) + 6\zeta'(3,a) + 2\zeta(3,a)$ (the $s=3$ layer) at the elementary denominators. The new constants entering are the *second* s-derivatives $\zeta''(2), \zeta''(3)$, $\beta''(s) = L''(s,\chi_{-4})$ and $L''(s,\chi_{-3})$; the values $L(3,\chi_{-4}) = \pi^3/32$ and $L(3,\chi_{-3}) = 4\pi^3/(81\sqrt{3})$ are elementary. As at the first-derivative layers, the odd-character block flips sign as a single unit under $p \mapsto q-p$ (including any elementary π-power value-part it carries). All entries are exact and were verified to $< 10^{-70}$.

Second constant, first parameter-derivative.

$$\begin{aligned}
\gamma_2'(1) &= -\zeta''(2) - 2\zeta'(2),\\
\gamma_2'(\tfrac12) &= -3\zeta''(2) - (6+8\ln 2)\zeta'(2) - \tfrac43\pi^2\ln 2 - \tfrac23\pi^2\ln^2 2,\\
\gamma_2'(\tfrac13) &= -4\zeta''(2) - (8+9\ln 3)\zeta'(2) - \tfrac32\pi^2\ln 3 - \tfrac34\pi^2\ln^2 3\\
&\quad - \Big[\tfrac92 L''(2,\chi_{-3}) + 9(1+\ln 3)L'(2,\chi_{-3}) + \big(9\ln 3 + \tfrac92\ln^2 3\big)L(2,\chi_{-3})\Big],\\
\gamma_2'(\tfrac14) &= -6\zeta''(2) - (12+28\ln 2)\zeta'(2) - \tfrac{14}{3}\pi^2\ln 2 - 5\pi^2\ln^2 2\\
&\quad - \Big[8\beta''(2) + 16(1+2\ln 2)\beta'(2) + \big(32\ln 2 + 32\ln^2 2\big)G\Big],
\end{aligned}$$

with $\gamma_2'(2/3), \gamma_2'(3/4)$ obtained by negating the bracketed odd-character block, and the $q = 6$ entries given in the source report.

Second constant, second parameter-derivative.

$$\begin{aligned}
\gamma_2''(1) &= 2\zeta''(3) + 6\zeta'(3) + 2\zeta(3),\\
\gamma_2''(\tfrac12) &= 14\zeta''(3) + (42+32\ln 2)\zeta'(3) + (14+48\ln 2+16\ln^2 2)\zeta(3),\\
\gamma_2''(\tfrac13) &= 26\zeta''(3) + (78+54\ln 3)\zeta'(3) + (26+81\ln 3+27\ln^2 3)\zeta(3)\\
&\quad + \Big[27L''(3,\chi_{-3}) + (81+54\ln 3)L'(3,\chi_{-3}) + (27+81\ln 3+27\ln^2 3)L(3,\chi_{-3})\Big],\\
\gamma_2''(\tfrac14) &= 56\zeta''(3) + (168+240\ln 2)\zeta'(3) + (56+360\ln 2+248\ln^2 2)\zeta(3)\\
&\quad + \Big[64\beta''(3) + (192+256\ln 2)\beta'(3) + (2+12\ln 2+8\ln^2 2)\pi^3\Big],
\end{aligned}$$

again with $\gamma_2''(2/3), \gamma_2''(3/4)$ the odd-block negations. The reducible fractions collapse correctly ($\frac24 = \frac36 = \frac12$, $\frac26 = \frac13$), an internal consistency check.

9.15 The derivative $\leftrightarrow$ negapolygamma duality

The functional equation pairs the layer $s = k+1$ of $\gamma_1^{(k)}$ with the layer $s = -k$, which is exactly where the negative-order polygamma $\psi^{(-(k+1))}$ lives (its ladder uses $\zeta'(-k,\cdot)$). The seam is a single uniform identity.

Theorem 9.70 (Harmonic-number bridge law)**.** *Let χ be a primitive character mod q for which $L(-k,\chi) \neq 0$. Then*

$$\frac{L'(k+1,\chi)}{L(k+1,\chi)} + \overline{\left(\frac{L'(-k,\chi)}{L(-k,\chi)}\right)} = \gamma + \ln\frac{2\pi}{q} - H_k. \tag{9.166}$$

The same H_k that sets the master coefficient in Theorem 9.1 sets the bridge constant; the case $k = 1$ recovers the generalized-Glaisher constant $C_q = \gamma + \ln(2\pi/q) - 1$.

Proof of the bridge. For a primitive nonprincipal character of parity $\chi(-1) = (-1)^\epsilon$, the completed function is $\Lambda(s,\chi) = (q/\pi)^{(s+\epsilon)/2}\Gamma((s+\epsilon)/2)L(s,\chi)$. Its functional equation gives $\Lambda'/\Lambda(s,\chi) = -\Lambda'/\Lambda(1-s,\bar\chi)$. At $s = k+1$ with no trivial zero at $-k$, the two digamma terms, using reflection and recurrence, sum to $2(H_k - \gamma - \log 2)$ before their factor $1/2$ is applied. Collecting the conductor factor proves (9.166). The conductor-one principal character requires the meromorphic completed zeta function; the same nonzero-value calculation is valid for $k \geq 1$ where neither point is a pole. □

Remark 9.71 (Trivial-zero bridge). For $k \geq 1$, $L(-k,\chi) = 0$ when $\epsilon \equiv k \pmod 2$. At a simple trivial zero, cancellation of the pole in the completed gamma factor gives instead

$$\frac{L'(k+1,\chi)}{L(k+1,\chi)} + \overline{\frac{L''(-k,\chi)}{2L'(-k,\chi)}} = \gamma + \log(2\pi/q) - H_k.$$

The factor 2 in the denominator is essential. This follows by writing $L(-k+u) = L'(-k)u + \frac{1}{2}L''(-k)u^2 + O(u^3)$ in the same logarithmic functional equation and taking its finite part.

At $z = \frac{1}{4}$ the governing character is χ_{-4} (odd). For k even there is no trivial zero, and $\psi^{(-(k+1))}(\frac{1}{4})$ carries the unreduced derivative coordinate $\beta'(k+1) = L'(k+1,\chi_{-4})$, so $\gamma_1^{(k)}(\frac{1}{4})$ is an exact finite combination of $\psi^{(-(k+1))}(\frac{1}{4})$ and the remaining zeta-value and zeta-derivative terms. The representative instance $k = 2$ is, after eliminating $\beta'(3)$ entirely,

$$\begin{aligned}\gamma_1''(\tfrac{1}{4}) = \; & -256\,\pi^3\,\psi^{(-3)}(\tfrac{1}{4}) - 56\,\zeta'(3) - 84\,\zeta(3) - 120\ln 2\,\zeta(3) + 35\pi\,\zeta(3) \\ & - 3\pi^3 - 4\ln 2\,\pi^3 + 64\ln \mathsf{A}\,\pi^3 - 2\gamma\,\pi^3 + 2\ln(2\pi)\,\pi^3,\end{aligned} \tag{9.167}$$

verified to $< 10^{-118}$. (The irrational coefficient $-256\pi^3$ on $\psi^{(-3)}(\frac{1}{4})$ is why a rational integer-relation search for this identity must take $\pi^3\psi^{(-3)}$, not $\psi^{(-3)}$, as the search atom.) For k odd the value-level duality *degenerates*: by Remark 9.71, $\psi^{(-(k+1))}(\frac{1}{4})$ then carries only the lower *value* atom (for $k = 1$, $\beta'(-1) = 2G/\pi$, a value rather than a derivative at the positive point; for $k = 3$, $\beta'(-3) \sim \beta(4)/\pi^3$), while the genuine derivative atom $\beta'(k+1)$ of $\gamma_1^{(k)}(\frac{1}{4})$ is tied instead to the second derivative $\beta''(-k)$, absent from $\psi^{(-(k+1))}$. Numerically the coefficient of $\beta'(k+1)$ in $\psi^{(-(k+1))}(\frac{1}{4})$ is zero to $\sim 10^{-163}$ for $k = 1, 3$. The clean odd-k value-level duality therefore lives at even-character points (e.g. $q = 5$), where the parity is reversed.

9.16 One normalized functional equation for all derivative bridges

The Stieltjes drafts connect derivatives of $L(s,\chi)$ at a positive integer with derivatives at its reflected negative integer. A single identity of analytic germs organizes all these formulas and treats trivial zeros without differentiating singular logarithms. It also makes the conjugation required for complex characters unavoidable. The input is the classical primitive Dirichlet functional equation and the reflection and duplication formulas for Γ [49]; the all-order normalized form below is a systematic continuation of the bridges used in the corpus.

9.16.1 Regularization and the normalized germ

Let χ be primitive of conductor q, allowing the trivial character of conductor 1, for which $L(s,\chi) = \zeta(s)$. Fix $k \geq 1$, and write

$$\delta = \frac{1-\chi(-1)}{2} \in \{0,1\}, \qquad \nu = \begin{cases} 1, & \chi(-1) = (-1)^k, \\ 0, & \chi(-1) \neq (-1)^k. \end{cases}$$

Define

$$B(t) = \frac{L(-k+t,\chi)}{t^\nu}, \qquad B^{\#}(z) = \overline{B(\overline{z})}, \tag{9.168}$$

with the removable value filled in when $\nu = 1$. The function $B^{\#}$ is holomorphic near zero; it is coefficientwise conjugation of the Taylor series of B.

Theorem 9.72 (Normalized positive–negative germ)**.** *The quantities $L(k+1,\chi)$ and $B(0)$ are nonzero. On a sufficiently small disk about $z=0$,*

$$\frac{L(k+1+z,\chi)}{L(k+1,\chi)} = K_\nu(z)\frac{B^{\#}(-z)}{B^{\#}(0)}, \tag{9.169}$$

where

$$\begin{aligned} K_\nu(z) &= \left(\frac{2\pi}{q}\right)^{z}\frac{\Gamma(k+1)}{\Gamma(k+1+z)}\,T_\nu(z),\\ T_0(z) &= \sec\frac{\pi z}{2}, \qquad T_1(z) = \frac{\pi z/2}{\sin(\pi z/2)}, \quad T_1(0)=1. \end{aligned} \tag{9.170}$$

All three normalized factors in (9.169) equal 1 at the origin.

Proof. The Euler product gives $L(k+1,\chi)\neq 0$, since $k+1>1$. Use the completed function

$$\Lambda(s,\chi) = \left(\frac{q}{\pi}\right)^{(s+\delta)/2}\Gamma\!\left(\frac{s+\delta}{2}\right)L(s,\chi)$$

and its functional equation

$$\Lambda(s,\chi) = \varepsilon_\chi\Lambda(1-s,\overline{\chi}), \qquad |\varepsilon_\chi| = 1. \tag{9.171}$$

At $s=-k$, the gamma factor is finite and nonzero if $\nu=0$, and has a simple pole if $\nu=1$. The reflected completed value at $k+1$ is finite and nonzero. Consequently $L(-k,\chi)\neq 0$ in the first case, and $L(s,\chi)$ has a simple zero at $-k$ in the second. In either case B is holomorphic and $B(0)\neq 0$.

Put

$$a = \frac{\delta-k}{2}, \qquad b = \frac{k+1+\delta}{2}.$$

Solving (9.171) at $s=k+1+z$ gives

$$L(k+1+z,\chi) = \varepsilon_\chi\left(\frac{q}{\pi}\right)^{-k-\frac{1}{2}-z}\frac{\Gamma(a-z/2)}{\Gamma(b+z/2)}L(-k-z,\overline{\chi}). \tag{9.172}$$

Since $L(-k-z,\overline{\chi}) = (-z)^\nu B^{\#}(-z)$, the numerator

$$F_\nu(z) = (-z)^\nu\Gamma(a-z/2)$$

has a nonzero removable value at the origin. Dividing (9.172) by its value at $z=0$ cancels the root number and all constant factors:

$$\frac{L(k+1+z,\chi)}{L(k+1,\chi)} = \left(\frac{q}{\pi}\right)^{-z}\frac{F_\nu(z)}{F_\nu(0)}\frac{\Gamma(b)}{\Gamma(b+z/2)}\frac{B^{\#}(-z)}{B^{\#}(0)}. \tag{9.173}$$

If $\nu=0$, then a is a half-integer. The gamma reflection formula and $\sin(\pi(a-z/2)) = \sin(\pi a)\cos(\pi z/2)$ give

$$\frac{F_0(z)}{F_0(0)} = \frac{\Gamma(1-a)}{\Gamma(1-a+z/2)}\sec\frac{\pi z}{2}. \tag{9.174}$$

If $\nu=1$, write $a=-m$, $m\geq 0$. Then $F_1(0) = 2(-1)^m/m!$, and the same reflection formula gives

$$\frac{F_1(z)}{F_1(0)} = \frac{\Gamma(1-a)}{\Gamma(1-a+z/2)}\frac{\pi z/2}{\sin(\pi z/2)}. \tag{9.175}$$

For example, the pole in $\Gamma(-m-z/2)$ is canceled by the explicit factor $-z$, so no logarithm of a function vanishing at zero has been taken.

The unordered pair $\{b, 1-a\}$ is $\{(k+1)/2, (k+2)/2\}$, for either value of δ. Duplication therefore yields

$$\frac{\Gamma(b)\Gamma(1-a)}{\Gamma(b+z/2)\Gamma(1-a+z/2)} = 2^z \frac{\Gamma(k+1)}{\Gamma(k+1+z)}. \tag{9.176}$$

Substituting (9.174) or (9.175), followed by (9.176), into (9.173) proves the theorem. □

Remark 9.73. The statement concerns analytic germs and local logarithms. It requires no claim about a global zero-free region. For a complex character the functional equation involves $\overline{\chi}$, and therefore the reflected Taylor coefficients are conjugated. Replacing $B^{\#}$ by B is justified for a real character but generally false for a complex one.

9.16.2 Closed logarithmic coefficients at every order

Choose the normalized local logarithms, each zero at the origin, of $L(k+1+z,\chi)/L(k+1,\chi)$ and $B(t)/B(0)$. For $r \geq 1$, set

$$\mathcal{L}_r = \frac{d^r}{ds^r} \log L(s,\chi)\Big|_{s=k+1}, \qquad b_r = \frac{d^r}{dt^r} \log B(t)\Big|_{t=0}.$$

These derivatives do not depend on a choice of logarithm branch. Write

$$H_k^{(r)} = \sum_{j=1}^{k} j^{-r}, \qquad H_k = H_k^{(1)}, \qquad \eta(r) = (1-2^{1-r})\zeta(r) \quad (r \geq 2).$$

Theorem 9.74 (All-order logarithmic bridges)**.** *For every $r \geq 1$,*

$$\mathcal{L}_r - (-1)^r \overline{b_r} = \kappa_r, \tag{9.177}$$

where

$$\kappa_r = \begin{cases} \gamma + \log\dfrac{2\pi}{q} - H_k, & r = 1, \\ (r-1)!\big(\zeta(r) - H_k^{(r)}\big), & r \geq 3 \text{ odd}, \\ (r-1)!\big(H_k^{(r)} + (-1)^{\nu}\eta(r)\big), & r \geq 2 \text{ even}. \end{cases} \tag{9.178}$$

Proof. Taking normalized logarithms of (9.169) gives

$$\log \frac{L(k+1+z,\chi)}{L(k+1,\chi)} = \log K_\nu(z) + \overline{\log \frac{B(-\overline{z})}{B(0)}}.$$

The r-th derivative of the last term at zero is $(-1)^r\overline{b_r}$. It remains to compute the derivatives of $\log K_\nu$.

The convergent gamma expansion, on $|z|<1$, is

$$\log \frac{\Gamma(k+1+z)}{\Gamma(k+1)} = (H_k - \gamma)z + \sum_{r\geq 2} \frac{(-1)^r}{r}\big(\zeta(r) - H_k^{(r)}\big)z^r. \tag{9.179}$$

The sine and cosine products give, in the same disk,

$$\log\sec\frac{\pi z}{2}=\sum_{m\ge1}\frac{(1-2^{-2m})\zeta(2m)}{m}z^{2m},\tag{9.180}$$

$$\log\frac{\pi z/2}{\sin(\pi z/2)}=\sum_{m\ge1}\frac{2^{-2m}\zeta(2m)}{m}z^{2m}.\tag{9.181}$$

These identities also follow directly by taking logarithms of the absolutely locally convergent products $\cos(\pi z/2)=\prod_{j\ge0}(1-z^2/(2j+1)^2)$ and $\sin(\pi z/2)/(\pi z/2)=\prod_{j\ge1}(1-z^2/(2j)^2)$.

The linear term of $\log K_\nu$ is the stated κ_1. Odd coefficients of order at least three come only from the negative of (9.179), giving the second case. For $r=2m$, the secant coefficient combines with the gamma coefficient to give

$$(r-1)!\left[H_k^{(r)}-\zeta(r)+2(1-2^{-r})\zeta(r)\right]=(r-1)!\big(H_k^{(r)}+\eta(r)\big).$$

Replacing the secant factor by the sine quotient instead gives

$$(r-1)!\left[H_k^{(r)}-\zeta(r)+2^{1-r}\zeta(r)\right]=(r-1)!\big(H_k^{(r)}-\eta(r)\big).$$

These are exactly the two even cases in (9.178). □

At a nonzero reflected value, $\nu=0$, the first two identities read

$$\frac{L'(k+1,\chi)}{L(k+1,\chi)}+\overline{\frac{L'(-k,\chi)}{L(-k,\chi)}}=\gamma+\log\frac{2\pi}{q}-H_k,\tag{9.182}$$

$$(\log L)''(k+1,\chi)-\overline{(\log L)''(-k,\chi)}=H_k^{(2)}+\frac{\pi^2}{12}.\tag{9.183}$$

At a trivial zero, $\nu=1$, the Taylor expansion

$$B(t)=L'(-k,\chi)+\frac{L''(-k,\chi)}{2}t+\frac{L'''(-k,\chi)}{6}t^2+\cdots$$

instead gives

$$\frac{L'(k+1,\chi)}{L(k+1,\chi)}+\frac12\overline{\frac{L''(-k,\chi)}{L'(-k,\chi)}}=\gamma+\log\frac{2\pi}{q}-H_k,\tag{9.184}$$

$$(\log L)''(k+1,\chi)-\overline{\left(\frac{L'''(-k,\chi)}{3L'(-k,\chi)}-\frac{L''(-k,\chi)^2}{4L'(-k,\chi)^2}\right)}=H_k^{(2)}-\frac{\pi^2}{12}.\tag{9.185}$$

These formulas preserve the familiar first-order harmonic bridge and supply a uniform extension to every derivative order and both character parities.

9.16.3 Raw jets and an exact triangular implementation

The normalized germ also gives direct formulas for ordinary derivatives, without first converting each order to logarithmic derivatives. Define K_m by

$$K_\nu(z)=\sum_{m\ge0}K_m\frac{z^m}{m!},\qquad K_0=1.$$

Since $\log K_\nu(z) = \sum_{r\ge1} \kappa_r z^r/r!$, these coefficients are the complete exponential Bell polynomials:

$$K_m = m! \sum_{\substack{m_1,\dots,m_m\ge0\\ m_1+2m_2+\cdots+mm_m=m}} \prod_{r=1}^{m} \frac{1}{m_r!}\left(\frac{\kappa_r}{r!}\right)^{m_r} \qquad (m\ge1). \tag{9.186}$$

Leibniz's rule in (9.169) proves

$$\frac{L^{(m)}(k+1,\chi)}{L(k+1,\chi)} = \sum_{j=0}^{m} \binom{m}{j} K_{m-j}(-1)^j \frac{\overline{B^{(j)}(0)}}{B(0)}. \tag{9.187}$$

The required regularized derivatives are explicitly

$$\frac{B^{(j)}(0)}{B(0)} = \frac{j!\,\nu!}{(j+\nu)!}\frac{L^{(j+\nu)}(-k,\chi)}{L^{(\nu)}(-k,\chi)}. \tag{9.188}$$

The diagonal coefficient in (9.187) is $(-1)^m$, so the transformation is triangular and invertible. Its coefficients belong to the ring generated by the explicit κ_r's. This supplies exact relation certificates at arbitrary order and avoids repeated symbolic differentiation across gamma poles.

Remark 9.75 (Correction of the complex-character formulas). The older derivative-relations report omits the conjugation in versions of the trivial-zero first bridge and the nonzero second bridge. Equations (9.184) and (9.183) give the required replacements. The two main Stieltjes articles already include conjugation in their displayed first bridges; those correct formulas should be retained. Exact source locators and numerical counterexamples to the omitted-conjugation versions appear in the correction register.

9.17 Experimental independence tests

The tables introduce $\zeta''(2), \zeta''(3), \beta''(2), \beta''(3)$ and $L''(2,\chi_{-3}), L''(3,\chi_{-3})$. Historical searches at 200 digits with planted controls recovered the known value reductions and found no accepted relation for the six derivative coordinates in the lower-order baskets. This supports using them as named coordinates in a reduction table. Their mutual independence, and the impossibility of reductions in larger baskets, remain open.

9.18 Elementary grids and the sixth-point completion

For $q = 3, 4, 6$ define

$$A_3(s) = \frac{3^s-1}{2}\zeta(s), \quad A_4(s) = \frac{4^s-2^s}{2}\zeta(s), \quad A_6(s) = \frac{(2^s-1)(3^s-1)}{2}\zeta(s),$$

$$B_3(s) = \frac{3^s}{2}L(s,\chi_{-3}), \quad B_4(s) = \frac{4^s}{2}\beta(s), \quad B_6(s) = \frac{6^s+3^s}{2}L(s,\chi_{-3}).$$

Residue-class decomposition proves $\zeta(s,1/q) = A_q(s) + B_q(s)$ and $\zeta(s,1-1/q) = A_q(s) - B_q(s)$ for these three q. Consequently all four derivative tables, including the omitted sixth-point rows in the later article, follow from the differential operators

$$\gamma_1'(a) = (\partial_s+1)\zeta(s,a)|_{s=2}, \qquad \gamma_1''(a) = -(2\partial_s+3)\zeta(s,a)|_{s=3},$$
$$\gamma_2'(a) = -(\partial_s^2+2\partial_s)\zeta(s,a)|_{s=2}, \qquad \gamma_2''(a) = (2\partial_s^2+6\partial_s+2)\zeta(s,a)|_{s=3}.$$

This supplies an exact symbolic recipe for every entry, without new fitted constants. Write $a = \log 2$, $b = \log 3$, and $L_j(s) = L^{(j)}(s, \chi_{-3})$. At the sixth points the first-constant rows are

$$\begin{aligned}
\gamma_1'(1/6),\ \gamma_1'(5/6) &= 12\zeta'(2) + 2\pi^2 + \frac{8}{3}\pi^2 a + \frac{9}{4}\pi^2 b \\
&\quad \pm \left[\frac{45}{2}L_1(2) + \left(\frac{45}{2} + 18a + \frac{45}{2}b\right) L_0(2)\right], \\
\gamma_1''(1/6),\ \gamma_1''(5/6) &= -182\zeta'(3) - (273 + 208a + 189b)\zeta(3) \\
&\quad \mp \left[243L_1(3) + \left(\frac{729}{2} + 216a + 243b\right) L_0(3)\right].
\end{aligned}$$

The second-constant rows are

$$\begin{aligned}
\gamma_2'(1/6),\ \gamma_2'(5/6) &= S_2 \mp O_2, \\
S_2 &= -12\zeta''(2) - (24 + 32a + 27b)\zeta'(2) \\
&\quad - \pi^2\left(\frac{16}{3}a + \frac{9}{2}b + \frac{8}{3}a^2 + \frac{9}{4}b^2 + 6ab\right), \\
O_2 &= \frac{45}{2}L_2(2) + (45 + 36a + 45b)L_1(2) \\
&\quad + \left(36a + 45b + 18a^2 + \frac{45}{2}b^2 + 36ab\right) L_0(2), \\
\gamma_2''(1/6),\ \gamma_2''(5/6) &= S_3 \pm O_3, \\
S_3 &= 182\zeta''(3) + (546 + 416a + 378b)\zeta'(3) \\
&\quad + (182 + 624a + 567b + 208a^2 + 189b^2 + 432ab)\zeta(3), \\
O_3 &= 243L_2(3) + (729 + 432a + 486b)L_1(3) \\
&\quad + (243 + 648a + 729b + 216a^2 + 243b^2 + 432ab)L_0(3).
\end{aligned}$$

The upper sign is the 1/6 entry. These formulas consolidate the tables from the first-derivative, second-derivative and general-order reports.

9.19 Fixed-coordinate periodic collision contacts

The differentiation anomaly of Theorem 9.3 already fixes the individual periodic factors. A second question remains: does convolution of those factors equal the coordinate finite-part extension of their separated correlation? The independent continuations [121] give the same answer at every index and derivative order. The comparison below keeps the coordinate x for each original factor and the independent coordinates a and $1 - a$ for the shift. In this convention there is exactly one contact term. Its coefficient is distinct from the scalar constant in an off-point collision expansion.

Write $T_{m,p} = F_{m,p}$, with $T_{m,0} = G_m$, in the notation of (9.4). Reflection is denoted by a check, so $\widehat{\check{U}}(k) = \widehat{U}(-k)$. Set

$$P_p(u) = \frac{(1+u)_p}{p!} = \prod_{j=1}^{p}(1 + u/j), \qquad P_0 = 1, \qquad H_p^{(j)} = \sum_{l=1}^{p} l^{-j}.$$

All indices m, n, p, q are nonnegative integers; put $r = p + q$ and $w = u + v$. The scalar Stieltjes

germ and its argument derivatives are

$$g(u,x)=\zeta(1+u,x)-1/u=\sum_{m\ge0}(-1)^m\gamma_m(x)u^m/m!, \tag{9.189}$$

$$G_p(u,x)=\partial_x^p g(u,x)=(-1)^p(1+u)_p\zeta(1+p+u,x)-\delta_{p0}/u. \tag{9.190}$$

The apparent pole in g is removable for each $x>0$.

9.19.1 Whole resonant numerators and Fourier input

Let $\mathcal{Z}_s=Z(s)$ be the meromorphic family of Lemma 8.60. Its Fourier coefficients are

$$\widehat{\mathcal{Z}_s}(0)=0,\qquad \widehat{\mathcal{Z}_s}(k)=\Gamma(1-s)(2\pi\mathrm{i}k)^{s-1}\quad(k\neq0), \tag{9.191}$$

on the branch (8.60).

Lemma 9.76 (Full resonant completion)**.** *The germ generating the unit-coordinate finite parts is*

$$T_p(u):=\sum_{m\ge0}\frac{(-1)^m}{m!}T_{m,p}u^m=\mathrm{D}^p\mathcal{Z}_{1+u}-\frac{\delta_{p0}}{u}\,1+\frac{P_p(u)}{u}\delta_0^{(p)}. \tag{9.192}$$

It is holomorphic near zero, and every $T_{m,p}$ has mean zero.

Proof. The local singular part of G_p is $c_p(u)x^{-p-1-u}$, with $c_p(u)=(-1)^p p!P_p(u)$; its remaining shifted Hurwitz part is jointly holomorphic. Subtract the test-function Taylor polynomial through degree p. Its indices $j<p$ contribute $c_p(u)\varphi^{(j)}(0)/(j!(j-p-u))$, holomorphic near zero. The index p contributes

$$-\frac{c_p(u)\varphi^{(p)}(0)}{p!u}=-\frac{P_p(u)}{u}\langle\delta_0^{(p)},\varphi\rangle.$$

Every coefficient of this resonant monomial is proportional to $x^{-1}\log^j x$, whose unit-coordinate finite-part integral on $(0,1)$ is zero. Thus its whole numerator must be removed, including its regular Taylor coefficients. This gives (9.192), using $\mathrm{D}^p\mathcal{Z}_{1+u}=(-1)^p(1+u)_p\mathcal{Z}_{1+p+u}$. After subtraction the local remainder, with any fixed number of spectral derivatives, is dominated by $Cx^{-\eta}(1+|\log x|^M)$, $\eta<1$. Consequently pairing and coefficient extraction are holomorphic and give exactly the cutoff finite parts. The nonlocal zero Fourier mode is zero; for $p=0$ the last two terms cancel in mean, and for $p>0$ both have mean zero. □

The existing differentiation anomaly becomes

$$T_{m,p}=\mathrm{D}^pG_m+\kappa_{m,p}\delta_0^{(p)},\qquad \kappa_{m,p}=(-1)^m m![u^{m+1}]P_p(u). \tag{9.193}$$

This is the earlier elementary-symmetric formula, since $[u^{m+1}]P_p=e_{m+1}(p)$. It also follows immediately by subtracting $\mathrm{D}^pT_0(u)$ from (9.192). In particular $\kappa_{0,p}=H_p$ and $\kappa_{m,p}=0$ when $m\ge p$.

For a scalar complete exponential Bell polynomial B_j, define

$$b_m(z)=\frac{(-1)^{m+1}}{m+1}B_{m+1}(z,1!\zeta(2),2!\zeta(3),\dots,m!\zeta(m+1)). \tag{9.194}$$

Equivalently, B_j is defined by $\exp(\sum_{l\ge1}x_lu^l/l!)=\sum_{j\ge0}B_ju^j/j!$. For $k\neq0$, put $L_k=\gamma+\log(2\pi\mathrm{i}k)$. Then

$$\widehat{T}_{m,p}(k)=(2\pi\mathrm{i}k)^p\big(b_m(L_k)+\kappa_{m,p}\big),\qquad \widehat{T}_{m,p}(0)=0. \tag{9.195}$$

Indeed (9.192) gives the nonzero-mode germ $(2\pi\mathrm{i}k)^p\{\Gamma(-u)(2\pi\mathrm{i}k)^u + P_p(u)/u\}$. Insert $\Gamma(-u) = -\Gamma(1-u)/u$ and

$$\Gamma(1-u)(2\pi\mathrm{i}k)^u = \exp\left(L_k u + \sum_{j\geq 2}\zeta(j)u^j/j\right),$$

the classical gamma expansion [122]. This proves (9.195); for example $b_0(z) = -z$ and $b_1(z) = (z^2+\zeta(2))/2$.

The lowest-index comparison, proved in the general theorem below, is

$$\check{G}_0 * G_0 = \mathrm{FP}_{\mathbb{T},a}\left[\gamma_1(a) + \gamma_1(1-a) - \pi^2/3\right] + \frac{\pi^2}{3}\delta_0. \tag{9.196}$$

Both factors have zero mean. The atom restores that same mean in the convolution; it is invisible in the punctured-circle function.

9.20 The off-collision kernel and its second finite part

For $0 < a < 1$, set $b = 1-a$ and define, in the original integration variable,

$$J^{p,q}_{mn}(a) = \mathrm{FP}\int_0^1 \gamma_m^{(p)}(x)\gamma_n^{(q)}(\{x+a\})\,\mathrm{d}x. \tag{9.197}$$

There are two independent right-hand singularities, at $x = 0+$ and $x = b+$; the second coordinate is $x-b$. This is the convention of [121]. Define the distinct distribution

$$\mathsf{J}^{p,q}_{mn} = \mathrm{FP}_{\mathbb{T},a}\, J^{p,q}_{mn}(a). \tag{9.198}$$

The second finite part is taken in the shift variable, at both $a = 0+$ and $a = 1-$.

We recall the off-point generating kernel, deriving the ingredients needed for the distributional comparison. Put

$$\mathcal{E}(u,v) = \frac{\Gamma(1-u)\Gamma(1+u+v)}{\Gamma(1+v)}, \tag{9.199}$$

$$\mathcal{A}(u,v) = \frac{\Gamma(1-u)\Gamma(1-v)}{\Gamma(1-u-v)}\frac{\cos(\pi(u-v)/2)}{\cos(\pi(u+v)/2)}. \tag{9.200}$$

These are holomorphic near the origin and satisfy

$$v\mathcal{E}(u,v) + u\mathcal{E}(v,u) = (u+v)\mathcal{A}(u,v), \qquad \mathcal{A}(u,0) = \mathcal{A}(0,v) = 1. \tag{9.201}$$

The identity follows by inserting the gamma reflection formula. Equivalently, it follows by comparing the two Hurwitz forms of the classical spectral correlation.

For $r \geq 1$, write

$$W_r(t,x) = (1+t)_r\zeta(1+r+t,x) = (-1)^r G_r(t,x).$$

The off-point germ whose coefficients are $(-1)^{m+n}J^{p,q}_{mn}/(m!n!)$ is

$$\begin{aligned}\mathcal{J}_{pq}(u,v;a) = (-1)^q&\frac{P_p(u)W_r(v,a) - \mathcal{E}(u,v)W_r(w,a)}{u}\\ &+ (-1)^p\frac{P_q(v)W_r(u,b) - \mathcal{E}(v,u)W_r(w,b)}{v}.\end{aligned} \tag{9.202}$$

For $p = q = 0$, the corresponding expression is

$$\mathcal{J}_{00}(u,v;a) = \frac{g(v,a) - \mathcal{E}(u,v)g(w,a)}{u} + \frac{g(u,b) - \mathcal{E}(v,u)g(w,b)}{v} + \frac{1 - \mathcal{A}(u,v)}{uv}. \tag{9.203}$$

Each divided difference is holomorphic: its numerator vanishes on the relevant spectral axis.

To verify these formulas, start where all ordinary integrals converge and use the Hurwitz Fourier series. The shifted spectral kernel is

$$C(s,t;a) = B(1-s, s+t-1)\zeta(s+t-1,a) + B(1-t, s+t-1)\zeta(s+t-1, 1-a). \tag{9.204}$$

It follows either by gamma reflection from the product of Fourier coefficients, or from the classical polylogarithmic expression. This product-kernel method is classical [28]; the shifted specialization and its Stieltjes completion are already used in [121]. Multiply by $(-1)^{p+q}(1+u)_p(1+v)_q$, set $s = 1+p+u, t = 1+q+v$, and remove the *full* resonant Taylor contribution at each of the two original singularities. The gamma identity

$$(-1)^p(1+u)_p\Gamma(-p-u) = \Gamma(-u)$$

then gives (9.202); the subtractions of $1/u$ and $1/v$ give (9.203). The same finite Taylor-subtraction argument as in Lemma 9.76 justifies holomorphy and coefficient extraction.

The exact Hurwitz shift shows that, near $a = 0+$,

$$J^{p,q}_{mn}(a) = a^{-r-1}\mathcal{Q}^{p,q}_{mn}(\log a) + \text{an analytic function of } a,$$

where $\deg \mathcal{Q}^{p,q}_{mn} \le m+n+1$. The other endpoint is obtained by interchanging (m,p) and (n,q) and reflecting. Consequently (9.198) is well defined, with only finitely many test-function Taylor subtractions at each endpoint.

Taking the shift-variable finite part in (9.202) yields, for $r \ge 1$,

$$\mathsf{J}_{pq}(u,v) = (-1)^p\frac{P_p(u)T_r(v) - \mathcal{E}(u,v)T_r(w)}{u} + (-1)^q\frac{P_q(v)\check{T}_r(u) - \mathcal{E}(v,u)\check{T}_r(w)}{v}. \tag{9.205}$$

For $r = 0$, use the same formula with $P_0 = 1$ and add $(1-\mathcal{A})/(uv)$ times the constant distribution 1. This expression specifies the full second finite part; it is not merely a choice of extension having the correct off-point values.

9.21 The all-index collision-contact theorem

Define the convolution distribution

$$\mathsf{C}^{p,q}_{mn} = \check{T}_{m,p} * T_{n,q}. \tag{9.206}$$

Its restriction to the punctured circle is (9.197). Indeed, for a nonzero shift the singular supports of the two factors are disjoint. Multiplication by the other, smooth factor near each singular point gives exactly the two original coordinate finite parts. A partition of unity in x makes this assertion a direct consequence of the distribution definitions. It does not involve defining a pointwise product of two distributions at a coincident singularity.

Theorem 9.77 (Complete periodic contact correction)**.** *For all nonnegative m, n, p, q, with $r = p+q$,*

$$\boxed{\mathsf{C}^{p,q}_{mn} = \mathsf{J}^{p,q}_{mn} + \Delta^{p,q}_{mn}\delta_0^{(r)}.} \tag{9.207}$$

The coefficient is

$$\Delta^{p,q}_{mn} = (-1)^{m+n+p} m!n![u^{m+1}v^{n+1}]\mathcal{N}_{pq}(u,v), \tag{9.208}$$

$$\begin{aligned}\mathcal{N}_{pq}(u,v) = {}& \mathcal{A}(u,v)P_r(u+v) - P_p(u)P_r(v) \\ & - P_q(v)P_r(u) + P_p(u)P_q(v).\end{aligned} \tag{9.209}$$

Equivalently, the spectral generating coefficient is the holomorphic germ

$$\Delta_{pq}(u,v) = \frac{(-1)^p}{uv}\mathcal{N}_{pq}(u,v). \tag{9.210}$$

There are no additional lower-order delta derivatives in this fixed convention.

Proof. First let $r \geq 1$. Work with meromorphic distribution germs before taking finite parts. Set $U_p(u) = \mathrm{D}^p \mathcal{Z}_{1+u}$. By (9.204) and the gamma cancellation preceding (9.205), the raw correlation is

$$\check{U}_p(u) * U_q(v) = -\frac{(-1)^p \mathcal{E}(u,v)}{u}\mathrm{D}^r \mathcal{Z}_{1+w} - \frac{(-1)^q \mathcal{E}(v,u)}{v}\mathrm{D}^r \check{\mathcal{Z}}_{1+w}.$$

This identity can also be checked at each nonzero Fourier mode from (9.191). It holds as a meromorphic distribution identity because it holds in an ordinary convergence region and all its Fourier coefficients continue meromorphically.

Insert (9.192) in $\check{T}_p(u) * T_q(v)$. Constant-distribution terms vanish when $r \geq 1$, since at least one factor has zero mean and a positive derivative. The two cross terms and the delta–delta term give

$$\begin{aligned}\mathsf{C}_{pq}(u,v) = {}& (-1)^p \frac{P_p(u)\mathrm{D}^r \mathcal{Z}_{1+v} - \mathcal{E}(u,v)\mathrm{D}^r \mathcal{Z}_{1+w}}{u} \\ & + (-1)^q \frac{P_q(v)\mathrm{D}^r \check{\mathcal{Z}}_{1+u} - \mathcal{E}(v,u)\mathrm{D}^r \check{\mathcal{Z}}_{1+w}}{v} + \frac{(-1)^p P_p(u)P_q(v)}{uv}\delta_0^{(r)}.\end{aligned}$$

Replace $\mathrm{D}^r \mathcal{Z}_{1+t}$ by $T_r(t) - P_r(t)\delta_0^{(r)}/t$ and use $\check{\delta}_0^{(r)} = (-1)^r \delta_0^{(r)}$. The nonlocal terms are exactly (9.205). The remaining coefficient of $\delta_0^{(r)}$, after factoring $(-1)^p$, is

$$-\frac{P_p(u)P_r(v)}{uv} - \frac{P_q(v)P_r(u)}{uv} + \frac{P_p(u)P_q(v)}{uv} + \frac{P_r(w)}{w}\left(\frac{\mathcal{E}(u,v)}{u} + \frac{\mathcal{E}(v,u)}{v}\right).$$

Equation (9.201) gives (9.210).

For $p = q = 0$, keep the zero-mode terms explicitly. Since $T_0(u) = \mathcal{Z}_{1+u} + (\delta_0 - 1)/u$, their convolution is

$$\mathsf{C}_{00}(u,v) = \frac{\mathcal{Z}_{1+v} - \mathcal{E}(u,v)\mathcal{Z}_{1+w}}{u} + \frac{\check{\mathcal{Z}}_{1+u} - \mathcal{E}(v,u)\check{\mathcal{Z}}_{1+w}}{v} + \frac{\delta_0 - 1}{uv}.$$

Replace $\mathcal{Z}_{1+t} = T_0(t) + (1-\delta_0)/t$ and apply (9.201). The result is

$$\mathsf{C}_{00}(u,v) = \mathsf{J}_{00}(u,v) + \frac{\mathcal{A}(u,v) - 1}{uv}\delta_0,$$

which is exactly (9.210) with all P's equal to one.

The numerator (9.209) vanishes identically on $u = 0$ and on $v = 0$, because $P_j(0) = 1$ and $\mathcal{A}$ equals one on both axes. Therefore it is divisible by uv as a holomorphic germ. Taking coefficients proves (9.207). The displayed calculation exhausts the difference, and produces only $\delta_0^{(r)}$; hence there are no omitted lower contacts. □

9.21.1 Why the correction has no Stieltjes constants

The logarithm of the gamma–trigonometric factor is

$$\log \mathcal{A}(u,v) = \sum_{k\geq 2} \frac{\zeta(k)}{k}\big(u^k + v^k - (u+v)^k\big) - \sum_{j\geq 1} \frac{(1-2^{-2j})\zeta(2j)}{j}\big((u-v)^{2j} - (u+v)^{2j}\big). \tag{9.211}$$

Euler's constant cancels from the gamma ratio. It follows that

$$\Delta_{mn}^{p,q} \in \mathbb{Q}[\zeta(2), \zeta(3), \ldots],$$

with the finite harmonic data encoded in rational coefficients of P_p, P_q, P_r. In particular there are no ordinary or generalized Stieltjes constants and no spectral derivatives of zeta in the contact coefficient. This is an assertion about the explicit expression, not an arithmetic independence theorem.

The reflection law is

$$\Delta_{nm}^{q,p} = (-1)^r \Delta_{mn}^{p,q}. \tag{9.212}$$

It follows both from (9.209) and from reflecting the convolution and its coordinate extension. Reflection changes $\delta_0^{(r)}$ by precisely the necessary sign.

9.21.2 All derivative orders at Stieltjes index zero

Corollary 9.78 (Polygamma contact formula)**.** *For $r = p + q$, including $r = 0$,*

$$\boxed{\Delta_{00}^{p,q} = (-1)^p \left[\frac{\pi^2}{3} - H_r^{(2)} + (H_r - H_p)(H_r - H_q)\right].} \tag{9.213}$$

Proof. The coefficient of uv in $\mathcal{A}$ is $2\zeta(2)$, and its linear part is zero. The coefficient of uv in $P_r(u+v)$ is $H_r^2 - H_r^{(2)}$. Extracting the remaining products in (9.209) gives

$$2\zeta(2) + H_r^2 - H_r^{(2)} - (H_p + H_q)H_r + H_p H_q,$$

which is (9.213). □

Some low-order values are

(p,q)	r	$\Delta_{00}^{p,q}$
$(0,0)$	0	$\pi^2/3$
$(0,1)$	1	$\pi^2/3 - 1$
$(1,0)$	1	$1 - \pi^2/3$
$(1,1)$	2	$1 - \pi^2/3$
$(0,2)$	2	$\pi^2/3 - 5/4$
$(1,2)$	3	$13/12 - \pi^2/3$

These are corrections to a *distribution in the shift*, not the values of the common-point product integrals studied in [121].

9.21.3 A closed parity law and derivative-split stabilization

Corollary 9.79 (The complete first row)**.** *For every $m \geq 0$,*

$$\Delta_{m0}^{0,0} = m!\zeta(m+2)\begin{cases} 3-2^{-m}, & m \text{ even}, \\ 1, & m \text{ odd}. \end{cases} \tag{9.214}$$

Proof. Differentiating (9.200) in v and setting $v = 0$ gives

$$\partial_v \mathcal{A}(u,0) = \psi(1-u) - \psi(1) + \pi \tan\frac{\pi u}{2}. \tag{9.215}$$

The coefficient of u^k is $-\zeta(k+1)$ for even $k \geq 2$, and $(3-2^{1-k})\zeta(k+1)$ for odd $k \geq 1$. This follows from the gamma series and the cosine product, or by differentiating (9.211). Substitute $k = m+1$ in (9.208). □

For example,

$$\begin{aligned} &\Delta_{10}^{0,0} = \zeta(3), && \Delta_{20}^{0,0} = \tfrac{11}{2}\zeta(4), && \Delta_{30}^{0,0} = 6\zeta(5), \\ &\Delta_{11}^{0,0} = \tfrac{7}{2}\zeta(4), && \Delta_{21}^{0,0} = 4\zeta(5) + 4\zeta(2)\zeta(3), && \Delta_{22}^{0,0} = 4\zeta(3)^2 + \tfrac{83}{2}\zeta(6). \end{aligned}$$

Corollary 9.80 (Stabilization of the derivative split)**.** *If $m \geq r$ or $n \geq r$, then*

$$\Delta_{mn}^{p,q} = (-1)^p \Delta_{mn}^{0,r}. \tag{9.216}$$

Proof. Every product in (9.209) other than $\mathcal{A}P_r(u+v)$ has degree at most r in each variable. The indicated coefficient therefore vanishes in those products as soon as either $m+1 > r$ or $n+1 > r$. The remaining expression depends only on r, apart from the factor $(-1)^p$. □

An explicit first-row generator for arbitrary derivative orders is also useful:

$$\begin{aligned} \Delta_{m0}^{p,q} = (-1)^{m+p} m![u^{m+1}]\big[&P_r(u)\partial_v \mathcal{A}(u,0) + P_r'(u) \\ &- H_q P_r(u) + (H_q - H_r)P_p(u)\big]. \end{aligned} \tag{9.217}$$

For $m \geq r$, the last three polynomial terms disappear. Formula (9.215) then gives a finite sum of zeta values with indices from $m+2-r$ through $m+2$, weighted by elementary symmetric polynomials in reciprocal integers.

9.22 Exact convergent integral identities from finite Fourier data

The distributional result produces ordinary integrals once the test function vanishes to sufficient order at the collision. Put

$$V_{M,k}(a) = (1-\cos(2\pi k a))^M, \qquad M, k \in \mathbb{N}_{>0}. \tag{9.218}$$

Its finite Fourier expansion is

$$V_{M,k}(a) = 2^{-M} \sum_{j=-M}^{M} (-1)^j \binom{2M}{M+j} e^{2\pi \mathrm{i} j k a}. \tag{9.219}$$

This follows by expanding $2^{-M}(2 - e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta})^M$. It vanishes to order $2M$ at both identified endpoints.

Theorem 9.81 (Ordinary all-index correlation integrals)**.** *If $2M > r = p + q$, then*

$$\int_0^1 V_{M,k}(a) J_{mn}^{p,q}(a)\,\mathrm{d}a = 2^{1-M} \sum_{j=1}^{M} (-1)^j \binom{2M}{M+j} \operatorname{Re}\Big(\widehat{T}_{m,p}(-jk)\widehat{T}_{n,q}(jk)\Big). \tag{9.220}$$

The integral is ordinary and absolutely convergent. The right-hand side is a finite exact expression given by (9.195)*.*

Proof. The endpoint expansion in Section 9.20 bounds the weighted integrand by a constant times $a^{2M-r-1}(1 + |\log a|^{m+n+1})$ near zero, and similarly at one. Since $2M > r$, the integral converges. All derivatives of $V_{M,k}$ through degree r vanish at zero, so the contact in (9.207) pairs to zero. Pair the convolution distribution with (9.219); its zero Fourier coefficient is zero. Realness of the original distributions combines the positive and negative modes into the displayed real part. □

At Stieltjes index zero, put $\lambda_j = \gamma + \log(2\pi jk)$. Then the Fourier product in (9.220) is

$$(-1)^p (2\pi \mathrm{i} jk)^r \left[(\lambda_j - H_p)(\lambda_j - H_q) + \frac{\pi^2}{4} + \frac{\pi \mathrm{i}}{2}(H_q - H_p)\right]. \tag{9.221}$$

For even $r = 2d$ its real part is

$$(-1)^{p+d} (2\pi jk)^r \left[(\lambda_j - H_p)(\lambda_j - H_q) + \frac{\pi^2}{4}\right],$$

whereas for odd $r = 2d + 1$ it is

$$(-1)^{p+d+1} (2\pi jk)^r \frac{\pi}{2}(H_q - H_p).$$

Thus all logarithms disappear from these symmetric weighted polygamma correlations when the total derivative degree is odd.

At the first index, with $\lambda_k = \gamma + \log(2\pi k)$,

$$\int_0^1 (1 - \cos(2\pi ka)) \left[\gamma_1(a) + \gamma_1(1-a) - \frac{\pi^2}{3}\right] \mathrm{d}a = -\lambda_k^2 - \frac{\pi^2}{4}. \tag{9.222}$$

The corresponding single-factor consequences of (9.195) are

$$\int_0^1 (1 - \cos(2\pi kx))[-\psi(x)]\,\mathrm{d}x = \lambda_k, \tag{9.223}$$

$$\int_0^1 (1 - \cos(2\pi kx))\gamma_1(x)\,\mathrm{d}x = \frac{\pi^2}{24} - \frac{1}{2}\lambda_k^2. \tag{9.224}$$

These elementary Fourier specializations are included as checks and applications of the conventions, not as claims of priority over the classical Fourier theory.

9.23 A Hurwitz kernel for all positive-integer spectral jets

A second route makes the spectral zeta derivatives in the weighted identities explicit. Define the finite exponential sum

$$S_M(t) = \sum_{j=1}^{M} (-1)^j \binom{2M}{M+j} j^t, \qquad j^t = e^{t \log j}. \tag{9.225}$$

It is entire, and $S_M^{(h)}(t)$ is the same finite sum with a factor $\log^h j$.

Theorem 9.82 (Weighted Hurwitz kernel)**.** *For* $\operatorname{Re} s < 2M+1$, $s \neq 1$,

$$\int_0^1 V_{M,k}(x)\zeta(s,x)\,\mathrm{d}x = 2^{1-M}\Gamma(1-s)(2\pi k)^{s-1}\sin\frac{\pi s}{2}\,S_M(s-1). \tag{9.226}$$

At $s = 2,\dots,2M$*, the right-hand side is interpreted by its removable limit. The identity can be spectrally differentiated to every finite order in this domain, except at the genuine pole* $s = 1$*.*

Proof. For $\operatorname{Re} s < 0$, pair the absolutely convergent Hurwitz Fourier expansion with (9.219). Its zero mode vanishes. The two signs of each nonzero frequency combine to $2\Gamma(1-s)(2\pi jk)^{s-1}\cos(\pi(s-1)/2)$, and the cosine equals $\sin(\pi s/2)$. This gives (9.226).

For the larger domain, use $\zeta(s,x) = x^{-s} + \zeta(s,1+x)$. The weight is $O(x^{2M})$, so the first term and each of its spectral derivatives are integrable when $\operatorname{Re} s < 2M+1$. The shifted factor is holomorphic in s except for its usual pole at one and locally bounded in x. Analytic continuation proves the formula on the stated connected domain and shows that the apparent poles at the other displayed integers are removable. □

The cancellations are finite identities, not numerical limiting phenomena:

$$S_M(2d) = 0 \qquad (1 \le d < M). \tag{9.227}$$

Indeed, differentiate (9.219) $2d$ times at zero and use the order-$2M$ zero of the weight. At even spectral integers the sine factor instead cancels the gamma pole.

9.23.1 Complete holomorphic resonance germs

The next formulas give every derivative, not just the value at a resonance. Let h be a small spectral perturbation. For an odd integer $s_0 = 2d+1$, with $1 \le d < M$,

$$\int_0^1 V_{M,k}(x)\zeta(2d+1+h,x)\,\mathrm{d}x = \frac{(-1)^{d+1}2^{1-M}(2\pi k)^{2d}}{(2d)!} \times \frac{\Gamma(1-h)(2\pi k)^h\cos(\pi h/2)}{P_{2d}(h)}\,\frac{S_M(2d+h)}{h}. \tag{9.228}$$

For an even integer $s_0 = 2d$, with $1 \le d \le M$,

$$\int_0^1 V_{M,k}(x)\zeta(2d+h,x)\,\mathrm{d}x = \frac{(-1)^d 2^{1-M}(2\pi k)^{2d-1}}{(2d-1)!} \times \frac{\Gamma(1-h)(2\pi k)^h}{P_{2d-1}(h)}\,\frac{\sin(\pi h/2)}{h}\,S_M(2d-1+h). \tag{9.229}$$

Both right-hand sides are holomorphic. These formulas follow directly from

$$\Gamma(-n-h) = \frac{(-1)^{n+1}\Gamma(1-h)}{n!\,hP_n(h)}.$$

Consequently, the integral with $\zeta^{(j)}(s_0,x)$ is exactly $j!$ times the coefficient of h^j in the corresponding right-hand side. The logarithm of its common gamma–harmonic factor is

$$\log\frac{\Gamma(1-h)(2\pi k)^h}{P_n(h)} = (\gamma + \log(2\pi k) - H_n)h + \sum_{j\ge 2}\frac{\zeta(j)+(-1)^j H_n^{(j)}}{j}h^j. \tag{9.230}$$

This is a finite coefficient algorithm in gamma, harmonic numbers, ordinary zeta values, and logarithms of the integers $1,\dots,M$. No derivative of an infinite special-function series has to be fitted numerically.

At the pole $s=1$, put $c_M = 2^{-M}\binom{2M}{M}$, the mean of the weight. The Stieltjes counterpart is the holomorphic identity

$$\int_0^1 V_{M,k}(x)g(h,x)\,\mathrm{d}x = \frac{-2^{1-M}\Gamma(1-h)(2\pi k)^h\cos(\pi h/2)S_M(h) - c_M}{h}. \tag{9.231}$$

Its numerator vanishes at zero because $S_M(0) = -\binom{2M}{M}/2$. Extracting $(-1)^m m![h^m]$ evaluates the ordinary weighted integral of $\gamma_m(x)$ at every index. For $M=1$, this recovers (9.223) and (9.224).

9.23.2 Explicit logarithmic identities at order three

For every positive integer k, put $\lambda_k = \gamma + \log(2\pi k)$. Taking $M=2, d=1$ in (9.228), where $S_2(t) = -4 + 2^t$, gives

$$\boxed{\int_0^1 (1-\cos(2\pi kx))^2\zeta(3,x)\,\mathrm{d}x = 4\pi^2k^2\log 2.} \tag{9.232}$$

$$\boxed{\int_0^1 (1-\cos(2\pi kx))^2\zeta'(3,x)\,\mathrm{d}x = 4\pi^2k^2\log 2\left(\lambda_k - \frac{3}{2} + \frac{\log 2}{2}\right).} \tag{9.233}$$

Here ζ' denotes the spectral derivative. To see the derivative constant explicitly, the common factor in (9.228) has logarithmic derivative $\lambda_k - H_2 = \lambda_k - 3/2$ at zero, while

$$\frac{S_2(2+h)}{h} = 4\log 2 + 2\log^2 2\,h + O(h^2).$$

The local integrands behave like x and $x\log x$, respectively, so these are ordinary absolutely convergent integrals. Their polygamma form follows from $\psi^{(2)}(x) = -2\zeta(3,x)$.

The correlation identity in Theorem 9.81, at $(p,q)=(1,1)$, gives the compatible combination

$$\int_0^1 (1-\cos(2\pi kx))^2\big[\zeta(3,x) + 2\zeta'(3,x)\big]\,\mathrm{d}x = 4\pi^2k^2\big[2\log 2(\lambda_k - 1) + \log^2 2\big]. \tag{9.234}$$

This agreement checks the derivative signs and the harmonic constants by two exact routes: the shifted-contact calculus and the single-Hurwitz spectral kernel.

9.24 Polylogarithm boundary values and invisible local terms

For $\sigma \in \{+1,-1\}$, let

$$\mathfrak{L}_s^\sigma(a) = \sum_{k\ge 1}\frac{e^{\sigma 2\pi i k a}}{k^s} \tag{9.235}$$

be interpreted as a periodic distribution. Its Fourier coefficients have polynomial growth on every compact spectral set, so this is an entire distribution-valued function of s. It is also the Abel limit of $\mathrm{Li}_s(\rho e^{\sigma 2\pi i a})$ as $\rho\uparrow 1$. Spectral derivatives multiply the coefficients by powers of $-\log k$. Off the integer points these distributions agree with the usual analytic boundary values of polylogarithms [96].

Put $c=\gamma+\log(2\pi)$ and expand the finite polynomial

$$(-1)^p(\sigma 2\pi\mathrm{i})^r\big[b_m(c+X-\sigma\mathrm{i}\pi/2)+\kappa_{m,p}\big]\big[b_n(c+X+\sigma\mathrm{i}\pi/2)+\kappa_{n,q}\big]=\sum_{j=0}^{m+n+2}a_{j,\sigma}X^j. \tag{9.236}$$

Equation (9.195) proves the all-index polylogarithmic identity

$$\mathsf{C}^{p,q}_{mn}=\sum_{\sigma=\pm1}\sum_{j=0}^{m+n+2}a_{j,\sigma}(-1)^j\,\partial_s^j\mathfrak{L}^\sigma_s\big|_{s=-r}. \tag{9.237}$$

This is a finite formula; its equality is checked at every Fourier mode, including the zero mode, which vanishes on both sides. Together with Theorem 9.77, it identifies the exact delta derivative that must be added to the coordinate extension of the corresponding off-point polylogarithm identity.

The elementary cautionary example is

$$\mathfrak{L}^+_0+\mathfrak{L}^-_0=\delta_0-1. \tag{9.238}$$

For $a\notin\mathbb{Z}$, the ordinary rational values satisfy $\mathrm{Li}_0(e^{2\pi\mathrm{i}a})+\mathrm{Li}_0(e^{-2\pi\mathrm{i}a})=-1$. The missing delta cannot be read from that pointwise identity. Similarly, for $r\geq1$,

$$\mathfrak{L}^+_{-r}+(-1)^r\mathfrak{L}^-_{-r}=\frac{\delta_0^{(r)}}{(2\pi\mathrm{i})^r}. \tag{9.239}$$

These identities follow immediately by adding positive and negative Fourier modes, or by differentiating (9.238). They explain why substituting rational negative-order polylogarithms before fixing distributional boundary values can erase precisely the terms computed by the contact theorem.

9.25 Exact primitives and the log-gamma bridge

Define the mean-zero periodic inverse derivative by

$$\widehat{\mathcal{I}U}(k)=\frac{\widehat{U}(k)}{2\pi\mathrm{i}k}\quad(k\neq0),\qquad\widehat{\mathcal{I}U}(0)=0.$$

It satisfies $\mathrm{D}\mathcal{I}U=U-\widehat{U}(0)1$. Let B_h be the Bernoulli polynomial, periodically interpreted when necessary. Its standard Fourier coefficients, obtainable also from the negative-integer Hurwitz values [93], give

$$\mathcal{I}^h\delta_0^{(r)}=\begin{cases}\delta_0^{(r-h)}, & h<r,\\ \delta_0-1, & h=r\geq1,\\ -B_{h-r}(\{a\})/(h-r)!, & h>r.\end{cases} \tag{9.240}$$

For $h=r=0$, the left side is simply δ_0 and the middle rule is not used.

Corollary 9.83 (All normalized contact primitives)**.** *Apply $\mathcal{I}^h$ to Theorem 9.77. For $h>r$ the contact correction is exactly*

$$\mathcal{I}^h\mathsf{C}^{p,q}_{mn}-\mathcal{I}^h\mathsf{J}^{p,q}_{mn}=-\Delta^{p,q}_{mn}\frac{B_{h-r}(\{a\})}{(h-r)!}. \tag{9.241}$$

There is no undetermined polynomial or integration constant: the zero Fourier mode is fixed by definition.

For $h \geq r+2$, the integrated convolution has an absolutely convergent Fourier series because its coefficients are $O(|k|^{r-h}(1+\log^{m+n+2}|k|))$. Thus sufficiently integrated identities are equalities of continuous ordinary functions, even though their derivatives contain contact distributions.

For a concrete gamma example, set

$$\ell(x) = \log\Gamma(x) - \frac{1}{2}\log(2\pi).$$

Raabe's mean and the Hurwitz identity for $\zeta'(0,x)$ give $\int_0^1 \ell(x)\,\mathrm{d}x = 0$. Distributionally $\mathrm{D}\ell = -T_{0,0}$; no extra delta occurs because the constant term in the endpoint expansion of $\log\Gamma(x)$ is zero and $\log\Gamma(1) = 0$. Hence $\ell = -\mathcal{I}T_{0,0}$ and

$$K_\Gamma(a) := \int_0^1 \ell(x)\ell(\{x+a\})\,\mathrm{d}x = -\mathcal{I}^2\mathsf{C}_{00}^{0,0}. \tag{9.242}$$

The integral exists also at $a = 0$ because logarithmic endpoint singularities are square integrable. The contact theorem gives the normalized form

$$K_\Gamma(a) = -\mathcal{I}^2\mathsf{J}_{00}^{0,0}(a) + \frac{\pi^2}{6}B_2(\{a\}). \tag{9.243}$$

The Bernoulli term is the integrated image of the $\pi^2\delta_0/3$ term in (9.196).

The ordinary Fourier representation is already (8.144), after subtracting the Raabe mean. The additional information here is its fixed-coordinate Bernoulli correction: it is the integrated image of the collision contact, which a punctured-circle polylogarithm identity cannot determine.

Evidence and remaining scope. The two independent contact generators agree after aligning their derivative signs and unit-coordinate conventions. Their finite exact tables and independent subtracted Fourier-mode quadratures test those normalizations; the analytic comparison above proves the all-index law. This completes the specified periodic extension problem. Cubic geometric collision limits, ordered higher-depth germs and canonical changes of subtraction coordinates retain their separate audit status. The contact coefficient does not assert independence of its displayed zeta coordinates.

Chapter 10

Positive kernels and the geometry of Stieltjes zeros

The derivative tower gives exact spectral coordinates. Its Mellin form does more: a reciprocal-gamma polynomial becomes a sign-changing kernel inside a positive Laplace transform. Root preservation bounds the total number of positive zeros, gamma concentration explains their motion, and outward rational computation certifies concrete base cases. This chapter unites the fixed-index zero geometry with the complementary spectral regime in which the Laurent index grows with the derivative order.

10.1 A unified Stieltjes–Lerch derivative family

For $a > 0$, write

$$\zeta(s,a) = \frac{1}{s-1} + \sum_{n\geq 0} \frac{(-1)^n}{n!}\gamma_n(a)(s-1)^n. \tag{10.1}$$

The superscript in $\gamma_n^{(k)}(a)$ means differentiation k times in a. In contrast, $\zeta^{(j)}(s,a)$ denotes differentiation j times in s. Set

$$P_k(t) = \prod_{j=1}^{k}\left(1+\frac{t}{j}\right) = \sum_{i=0}^{k} e_i(k)t^i, \qquad H_k^{(r)} = \sum_{j=1}^{k} j^{-r}, \quad H_k = H_k^{(1)}. \tag{10.2}$$

We take $H_0^{(r)} = 0$.

For $0 \leq \rho < 1$, define the convergent Lerch coefficients

$$\ell_n(\rho,a) = \sum_{m\geq 0} \frac{\rho^m \log^n(a+m)}{a+m} = (-1)^n\, \partial_s^n \Phi(\rho,s,a)|_{s=1}\,, \qquad \Phi(\rho,s,a) = \sum_{m\geq 0}\frac{\rho^m}{(a+m)^s}. \tag{10.3}$$

For $k \geq 1$ put

$$\mathcal{D}^{\rho}_{n,k}(a) = \begin{cases} \partial_a^k \ell_n(\rho,a), & 0 \leq \rho < 1, \\ \gamma_n^{(k)}(a), & \rho = 1. \end{cases} \tag{10.4}$$

Only the derivative family is unified at $\rho = 1$: the series defining ℓ_n generally diverges there. For $0 < \rho < 1$ the specialization $a = 1$ is directly polylogarithmic,

$$\ell_n(\rho,1) = \frac{(-1)^n}{\rho}\, \partial_s^n \operatorname{Li}_s(\rho)|_{s=1}\,. \tag{10.5}$$

At $\rho = 0$ it is simply $\log^n a/a$.

Proposition 10.1 (The classical master identity)**.** *For $k \geq 1$, $n \geq 0$, and $a > 0$,*

$$\gamma_n^{(k)}(a) = (-1)^{n+k} k! n! \sum_{i+j=n} e_i(k) \frac{\zeta^{(j)}(k+1, a)}{j!}. \tag{10.6}$$

The same formula with $\Phi(\rho, \cdot, a)$ in place of $\zeta(\cdot, a)$ holds for $\mathcal{D}_{n,k}^{\rho}$ when $\rho < 1$.

Proof. The normally convergent defining series gives $\partial_a^k \zeta(s, a) = (-1)^k (s)_k \zeta(s+k, a)$ first in a right half-plane, and then by meromorphic continuation. Apply it to (10.1), noting that the pole term is independent of a. Substituting $s = 1 + t$ and using $(1+t)_k = k! P_k(t)$ gives

$$\sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n^{(k)}(a) t^n = (-1)^k k! P_k(t) \zeta(k + 1 + t, a). \tag{10.7}$$

Coefficient comparison proves (10.6). For $\rho < 1$ the same calculation is justified directly by locally uniform convergence of the Lerch series. This is Coffey's Stirling-number formula in the harmonic/symmetric-function coordinates used by the source draft. □

Definition 10.2 (Reciprocal-gamma Appell polynomials)**.** *Define the monic polynomials Q_n by*

$$\frac{e^{xt}}{\Gamma(1+t)} = \sum_{n \geq 0} Q_n(x) \frac{t^n}{n!}. \tag{10.8}$$

Set $W_\rho(x) = (1 - \rho e^{-x})^{-1}$ for $x > 0$ and $0 \leq \rho \leq 1$.

Writing $y = x + \gamma$ and $\zeta_r = \zeta(r)$, the first polynomials are

$$\begin{aligned} Q_0(x) &= 1, & Q_1(x) &= y, \\ Q_2(x) &= y^2 - \zeta_2, & Q_3(x) &= y^3 - 3\zeta_2 y + 2\zeta_3, \\ Q_4(x) &= y^4 - 6\zeta_2 y^2 + 8\zeta_3 y + 3\zeta_2^2 - 6\zeta_4. \end{aligned} \tag{10.9}$$

They satisfy $Q_n' = n Q_{n-1}$ and the explicit recurrence

$$Q_{n+1}(x) = (x + \gamma) Q_n(x) + \sum_{r=1}^{n} (-1)^r \binom{n}{r} r! \zeta(r+1) Q_{n-r}(x). \tag{10.10}$$

This follows by differentiating (10.8) in t and using $-\psi(1+t) = \gamma + \sum_{r \geq 1} (-1)^r \zeta(r+1) t^r$ near zero.

Theorem 10.3 (Appell–Laplace representation)**.** *For all $n \geq 0$, $k \geq 1$, $a > 0$, and $0 \leq \rho \leq 1$,*

$$F_{n,k}^{\rho}(a) := (-1)^{n+k} \mathcal{D}_{n,k}^{\rho}(a) = \int_0^\infty x^k e^{-ax} W_\rho(x) Q_n(\log x) \, \mathrm{d}x. \tag{10.11}$$

The integral and every fixed number of its a-derivatives converge locally uniformly for $a > 0$, uniformly in $\rho \in [0, 1]$.

Proof. The Hurwitz and Lerch Mellin integrals give, for $|t|$ sufficiently small,

$$k! P_k(t) \Phi(\rho, k + 1 + t, a) = \frac{1}{\Gamma(1+t)} \int_0^\infty x^{k+t} e^{-ax} W_\rho(x) \, \mathrm{d}x,$$

where $\Phi(1, s, a)$ means $\zeta(s, a)$ in its convergent half-plane. The cancellation of $\Gamma(k + 1 + t)$ is the reason for introducing Q_n. Extract the coefficient of t^n and apply (10.7).

For justification, $W_\rho(x) \leq W_1(x) = O(x^{-1})$ at zero and $W_\rho(x) = O(1)$ at infinity, uniformly in ρ. On $|t| \leq \eta < k$, the integrand near zero is bounded by a constant times $x^{k-1-\eta}$, and its coefficient derivatives introduce only powers of $|\log x|$. At infinity, a fixed positive lower bound on a supplies exponential domination. Multiplication by any fixed power of x handles the a-derivatives. These estimates also prove continuity at $\rho = 1$. □

10.2 Real-rootedness and a global zero bound

Lemma 10.4 (A real-root preserving operator)**.** *If a real polynomial f of degree n has only real roots and $c \neq 0$ is real, then $f + cf'$ has only real roots. An inherited root of multiplicity m has multiplicity $m-1$ if $m \geq 2$, and every other root of $f + cf'$ is simple. Consequently, after $n-1$ such operators, any real-rooted polynomial of degree n becomes strictly real-rooted.*

Proof. Away from the distinct roots r_i of f, of multiplicities m_i, the new roots solve

$$\frac{f'(x)}{f(x)} = \sum_i \frac{m_i}{x - r_i} = -\frac{1}{c}.$$

The left side is strictly decreasing on each complementary interval because its derivative is $-\sum_i m_i/(x-r_i)^2 < 0$. It has one solution between each consecutive pair of distinct roots and one further solution in the appropriate exterior interval. All these solutions are simple. Factoring at r_i proves the inherited multiplicity assertion, and counting degrees accounts for all roots. Repeating reduces every possible multiple root at each step; new roots never introduce multiplicity. □

Theorem 10.5 (Strictly real-rooted reciprocal-gamma polynomials)**.** *For every $n \geq 1$, Q_n has exactly n distinct real roots. The roots of Q_{n-1} strictly interlace those of Q_n.*

Proof. The classical Weierstrass product [44] is

$$\frac{1}{\Gamma(1+t)} = e^{\gamma t} \prod_{m=1}^{\infty} (1 + t/m) e^{-t/m}, \tag{10.12}$$

with locally uniform convergence. For a finite product, extracting $n![t^n]$ after multiplication by e^{xt} is equivalent to applying operators $1 + \partial_x/m$ and real translations to x^n. Lemma 10.4 therefore proves real-rootedness of each finite-product polynomial. Coefficient convergence and closure of the set of monic real-rooted polynomials prove real-rootedness in the limit.

A limit of simple-rooted polynomials alone would not prove simplicity. To obtain it, remove the first $n-1$ factors $(1+t/m)e^{-t/m}$ from (10.12). The Appell polynomial associated with the remaining tail is still real-rooted by the same finite-product argument. Restore the removed factors. The translations preserve multiplicity and commute with the differential operators; the $n-1$ operators $1 + \partial_x/m$ make all roots simple by Lemma 10.4. Finally $Q_n' = nQ_{n-1}$ and Rolle's theorem give strict interlacing. □

Lemma 10.6 (Laplace variation bound, with multiplicity)**.** *Let $g : (0, \infty) \to \mathbb{R}$ have exactly N sign changes, with constant nonzero sign between the sign-change points, up to null sets. Suppose all exponentially weighted polynomial moments needed below are absolutely integrable. Its Laplace transform $G(a) = \int_0^\infty e^{-ax} g(x)\,\mathrm{d}x$ is not identically zero and has at most N zeros on $(0, \infty)$, counted with multiplicity.*

Proof. For $N = 0$ the transform has a fixed strict sign. For $N > 0$, choose a sign-change point x_0. Multiplying g by $x_0 - x$ removes that sign change and introduces no others. The transform of the new function is

$$(\partial_a + x_0)G(a) = e^{-x_0 a} \frac{\mathrm{d}}{\mathrm{d}a}\big(e^{x_0 a} G(a)\big).$$

By induction it has at most $N-1$ zeros. If G has any finite collection of zeros of total multiplicity M, Rolle's theorem with multiplicity gives at least $M-1$ zeros of the displayed derivative between

the first and last zero, including inherited multiplicities at the endpoints when appropriate. Thus $M \le N$. This bounds every finite collection, and hence all zeros. The induction hypothesis also says that the transform of the new kernel is not identically zero. If G were identically zero, its displayed transformed derivative would be identically zero as well, a contradiction. This completes the induction including nonvanishing. □

Theorem 10.7 (Global bound on positive zeros)**.** *For every $n \ge 0$, $k \ge 1$, and $0 \le \rho \le 1$, the function $\mathcal{D}^{\rho}_{n,k}(a)$ has at most n positive zeros, counted with multiplicity. In particular, this holds for $\gamma^{(k)}_n(a)$.*

Proof. The factor $x^k W_\rho(x)$ in (10.11) is strictly positive. By Theorem 10.5, $Q_n(\log x)$ has precisely n simple sign changes on $(0,\infty)$. The estimates in Theorem 10.3 justify every step of Lemma 10.6. The transform is not identically zero: as $a \downarrow 0$, scaling $x = u/a$ gives

$$F^{\rho}_{n,k}(a) \sim k! a^{-k-1} \log^n(1/a), \tag{10.13}$$

uniformly in ρ, with the convention $\log^0(1/a) = 1$. This follows from monicity of Q_n and $W_\rho(u/a) \to 1$, using the same integrable bounds after splitting at $u = a$. The variation bound now applies. □

Remark 10.8 (The elementary endpoint explains possible multiplicity). At $\rho = 0$ one has

$$F^{0}_{n,k}(a) = k! a^{-k-1}\, n! [t^n] P_k(t) e^{-t \log a}.$$

The polynomial in $x = -\log a$ is $\prod_{j=1}^{k}(1 + \partial_x/j)x^n$. It has n real roots counted with multiplicity by Lemma 10.4. When $n > k$, the root $x = 0$ has multiplicity exactly $n - k$. Thus all zeros cannot be declared simple uniformly in ρ at small k. Any uniform simplicity threshold for $n \ge 2$ must be at least $n - 1$.

10.3 An all-order concentration expansion

The scale on which the zeros move is $a = ck$, with c bounded away from zero and infinity. Set

$$\mathcal{T}^{\rho}_{n,k}(c) = \frac{(-1)^{n+k}(ck)^{k+1}}{k!} \mathcal{D}^{\rho}_{n,k}(ck), \qquad h_{n,\rho}(x) = W_\rho(x) Q_n(\log x). \tag{10.14}$$

Let Y_k have gamma density

$$\frac{k^{k+1}}{k!} y^k e^{-ky}, \qquad y > 0.$$

Then $\mathbb{E}Y_k = 1 + 1/k$, and (10.11) becomes the exact identity

$$\mathcal{T}^{\rho}_{n,k}(c) = \mathbb{E}\, h_{n,\rho}(Y_k/c). \tag{10.15}$$

The shape is $k+1$, not k; the resulting $1/k$ shift matters for the constant term of each zero.

Define polynomials C_j by the formal generating function

$$\sum_{j \ge 0} C_j(z) w^j = \exp\left\{ \sum_{\ell \ge 1} w^\ell \left(\frac{z^\ell}{\ell} + \frac{z^{\ell+1}}{\ell+1} \right) \right\}. \tag{10.16}$$

For $C_j(z) = \sum_r c_{j,r} z^r$, use the *normal-ordered* differential operator

$$(A_j h)(x) = \sum_r c_{j,r} x^r h^{(r)}(x). \tag{10.17}$$

This notation does not mean $C_j(x\partial_x)$: the powers of x stand to the left of the derivatives. For example,

$$\begin{aligned} A_0h &= h, \\ A_1h &= xh' + \tfrac{1}{2}x^2h'', \\ A_2h &= x^2h'' + \tfrac{5}{6}x^3h''' + \tfrac{1}{8}x^4h^{(4)}. \end{aligned} \tag{10.18}$$

Theorem 10.9 (Uniform complete expansion)**.** *Fix $n, J, r \geq 0$ and a compact interval $I \subset (0, \infty)$. As $k \to \infty$,*

$$\mathcal{T}^{\rho}_{n,k}(c) = \sum_{j=0}^{J} \frac{(A_j h_{n,\rho})(1/c)}{k^j} + O_{n,J,r,I}(k^{-J-1}), \tag{10.19}$$

uniformly for $c \in I$ and $0 \leq \rho \leq 1$, also after any number of c-derivatives up to r.

Proof. The centered moment generating function is

$$\mathbb{E}e^{z(Y_k-1)} = e^{-z}(1 - z/k)^{-k-1}.$$

For fixed z near zero its logarithm is exactly the convergent expansion

$$\log \mathbb{E}e^{z(Y_k-1)} = \sum_{\ell \geq 1} k^{-\ell} \left(\frac{z^\ell}{\ell} + \frac{z^{\ell+1}}{\ell+1} \right). \tag{10.20}$$

Consequently the coefficient of k^{-j} in $\mathbb{E}(Y_k - 1)^m/m!$ is $[z^m]C_j(z)$. For fixed m, these moments are finite polynomials in $1/k$, since only finitely many terms of the exponential can contribute to z^m. The degree of C_j is at most $2j$.

Here are the estimates needed to turn coefficient comparison into a uniform asymptotic statement. On $1/2 \leq y \leq 3/2$, all derivatives of $h_{n,\rho}(y/c)$ of any fixed order in y and c are uniformly bounded for $c \in I$ and $\rho \in [0, 1]$. Indeed y/c stays in a compact subinterval of $(0, \infty)$ and $1 - \rho e^{-y/c}$ is uniformly positive there. Taylor's theorem at $y = 1$, through degree $2J + 1$, has remainder bounded by $C|y - 1|^{2J+2}$. Formula (10.20) gives

$$\mathbb{E}|Y_k - 1|^{2J+2} = O(k^{-J-1}).$$

Replacing truncated central moments by full central moments changes the answer only exponentially, as we now verify.

For each fixed derivative order, the functions to be averaged admit bounds

$$|\partial_c^u h_{n,\rho}(y/c)| \leq C\big(y^{-B} + y^B\big)(1 + |\log y|^n)$$

for some fixed B and C, uniformly in the stated parameters. These follow from $W_1(x) = O(1 + x^{-1})$ and its differentiated forms; enlarging B absorbs the logarithm if convenient. For a fixed real exponent v,

$$\mathbb{E}\big[Y_k^v \mathbf{1}_{Y_k \notin [1/2,3/2]}\big] = \frac{\Gamma(k+1+v)}{\Gamma(k+1)k^v} \Pr\{Z_{k,v} \notin [1/2, 3/2]\},$$

where $Z_{k,v}$ has shape $k + 1 + v$ and rate k, for all sufficiently large k. The prefactor is bounded, and a Chernoff bound obtained from the gamma moment generating function makes the probability $O(e^{-\eta k})$ for some $\eta > 0$. The same holds for the finitely many polynomial and inverse-polynomial bounds just used. This proves exponentially small tails for the Taylor remainder and all retained moments.

Taylor expansion in (10.15), followed by (10.20), therefore gives exactly (10.19) and the coefficients (10.17). Applying the same argument to $\partial_c^u h_{n,\rho}(y/c)$ proves the stated derivative uniformity. No differentiation of an uncontrolled big-O term is used. □

Remark 10.10 (What uniformity does not say). The constants in Theorem 10.9 may depend on n and on the compact c-interval. The theorem does not supply estimates uniform when n grows with k, nor when $c \downarrow 0$ or $c \to \infty$. These are separate problems, especially because the extreme roots of Q_n then control very different scales.

10.4 Exactly n eventual zeros and their locations

Order the Appell roots and define the corresponding slopes by

$$x_{n,1} > x_{n,2} > \cdots > x_{n,n}, \qquad c_{n,j} = e^{-x_{n,j}}, \quad 0 < c_{n,1} < \cdots < c_{n,n}. \tag{10.21}$$

Theorem 10.11 (Uniform eventual zero theorem)**.** *For every fixed $n \geq 1$ there is an integer K_n such that, for every $k \geq K_n$ and every $\rho \in [0,1]$, $\mathcal{D}^{\rho}_{n,k}$ has exactly n positive zeros, all simple. Number them increasingly as $a^{\rho}_{n,j,k}$. Uniformly in ρ,*

$$a^{\rho}_{n,j,k} = kc_{n,j} + d_{n,j}(\rho) + O_n(k^{-1}), \tag{10.22}$$

where

$$d_{n,j}(\rho) = \frac{c_{n,j}}{2}\left(1 + \frac{Q_n''(x_{n,j})}{Q_n'(x_{n,j})}\right) - \frac{\rho}{\exp(1/c_{n,j}) - \rho}. \tag{10.23}$$

More generally, for every fixed $J \geq 0$ there are coefficients $b_{n,j,\ell}(\rho)$ such that

$$a^{\rho}_{n,j,k} = kc_{n,j} + d_{n,j}(\rho) + \sum_{\ell=1}^{J} b_{n,j,\ell}(\rho)k^{-\ell} + O_{n,J}(k^{-J-1}), \tag{10.24}$$

uniformly for $\rho \in [0,1]$. These coefficients are recursively computable from (10.16).

Proof. The leading term in Theorem 10.9 is

$$G_0(c,\rho) = W_\rho(1/c)Q_n(-\log c).$$

Its zeros are exactly the $c_{n,j}$, and its derivative at such a zero is

$$\partial_c G_0(c_{n,j},\rho) = -\frac{W_\rho(1/c_{n,j})}{c_{n,j}}Q_n'(x_{n,j}) \neq 0. \tag{10.25}$$

Choose disjoint closed intervals around these slopes on whose endpoints $Q_n(-\log c)$ has opposite signs. Since $1 \leq W_\rho(1/c) \leq W_1(1/c)$ on their union, the endpoint absolute values are bounded away from zero uniformly in ρ. The uniform $J = 0$ approximation gives one zero in each interval for all sufficiently large k, uniformly in ρ. The global bound of Theorem 10.7 excludes any further zero and, because it counts multiplicity, makes every one of the n zeros simple.

Put $c = c_{n,j} + d/k + O(k^{-2})$. The k^{-1} coefficient of (10.19) gives

$$d = -\frac{G_1(c_{n,j},\rho)}{\partial_c G_0(c_{n,j},\rho)}, \qquad G_1(c,\rho) = (A_1 h_{n,\rho})(1/c). \tag{10.26}$$

At $x = 1/c_{n,j}$, write $Q' = Q_n'(x_{n,j})$ and $Q'' = Q_n''(x_{n,j})$. Because $Q_n(\log x) = 0$,

$$xh' + \tfrac{1}{2}x^2h'' = \tfrac{1}{2}W_\rho(x)(Q' + Q'') + xW_\rho'(x)Q'.$$

Also

$$\frac{W'_\rho(x)}{W_\rho(x)} = -\frac{\rho}{e^x - \rho}.$$

Substitution in (10.26) proves (10.23).

For completeness, the all-order assertion does not require a convergent power series in $1/k$. Apply Theorem 10.9 to one additional order at each step. Substitute a finite ansatz for c into the finite expansion. At every new order the unknown coefficient occurs linearly with the nonzero multiplier (10.25); hence it is uniquely determined. This derivative is bounded away from zero uniformly in ρ. The residual of a truncation, divided by a uniform lower derivative bound on a smaller root interval, bounds the error in the true zero by the mean value theorem. Expanding one order farther than the desired expansion of $a = kc$ gives (10.24) with its stated uniform remainder. □

10.4.1 Examples and the first zero

For $n = 1$, $x_{1,1} = -\gamma$ and $c_{1,1} = e^\gamma$. Thus the unique zero of $\gamma_1^{(k)}$ satisfies

$$a^1_{1,1,k} = e^\gamma k + \frac{e^\gamma}{2} - \frac{1}{\exp(e^{-\gamma}) - 1} + O(k^{-1}). \tag{10.27}$$

The constant term is approximately -0.4370805068. At $\rho = 0$ the zero is exactly e^{H_k}, whose constant term is $e^\gamma/2$; the Lerch kernel therefore changes the constant term, not the slope. A fully explicit refinement, obtained from Appendix 10.5, is available throughout the deformation. Put $c = e^\gamma$ and $g = \rho/(e^{1/c} - \rho)$. Then, uniformly in $\rho \in [0, 1]$,

$$a_k(\rho) = ck + \frac{c}{2} - g + \frac{1}{k}\left\{\frac{c}{24} + \frac{g(g+1)}{c} - \frac{g(g+1)(2g+1)}{2c^2}\right\} + O(k^{-2}). \tag{10.28}$$

At $\rho = 1$, the coefficient of k^{-1} is approximately 0.0288707254211. At $\rho = 0$ it is $c/24$, agreeing with the expansion of the exact zero e^{H_k}.

For $n = 2$ the roots are $-\gamma \pm \sqrt{\zeta(2)}$, and hence

$$c_{2,1} = e^{\gamma - \pi/\sqrt{6}}, \qquad c_{2,2} = e^{\gamma + \pi/\sqrt{6}}. \tag{10.29}$$

The identity $Q''_n(x_j)/Q'_n(x_j) = 2\sum_{i \neq j}(x_j - x_i)^{-1}$ gives an alternative expression for every offset entirely in terms of the Appell roots.

10.4.2 Nonasymptotic information at $n = 1$

Theorem 10.12 (Harmonic brackets and monotonicity)**.** *For each $k \geq 1$ and $\rho \in [0, 1]$, $\mathcal{D}^\rho_{1,k}$ has exactly one positive zero, denoted $a_k(\rho)$. It is simple, and*

$$e^{H_{k-1}} < a_k(\rho) \leq e^{H_k}. \tag{10.30}$$

Equality at the upper endpoint occurs exactly when $\rho = 0$. The function $\rho \mapsto a_k(\rho)$ is continuous on $[0, 1]$ and strictly decreasing. In particular, $a_k(1) < a_k(\rho) < e^{H_k}$ for $0 < \rho < 1$.

Proof. Normalize the positive density proportional to $x^k e^{-ax} W_\rho(x)$ and write $m(a, \rho)$ for its expectation of $\log X + \gamma$. The sign of $F^\rho_{1,k}(a)$ is the sign of m. Under an unweighted gamma density of shape $k + 1$ and rate a, this expectation equals $H_k - \log a$. Weighting by the decreasing function

n	j	slope $c_{n,j}$	offset $d_{n,j}(1)$
1	1	1.781072417990	−0.437080506782
2	1	0.493943487920	0.287385687369
2	2	6.422230550060	−5.227782104747
3	1	0.261424326196	0.354674627668
3	2	1.064376550729	−0.506464646434
3	3	20.305020815096	−21.208385701520
4	1	0.172347335379	0.355378064541
4	2	0.465599872096	0.040510911109
4	3	2.062627326998	−2.178710104429
4	4	60.797835737559	−70.712693097880

Table 10.1: Numerical evaluations of the exact slope and offset formulas. These decimals are not isolating intervals. The accompanying CSV records zero computations at $k = 10, 40, 160$.

W_ρ lowers the expectation of the increasing function $\log X$; the decrease is strict when $\rho > 0$. The elementary covariance identity

$$2\operatorname{Cov}(f(X), g(X)) = \mathbb{E}\big[(f(X) - f(X'))(g(X) - g(X'))\big]$$

for independent identically distributed X, X' justifies this comparison without an external stochastic-order theorem. Hence $m(e^{H_k}, \rho) \le 0$, with equality exactly at $\rho = 0$.

Alternatively use a gamma density of shape k and rate a and weight it by $xW_\rho(x)$. This weight is strictly increasing, since

$$\frac{\mathrm{d}}{\mathrm{d}x}\frac{x}{1-\rho e^{-x}} = \frac{1-\rho(1+x)e^{-x}}{(1-\rho e^{-x})^2} > 0 \qquad (x > 0).$$

The unweighted expectation is $H_{k-1} - \log a$, so $m(e^{H_{k-1}}, \rho) > 0$. A zero exists between the brackets, and Theorem 10.7 proves uniqueness and simplicity.

For fixed ρ, exponential tilting gives $\partial_a m = -\operatorname{Cov}(\log X, X) < 0$. For $\rho_2 > \rho_1$, the ratio

$$\frac{W_{\rho_2}(x)}{W_{\rho_1}(x)} = \frac{1-\rho_1 e^{-x}}{1-\rho_2 e^{-x}}$$

is strictly decreasing in x. The same covariance comparison therefore gives $m(a, \rho_2) < m(a, \rho_1)$. Together with strict decrease in a, this implies $a_k(\rho_2) < a_k(\rho_1)$. Continuity follows from locally uniform dominated convergence in (10.11), uniqueness of the root, and the common compact bracket. No differentiability in ρ at the endpoint $\rho = 1$ is assumed. □

10.4.3 Propagation and interlacing in the derivative order

Corollary 10.13 (Persistence and rightward interlacing)**.** *Fix $n \ge 1$ and $\rho \in [0, 1]$. Once $F^\rho_{n,k}$ has n distinct positive zeros, the same is true at every larger derivative order. At consecutive such orders,*

$$a^\rho_{n,j,k} < a^\rho_{n,j,k+1} < a^\rho_{n,j+1,k} \quad (1 \le j < n), \qquad a^\rho_{n,n,k} < a^\rho_{n,n,k+1}. \tag{10.31}$$

Thus the unique $n = 1$ zero moves strictly rightward for every $k \ge 1$.

Proof. Differentiating the integral gives $F^\rho_{n,k+1} = -\partial_a F^\rho_{n,k}$. Rolle's theorem supplies a zero between each pair of consecutive simple zeros. Also $F^\rho_{n,k}(a) \to 0$ as $a \to \infty$, and it has a constant nonzero

sign after its last zero. There must be a stationary point after that zero: otherwise its nonzero derivative would have constant sign and could not both depart from zero and tend back to zero. Hence there are at least n distinct zeros at the next order. Theorem 10.7 gives exactly those zeros, all simple, and excludes any zero before the first interval or a coincident zero at an old simple root. Induction proves persistence. □

10.5 A recurrence for the next zero coefficient

For a particular root, abbreviate $c_0 = c_{n,j}$ and $G_m(c,\rho) = (A_m h_{n,\rho})(1/c)$. Suppose

$$a^{\rho}_{n,j,k} = kc_0 + d + \frac{b_1}{k} + O(k^{-2}).$$

Taylor expansion of (10.19) gives

$$b_1 = -\frac{\frac{1}{2}\,\partial_c^2 G_0(c_0,\rho)d^2 + \partial_c G_1(c_0,\rho)d + G_2(c_0,\rho)}{\partial_c G_0(c_0,\rho)}. \tag{10.32}$$

Here d is (10.23), G_2 is explicit from (10.18), and the denominator is nonzero by (10.25). This is often more usable than a long expanded expression involving several root separations. It also explains why the quantity $k(a - kc_0 - d)$ tends to a finite limit. Higher coefficients follow by the same triangular recursion.

10.6 The classical half-unit bracket

For this specialization abbreviate $h_n = h_{n,1}$, and put $t_0 = e^{-\gamma}$.

Theorem 10.14 (Unique zero and an explicit bracket)**.** *For every integer $k \geq 1$, $\gamma_1^{(k)}(a)$ has a unique positive zero a_k, which is simple. With $H_0 = 0$,*

$$e^{H_{k-1}} < a_k < e^{H_{k-1}} + \tfrac{1}{2}. \tag{10.33}$$

The zeros are strictly increasing in k. Their asymptotic expansion starts with

$$a_k = e^{\gamma}\left(k + \tfrac{1}{2}\right) - \frac{1}{e^{e^{-\gamma}} - 1} + O(k^{-1}). \tag{10.34}$$

Formula (10.32) supplies the next coefficient and an $O(k^{-2})$ remainder.

Proof. The endpoint signs, the Laplace representation above, and the bound of one zero prove existence, uniqueness, and simplicity. To obtain the bracket, normalize the positive density $t^k e^{-at}/(1 - e^{-t})$. The sign of $F_{1,k}(a)$ is the sign of $\mathbb{E}(\log T + \gamma)$ under this density.

Relative to a gamma density of shape k and rate a, the tilting factor is $r(t) = t/(1 - e^{-t})$, which is strictly increasing: its derivative has numerator $1 - e^{-t}(1 + t) > 0$. A strictly increasing tilt strictly increases the expectation of the strictly increasing function $\log t$. This elementary fact follows from

$$2\,\mathrm{Cov}(u(X), v(X)) = \mathbb{E}[(u(X) - u(Y))(v(X) - v(Y))] > 0$$

for independent X, Y with the same continuous distribution and strictly increasing u, v. Consequently

$$\mathbb{E}\log T > \psi(k) - \log a = H_{k-1} - \gamma - \log a.$$

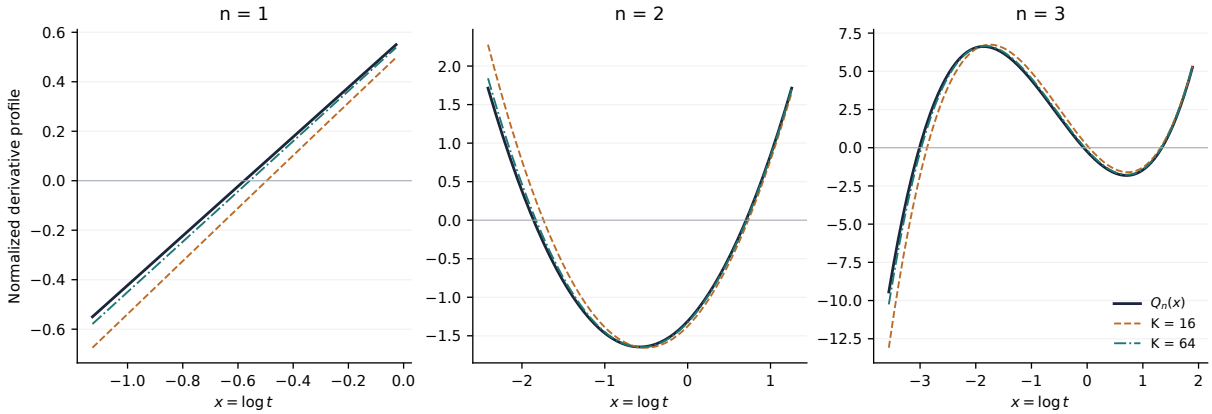


Figure 10.1: Normalized derivative profiles $(1-e^{-e^x})\mathcal{H}_{n,K}(e^x)$ approach $Q_n(x)$ as $K = k+1$ increases. Here $\mathcal{H}_{n,K}(t) = \mathbb{E}[Q_n(\log(tY/K))/(1-e^{-tY/K})]$ for $Y \sim \text{Gamma}(K,1)$. The plots illustrate $n = 1, 2, 3$. Quadrature produces the curves; Theorems 10.7 and 10.11, rather than the plots, prove the zero counts and asymptotic completeness.

At the zero this gives $a_k > e^{H_{k-1}} \geq 1$.

For $a > 1/2$, compare instead with a gamma density of shape k and rate $a - 1/2$. The tilting factor is $s(t) = t/(2\sinh(t/2))$, which is strictly decreasing because $(t/2)\cosh(t/2) > \sinh(t/2)$. The covariance inequality reverses:

$$\mathbb{E}\log T < \psi(k) - \log(a - 1/2).$$

Applying it at $a_k > 1$ gives $a_k < e^{H_{k-1}} + 1/2$.

Finally, at a zero the signed kernel $f(t)$ in (10.11) satisfies $\int e^{-a_k t} f(t)\,dt = 0$ and $(t-t_0)f(t) > 0$ except at t_0. Thus

$$F_{1,k+1}(a_k) = \int_0^\infty (t-t_0)e^{-a_k t} f(t)\,dt > 0.$$

Since $F_{1,k+1}$ crosses from positive to negative at its unique zero, $a_{k+1} > a_k$. The root of Q_1 is $-\gamma$, so (10.34) is Theorem 10.11. □

The interval in (10.33) is an analytic certificate for bracketing a unique root, valid even at $k = 1$. Its width is uniformly $1/2$; no assertion that this is the best possible width is made. The accompanying computation uses this bracket rather than guessing an initial zero from a graph.

10.7 Exact low-index zero counts

The general theorem proves eventual saturation. At the classical endpoint $\rho = 1$, a finite exact certificate supplies a stronger statement for $1 \leq n \leq 4$ at every $k \geq 1$. Define the sign-preserving normalization

$$\mathcal{V}_{n,k}(a) = \frac{a^{k+1}(-1)^{n+k}}{k!}\gamma_n^{(k)}(a).$$

10.7.1 A reproducible Euler–Maclaurin sign certificate

The finite certificates above can be checked without floating-point special functions. Put $p = k + 1$, and define

$$R_{n,\ell}(v) = n! \sum_{j=0}^{n} e_{n-j}(\ell) \frac{(-v)^j}{j!}.$$

The master identity gives the absolutely convergent spectral representation

$$\mathcal{V}_{n,k}(a) = \sum_{m \geq 0} \left(\frac{a}{a+m} \right)^p R_{n,k}(\log(a+m)).$$

For positive rational a, integers $M, R \geq 1$, $A = a + M > 1$ and $L = k + 2R$, let

$$\begin{aligned} \mathcal{E}_{M,R} = & \sum_{m=0}^{M-1} \left(\frac{a}{a+m} \right)^p R_{n,k}(\log(a+m)) \\ & + \left(\frac{a}{A} \right)^p \left[\frac{A}{k} R_{n,k-1}(\log A) + \frac{1}{2} R_{n,k}(\log A) \right] \\ & + \left(\frac{a}{A} \right)^p \sum_{r=1}^{R} \frac{B_{2r}}{(2r)!} \left(\prod_{j=1}^{2r-1} \frac{k+j}{A} \right) R_{n,k+2r-1}(\log A). \end{aligned}$$

Proposition 10.15 (An explicit rational-certifiable remainder)**.** *One has*

$$|\mathcal{V}_{n,k}(a) - \mathcal{E}_{M,R}| \leq \left(\frac{a}{A} \right)^p \frac{|B_{2R}|}{(2R)!} \left(\prod_{j=1}^{2R-1} \frac{k+j}{A} \right) \mathcal{B}_{n,L}(\log A),$$

$$\mathcal{B}_{n,L}(v) = \sum_{i=0}^{\min(n,L)} \sum_{j=0}^{n-i} \frac{n! e_i(L)}{(n-i-j)! L^j} v^{n-i-j}.$$

Every factor in the bound is nonnegative.

Proof. Apply Euler–Maclaurin through the Rth Bernoulli term to the spectral sum, or equivalently apply it to the Hurwitz series before extracting the nth spectral coefficient. The integral and half-endpoint terms give the second line of $\mathcal{E}_{M,R}$. The identity $(1+u)_k (k+1+u)_{2r-1} = (1+u)_{k+2r-1}$ gives its correction terms.

The periodic Bernoulli remainder has absolute kernel bound $|B_{2R}|$. After differentiation, bound the absolute value of $R_{n,L}(\log t)$ by $n! \sum_i e_i(L) (\log t)^{n-i}/(n-i)!$ for $t \geq A > 1$. For each $d \geq 0$,

$$\int_A^\infty t^{-L-1} \log^d t \, dt = \frac{A^{-L}}{L} \sum_{j=0}^{d} \frac{d!}{(d-j)! L^j} \log^{d-j} A.$$

The last factor in $(k+1)_{2R}$ cancels the leading L in this integral. Substitution yields precisely the displayed bound. □

Logarithms of positive rationals have elementary rational enclosures. Reduce their argument to $1 \leq y < 2$ by a power of two, put $u = (y-1)/(y+1)$, and use

$$0 \leq \log y - 2 \sum_{j=0}^{T-1} \frac{u^{2j+1}}{2j+1} \leq \frac{2u^{2T+1}}{(2T+1)(1-u^2)}.$$

The interval verifier uses integer endpoints scaled by 10^{90} and rounds every operation outward. The Bernoulli numbers and $e_i(\ell)$ are exact rationals. A final enclosure lying strictly on one side of zero is therefore a sign certificate; the proof of the analytic remainder and the rational arithmetic play separate roles.

The fresh replay verified fourteen base signs for $n \le 4$ and sixteen endpoint signs isolating the unique $n = 1$ roots for $k = 1, 2, 5, 10, 20, 50, 100, 200$. The endpoint pairs have width 10^{-20}. These are global root certificates because the zero-count theorem proves uniqueness, rather than merely sampled sign changes near a candidate root.

The following enclosures use integer interval arithmetic with outward rounding, rational logarithm bounds and a proved Euler–Maclaurin remainder. They have been replayed in this consolidation. The factor $a^{k+1}/k!$ is part of the tabulated value.

n	a	Certified enclosure for $\mathcal{V}_{n,1}(a)$	Sign
1	1	$[0.70738581,\ 0.70738582]$	$+$
1	1.5	$[-0.20183770,\ -0.20183769]$	$-$
2	1	$[0.11418372,\ 0.11418373]$	$+$
2	1.3	$[-0.12008911,\ -0.12008910]$	$-$
2	2	$[0.45673490,\ 0.45673491]$	$+$
3	0.8	$[0.16385907,\ 0.16385908]$	$+$
3	1.1	$[-0.04095264,\ -0.04095263]$	$-$
3	1.5	$[0.05603756,\ 0.05603757]$	$+$
3	2.1	$[-0.24542536,\ -0.24542535]$	$-$
4	0.8	$[0.03897530,\ 0.03897531]$	$+$
4	0.94	$[-0.00338878,\ -0.00338877]$	$-$
4	1.2	$[0.02176370,\ 0.02176371]$	$+$
4	1.7	$[-0.03656822,\ -0.03656821]$	$-$
4	2.2	$[0.15091738,\ 0.15091739]$	$+$

Corollary 10.16 (Saturation at every derivative order for $n \le 4$)**.** *For $1 \le n \le 4$ and every $k \ge 1$, $\gamma_n^{(k)}$ has exactly n positive zeros, all simple. They interlace to the right as k increases.*

Proof. At $k = 1$ each row group gives n sign-changing disjoint intervals, so there are at least n distinct zeros. The global bound with multiplicity makes them the only zeros and makes each simple. Persistence and interlacing follow from Corollary 10.13. □

This finite certification does not settle every small-k case for $n \ge 5$, and it does not assert saturation at every small order uniformly in the Lerch deformation.

10.8 High parameter derivatives of Stieltjes constants

The parameter-derivative tower admits a useful separation into a finite polynomial coefficient and an exponentially small spectral tail. This gives a different asymptotic regime from the frequently studied limit of a large Stieltjes index with derivative order fixed. Here the derivative order k tends to infinity, while the index n may grow proportionally to $\log k$. The resulting profile is the reciprocal gamma function.

10.8.1 An exact spectral expansion

Fix the Laurent convention

$$\zeta(s,a) = \frac{1}{s-1} + \sum_{n=0}^{\infty} \frac{(-1)^n}{n!}\gamma_n(a)(s-1)^n, \qquad a > 0. \tag{10.35}$$

The superscript $\gamma_n^{(k)}(a)$ always denotes differentiation with respect to a, not to the index or the zeta variable. Define

$$P_{n,k}(x) = n![z^n]\left(e^{xz}\prod_{j=1}^{k}\left(1+\frac{z}{j}\right)\right). \tag{10.36}$$

This is a degree-n polynomial in x; its coefficients are rational. In particular $P_{0,k} = 1$ and $P_{1,k}(x) = x + H_k$.

Theorem 10.17 (Spectral expansion and a finite-tail certificate)**.** *For $a > 0$, integers $k \geq 1$, and $n \geq 0$,*

$$\frac{(-1)^{n+k}}{k!}\gamma_n^{(k)}(a) = \sum_{m=0}^{\infty} \frac{P_{n,k}(-\log(a+m))}{(a+m)^{k+1}}. \tag{10.37}$$

The series converges absolutely, locally uniformly for $a > 0$. If the sum is truncated before $m = M$, where $b = a + M \geq 1$, then for every real $0 < R < k$ the magnitude of the omitted part is at most

$$\frac{n!}{R^n}\prod_{j=1}^{k}\left(1+\frac{R}{j}\right)\left(b^{-k-1+R} + \frac{b^{-k+R}}{k-R}\right). \tag{10.38}$$

Proof. The Hurwitz-zeta parameter identity

$$\partial_a^k \zeta(s,a) = (-1)^k (s)_k \zeta(s+k,a)$$

follows initially by differentiating its absolutely convergent Dirichlet series and then by analytic continuation; here $(s)_k = s(s+1)\cdots(s+k-1)$. The pole in (10.35) is independent of a. Put $s = 1+z$, use $(1+z)_k = k!\prod_{j=1}^{k}(1+z/j)$, and compare coefficients. For $|z| < k$, the series for $\zeta(k+1+z,a)$ converges normally. This gives (10.37) and justifies the coefficient interchange. Alternatively, absolute convergence follows directly because each summand is a linear combination of $(a+m)^{-k-1}\log^{\ell}(a+m)$, $0 \leq \ell \leq n$. The parameter identity is also the starting point of Coffey's all-order formulas [25].

On the circle $|z| = R$, and for $x \geq b \geq 1$,

$$\left|e^{-z\log x}\prod_{j=1}^{k}(1+z/j)\right| \leq x^R \prod_{j=1}^{k}(1+R/j).$$

Cauchy's coefficient estimate bounds the m-th omitted summand by $n!R^{-n}\prod_j(1+R/j)(a+m)^{-k-1+R}$. For the positive decreasing power, comparison with its integral gives

$$\sum_{m=M}^{\infty}(a+m)^{-k-1+R} \leq b^{-k-1+R} + \int_b^{\infty} x^{-k-1+R}\,dx.$$

This is (10.38). □

The condition $b \geq 1$ belongs to the stated majorant, rather than to the exact expansion. It always holds for the tail after the first term, because then $b = a + 1 > 1$. The choice of R can be optimized for a given n, k, M; there is no need to differentiate a highly oscillatory numerical representation of γ_n.

10.8.2 Exact rational enclosures at parameter one

Write $\begin{bmatrix} v \\ u \end{bmatrix}$ for the unsigned Stirling number of the first kind. The elementary identity

$$\prod_{j=1}^{k}(1+z/j) = \frac{1}{k!}\sum_{n=0}^{k}\begin{bmatrix} k+1 \\ n+1 \end{bmatrix} z^n \tag{10.39}$$

turns the first spectral term at $a = 1$ into an exact rational number.

Corollary 10.18 (Rational certificates)**.** *Let $k \geq 2$, $n \geq 0$, and*

$$e_n(k) = \begin{cases} \begin{bmatrix} k+1 \\ n+1 \end{bmatrix}/k!, & 0 \leq n \leq k, \\ 0, & n > k. \end{cases}$$

For any integer $1 \leq h < k$, set

$$B_{k,n,h} = h^{-n}\binom{k+h}{h}\left(2^{-k-1+h} + \frac{2^{-k+h}}{k-h}\right). \tag{10.40}$$

Then the following is an exact inclusion between real numbers:

$$\frac{(-1)^{n+k}\gamma_n^{(k)}(1)}{k!\,n!} \in [e_n(k) - B_{k,n,h},\, e_n(k) + B_{k,n,h}]. \tag{10.41}$$

One may take the minimum radius over all $h = 1, \ldots, k-1$. In particular, the simple choice $h = 1$ gives

$$B_k = (k+1)\left(2^{-k} + \frac{2^{-k+1}}{k-1}\right), \tag{10.42}$$

independently of n.

Proof. Apply Theorem 10.17 with $a = M = 1$, $b = 2$, and $R = h$, and divide its bound by $n!$. The product in that bound is $\prod_{j=1}^{k}(1+h/j) = \binom{k+h}{h}$. Equation (10.39) supplies the center. All quantities in the resulting endpoints are rational. □

Thus the term *certificate* has its literal meaning here: integer arithmetic constructs the endpoints, and the proved tail bound supplies the inclusion. Floating-point agreement is not being renamed a proof. For example, positivity of an endpoint $e_n(k) - B_{k,n,h}$ rigorously proves $(-1)^{n+k}\gamma_n^{(k)}(1) > 0$. The accompanying script emits centers, radii, and both endpoints as numerator–denominator pairs, and checks their arithmetic exactly.

10.8.3 A uniform coefficient lemma

The main asymptotic step is an elementary coefficient estimate. It is convenient to set $n^{\underline{j}} = n(n-1)\cdots(n-j+1)$ for $j \leq n$, and $n^{\underline{j}} = 0$ for $j > n$.

Lemma 10.19 (Two uniform coefficient terms)**.** *Suppose f is analytic on a neighborhood of the closed disk $|z| \leq R$, with $|f(z)| \leq M$ there. Let $L > 0$, $n \geq 1$, and $r = n/L < R$. Put*

$$Q_n(L; f) = \frac{n!}{L^n}[z^n]e^{Lz}f(z).$$

Then

$$|Q_n(L;f) - f(r)| \le \frac{Mn}{L^2R^2(1-r/R)^3}, \tag{10.43}$$

$$\left|Q_n(L;f) - f(r) + \frac{n}{2L^2}f''(r)\right| \le \frac{3Mn}{L^3R^3(1-r/R)^5}. \tag{10.44}$$

For $n = 0$, $Q_0(L;f) = f(0)$ exactly.

Proof. Write $f(z) = \sum_{j\ge0} c_j z^j$. Since $|c_j| \le MR^{-j}$,

$$Q_n(L;f) = \sum_{j\ge0} c_j \frac{n^{\underline{j}}}{L^j}$$

is a finite sum, which we compare with the absolutely convergent Taylor series at r.

For completeness, the needed inequalities remain valid when $j > n$. Choose j independent uniform elements of a set of n elements. The probability of no repeated choice is $n^{\underline{j}}/n^j$, including its value zero when $j > n$. For the $K = \binom{j}{2}$ pair-collision events, every event has probability $1/n$, and every intersection of two distinct events has probability $1/n^2$, even if the pairs share a position. The union bound and the first two Bonferroni inequalities give

$$0 \le n^j - n^{\underline{j}} \le \binom{j}{2} n^{j-1},$$

$$0 \le n^{\underline{j}} - n^j + \binom{j}{2} n^{j-1} \le \binom{\binom{j}{2}}{2} n^{j-2}.$$

At $j = 0, 1, 2$, the second remainder is zero, so no negative-power exception is needed. Now use

$$\sum_{j\ge2} \binom{j}{2} x^{j-2} = (1-x)^{-3}, \qquad \sum_{j\ge3} \binom{\binom{j}{2}}{2} x^{j-3} = 3(1-x)^{-5}.$$

The first inequality gives (10.43). In the second comparison,

$$\sum_{j\ge2} c_j \binom{j}{2} \frac{n^{j-1}}{L^j} = \frac{n}{2L^2} f''(r).$$

This proves (10.44). □

10.8.4 The reciprocal-gamma profile

Theorem 10.20 (A joint logarithmic-range asymptotic)**.** *Fix $0 < a_0 \le a_1 < \infty$ and $C > 0$. For $a \in [a_0, a_1]$, put*

$$L = \log(k/a), \qquad r = \frac{n}{L}, \qquad g(r) = \frac{1}{\Gamma(1+r)}, \qquad \rho = \frac{a_1}{a_1+1} < 1.$$

Uniformly over integers $0 \le n \le CL$, as $k \to \infty$,

$$\frac{(-1)^{n+k} a^{k+1} \gamma_n^{(k)}(a)}{k!\, L^n} = g(r) - \frac{n}{2L^2} g''(r) + O_{a_0,a_1,C}\left(\frac{n}{L^3} + k^{C+1}\rho^k\right). \tag{10.45}$$

Equivalently, the two displayed terms are

$$\frac{1}{\Gamma(1+r)}\left[1+\frac{n}{2L^2}\left\{\psi^{(1)}(1+r)-\psi(1+r)^2\right\}\right]. \tag{10.46}$$

The formula includes $n=0$, for which the first spectral term is exactly one. In particular, if $n/\log(k/a)\to c\ge 0$ in a bounded range, then the left side of (10.45) tends to $1/\Gamma(1+c)$.

Proof. Separate the $m=0$ term of (10.37). Its generating function factors as

$$e^{-z\log a}\prod_{j=1}^{k}(1+z/j)=e^{Lz}f_k(z),\qquad f_k(z)=k^{-z}\prod_{j=1}^{k}(1+z/j).$$

The gamma product gives

$$f_k(z)=\frac{\Gamma(k+1+z)}{\Gamma(k+1)k^z\Gamma(1+z)}=g(z)e^{D_k(z)}.$$

On $|z|<k+1$, define the analytic function

$$D_k(z)=z(H_k-\log k-\gamma)+\sum_{\ell=2}^{\infty}\frac{(-1)^\ell z^\ell}{\ell}\big(\zeta(\ell)-H_k^{(\ell)}\big). \tag{10.47}$$

The factorization follows first on a small disk from the Weierstrass product and then throughout $|z|<k+1$ by analytic continuation. Defining D_k by its series avoids taking the logarithm of g at its zeros.

The elementary bounds

$$0<H_k-\log k-\gamma<\frac{1}{2k},\qquad \sum_{j>k}j^{-2}\le\frac{1}{k}$$

imply, for $|z|\le R<k+1$,

$$|D_k(z)|\le\frac{R}{2k}+\frac{R^2}{2k(1-R/(k+1))}. \tag{10.48}$$

Indeed, for $\ell\ge 2$, bound the tail of $\zeta(\ell)$ by $(k+1)^{2-\ell}/k$, and use $1/\ell\le 1/2$. Consequently f_k is uniformly bounded on every fixed disk, $f_k-g=O_R(k^{-1})$, and Cauchy's estimate gives $f_k''-g''=O_C(k^{-1})$ on $[0,C]$. The same estimate with $R=r$, or the real product, also gives the slightly sharper

$$f_k(r)-g(r)=O_C(r/k)\qquad(0\le r\le C).$$

It is this factor r that preserves the exact $n=0$ first term.

Choose $R=C+1$ in Lemma 10.19. For $n\ge 1$ the normalized first spectral term is

$$Q_n(L;f_k)=f_k(r)-\frac{n}{2L^2}f_k''(r)+O_C(n/L^3).$$

Replacing f_k by g costs $O_C(n/(kL)+n/(kL^2))$, which is absorbed in $O(n/L^3)$ because $L^2\le k$ for all sufficiently large k, uniformly on the compact parameter interval.

It remains to control the other spectral terms. Apply (10.38) with $M=1$, and multiply the bound by a^{k+1}/L^n. The result is

$$\frac{n!}{(RL)^n}\prod_{j=1}^{k}(1+R/j)\left(\frac{a}{a+1}\right)^{k+1}(a+1)^R\left(1+\frac{a+1}{k-R}\right).$$

The first factor is at most one: for $n \geq 1$, use $n! \leq n^n$ and $n/L \leq C < R$; for $n = 0$, it is one by convention. The product is $O_R(k^R)$, as follows already from (10.48) at $z = R$. The remaining factors are uniformly bounded apart from $(a/(a+1))^{k+1} \leq \rho^{k+1}$. Thus the normalized tail is $O_{a_0,a_1,C}(k^{C+1}\rho^k)$. The case $n = 0$ follows directly from $P_{0,k} = 1$. Finally, $g''(r) = g(r)\{\psi(1+r)^2 - \psi^{(1)}(1+r)\}$. □

Corollary 10.21 (A uniform eventual sign)**.** *For every compact interval $[a_0, a_1] \subset (0, \infty)$ and every $C > 0$, there is k_0 such that*

$$(-1)^{n+k}\gamma_n^{(k)}(a) > 0$$

for all $k \geq k_0$, $a \in [a_0, a_1]$, and integers $0 \leq n \leq C \log(k/a)$.

Proof. The continuous function g is strictly positive on $[0, C]$, so its minimum there is positive. Both the displayed correction and the remainder in Theorem 10.20 tend uniformly to zero. The other factors in the normalization are positive. □

The corollary is restricted to its stated range. It gives no sign theorem when n grows arbitrarily quickly compared with $\log k$. For a concrete finite pair n, k at $a = 1$, the rational interval in Corollary 10.18 is a separate, effective certificate.

10.8.5 Fixed-index expansions and the classical mechanism

For a fixed n, it is also useful to retain the polynomial in $L = \log(k/a)$. Define Appell polynomials

$$A_n(L) = n![z^n]\frac{e^{Lz}}{\Gamma(1+z)}.$$

The first few are

$$\begin{aligned} A_0(L) &= 1, \\ A_1(L) &= L + \gamma, \\ A_2(L) &= (L+\gamma)^2 - \zeta(2), \\ A_3(L) &= (L+\gamma)^3 - 3\zeta(2)(L+\gamma) + 2\zeta(3). \end{aligned}$$

The standard gamma-ratio expansion [49] gives, uniformly on fixed z-disks,

$$e^{D_k(z)} = 1 + \frac{z+z^2}{2k} + \frac{3z^4+2z^3-3z^2-2z}{24k^2} + O(k^{-3}).$$

Consequently, for fixed $n \geq 1$ and a in a compact positive interval,

$$\begin{aligned} \frac{(-1)^{n+k}a^{k+1}\gamma_n^{(k)}(a)}{k!} &= A_n(L) + \frac{nA_{n-1}(L) + n(n-1)A_{n-2}(L)}{2k} \\ &\quad + O_n\left(\frac{(1+L)^{n-1}}{k^2}\right) + O_{a_0,a_1,n}(k^2\rho^k). \end{aligned} \tag{10.49}$$

Terms with a negative Appell index are omitted. To see the stated polynomial error, note that the analytic remainder after the first correction vanishes at $z = 0$; its coefficient convolution with $e^{Lz}g(z)$ therefore has degree at most $n-1$ in L. For $n = 0$ the first term is exact and only the

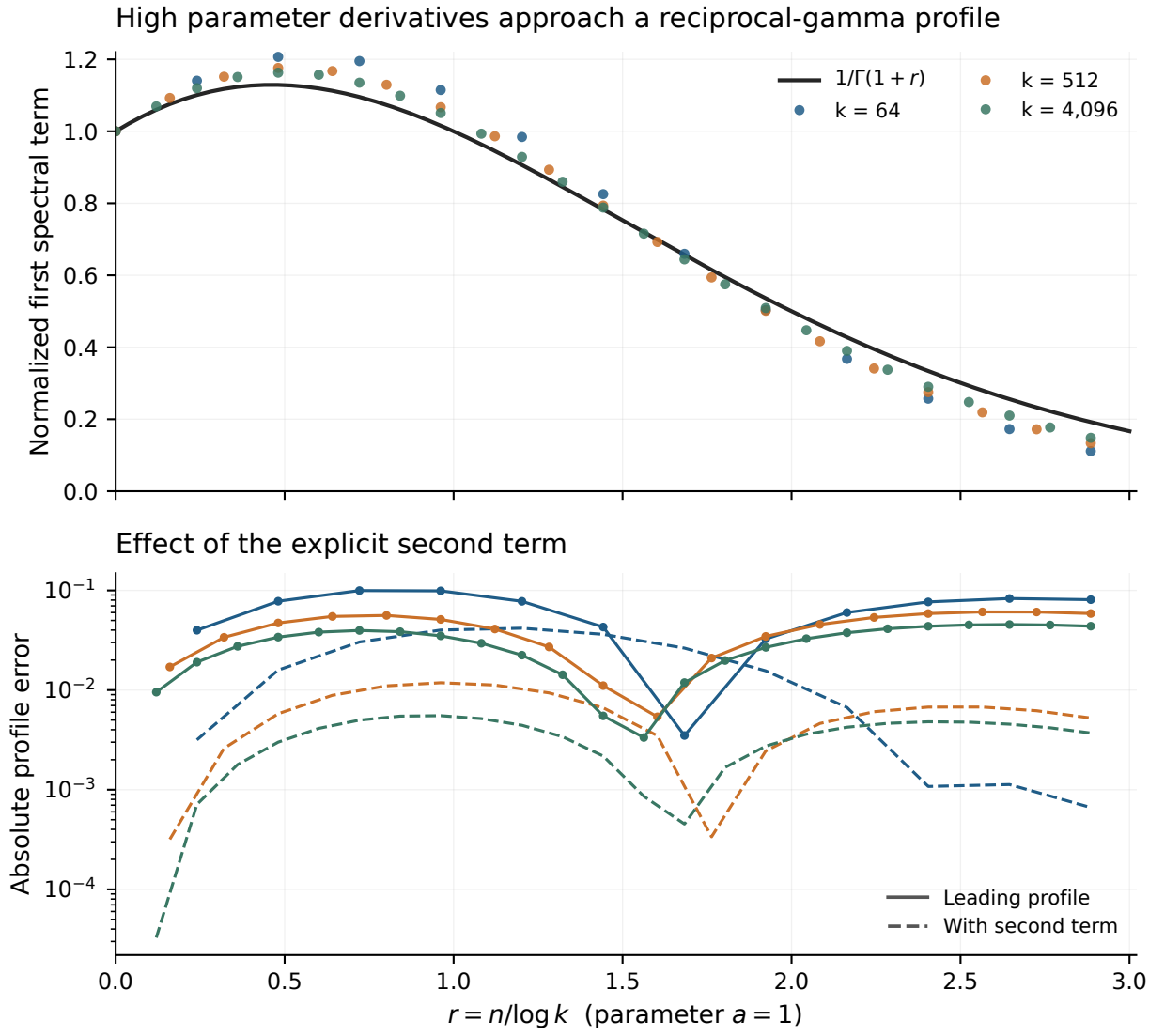


Figure 10.2: The reciprocal-gamma profile at $a = 1$. The first spectral term is computed from the elementary-symmetric coefficient recurrence; its rigorously bounded spectral error is smaller than the plotted resolution. The lower panel compares its deviation from the limiting profile with its deviation after the explicit second term is included. These plotted arithmetic evaluations are floating point; the separate rational certificates do not depend on them.

spectral error remains. Higher powers of k^{-1} follow by the same coefficient procedure. For example the k^{-2} numerator is

$$\frac{1}{24}\left(3n^{\underline{4}}A_{n-4} + 2n^{\underline{3}}A_{n-3} - 3n^{\underline{2}}A_{n-2} - 2nA_{n-1}\right).$$

At $a = 1$, (10.39) identifies the first spectral term with first-kind Stirling numbers. These are also the cycle-count coefficients of permutations. The reciprocal-gamma factor is thus a classical mechanism, closely related to mod-Poisson asymptotics [56]. The contribution to this programme is the explicit spectral transfer, the uniform second coefficient estimate, and the exact rational enclosure, not a claim that the reciprocal-gamma phenomenon itself is new.

10.9 The first classical failure of saturation: the fifth index

The first four Stieltjes indices in the preceding discussion have the maximum possible number of positive parameter-derivative zeros at every derivative order. That pattern fails at the next index. The failure is not restricted to a Lerch deformation: it occurs at the classical Hurwitz endpoint itself. We prove the exact transition

$$\#\{a > 0 : \gamma_5^{(k)}(a) = 0\} = \begin{cases} 3, & k = 1, 2, \\ 5, & k \geq 3, \end{cases} \tag{10.50}$$

and every zero in this display is simple. Thus the least derivative order after which fifth-index saturation persists is exactly three.

Two ingredients distinguish this conclusion from a numerical zero scan. A general upper bound of five controls the entire positive axis at order three. Certified signs at its stationary points then count the zeros at orders two and one by monotonicity. The large positive zeros near 148.9122 and 319.6184 are essential to this argument; a scan confined to a small neighborhood of $a = 1$ misses them.

10.9.1 Endpoint signs, parity, and monotonicity of zero counts

Retain the Laurent convention

$$\zeta(s, a) = \frac{1}{s-1} + \sum_{n\geq 0} \frac{(-1)^n}{n!}\gamma_n(a)(s-1)^n, \qquad a > 0.$$

For the classical family write

$$U_{n,k}(a) = (-1)^{n+k}\gamma_n^{(k)}(a), \qquad k \geq 1.$$

Then $U'_{n,k} = -U_{n,k+1}$. The established Appell–Laplace formula is

$$U_{n,k}(a) = \int_0^\infty \frac{t^k e^{-at}}{1 - e^{-t}}\, Q_n(\log t)\, dt, \qquad \frac{e^{xz}}{\Gamma(1+z)} = \sum_{n\geq 0} Q_n(x)\frac{z^n}{n!}. \tag{10.51}$$

The reciprocal-gamma product makes Q_n strictly real-rooted, and Laplace variation diminution therefore bounds the positive zero count by n, including multiplicities. These are the global zero theorem and its proof mechanism, reproduced in Section 10.2 for completeness. The classical parameter-derivative identity underlying (10.51) is also contained in Coffey's all-order formulas [25].

For fixed $n \geq 0$ and $k \geq 1$ the endpoint expansions are

$$U_{n,k}(a) \sim k! a^{-k-1} \log^n(1/a), \qquad a \downarrow 0, \tag{10.52}$$

$$U_{n,k}(a) \sim (-1)^n (k-1)! a^{-k} \log^n a, \qquad a \to \infty. \tag{10.53}$$

When $n = 0$, the logarithmic powers mean one. The first formula follows either by scaling the integral or by differentiating the shift term $\log^n a/a$ in $\gamma_n(a) = \log^n a/a + \gamma_n(a+1)$. For the second, set $t = u/a$ in (10.51) and use $(1-e^{-u/a})^{-1} = a/u + O(1+u/a)$ together with monicity of Q_n. The bound $(1-e^{-v})^{-1} \leq 1 + v^{-1}$ supplies an integrable majorant after expanding the fixed polynomial in $\log u - \log a$. In particular, every $U_{n,k}$ tends to zero at infinity and has fixed strict signs near both endpoints.

Proposition 10.22 (Parity and persistence before saturation)**.** *Let $N_{n,k}$ be the total multiplicity of the positive zeros of $\gamma_n^{(k)}$. Then*

$$0 \leq N_{n,k} \leq n, \qquad N_{n,k} \equiv n \pmod 2, \qquad N_{n,k+1} \geq N_{n,k}. \tag{10.54}$$

Consequently every strict increase in the total zero count is by an even integer. The same assertions hold, at each fixed $0 \leq \rho \leq 1$, for the Stieltjes–Lerch family $D_{n,k}^{\rho}$.

Proof. The upper bound is the global variation theorem. The endpoint signs in (10.52)–(10.53) show that the total multiplicity has parity n: a real analytic zero reverses sign exactly when its multiplicity is odd.

Suppose the distinct zeros of $U_{n,k}$ are $r_1 < \cdots < r_q$, with multiplicities $m_1, \ldots, m_q$ and total $N = \sum_i m_i > 0$. Its derivative has inherited multiplicity $m_i - 1$ at each r_i, and Rolle's theorem supplies a zero between each successive pair. These account for at least $(N-q) + (q-1) = N - 1$ derivative zeros counted with multiplicity. There is one more stationary point strictly after r_q: a nonzero function leaving zero at r_q, with constant sign afterwards and limit zero at infinity, must attain an interior extremum. Thus $U_{n,k+1} = -U'_{n,k}$ has at least N zeros counted with multiplicity. The case $N = 0$ is immediate.

For fixed $\rho < 1$, the analogous Mellin weight $(1-\rho e^{-t})^{-1}$ is positive and tends to $(1-\rho)^{-1}$ at zero. The large-a leading term is then $(-1)^n k!(1-\rho)^{-1} a^{-k-1} \log^n a$; the small-$a$ leading term is unchanged. The same endpoint and Rolle arguments apply, including the possible multiple zeros at $\rho = 0$. □

Proposition 10.23 (A classical lower bound from matching endpoint data)**.** *For every $k \geq 1$, $\gamma_n^{(k)}$ has at least two distinct positive zeros if $n \geq 2$ is even, and at least three distinct positive zeros if $n \geq 3$ is odd. In particular, saturation for $n = 2, 3$ at every derivative order follows analytically, without a finite sign certificate.*

Proof. For $n \geq 2$, the shift identity at $a = 1$ and its first derivative give

$$\gamma_n(1) = \gamma_n(2), \qquad \gamma_n'(1) = \gamma_n'(2), \qquad \int_1^2 \gamma_n'(a)\, da = 0.$$

Put $g = \gamma_n'$. This is a nonzero real analytic function. If its common endpoint value is positive, its integral forces a negative interior value; hence there are at least two sign-changing zeros in $(1, 2)$. The same conclusion holds with signs reversed when the common endpoint value is negative. If the common endpoint value is zero, the integral forces both signs in the interior, and there are at least three distinct zeros in $[1, 2]$, including the endpoints.

For odd n in the nonzero-endpoint case, the two interior sign-changing zeros have odd multiplicities. The odd total parity from Proposition 10.22 forces an additional odd-multiplicity zero, distinct from them. This proves the asserted lower bounds at $k = 1$. Rolle's theorem between distinct zeros, together with the stationary point after the last zero used above, preserves a lower bound on the number of distinct zeros at every subsequent derivative order. The global upper bound makes the two or three zeros simple when $n = 2$ or $n = 3$. □

Thus the count three in (10.50) is the smallest possible classical fifth-index count. Passing from order two to order three raises the positive zero count by exactly two.

10.9.2 A finite certificate that controls the whole positive axis

For the remainder of the proof abbreviate

$$U_k = U_{5,k}, \qquad V_k(a) = \frac{a^{k+1}}{k!} U_k(a).$$

The normalization V_k preserves signs, but its stationary points are not the zeros of U_{k+1}. All monotonicity arguments below are applied to U_k, while V_k is used only for stable sign enclosures.

Let $\varepsilon = 10^{-16}$. Each row of Table 10.2 specifies a rational interval $[\ell, \ell + \varepsilon]$. The stated endpoint signs of V_k are certified by the rational Euler–Maclaurin enclosure already derived above; the displayed error bound is valid for every fixed nonnegative index.

k	j	ℓ	sign at ℓ	sign at $\ell + 10^{-16}$
3	1	0.9409528529131887	+	−
3	2	1.1292935180347063	−	+
3	3	1.6357795548293846	+	−
3	4	6.1361342793896981	−	+
3	5	319.6184025568026870	+	−
2	1	1.3713298168772550	+	−
2	2	1.9113619446041762	−	+
2	3	148.9122045692035618	+	−
1	1	1.1378900542449414	+	−
1	2	1.6389372768448503	−	+
1	3	2.1249028498579805	+	−

Table 10.2: Rational isolating intervals of width 10^{-16}. These endpoint signs and the critical-value signs in the next table are finite certificates. The proof establishes uniqueness and completeness.

Denote the five third-order roots by $c_1 < \cdots < c_5$ and the three second-order roots by $d_1 < d_2 < d_3$. The interval computation also encloses the preceding-order function uniformly on every indicated root interval. Table 10.3 lists rational enclosures with outward rounding; they apply in particular at the true roots.

Theorem 10.24 (The exact fifth-index transition)**.** *The positive zero counts are exactly those in* (10.50)*. All zeros are simple. The eleven intervals in Table 10.2 isolate all zeros at derivative orders one, two, and three. For every $k \geq 3$, the five zeros at order $k + 1$ strictly interlace to the right of those at order k.*

Quantity	Certified rational enclosure
$V_2(c_1)$	$[0.006154632340,\ 0.006154632341]$
$V_2(c_2)$	$[0.041073722584,\ 0.041073722585]$
$V_2(c_3)$	$[-0.097090480335,\ -0.097090480334]$
$V_2(c_4)$	$[106.545043234228,\ 106.545043234229]$
$V_2(c_5)$	$[-135615.425337743697,\ -135615.425337743695]$
$V_1(d_1)$	$[-0.014754703693,\ -0.014754703692]$
$V_1(d_2)$	$[0.028710652775,\ 0.028710652776]$
$V_1(d_3)$	$[-466915.576951780840,\ -466915.576951780837]$

Table 10.3: Signs of the lower-order function at every critical point. The verifier encloses the function on the entire rational root bracket, so these signs do not presume a floating-point approximation to a root.

Proof. The five third-order interval signs give at least five distinct zeros of U_3. The global upper bound of five, counting multiplicity, shows that they are all its positive zeros and that each is simple. Starting with the positive left endpoint sign, the signs of U_3 on the six complementary intervals are

$$+,\ -,\ +,\ -,\ +,\ -.$$

Consequently $U_2' = -U_3$ has constant nonzero sign on each such interval. Its values at the five critical points have signs

$$\operatorname{sgn}(U_2(c_1), \dots, U_2(c_5)) = (+, +, -, +, -).$$

Therefore there is exactly one second-order zero in each of (c_2, c_3), (c_3, c_4), and (c_4, c_5), and no zero in $(0, c_1)$ or (c_1, c_2). After c_5, the function U_2 is strictly increasing from a negative value to its limit zero, so there is no zero there either. The three zeros lie away from the critical points, so each is simple. Their brackets in Table 10.2 consequently identify them as d_1, d_2, d_3.

The signs of U_2 are now $+, -, +, -$ on the four intervals cut out by the d_j, and $U_1' = -U_2$. The critical-value signs are $-, +, -$. Since U_1 is positive near zero, strict monotonicity gives exactly one zero in each of $(0, d_1)$, (d_1, d_2), and (d_2, d_3). After d_3, U_1 increases from a negative value to zero, so it has no further zero. Again all three crossings are simple.

Finally, five simple zeros at order three force at least five distinct zeros at the next order: Rolle supplies four between adjacent roots and the vanishing tail supplies one after the last root. The global upper bound gives exactly five simple zeros, with no additional zero before the first interval. Repeating proves saturation and strict right interlacing for every $k \geq 3$. □

This proof uses no claim that a dense set of sampled values accounts for every zero. Completeness at the highest checked order comes from a global analytic theorem; completeness at lower orders comes from the complete list of stationary points and their certified signs.

The fresh replay supplies twenty-two root endpoint signs and eight signs on whole critical-point intervals, using denominator 10^{80}, 120 logarithm-series terms, 24 initial Hurwitz summands and 16 Bernoulli corrections. The generic remainder proof in the preceding certificate subsection applies with $n = 5$. No floating-point sign enters acceptance. The data and executable verifier remain with the full incoming package.

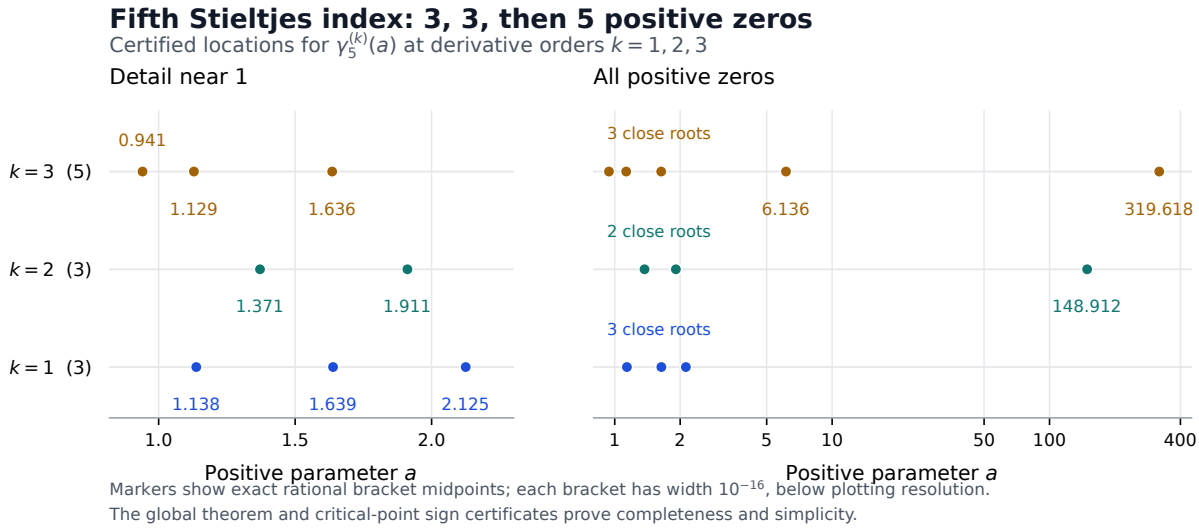


Figure 10.3: The eleven certified zero locations. The local panel separates the near-unit roots; the logarithmic panel includes the large roots. The brackets have width 10^{-16}, much smaller than a plotted marker.

10.10 Exact sixth- and seventh-index saturation thresholds

The incoming fifth-index method suggests a reproducible extension of the classical zero classification. Its next two first-derivative counts were reported only as numerical conjectures. Exact critical-value descent now settles both indices at every derivative order, not only the sampled first order. The argument retains the existing global bound and adds finite rational signs; it does not infer completeness from an approximate scan.

Theorem 10.25 (The next two classical transitions)**.** *Let $N_{n,k}$ count the positive zeros of $\gamma_n^{(k)}$, with multiplicity. For every integer $k \geq 1$,*

$$N_{6,k} = \begin{cases} 4, & 1 \leq k \leq 3, \\ 6, & k \geq 4, \end{cases} \qquad N_{7,k} = \begin{cases} 5, & 1 \leq k \leq 4, \\ 7, & k \geq 5. \end{cases}$$

Every zero is simple. Thus the least classical saturation thresholds are $K_6^{\mathrm{cl}} = 4$ and $K_7^{\mathrm{cl}} = 5$. At and above each threshold, the next-order zeros strictly interlace to the right of the current ones.

Retain $U_{n,k} = (-1)^{n+k}\gamma_n^{(k)}$ and the sign-preserving normalization $V_{n,k} = a^{k+1}U_{n,k}/k!$ of the previous section. The endpoint expansions give $U_{n,k} > 0$ near zero, $\operatorname{sgn} U_{n,k} = (-1)^n$ near infinity, and $U_{n,k} \to 0$ there. Also $U'_{n,k} = -U_{n,k+1}$.

Lemma 10.26 (Complete descent from critical-value signs)**.** *Suppose all positive zeros $c_1 < \cdots < c_r$ of $U_{n,k+1}$ are simple, and $U_{n,k}(c_j)$ has a nonzero certified sign s_j at each one. Put $s_0 = 1$. Then every positive zero of $U_{n,k}$ is simple and*

$$N_{n,k} = \#\{0 \leq j < r : s_j \neq s_{j+1}\}.$$

There is no zero on the final interval (c_r, ∞).

Proof. On each open interval cut out by the complete critical-point list, $U'_{n,k} = -U_{n,k+1}$ is nonzero and has constant sign. Hence $U_{n,k}$ is strictly monotone there. Its limits or endpoint values show one crossing exactly when the stated signs differ. It cannot vanish at a critical point, since each critical value is nonzero. On the final interval it is strictly monotone with limit zero; a nonzero strictly monotone function tending to zero cannot cross zero there. Every crossing lies where the derivative is nonzero, so it is simple. These intervals cover the whole positive axis. □

n	k	j	Lower endpoint ℓ	$V_{n,k}(\ell)$	$V_{n,k}(\ell + 10^{-14})$
6	4	1	0.95661907907155	+	−
6	4	2	1.07179934147775	−	+
6	4	3	1.51367699674930	+	−
6	4	4	3.14848332199473	−	+
6	4	5	16.34597601190147	+	−
6	4	6	1389.12290266995208	−	+
7	5	1	0.97564797285268	+	−
7	5	2	1.03011037496236	−	+
7	5	3	1.39376738084727	+	−
7	5	4	2.31266214479726	−	+
7	5	5	5.93513467415295	+	−
7	5	6	38.92782595180997	−	+
7	5	7	5297.39431175614964	+	−

Table 10.4: All saturated-order root brackets. Each displayed decimal is an exact rational number, and each bracket has width 10^{-14}. The signs are determined by outward integer intervals, rather than floating-point root approximations.

n	Known critical order	Predecessor order	Signs at all critical points	Count
6	4	3	$+,+,-,+,-,+$	4
6	3	2	$-,+,-,+$	4
6	2	1	$-,+,-,+$	4
7	5	4	$+,+,-,+,-,+,-$	5
7	4	3	$-,+,-,+,-$	5
7	3	2	$-,+,-,+,-$	5
7	2	1	$-,+,-,+,-$	5

Table 10.5: The complete critical-value descent. Each predecessor sign is certified uniformly over the entire rational root bracket of its derivative.

Proof of Theorem 10.25. The terminal brackets in Table 10.4 have opposite strict endpoint signs, so give six distinct zeros at $(n, k) = (6, 4)$ and seven at $(7, 5)$. The global bound with multiplicity, Theorem 10.7, makes these complete lists and all their zeros simple. On every terminal bracket the preceding derivative has the sign recorded in the first applicable row of Table 10.5. Thus Lemma 10.26 gives exactly four sixth-index roots at order three and five seventh-index roots at order four.

The same exact certificate file supplies disjoint brackets with opposite endpoint signs for every one of these predecessor roots. Since the preceding descent already proves the total count, these brackets form the complete critical-point lists for the next step. Their whole-interval predecessor

signs are the next rows of Table 10.5. Iterating the lemma reaches order one, with the counts displayed in the theorem and simple zeros throughout. This use of lower-order brackets has no circular completeness assumption: their completeness is established before the next descent step.

At a saturated order there are n distinct simple zeros. Rolle's theorem supplies $n-1$ next-order zeros between adjacent ones and one after the last, because the function has a nonzero sign there and tends back to zero. The same global bound makes these exactly n simple zeros, with strict right interlacing. Induction covers every larger order. The lower counts prove minimality of the two thresholds. □

The exact evidence and its analytic contract. The finite proof data contain 45 rational root brackets: 18 for the sixth index and 27 for the seventh. They require **90** endpoint enclosures and **36** whole critical-interval enclosures. The first implementation uses the generic Euler–Maclaurin bound already proved in this chapter, with fixed denominator 10^{80}, 120 logarithm-series terms, 24 initial Hurwitz terms and 16 Bernoulli corrections. Its elementary symmetric coefficient table is rebuilt through degree seven; this changes no analytic hypothesis in the index-uniform remainder formula.

A second implementation independently extracts rising-factorial coefficients and encloses all 126 signs. It uses denominator 10^{100}, 128 logarithm-series terms, 32 initial Hurwitz terms and 16 Bernoulli corrections. Its precision was increased because the delivered 55-digit default could not resolve some new narrow brackets after very small powers were multiplied by large factorial coefficients. Unresolved intervals were never accepted as signs. Both replays use exact integer and fraction arithmetic throughout acceptance. The complete endpoints, analytic error budgets, source digests and descent signs are retained in `verification/higher-zero-certificates.json` and `verification/higher-zero-independent.json`.

The resulting first thresholds are

$$(K_1^{\mathrm{cl}}, \dots, K_7^{\mathrm{cl}}) = (1, 1, 1, 1, 3, 4, 5).$$

This finite sequence suggests further threshold questions; it does not prove a formula in the index. In particular no claim that $K_n^{\mathrm{cl}} = n - 2$ for every $n \geq 5$ is made. Extending the terminating critical-point method with effective saturation estimates is a concrete next research problem.

Bibliography

[1] L. Lewin, *Polylogarithms and Associated Functions*, North-Holland, 1981; and (ed.) *Structural Properties of Polylogarithms*, AMS, 1991. (The trigamma Clausen-DFT law; cited via MathWorld, *Trigamma Function*, eq. (14).)

[2] Wolfram Research, *The Wolfram Functions Site*, `functions.wolfram.com`, PolyLog identity 10.08.03.0007.01. (The forward cyclotomic formula.)

[3] E. E. Kummer, *Beitrag zur Theorie der Function* $\Gamma(x)$, J. reine angew. Math. **35** (1847), 1–4. (The Fourier series of $\log\Gamma$.)

[4] C. J. Malmsten, *De integralibus quibusdam definitis, seriebusque infinitis*, J. reine angew. Math. **38** (1849), 1–39 (dated 1846). (The log-log integrals and the reflection identity for γ_1.)

[5] I. Vardi, *Integrals, an introduction to analytic number theory*, Amer. Math. Monthly **95** (1988), 308–315.

[6] I. V. Blagouchine, *A theorem for the closed-form evaluation of the first generalized Stieltjes constant at rational arguments and some related summations*, J. Number Theory **148** (2015), 537–592. (Vendored in the repository at `src/PolyLog/docs/pdfs/1401.3724v3/`.)

[7] O. Espinosa and V. Moll, *On some integrals involving the Hurwitz zeta function, Part 1*, Ramanujan J. **6** (2002), 159–188. (The second log-gamma moment.)

[8] E. T. Whittaker and G. N. Watson, *A Course of Modern Analysis*, 4th ed., Cambridge University Press, 1927. (Binet's formulas; the Hurwitz-series ladder.)

[9] C. Deninger, *On the analogue of the formula of Chowla and Selberg for real quadratic fields*, J. reine angew. Math. **351** (1984), 171–191. (The $\zeta''(0,a)$ layer.)

[10] N. Koblitz and A. Ogus, *Algebraicity of some products of values of the* Γ *function*, appendix (pp. 343–346) to P. Deligne, *Valeurs de fonctions L et périodes d'intégrales*, in *Automorphic Forms, Representations and L-Functions* (Corvallis, 1977), Proc. Sympos. Pure Math. **33**, part 2, Amer. Math. Soc., 1979, 313–346.

[11] C. F. Gauss, *Disquisitiones generales circa seriem infinitam*, Comment. Soc. Regiae Sci. Gottingensis Rec. **2** (1813), 3–46; Werke **3**, 265–327. (The multiplication theorem.)

[12] V. Reshetnikov, *FunctionTransformations* (Wolfram Language package), 2016, https://github.com/VladimirReshetnikov/FunctionTransformations. (The 437-rule polylog and 670-rule Γ reduction tables anchoring the corpus.)

[13] A. Selberg and S. Chowla, *On Epstein's zeta-function*, J. reine angew. Math. **227** (1967), 86–110.

[14] R. M. Damerell, *L-functions of elliptic curves with complex multiplication, I*, Acta Arith. **17** (1970), 287–301.

[15] A. Hurwitz, *Über die Entwicklungskoeffizienten der lemniskatischen Funktionen*, Math. Ann. **51** (1899), 196–226.

[16] D. A. Cox, *Primes of the Form $x^2 + ny^2$: Fermat, Class Field Theory, and Complex Multiplication*, Wiley, 1989. (Genus theory; class invariants; the $3 \mid D$ obstruction for γ_2.)

[17] E. W. Weisstein, *Digamma Function*, MathWorld — A Wolfram Web Resource. (Source of the FoxTrot identity.)

[18] V. S. Adamchik, *Polygamma functions of negative order*, J. Comput. Appl. Math. **100** (1998), 191–199.

[19] T. M. Apostol, *Formulas for higher derivatives of the Riemann zeta function*, Math. Comp. **44** (1985), 223–232.

[20] T. M. Apostol, *Introduction to Analytic Number Theory*, Springer, 1976.

[21] B. C. Berndt, *Ramanujan's Notebooks, Part I*, Springer, 1985.

[22] I. V. Blagouchine, *Rediscovery of Malmsten's integrals, their evaluation by contour integration methods and some related results*, Ramanujan J. **35** (2014), 21–110; Erratum–Addendum: **42** (2017), 777–781.

[23] K. Chakraborty, S. Kanemitsu and T. Kuzumaki, *Finite expressions for higher derivatives of the Dirichlet L-function and the Deninger R-function*, Hardy–Ramanujan J. **32** (2009), 38–53.

[24] B. K. Choudhury, *The Riemann zeta-function and its derivatives*, Proc. R. Soc. Lond. A **450** (1995), 477–499.

[25] M. W. Coffey, *Series representations for the Stieltjes constants*, Rocky Mountain J. Math. **44** (2014), 443–477 (arXiv:0905.1111).

[26] D. F. Connon, *On some new formulae involving the Stieltjes constants*, arXiv:1902.00510 (2019).

[27] O. Espinosa and V. H. Moll, *A generalized polygamma function*, Integral Transforms Spec. Funct. **15** (2004), 101–115.

[28] O. Espinosa and V. H. Moll, *On some integrals involving the Hurwitz zeta function: Part 1*, Ramanujan J. **6** (2002), 159–188.

[29] O. Espinosa and V. H. Moll, *On some integrals involving the Hurwitz zeta function: Part 2*, Ramanujan J. **6** (2002), 449–468.

[30] O. Kouba, *Generalized Stieltjes constants and integrals involving the log–log function: Kummer's theorem in action*, J. Classical Anal. **9** (2016), no. 2, 79–88 (arXiv:1702.05439).

[31] J. Miller and V. S. Adamchik, *Derivatives of the Hurwitz zeta function for rational arguments*, J. Comput. Appl. Math. **100** (1998), 201–206.

[32] D. H. Bailey and J. M. Borwein, *Computation and structure of character polylogarithms with applications to character Mordell–Tornheim–Witten sums*, Math. Comp. **85** (2016), 295–324.

[33] E. Panzer, *The parity theorem for multiple polylogarithms*, J. Number Theory **172** (2017), 93–113. https://arxiv.org/abs/1512.04482.

[34] T. Rivoal and J. Roques, *Hadamard products of algebraic functions*, Section 3 (gamma relations and Rohrlich's conjecture). https://rivoal.perso.math.cnrs.fr/articles/algcoef.pdf.

[35] D. Radchenko and D. Zagier, *Arithmetic properties of the Herglotz function*. https://arxiv.org/abs/2012.15805. Author version: https://people.mpim-bonn.mpg.de/zagier/files/preprints/HerglotzFunction.pdf.

[36] M. Abouzahra and L. Lewin, *The polylogarithm in algebraic number fields*, Journal of Number Theory **21** (1985), no. 2, 214–244. doi:10.1016/0022-314X(85)90052-6.

[37] M. Abouzahra, L. Lewin and H. Xiao, *Polylogarithms in the field of omega (a root of a given cubic): Functional equations and ladders*, Aequationes Mathematicae **33** (1987), 23–45. doi:10.1007/BF01836149.

[38] D. H. Bailey and D. J. Broadhurst, *A seventeenth-order polylogarithm ladder*, preprint, 1999. https://arxiv.org/abs/math/9906134.

[39] H. Gangl, *Functional equations and ladders for polylogarithms*, Communications in Number Theory and Physics **7** (2013), no. 3, 397–410. doi:10.4310/CNTP.2013.v7.n3.a1.

[40] H. Cohen, L. Lewin and D. Zagier, *A Sixteenth-order Polylogarithm Ladder*, Experimental Mathematics **1** (1992), no. 1. https://people.mpim-bonn.mpg.de/zagier/files/exp-math-1/fulltext.pdf. (The higher-order ladders in this paper are explicitly conjectural numerical identities.)

[41] C. Hakimoglu-Brown, *Analytic Dilogarithm Identities* (2025). https://arxiv.org/abs/2506.10206.

[42] E. Remiddi and J. A. M. Vermaseren, *Harmonic polylogarithms*, Int. J. Mod. Phys. A **15** (2000), 725–754. https://arxiv.org/abs/hep-ph/9905237.

[43] F. Brown, *Mixed Tate motives over* $\mathbb{Z}$, Annals of Mathematics **175** (2012), 949–976. https://arxiv.org/abs/1102.1312.

[44] NIST Digital Library of Mathematical Functions, §5.8, *Infinite Products*; reciprocal-gamma Weierstrass product. https://dlmf.nist.gov/5.8. Accessed 7 October 2026.

[45] D. Zagier, *The dilogarithm function*, in P. Cartier, B. Julia, P. Moussa, and P. Vanhove (eds.), *Frontiers in Number Theory, Physics, and Geometry II*, Springer, 2007, pp. 3–65. https://doi.org/10.1007/978-3-540-30308-4_1.

[46] J. Milnor, *Introduction to Algebraic K-Theory*, Annals of Mathematics Studies **72**, Princeton University Press, 1971. https://doi.org/10.1515/9781400881796.

[47] L. C. Washington, *Introduction to Cyclotomic Fields*, second edition, Graduate Texts in Mathematics **83**, Springer, 1997. https://doi.org/10.1007/978-1-4612-1934-7.

[48] R. Sharifi, *Iwasawa theory*, online lecture notes, Chapter 4 (cyclotomic units and Dirichlet L-values). https://www.math.ucla.edu/~sharifi/notes/iwasawa-ch04.html. Accessed 8 October 2026.

[49] NIST, *Digital Library of Mathematical Functions*, §§2.11, 5.11, and 25.15; in particular the shifted Stirling expansion and Dirichlet nonvanishing. https://dlmf.nist.gov/. Accessed 8 October 2026.

[50] G. Nemes, *Error bounds for the asymptotic expansion of the Hurwitz zeta function*, *Proc. R. Soc. A* **473** (2017), 20170363. https://doi.org/10.1098/rspa.2017.0363; https://arxiv.org/abs/1702.05316.

[51] B. Simon, *The classical moment problem as a self-adjoint finite difference operator*, *Adv. Math.* **137** (1998), 82–203. https://doi.org/10.1006/aima.1998.1728; https://arxiv.org/abs/math-ph/9906008.

[52] D. S. Kubert, *The universal ordinary distribution*, Bulletin de la Société Mathématique de France **107** (1979), 179–202. https://doi.org/10.24033/bsmf.1891.

[53] Y. Ouyang, *On the universal norm distribution*, Journal of the Ramanujan Mathematical Society **17** (2002), no. 4, 287–311. https://arxiv.org/abs/math/0210297.

[54] O. Espinosa and V. H. Moll, "On Some Integrals Involving the Hurwitz Zeta Function: Part 2," *Ramanujan Journal* **6** (2002), 449–468. https://www.math.tulane.edu/~vhm/papers_html/hurwitz2.pdf.

[55] J. S. Milne, *Algebraic Number Theory*, version 3.08, 19 July 2020, especially Theorem 4.3 and Proposition 7.55 with Remark 7.56. https://www.jmilne.org/math/CourseNotes/ANT.pdf.

[56] E. Kowalski and A. Nikeghbali, *Mod-Poisson convergence in probability and number theory*, author manuscript. https://people.math.ethz.ch/~kowalski/mod-poisson.pdf. The permutation-cycle example explains the classical reciprocal-gamma factor underlying the coefficient mechanism.

[57] D. Romik and R. Scherer, *Alternative summation orders for the Eisenstein series G_2 and Weierstrass $\wp$-function*, arXiv:1811.01523, version 2 (2020). https://arxiv.org/abs/1811.01523.

[58] S. Ramanujan, *On certain arithmetical functions*, Transactions of the Cambridge Philosophical Society **22** (1916), no. 9, 159–184. Transcription of the original paper: https://ramanujan.sirinudi.org/Volumes/published/ram18.html.

[59] C. Xu and W. Wang, *Dirichlet type extensions of Euler sums*, Comptes Rendus Mathématique **361** (2023), 979–1010. https://doi.org/10.5802/crmath.453. The specialization used here is equation (20), section 3.4, printed page 994.

[60] D. E. Radford, A natural ring basis for the shuffle algebra and an application to group schemes, *Journal of Algebra* **58** (1979), no. 2, 432–454. doi:10.1016/0021-8693(79)90171-6.

[61] R. Umezawa, *An explicit parity theorem for multiple polylogarithms*, arXiv:2508.02040, 2025. https://arxiv.org/abs/2508.02040.

[62] D. Zagier, Special values and functional equations of polylogarithms, in L. Lewin (ed.), *Structural Properties of Polylogarithms*, Mathematical Surveys and Monographs **37**, American Mathematical Society, Providence, RI, 1991, 377–400. Author copy: https://people.mpim-bonn.mpg.de/zagier/files/tex/LewinPolylogarithms/fulltext.pdf.

[63] H. Nakamura and D. Shiraishi, Landen's trilogarithm functional equation and ℓ-adic Galois multiple polylogarithms, in M. Morishita, H. Nakamura, and J. Ueki (eds.), *Low Dimensional Topology and Number Theory*, Springer Proceedings in Mathematics & Statistics **456**, Springer, Singapore, 2025, 237–262. doi:10.1007/978-981-97-3778-9_8. Author preprint: https://arxiv.org/abs/2210.17182.

[64] T. Amdeberhan, M. W. Coffey, O. Espinosa, C. Koutschan, D. V. Manna, and V. H. Moll, Integrals of powers of loggamma, *Proceedings of the American Mathematical Society* **139** (2011), no. 2, 535–545. doi:10.1090/S0002-9939-2010-10589-0. Author preprint: https://www.math.tulane.edu/~vhm/papers_html/lg-subm.pdf.

[65] D. H. Bailey, D. Borwein, and J. M. Borwein, On Eulerian log-gamma integrals and Tornheim–Witten zeta functions, *The Ramanujan Journal* **36** (2015), nos. 1–2, 43–68. doi:10.1007/s11139-012-9427-1. Author preprint: https://www.davidhbailey.com/dhbpapers/log-gamma.pdf. The version-specific auxiliary sign check concerns the author copy dated 27 June 2012: https://carmamaths.org/resources/jon/log-gamma.pdf, Remark 3, equations (63)–(65). The separately linked July copy has different equation numbering; the sign observation is not a journal-version audit.

[66] J. M. Borwein and A. Straub, *Relations for Nielsen polylogarithms*, Journal of Approximation Theory **193** (2015), 74–88. doi:10.1016/j.jat.2013.07.003. https://arminstraub.com/downloads/pub/nielsenrelations.pdf.

[67] J. M. Borwein, D. M. Bradley, D. J. Broadhurst, and P. Lisoněk, *Special values of multiple polylogarithms*, Transactions of the American Mathematical Society **353** (2001), 907–941. doi:10.1090/S0002-9947-00-02616-7. arXiv:math/9910045.

[68] J. Vollinga and S. Weinzierl, *Numerical evaluation of multiple polylogarithms*, Computer Physics Communications **167** (2005), 177–194. doi:10.1016/j.cpc.2004.12.009. arXiv:hep-ph/0410259.

[69] J. Zhao, *Standard relations of multiple polylogarithm values at roots of unity*, arXiv:0707.1459. https://arxiv.org/abs/0707.1459.

[70] W. Hintze, *Closed form of the sum* $s_4 = \sum_{n=1}^{\infty}(-1)^n H_n/(2n+1)^4$, Mathematics Stack Exchange question 4160026 and answer by pisco, 3 June 2021. https://math.stackexchange.com/questions/4160026/.

[71] M. H. Zhao, *MZV evaluation of hypergeometric series*, arXiv:2007.02508, version 10, 27 September 2020. https://arxiv.org/abs/2007.02508.

[72] J. Zhao, *Multiple polylogarithm values at roots of unity*, C. R. Math. Acad. Sci. Paris **346** (2008), 1029–1032; expanded version arXiv:0810.1064. https://arxiv.org/abs/0810.1064.

[73] G. Racinet, *Doubles mélanges des polylogarithmes multiples aux racines de l'unité*, Publications Mathématiques de l'IHÉS **95** (2002), 185–231. https://arxiv.org/abs/math/0202142.

[74] J. Aarts, R. Fokkink, and G. Kruijtzer, Morphic numbers, *Nieuw Archief voor Wiskunde*, series 5, **2**, no. 1 (2001), 56–58. https://www.nieuwarchief.nl/serie5/pdf/naw5-2001-02-1-056.pdf.

[75] W. Ljunggren, On the irreducibility of certain trinomials and quadrinomials, *Mathematica Scandinavica* **8** (1960), 65–70. https://doi.org/10.7146/math.scand.a-10593.

[76] H. Tverberg, On the irreducibility of the trinomials $x^n \pm x^m \pm 1$, *Mathematica Scandinavica* **8** (1960), 121–126. https://doi.org/10.7146/math.scand.a-10599. Original scanned text: https://journals.msp.org/mscand/article/download/2744/2743/2775.

[77] D. W. Boyd, G. Martin, and M. Thom, Squarefree values of trinomial discriminants, *LMS Journal of Computation and Mathematics* **18** (2015), 148–169, especially Lemma 4.1(a),(b). https://doi.org/10.1112/S1461157014000436.

[78] NIST Digital Library of Mathematical Functions, §2.3(iii), Laplace's method. https://dlmf.nist.gov/2.3#iii.

[79] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009. The smooth implicit-function schema and singularity transfer are the classical background for the late-coefficient analysis.

[80] NIST Digital Library of Mathematical Functions, §25.15, *Dirichlet L-functions*, including (25.15.4), (25.15.6), and the nonvanishing statement (25.15.9). Accessed 9 October 2026. https://dlmf.nist.gov/25.15.

[81] J. Zhao, *Multiple polylogarithm values at roots of unity*, C. R. Acad. Sci. Paris, Ser. I **346** (2008), 1029–1032. doi:10.1016/j.crma.2008.09.011. https://www.numdam.org/item/10.1016/j.crma.2008.09.011.pdf. Its standard-relation framework includes more relations and depths than the restricted presentation computed in this article.

[82] National Institute of Standards and Technology, *NIST Digital Library of Mathematical Functions*, Section 13.4, equation 13.4.4, and Section 13.8(iii), equations 13.8.11–16. https://dlmf.nist.gov/13.4.E4; https://dlmf.nist.gov/13.8.iii. Consulted 9–10 October 2026.

[83] N. M. Temme, *Uniform asymptotic expansions of a class of integrals in terms of modified Bessel functions, with application to confluent hypergeometric functions*, SIAM J. Math. Anal. **21** (1990), no. 1, 241–261. doi:10.1137/0521013.

[84] A. Barquero-Sanchez, L. Cadwallader, O. Cannon, T. Genao, and R. Masri, *Faltings heights of CM elliptic curves and special Gamma values*, Research in Number Theory **3** (2017), article 13. doi:10.1007/s40993-017-0077-7. Sections 3–5 give the analytic normalization of the Kronecker and Chowla–Selberg formulas, including nonmaximal quadratic orders.

[85] H. Cohen, *On the Argument of the Lerch, Chowla–Selberg Formula and CM Values of* $\eta(\tau)$, arXiv:2606.02904v1 (2026). https://arxiv.org/html/2606.02904v1. Theorem 1.3 states the period-product formula with conductor correction; Theorem 2.1 addresses the choice of representatives and eta phases.

[86] A. V. Sutherland, *Computing Hilbert class polynomials with the Chinese Remainder Theorem*, Mathematics of Computation **80** (2011), 501–538. Author's implementation and references.

[87] N. Yui and D. Zagier, *On the singular values of Weber modular functions*, Mathematics of Computation **66** (1997), 1645–1662. Author-hosted full text.

[88] W. Stein, *Modular Forms: A Computational Approach*, Graduate Studies in Mathematics, vol. 79, American Mathematical Society, 2007. https://wstein.org/books/modform/modform/level_one.html.

[89] R. Masri, *Kronecker's solution of Pell's equation for CM fields*, Annales de l'Institut Fourier **63** (2013), 2287–2306. https://aif.centre-mersenne.org/item/10.5802/aif.2830.pdf.

[90] W. Kohnen, *Transcendence of zeros of Eisenstein series and other modular functions*, Commentarii Mathematici Universitatis Sancti Pauli **52** (2003), no. 1, 55–57. https://doi.org/10.14992/00008713.

[91] L. Chiriac and A. Jorza, *The variation of zeros of the Miller basis*, arXiv:2605.09731v1 (2026), introduction and Section 7. https://arxiv.org/abs/2605.09731.

[92] D. S. Kubert, *The $\mathbb{Z}/2\mathbb{Z}$ cohomology of the universal ordinary distribution*, Bulletin de la Société Mathématique de France **107** (1979), 203–224. doi:10.24033/bsmf.1892.

[93] NIST Digital Library of Mathematical Functions, *Section 25.11: Hurwitz Zeta Function*, especially the defining series, meromorphic continuation and multiplication formulas. https://dlmf.nist.gov/25.11.

[94] ProveIt research contribution, *Gaussian Polylogarithms Beyond Parity: Exact relation modules, an octahedral proof of the mixed harmonic sum, and global angular-zero geometry*, October 2026. Preserved source: `reports/angular-zero-continuation/article.tex`.

[95] NIST Digital Library of Mathematical Functions, §5.9, *Integral Representations*, especially formula 5.9.13. https://dlmf.nist.gov/5.9.

[96] NIST Digital Library of Mathematical Functions, §25.12, *Polylogarithms*. https://dlmf.nist.gov/25.12.

[97] NIST Digital Library of Mathematical Functions, §4.36, *Infinite Products and Partial Fractions*, especially formula 4.36.3. https://dlmf.nist.gov/4.36.

[98] J.-G. Liu and R. L. Pego, *On generating functions of Hausdorff moment sequences*, Transactions of the American Mathematical Society **368** (2016), 8499–8518. The author's arXiv version 4 includes a corrigendum. https://arxiv.org/abs/1401.8052v4.

[99] ProveIt Contributors, *Sharp Gaussian Bounds, Cyclic Distribution Torsion, and Extremal Lerch Zeros*, research continuation, 10 October 2026. Prepared against repository revision `a0a90ef318`. Original source and certificates are preserved under `reports/formal-reductions-continuation/continuations/extremal-cyclic-descent/`.

[100] V. Reshetnikov, *A conjectured identity for tetralogarithms* Li_4, Math.StackExchange question 1373123 (2015). https://math.stackexchange.com/questions/1373123/.

[101] V. Reshetnikov, *Several conjectured identities for polylogarithms*, MathOverflow question 261408 (2017). https://mathoverflow.net/questions/261408/.

[102] ProveIt Contributors, *Polylogarithm Certificates and Sharp Euler Bounds*, research continuation, 10 October 2026, prepared against `4c173c06cc`. Preserved under `reports/uniform-transition-continuation/continuations/tetralogarithm-euler-polynomial-zeros/`.

[103] ProveIt Contributors, *Global Radial Monotonicity for Leading-Index-One Double Polylogarithms*, research continuation, 10 October 2026, prepared against `1539f353af`. Preserved under `reports/angular-zero-continuation/continuations/leading-one-global-radius/`.

[104] ProveIt Contributors, *Sharp Universal Euler Bounds*, research continuation, 10 October 2026. Preserved under `reports/real-order-threshold/continuations/sharp-universal-euler/`.

[105] ProveIt Contributors, *Polylogarithm Research: Euler Bounds, Golden Ladders and Cayley Reductions*, research continuation, 10 October 2026. Preserved under `reports/fractional-cayley-scaling/continuations/golden-cayley-double-turning/`.

[106] ProveIt Contributors, *Signed-Kernel Oscillation at Every Depth: Minimal Pick Compensation, Zero Geometry, and Certified Polylogarithmic Identities*, research continuation, 10 October 2026, prepared against `3447bc59b78d138a7f0d536b66ac6e5cc5d5e49f`. Original source and certificates are preserved under `reports/angular-zero-continuation/continuations/all-depth-signed-oscillation/`.

[107] Z.-H. Sun, *On the properties of invariant functions*, arXiv:2209.14625v1 (2022), Remark 3.2, printed p. 15. https://arxiv.org/abs/2209.14625.

[108] ProveIt Contributors, *A Convolution Algebra of Stieltjes Functions*, research continuation, 10 October 2026, prepared against `0efdbaa694`. Preserved under `reports/stieltjes-correlation-calculus/continuations/convolution-algebra/`.

[109] ProveIt Contributors, *Shifted Hurwitz-Zeta Correlations: Exact Stieltjes Closure, Polygamma Contact Terms, and Log-Gamma Antiderivatives*, research continuation, 10 October 2026, observed source `0e5ab2f60f`. Preserved under `reports/stieltjes-correlation-calculus/continuations/shifted-hurwitz-jets/`.

[110] ProveIt Contributors, *Bilinear Closure of Generalized Stieltjes Correlations*, research continuation, 10 October 2026. Individual source observations and scope are retained in the original provenance note. Preserved under `reports/stieltjes-correlation-calculus/continuations/bilinear-correlation-closure/`.

[111] ProveIt Contributors, *A Convolution Calculus for Stieltjes Functions*, research continuation, 10 October 2026, prepared against `570b0567f3`. Preserved under `reports/stieltjes-correlation-calculus/continuations/convolution-calculus/`.

[112] NIST Digital Library of Mathematical Functions, Eq. 15.4.20, Gauss summation of the hypergeometric function at one. https://dlmf.nist.gov/15.4.E20.

[113] J. Choi, Summation formulas involving binomial coefficients, harmonic numbers, and generalized harmonic numbers, *Abstract and Applied Analysis* **2014** (2014), Article 501906, printed p. 6, Conjectures 4 and 6 and Eqs. (40)–(46). https://doi.org/10.1155/2014/501906.

[114] ProveIt Contributors, *Exact Stieltjes and Harmonic Identities*, research continuation, 10 October 2026, prepared against `fc4d3bf80534ad7c901d3b8c9e71baf2df0064ed`. Preserved under `reports/stieltjes-harmonic-resonance/continuations/stieltjes-harmonic-identities/`.

[115] NIST Digital Library of Mathematical Functions, §25.14, Lerch's transcendent. https://dlmf.nist.gov/25.14.

[116] NIST Digital Library of Mathematical Functions, §25.15, Dirichlet L-functions. https://dlmf.nist.gov/25.15.

[117] S. Hu and M.-S. Kim, On the Stieltjes constants and gamma functions with respect to alternating Hurwitz zeta functions, *Journal of Mathematical Analysis and Applications* **509**(1) (2022), 125930. https://arxiv.org/abs/2106.14674.

[118] H. Zhu, S. Hu and M.-S. Kim, On the properties of alternating invariant functions, arXiv:2505.10985v3 (2025), Theorem 1.13. https://arxiv.org/abs/2505.10985.

[119] ProveIt Contributors, *Twisted Stieltjes Convolutions and Harmonic Laurent Identities*, research continuation, 10 October 2026, prepared against `e0d9463bdee9685dfb1dddb819059cc738540c57`. Preserved under `reports/stieltjes-correlation-calculus/continuations/twisted-harmonic-laurent/`.

[120] ProveIt external research continuation, *Exact Identities: Collision Calculus, Shifted Harmonic Jets and Weighted Arctangent Families*, 10 October 2026. Original package retained at `reports/stieltjes-correlation-calculus/continuations/collision-and-weighted-identities/`. The weighted real evaluations are proved in Chapter 3; the remaining claims retain separate intake and audit status. Global priority is not established.

[121] ProveIt research continuations, *Periodic Stieltjes Collisions: Exact Contact Terms, Finite-Part Fubini Laws, and Gamma–Polylogarithm Primitives* and *Periodic Stieltjes Contact Calculus: All-Index Collision Corrections, Harmonic Derivative Anomalies, and Exact Zeta-Jet Integrals*, 10 October 2026. Independent routes retained in the correlation spine as `continuations/periodic-collision-contacts/` and `continuations/periodic-stieltjes-contact/`; their immediate coincident source is preserved as `continuations/coincident-stieltjes/`. The fixed-coordinate periodic comparison is reconciled in Chapter 9; the derivative anomaly was already established there.

[122] NIST Digital Library of Mathematical Functions, §5.7, *Series Expansions of the Gamma Function*, especially formula 5.7.3. https://dlmf.nist.gov/5.7.E3. Inspected 10 October 2026.

[123] ProveIt research continuation, *Mixed Spectral Identities*, 10 October 2026, prepared against `dcd95baeb7c0e914b138bddef6829c5b1d52b726`. Preserved at `reports/stieltjes-harmonic-resonance/continuations/mixed-spectral-identities/`. The mixed Gamma identity is derived from the classical Rogers–Dougall sum in Chapter 8. Its further resonance, harmonic and collision claims retain separate audit scopes. No global priority is asserted.

[124] NIST Digital Library of Mathematical Functions, §16.4, *Rogers–Dougall Very Well-Poised Sum*, formula 16.4.9. https://dlmf.nist.gov/16.4.E9. Inspected 10 October 2026.

[125] ProveIt research contribution, *A certified obstruction in the weight-seven reduction search*, in the extremal cyclic-descent continuation, `article/s6_obstruction.tex`. Pinned repository commit `5a790187c8e186e41e2b990b4941cb7a1a3c7b6b`. Source article.

[126] ProveIt research contribution, *A Cayley normal form that preserves series depth*, in the polylogarithm–Stieltjes critical-transition continuation, `article/05_cayley_depth.tex`. Pinned repository commit `5a790187c8e186e41e2b990b4941cb7a1a3c7b6b`. Source article.

[127] D. E. Radford, A natural ring basis for the shuffle algebra and an application to group schemes, *Journal of Algebra* **58** (1979), no. 2, 432–454. doi:10.1016/0021-8693(79)90171-6.

[128] F. Patras, La décomposition en poids des algèbres de Hopf, *Annales de l'Institut Fourier* **43** (1993), no. 4, 1067–1087. doi:10.5802/aif.1365.

[129] ProveIt research continuation, *Exact Resonant Identities Beyond the Diagonal*, 11 October 2026. Original package preserved under `reports/stieltjes-harmonic-resonance/continuations/exact-resonant-identities/`. The complete-Cayley formal obstruction is audited in Chapter 12; the numerical S_6 conjecture remains open.

[130] ProveIt research continuation, *Harmonic Parity and Resolvent Identities*, 11 October 2026. Original package preserved under `reports/stieltjes-harmonic-resonance/continuations/harmonic-parity-resolvents/`. The centered trigamma-square moments and their polylogarithmic generator are audited in Chapter 8; the broader classification and resolvent claims retain separate scopes.

[131] ProveIt research continuations, *Higher Digamma Powers and Ordered Gamma Sums*, *Canonical Digamma Polynomials and the Fifth Harmonic Bridge*, and *Mixed Resolvent Products, Higher Digamma Powers, and Stieltjes Collisions*, 11 October 2026. Original packages are preserved as `reports/stieltjes-harmonic-resonance/continuations/higher-digamma-powers/`, `reports/stieltjes-harmonic-resonance/continuations/canonical-digamma-fifth/`, and `reports/stieltjes-correlation-calculus/continuations/mixed-resolvents-harmonic-powers/`. Their all-power and fifth-moment cores are reconciled in Chapter 8; other Gamma, resolvent, collision and ordered branches retain separate audit scopes.

[132] NIST Digital Library of Mathematical Functions, Sections 5.5 (digamma recurrence and reflection), 5.7 (Taylor and reciprocal-Gamma series), and 5.11 (sectorial asymptotics). In particular formulas 5.5.2, 5.5.4, 5.7.4 and 5.11.2. https://dlmf.nist.gov/5. Inspected 11 October 2026 UTC.

[133] K. Ihara, M. Kaneko and D. Zagier, Derivation and double shuffle relations for multiple zeta values, *Compositio Mathematica* **142** (2006), 307–338. https://doi.org/10.1112/S0010437X0500182X.

[134] P. T. Young, Series of height one multiple zeta functions, *Integers* **24** (2024), A43, published 20 May 2024. https://math.colgate.edu/~integers/y43/y43.pdf. https://doi.org/10.5281/zenodo.11221606. The coefficient-description check in Appendix A concerns the inspected publisher PDF, printed page 12; its ordinary Stieltjes formula is retained.

[135] M. Can, L. Kargın, M. Cenkci and A. Dil, *On certain harmonic zeta functions*, arXiv:2403.07123v1, 11 March 2024, printed page 1. https://arxiv.org/pdf/2403.07123v1. The initial-index correction in Appendix A is scoped to that version's introductory Stieltjes-limit display. Inspected 11 October 2026 UTC.

[136] ProveIt research continuations, *Bernoulli Determinants, Cubic Tornheim Jets, and Canonical Digamma Integrals*, *Differential Polygamma and Lerch Identities*, and *Exact Polylogarithm Parity, Harmonic-Tail Reconstruction, and Higher Polygamma Moments*, 11 October 2026. Original packages preserved under `reports/stieltjes-harmonic-resonance/continuations/bernoulli-tornheim-digamma/`, `reports/stieltjes-harmonic-resonance/continuations/differential-polygamma-lerch-jets/`, and `reports/stieltjes-harmonic-resonance/continuations/polygamma-tail-identities/`. Their common differential-polynomial analytic core is reconciled in Chapter 8; their remaining coordinate, rank and Lerch claims retain separate audit scopes.